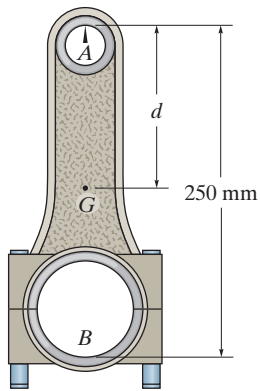
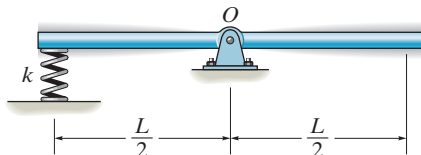


**\*22–12.** The connecting rod is supported by a knife edge at  $A$  and the period of vibration is measured as  $\tau_A = 3.38$  s. It is then removed and rotated  $180^\circ$  so that it is supported by the knife edge at  $B$ . In this case the period of vibration is measured as  $\tau_B = 3.96$  s. Determine the location  $d$  of the center of gravity  $G$ , and compute the radius of gyration  $k_G$ .



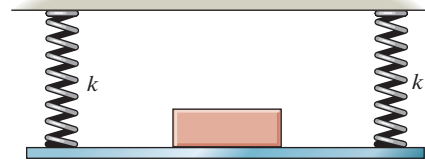
**Prob. 22–12**

**22–13.** Determine the natural period of vibration of the uniform bar of mass  $m$  when it is displaced downward slightly and released.



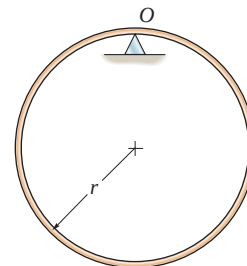
**Prob. 22–13**

**22–14.** A platform, having an unknown mass, is supported by *four* springs, each having the same stiffness  $k$ . When nothing is on the platform, the period of vertical vibration is measured as 2.35 s; whereas if a 3-kg block is supported on the platform, the period of vertical vibration is 5.23 s. Determine the mass of a block placed on the (empty) platform which causes the platform to vibrate vertically with a period of 5.62 s. What is the stiffness  $k$  of each of the springs?



**Prob. 22–14**

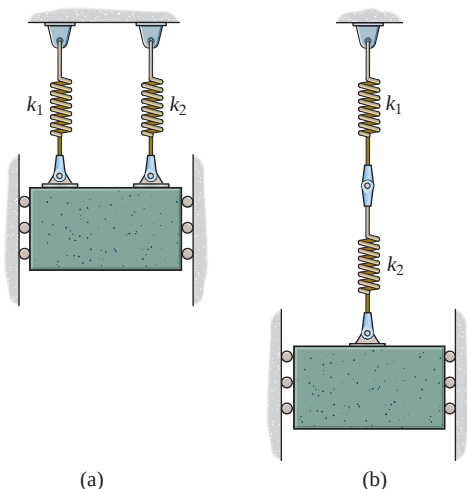
**22–15.** The thin hoop of mass  $m$  is supported by a knife-edge. Determine the natural period of vibration for small amplitudes of swing.



**Prob. 22–15**

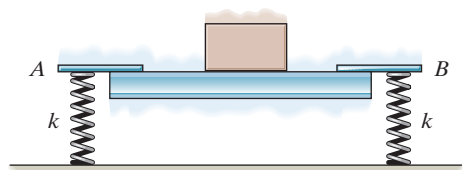
**\*22-16.** A block of mass  $m$  is suspended from two springs having a stiffness of  $k_1$  and  $k_2$ , arranged a) parallel to each other, and b) as a series. Determine the equivalent stiffness of a single spring with the same oscillation characteristics and the period of oscillation for each case.

**22-17.** The 15-kg block is suspended from two springs having different stiffnesses and arranged a) parallel to each other, and b) as a series. If the natural periods of oscillation of the parallel system and series system are observed to be 0.5 s and 1.5 s, respectively, determine the spring stiffnesses  $k_1$  and  $k_2$ .



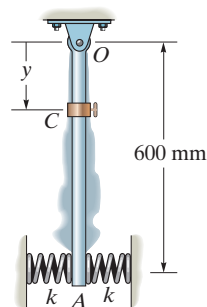
Probs. 22-16/17

**22-18.** The uniform beam is supported at its ends by two springs  $A$  and  $B$ , each having the same stiffness  $k$ . When nothing is supported on the beam, it has a period of vertical vibration of 0.83 s. If a 50-kg mass is placed at its center, the period of vertical vibration is 1.52 s. Compute the stiffness of each spring and the mass of the beam.



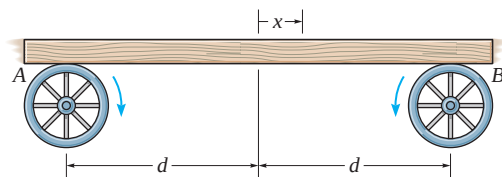
Prob. 22-18

**22-19.** The slender rod has a mass of 0.2 kg and is supported at  $O$  by a pin and at its end  $A$  by two springs, each having a stiffness  $k = 4$  N/m. The period of vibration of the rod can be set by fixing the 0.5-kg collar  $C$  to the rod at an appropriate location along its length. If the springs are originally unstretched when the rod is vertical, determine the position  $y$  of the collar so that the natural period of vibration becomes  $\tau = 1$  s. Neglect the size of the collar.



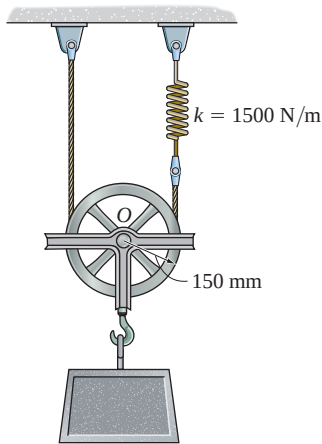
Prob. 22-19

**\*22-20.** A uniform board is supported on two wheels which rotate in opposite directions at a constant angular speed. If the coefficient of kinetic friction between the wheels and board is  $\mu$ , determine the frequency of vibration of the board if it is displaced slightly, a distance  $x$  from the midpoint between the wheels, and released.



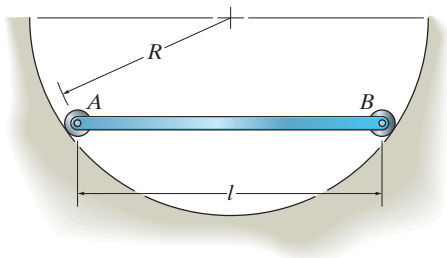
Prob. 22-20

**22-21.** The 50-kg block is suspended from the 10-kg pulley that has a radius of gyration about its center of mass of 125 mm. If the block is given a small vertical displacement and then released, determine the natural frequency of oscillation.



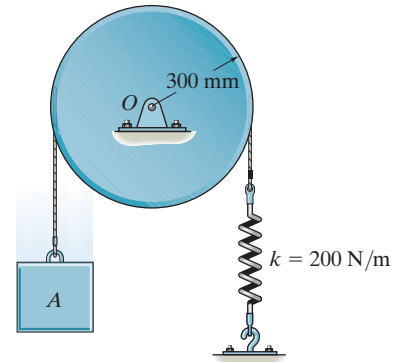
**Prob. 22-21**

**22-22.** The bar has a length  $l$  and mass  $m$ . It is supported at its ends by rollers of negligible mass. If it is given a small displacement and released, determine the natural frequency of vibration.



**Prob. 22-22**

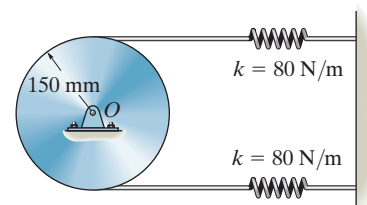
**22-23.** The 20-kg disk, is pinned at its mass center  $O$  and supports the 4-kg block  $A$ . If the belt which passes over the disk is not allowed to slip at its contacting surface, determine the natural period of vibration of the system.



**Prob. 22-23**

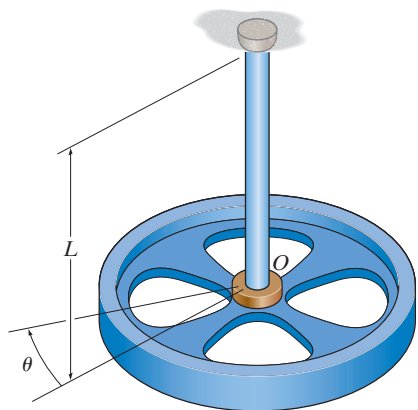
**\*22-24.** The 10-kg disk is pin connected at its mass center. Determine the natural period of vibration of the disk if the springs have sufficient tension in them to prevent the cord from slipping on the disk as it oscillates. *Hint:* Assume that the initial stretch in each spring is  $\delta_O$ .

**22-25.** If the disk in Prob. 22-24 has a mass of 10 kg, determine the natural frequency of vibration. *Hint:* Assume that the initial stretch in each spring is  $\delta_O$ .



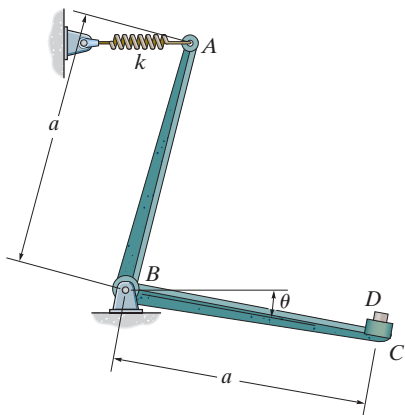
**Probs. 22-24/25**

**22–26.** A flywheel of mass  $m$ , which has a radius of gyration about its center of mass of  $k_O$ , is suspended from a circular shaft that has a torsional resistance of  $M = C\theta$ . If the flywheel is given a small angular displacement of  $\theta$  and released, determine the natural period of oscillation.



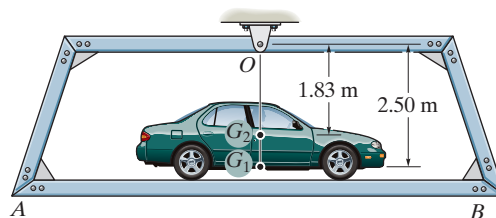
**Prob. 22–26**

**22–27.** If a block  $D$  of negligible size and of mass  $m$  is attached at  $C$ , and the bell crank of mass  $M$  is given a small angular displacement of  $\theta$ , the natural period of oscillation is  $\tau_1$ . When  $D$  is removed, the natural period of oscillation is  $\tau_2$ . Determine the bell crank's radius of gyration about its center of mass, pin  $B$ , and the spring's stiffness  $k$ . The spring is unstretched at  $\theta = 0^\circ$ , and the motion occurs in the horizontal plane.



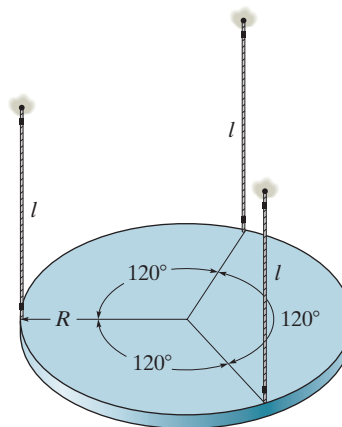
**Prob. 22–27**

**\*22–28.** The platform  $AB$  when empty has a mass of 400 kg, center of mass at  $G_1$ , and natural period of oscillation  $\tau_1 = 2.38$  s. If a car, having a mass of 1.2 Mg and center of mass at  $G_2$ , is placed on the platform, the natural period of oscillation becomes  $\tau_2 = 3.16$  s. Determine the moment of inertia of the car about an axis passing through  $G_2$ .



**Prob. 22–28**

**22–29.** The plate of mass  $m$  is supported by three symmetrically placed cords of length  $l$  as shown. If the plate is given a slight rotation about a vertical axis through its center and released, determine the natural period of oscillation.



**Prob. 22–29**

## \*22.2 Energy Methods

The simple harmonic motion of a body, discussed in the previous section, is due only to gravitational and elastic restoring forces acting on the body. Since these forces are *conservative*, it is also possible to use the conservation of energy equation to obtain the body's natural frequency or period of vibration. To show how to do this, consider again the block and spring model in Fig. 22–8. When the block is displaced  $x$  from the equilibrium position, the kinetic energy is  $T = \frac{1}{2}mv^2 = \frac{1}{2}m\dot{x}^2$  and the potential energy is  $V = \frac{1}{2}kx^2$ . Since energy is conserved, it is necessary that

$$T + V = \text{constant}$$

$$\frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 = \text{constant} \quad (22-17)$$

The differential equation describing the *accelerated motion* of the block can be obtained by *differentiating* this equation with respect to time; i.e.,

$$m\dot{x}\ddot{x} + kx\dot{x} = 0$$

$$\dot{x}(m\ddot{x} + kx) = 0$$

Since the velocity  $\dot{x}$  is not *always* zero in a vibrating system,

$$\ddot{x} + \omega_n^2 x = 0 \quad \omega_n = \sqrt{k/m}$$

which is the same as Eq. 22–1.

If the conservation of energy equation is written for a *system of connected bodies*, the natural frequency or the equation of motion can also be determined by time differentiation. It is *not necessary* to dismember the system to account for the internal forces because they do no work.

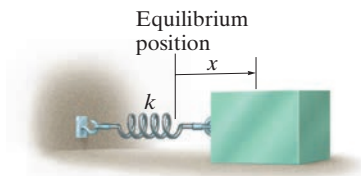


Fig. 22–8



The suspension of a railroad car consists of a set of springs which are mounted between the frame of the car and the wheel truck. This will give the car a natural frequency of vibration which can be determined.

## Procedure for Analysis

The natural frequency  $\omega_n$  of a body or system of connected bodies can be determined by applying the conservation of energy equation using the following procedure.

### Energy Equation.

- Draw the body when it is displaced by a *small amount* from its equilibrium position and define the location of the body from its equilibrium position by an appropriate position coordinate  $q$ .
- Formulate the conservation of energy for the body,  $T + V = \text{constant}$ , in terms of the position coordinate.
- In general, the kinetic energy must account for both the body's translational and rotational motion,  $T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2$ , Eq. 18-2.
- The potential energy is the sum of the gravitational and elastic potential energies of the body,  $V = V_g + V_e$ , Eq. 18-17. In particular,  $V_g$  should be measured from a datum for which  $q = 0$  (equilibrium position).

### Time Derivative.

- Take the time derivative of the energy equation using the chain rule of calculus and factor out the common terms. The resulting differential equation represents the equation of motion for the system. The natural frequency of  $\omega_n$  is obtained after rearranging the terms in the "standard form,"  $\ddot{q} + \omega_n^2 q = 0$ .

**EXAMPLE 22.5**

The thin hoop shown in Fig. 22–9a is supported by the peg at  $O$ . Determine the natural period of oscillation for small amplitudes of swing. The hoop has a mass  $m$ .

**SOLUTION**

**Energy Equation.** A diagram of the hoop when it is displaced a small amount ( $q =$ )  $\theta$  from the equilibrium position is shown in Fig. 22–9b. Using the table on the inside back cover and the parallel-axis theorem to determine  $I_O$ , the kinetic energy is

$$T = \frac{1}{2} I_O \omega_n^2 = \frac{1}{2} [mr^2 + mr^2] \dot{\theta}^2 = mr^2 \dot{\theta}^2$$

If a horizontal datum is placed through point  $O$ , then in the displaced position, the potential energy is

$$V = -mgr(\cos \theta)$$

The total energy in the system is

$$T + V = mr^2 \dot{\theta}^2 - mgr \cos \theta$$

**Time Derivative.**

$$mr^2(2\dot{\theta})\ddot{\theta} + mgr(\sin \theta)\dot{\theta} = 0$$

$$mr\dot{\theta}(2r\ddot{\theta} + g \sin \theta) = 0$$

Since  $\dot{\theta}$  is not always equal to zero, from the terms in parentheses,

$$\ddot{\theta} + \frac{g}{2r} \sin \theta = 0$$

For small angle  $\theta$ ,  $\sin \theta \approx \theta$ .

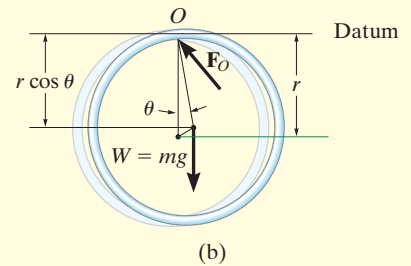
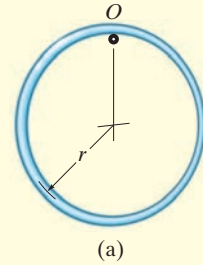
$$\ddot{\theta} + \frac{g}{2r} \theta = 0$$

$$\omega_n = \sqrt{\frac{g}{2r}}$$

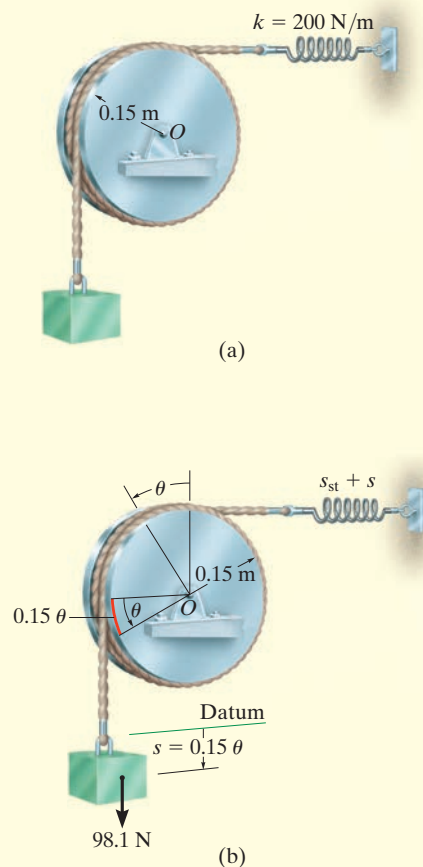
so that

$$\tau = \frac{2\pi}{\omega_n} = 2\pi \sqrt{\frac{2r}{g}}$$

*Ans.*



**Fig. 22–9**

**EXAMPLE 22.6****Fig. 22-10**

A 10-kg block is suspended from a cord wrapped around a 5-kg disk, as shown in Fig. 22-10a. If the spring has a stiffness  $k = 200 \text{ N/m}$ , determine the natural period of vibration for the system.

**SOLUTION**

**Energy Equation.** A diagram of the block and disk when they are displaced by respective amounts  $s$  and  $\theta$  from the equilibrium position is shown in Fig. 22-10b. Since  $s = (0.15 \text{ m})\theta$ , then  $v_b \approx \dot{s} = (0.15 \text{ m})\dot{\theta}$ . Thus, the kinetic energy of the system is

$$\begin{aligned} T &= \frac{1}{2} m_b v_b^2 + \frac{1}{2} I_O \omega_d^2 \\ &= \frac{1}{2} (10 \text{ kg}) [(0.15 \text{ m})\dot{\theta}]^2 + \frac{1}{2} \left[ \frac{1}{2} (5 \text{ kg}) (0.15 \text{ m})^2 \right] (\dot{\theta})^2 \\ &= 0.1406(\dot{\theta})^2 \end{aligned}$$

Establishing the datum at the equilibrium position of the block and realizing that the spring stretches  $s_{st}$  for equilibrium, the potential energy is

$$\begin{aligned} V &= \frac{1}{2} k(s_{st} + s)^2 - Ws \\ &= \frac{1}{2} (200 \text{ N/m}) [s_{st} + (0.15 \text{ m})\theta]^2 - 98.1 \text{ N} [(0.15 \text{ m})\theta] \end{aligned}$$

The total energy for the system is therefore,

$$T + V = 0.1406(\dot{\theta})^2 + 100(s_{st} + 0.15\theta)^2 - 14.715\theta$$

**Time Derivative.**

$$0.28125(\dot{\theta})\ddot{\theta} + 200(s_{st} + 0.15\theta)0.15\dot{\theta} - 14.72\dot{\theta} = 0$$

Since  $s_{st} = 98.1/200 = 0.4905 \text{ m}$ , the above equation reduces to the “standard form”

$$\ddot{\theta} + 16\theta = 0$$

so that

$$\omega_n = \sqrt{16} = 4 \text{ rad/s}$$

Thus,

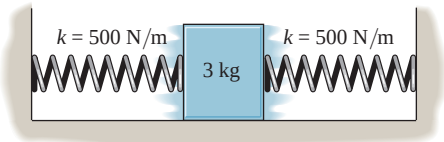
$$\tau = \frac{2\pi}{\omega_n} = \frac{2\pi}{4} = 1.57 \text{ s}$$

*Ans.*



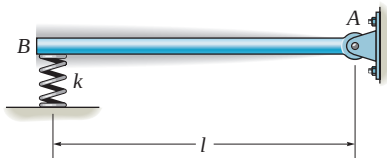
## PROBLEMS

**22–30.** Determine the differential equation of motion of the 3-kg block when it is displaced slightly and released. The surface is smooth and the springs are originally unstretched.



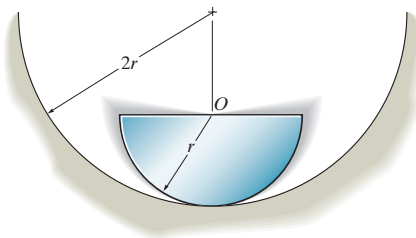
Prob. 22–30

**22–31.** The uniform rod of mass  $m$  is supported by a pin at  $A$  and a spring at  $B$ . If the end  $B$  is given a small downward displacement and released, determine the natural period of vibration.



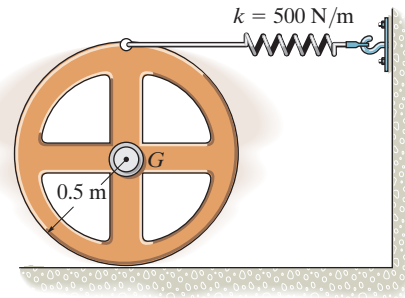
Prob. 22–31

**\*22–32.** The semicircular disk has a mass  $m$  and radius  $r$ , and it rolls without slipping in the semicircular trough. Determine the natural period of vibration of the disk if it is displaced slightly and released. *Hint:*  $I_O = \frac{1}{2}mr^2$ .



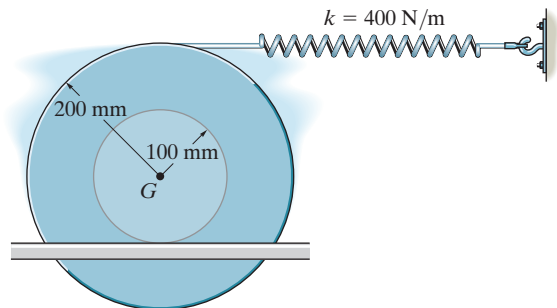
Prob. 22–32

**22–33.** If the 20-kg wheel is displaced a small amount and released, determine the natural period of vibration. The radius of gyration of the wheel is  $k_G = 0.36$  m. The wheel rolls without slipping.



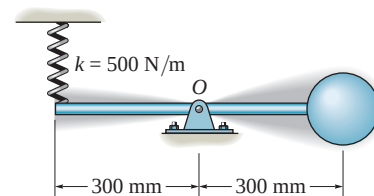
Prob. 22–33

**22–34.** Determine the differential equation of motion of the 3-kg spool. Assume that it does not slip at the surface of contact as it oscillates. The radius of gyration of the spool about its center of mass is  $k_G = 125$  mm.



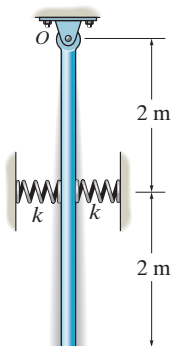
Prob. 22–34

**22–35.** Determine the natural period of vibration of the 3-kg sphere. Neglect the mass of the rod and the size of the sphere.



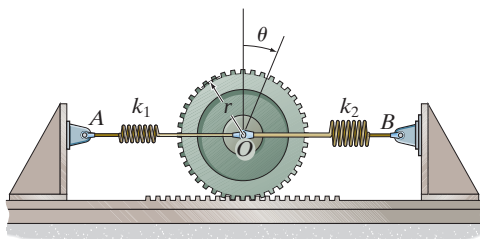
Prob. 22–35

**\*22–36.** If the lower end of the 6-kg slender rod is displaced a small amount and released from rest, determine the natural frequency of vibration. Each spring has a stiffness of  $k = 200 \text{ N/m}$  and is unstretched when the rod is hanging vertically.



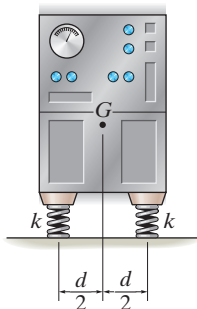
**Prob. 22–36**

**22–37.** The gear of mass  $m$  has a radius of gyration about its center of mass  $O$  of  $k_o$ . The springs have stiffnesses of  $k_1$  and  $k_2$ , respectively, and both springs are unstretched when the gear is in an equilibrium position. If the gear is given a small angular displacement of  $\theta$  and released, determine its natural period of oscillation.



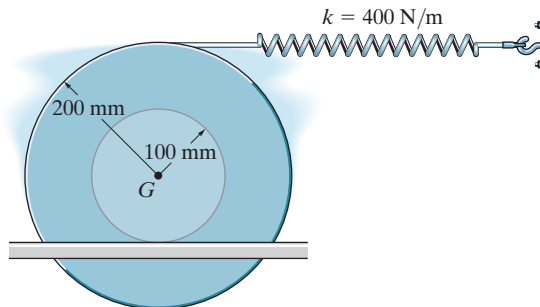
**Prob. 22–37**

**22–38.** The machine has a mass  $m$  and is uniformly supported by four springs, each having a stiffness  $k$ . Determine the natural period of vertical vibration.



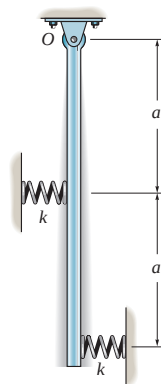
**Prob. 22–38**

**22–39.** Determine the differential equation of motion of the 3-kg spool. Assume that it does not slip at the surface of contact as it oscillates. The radius of gyration of the spool about its center of mass is  $k_G = 125 \text{ mm}$ .



**Prob. 22–39**

**\*22–40.** The slender rod has a mass  $m$  and is pinned at its end  $O$ . When it is vertical, the springs are unstretched. Determine the natural period of vibration.



**Prob. 22–40**

## \*22.3 Undamped Forced Vibration

Undamped forced vibration is considered to be one of the most important types of vibrating motion in engineering. Its principles can be used to describe the motion of many types of machines and structures.

**Periodic Force.** The block and spring shown in Fig. 22–11a provide a convenient model which represents the vibrational characteristics of a system subjected to a periodic force  $F = F_0 \sin \omega_0 t$ . This force has an amplitude of  $F_0$  and a *forcing frequency*  $\omega_0$ . The free-body diagram for the block when it is displaced a distance  $x$  is shown in Fig. 22–11b. Applying the equation of motion, we have

$$\rightarrow \Sigma F_x = ma_x; \quad F_0 \sin \omega_0 t - kx = m\ddot{x}$$

or

$$\ddot{x} + \frac{k}{m}x = \frac{F_0}{m} \sin \omega_0 t \quad (22-18)$$

This equation is a nonhomogeneous second-order differential equation. The general solution consists of a complementary solution,  $x_c$ , *plus* a particular solution,  $x_p$ .

The *complementary solution* is determined by setting the term on the right side of Eq. 22–18 equal to zero and solving the resulting homogeneous equation. The solution is defined by Eq. 22–9, i.e.,

$$x_c = C \sin(\omega_n t + \phi) \quad (22-19)$$

where  $\omega_n$  is the natural frequency,  $\omega_n = \sqrt{k/m}$ , Eq. 22–2.

Since the motion is periodic, the *particular solution* of Eq. 22–18 can be determined by assuming a solution of the form

$$x_p = X \sin \omega_0 t \quad (22-20)$$

where  $X$  is a constant. Taking the second time derivative and substituting into Eq. 22–18 yields

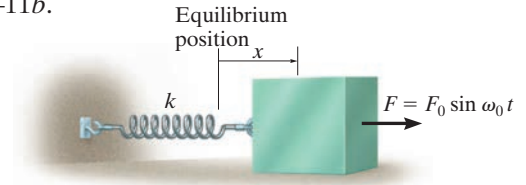
$$-X\omega_0^2 \sin \omega_0 t + \frac{k}{m}(X \sin \omega_0 t) = \frac{F_0}{m} \sin \omega_0 t$$

Factoring out  $\sin \omega_0 t$  and solving for  $X$  gives

$$X = \frac{F_0/m}{(k/m) - \omega_0^2} = \frac{F_0/k}{1 - (\omega_0/\omega_n)^2} \quad (22-21)$$

Substituting into Eq. 22–20, we obtain the particular solution

$$x_p = \frac{F_0/k}{1 - (\omega_0/\omega_n)^2} \sin \omega_0 t \quad (22-22)$$



(a)

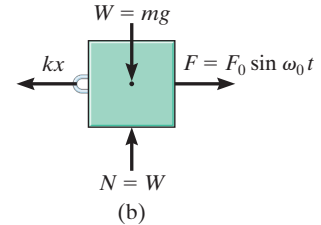
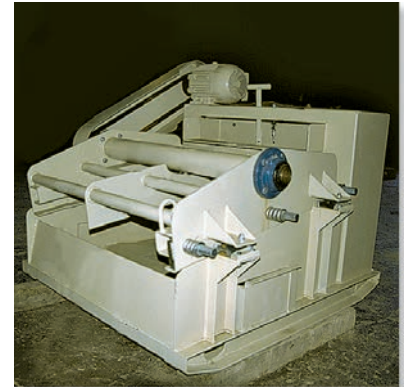


Fig. 22–11



Shaker tables provide forced vibration and are used to separate out granular materials.

The *general solution* is therefore the sum of two sine functions having different frequencies.

$$x = x_c + x_p = C \sin(\omega_n t + \phi) + \frac{F_0/k}{1 - (\omega_0/\omega_n)^2} \sin \omega_0 t \quad (22-23)$$

The *complementary solution*  $x_c$  defines the *free vibration*, which depends on the natural frequency  $\omega_n = \sqrt{k/m}$  and the constants  $C$  and  $\phi$ . The *particular solution*  $x_p$  describes the *forced vibration* of the block caused by the applied force  $F = F_0 \sin \omega_0 t$ . Since all vibrating systems are subject to *friction*, the free vibration,  $x_c$ , will in time dampen out. For this reason the free vibration is referred to as *transient*, and the forced vibration is called *steady-state*, since it is the only vibration that remains.

From Eq. 22-21 it is seen that the *amplitude* of forced or steady-state vibration depends on the *frequency ratio*  $\omega_0/\omega_n$ . If the *magnification factor* MF is defined as the ratio of the amplitude of steady-state vibration,  $X$ , to the static deflection,  $F_0/k$ , which would be produced by the amplitude of the periodic force  $F_0$ , then, from Eq. 22-21,



The soil compactor operates by forced vibration developed by an internal motor. It is important that the forcing frequency not be close to the natural frequency of vibration of the compactor, which can be determined when the motor is turned off; otherwise resonance will occur and the machine will become uncontrollable.

$$MF = \frac{X}{F_0/k} = \frac{1}{1 - (\omega_0/\omega_n)^2} \quad (22-24)$$

This equation is graphed in Fig. 22-12. Note that if the force or displacement is applied with a frequency close to the natural frequency of the system, i.e.,  $\omega_0/\omega_n \approx 1$ , the amplitude of vibration of the block becomes extremely large. This occurs because the force  $\mathbf{F}$  is applied to the block so that it always follows the motion of the block. This condition is called *resonance*, and in practice, resonating vibrations can cause tremendous stress and rapid failure of parts.\*

**Periodic Support Displacement.** Forced vibrations can also arise from the periodic excitation of the support of a system. The model shown in Fig. 22-13a represents the periodic vibration of a block which is caused by harmonic movement  $\delta = \delta_0 \sin \omega_0 t$  of the support. The free-body diagram for the block in this case is shown in Fig. 22-13b. The displacement  $\delta$  of the support is measured from the point of zero displacement, i.e., when the radial line  $OA$  coincides with  $OB$ . Therefore, general deformation of the spring is  $(x - \delta_0 \sin \omega_0 t)$ . Applying the equation of motion yields

$$\pm F_x = ma_x; \quad -k(x - \delta_0 \sin \omega_0 t) = m\ddot{x}$$

or

$$\ddot{x} + \frac{k}{m}x = \frac{k\delta_0}{m} \sin \omega_0 t \quad (22-25)$$

By comparison, this equation is identical to the form of Eq. 22-18, provided  $F_0$  is replaced by  $k\delta_0$ . If this substitution is made into the solutions defined by Eqs. 22-21 to 22-23, the results are appropriate for describing the motion of the block when subjected to the support displacement  $\delta = \delta_0 \sin \omega_0 t$ .

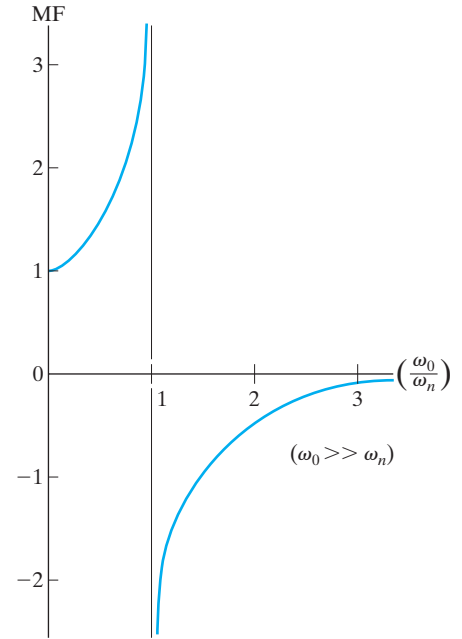


Fig. 22-12

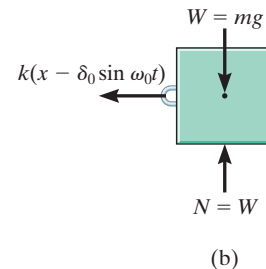
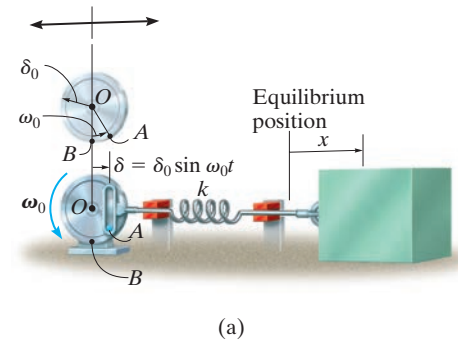
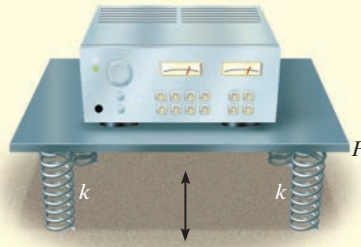


Fig. 22-13

\*A swing has a natural period of vibration, as determined in Example 22.1. If someone pushes on the swing only when it reaches its highest point, neglecting drag or wind resistance, resonance will occur since the natural and forcing frequencies are the same.

**EXAMPLE 22.7**

The instrument shown in Fig. 22–14 is rigidly attached to a platform  $P$ , which in turn is supported by *four* springs, each having a stiffness  $k = 800 \text{ N/m}$ . If the floor is subjected to a vertical displacement  $\delta = 10 \sin(8t) \text{ mm}$ , where  $t$  is in seconds, determine the amplitude of steady-state vibration. What is the frequency of the floor vibration required to cause resonance? The instrument and platform have a total mass of  $20 \text{ kg}$ .

**Fig. 22–14****SOLUTION**

The natural frequency is

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{4(800 \text{ N/m})}{20 \text{ kg}}} = 12.65 \text{ rad/s}$$

The amplitude of steady-state vibration is found using Eq. 22–21, with  $k\delta_0$  replacing  $F_0$ .

$$X = \frac{\delta_0}{1 - (\omega_0/\omega_n)^2} = \frac{10}{1 - [(8 \text{ rad/s})/(12.65 \text{ rad/s})]^2} = 16.7 \text{ mm} \quad \text{Ans.}$$

Resonance will occur when the amplitude of vibration  $X$  caused by the floor displacement approaches infinity. This requires

$$\omega_0 = \omega_n = 12.6 \text{ rad/s} \quad \text{Ans.}$$

## \*22.4 Viscous Damped Free Vibration

The vibration analysis considered thus far has not included the effects of friction or damping in the system, and as a result, the solutions obtained are only in close agreement with the actual motion. Since all vibrations die out in time, the presence of damping forces should be included in the analysis.

In many cases damping is attributed to the resistance created by the substance, such as water, oil, or air, in which the system vibrates. Provided the body moves slowly through this substance, the resistance to motion is directly proportional to the body's speed. The type of force developed under these conditions is called a *viscous damping force*. The magnitude of this force is expressed by an equation of the form

$$F = c\dot{x} \quad (22-26)$$

where the constant  $c$  is called the *coefficient of viscous damping* and has units of  $\text{N} \cdot \text{s}/\text{m}$ .

The vibrating motion of a body or system having viscous damping can be characterized by the block and spring shown in Fig. 22-15a. The effect of damping is provided by the *dashpot* connected to the block on the right side. Damping occurs when the piston  $P$  moves to the right or left within the enclosed cylinder. The cylinder contains a fluid, and the motion of the piston is retarded since the fluid must flow around or through a small hole in the piston. The dashpot is assumed to have a coefficient of viscous damping  $c$ .

If the block is displaced a distance  $x$  from its equilibrium position, the resulting free-body diagram is shown in Fig. 22-15b. Both the spring and damping force oppose the forward motion of the block, so that applying the equation of motion yields

$$\rightarrow \Sigma F_x = m\ddot{x}; \quad -kx - c\dot{x} = m\ddot{x}$$

or

$$m\ddot{x} + c\dot{x} + kx = 0 \quad (22-27)$$

This linear, second-order, homogeneous, differential equation has a solution of the form

$$x = e^{\lambda t}$$

where  $e$  is the base of the natural logarithm and  $\lambda$  (lambda) is a constant. The value of  $\lambda$  can be obtained by substituting this solution and its time derivatives into Eq. 22-27, which yields

$$m\lambda^2 e^{\lambda t} + c\lambda e^{\lambda t} + ke^{\lambda t} = 0$$

or

$$e^{\lambda t}(m\lambda^2 + c\lambda + k) = 0$$

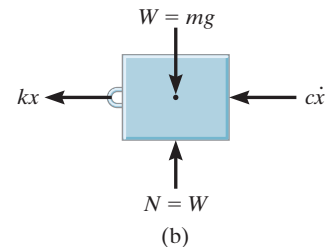
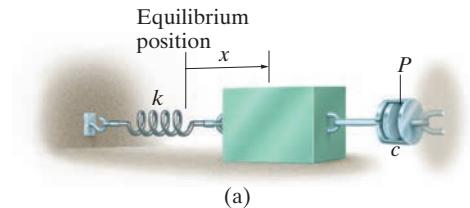


Fig. 22-15

Since  $e^{\lambda t}$  can never be zero, a solution is possible provided

$$m\lambda^2 + c\lambda + k = 0$$

Hence, by the quadratic formula, the two values of  $\lambda$  are

$$\begin{aligned}\lambda_1 &= -\frac{c}{2m} + \sqrt{\left(\frac{c}{2m}\right)^2 - \frac{k}{m}} \\ \lambda_2 &= -\frac{c}{2m} - \sqrt{\left(\frac{c}{2m}\right)^2 - \frac{k}{m}}\end{aligned}\quad (22-28)$$

The general solution of Eq. 22-27 is therefore a combination of exponentials which involves both of these roots. There are three possible combinations of  $\lambda_1$  and  $\lambda_2$  which must be considered. Before discussing these combinations, however, we will first define the *critical damping coefficient*  $c_c$  as the value of  $c$  which makes the radical in Eqs. 22-28 equal to zero; i.e.,

$$\left(\frac{c_c}{2m}\right)^2 - \frac{k}{m} = 0$$

or

$$c_c = 2m\sqrt{\frac{k}{m}} = 2m\omega_n \quad (22-29)$$

**Overdamped System.** When  $c > c_c$ , the roots  $\lambda_1$  and  $\lambda_2$  are both real. The general solution of Eq. 22-27 can then be written as

$$x = Ae^{\lambda_1 t} + Be^{\lambda_2 t} \quad (22-30)$$

Motion corresponding to this solution is *nonvibrating*. The effect of damping is so strong that when the block is displaced and released, it simply creeps back to its original position without oscillating. The system is said to be *overdamped*.

**Critically Damped System.** If  $c = c_c$ , then  $\lambda_1 = \lambda_2 = -c_c/2m = -\omega_n$ . This situation is known as *critical damping*, since it represents a condition where  $c$  has the smallest value necessary to cause the system to be nonvibrating. Using the methods of differential equations, it can be shown that the solution to Eq. 22-27 for critical damping is

$$x = (A + Bt)e^{-\omega_n t} \quad (22-31)$$



**Underdamped System.** Most often  $c < c_c$ , in which case the system is referred to as *underdamped*. In this case the roots  $\lambda_1$  and  $\lambda_2$  are complex numbers, and it can be shown that the general solution of Eq. 22-27 can be written as

$$x = D[e^{-(c/2m)t} \sin(\omega_d t + \phi)] \quad (22-32)$$

where  $D$  and  $\phi$  are constants generally determined from the initial conditions of the problem. The constant  $\omega_d$  is called the *damped natural frequency* of the system. It has a value of

$$\omega_d = \sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2} = \omega_n \sqrt{1 - \left(\frac{c}{c_c}\right)^2} \quad (22-33)$$

where the ratio  $c/c_c$  is called the *damping factor*.

The graph of Eq. 22-32 is shown in Fig. 22-16. The initial limit of motion,  $D$ , diminishes with each cycle of vibration, since motion is confined within the bounds of the exponential curve. Using the damped natural frequency  $\omega_d$ , the period of damped vibration can be written as

$$\tau_d = \frac{2\pi}{\omega_d} \quad (22-34)$$

Since  $\omega_d < \omega_n$ , Eq. 22-33, the period of damped vibration,  $\tau_d$ , will be greater than that of free vibration,  $\tau = 2\pi/\omega_n$ .



Please refer to the Companion Website for the animation: *Damped Spring-Mass Oscillation*

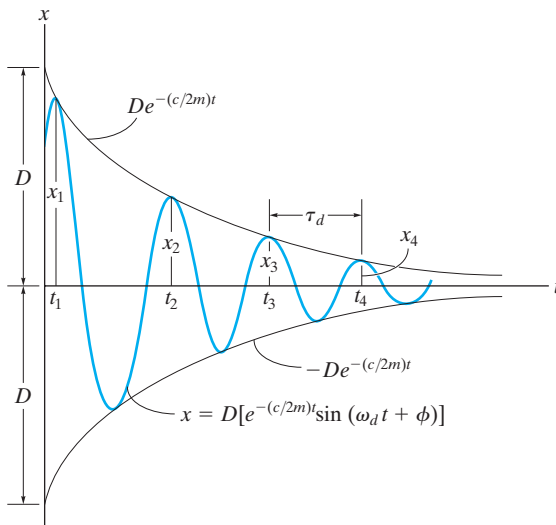


Fig. 22-16

## \*22.5 Viscous Damped Forced Vibration

The most general case of single-degree-of-freedom vibrating motion occurs when the system includes the effects of forced motion and induced damping. The analysis of this particular type of vibration is of practical value when applied to systems having significant damping characteristics.

If a dashpot is attached to the block and spring shown in Fig. 22-11a, the differential equation which describes the motion becomes

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega_0 t \quad (22-35)$$

A similar equation can be written for a block and spring having a periodic support displacement, Fig. 22-13a, which includes the effects of damping. In that case, however,  $F_0$  is replaced by  $k\delta_0$ . Since Eq. 22-35 is nonhomogeneous, the general solution is the sum of a complementary solution,  $x_c$ , and a particular solution,  $x_p$ . The complementary solution is determined by setting the right side of Eq. 22-35 equal to zero and solving the homogeneous equation, which is equivalent to Eq. 22-27. The solution is therefore given by Eq. 22-30, 22-31, or 22-32, depending on the values of  $\lambda_1$  and  $\lambda_2$ . Because all systems are subjected to friction, then this solution will dampen out with time. Only the particular solution, which describes the *steady-state vibration* of the system, will remain. Since the applied forcing function is harmonic, the steady-state motion will also be harmonic. Consequently, the particular solution will be of the form

$$X_p = X' \sin(\omega_0 t - \phi') \quad (22-36)$$

The constants  $X'$  and  $\phi'$  are determined by taking the first and second time derivatives and substituting them into Eq. 22-35, which after simplification yields

$$\begin{aligned} -X'm\omega_0^2 \sin(\omega_0 t - \phi') + \\ X'c\omega_0 \cos(\omega_0 t - \phi') + X'k \sin(\omega_0 t - \phi') = F_0 \sin \omega_0 t \end{aligned}$$

Since this equation holds for all time, the constant coefficients can be obtained by setting  $\omega_0 t - \phi' = 0$  and  $\omega_0 t - \phi' = \pi/2$ , which causes the above equation to become

$$\begin{aligned} X'c\omega_0 &= F_0 \sin \phi' \\ -X'm\omega_0^2 + X'k &= F_0 \cos \phi' \end{aligned}$$

The amplitude is obtained by squaring these equations, adding the results, and using the identity  $\sin^2\phi' + \cos^2\phi' = 1$ , which gives

$$X' = \frac{F_0}{\sqrt{(k - m\omega_0^2)^2 + c^2\omega_0^2}} \quad (22-37)$$

Dividing the first equation by the second gives

$$\phi' = \tan^{-1} \left[ \frac{c\omega_0}{k - m\omega_0^2} \right] \quad (22-38)$$

Since  $\omega_n = \sqrt{k/m}$  and  $c_c = 2m\omega_n$ , then the above equations can also be written as

$$X' = \frac{F_0/k}{\sqrt{[1 - (\omega_0/\omega_n)^2]^2 + [2(c/c_c)(\omega_0/\omega_n)]^2}} \quad (22-39)$$

$$\phi' = \tan^{-1} \left[ \frac{2(c/c_c)(\omega_0/\omega_n)}{1 - (\omega_0/\omega_n)^2} \right]$$

The angle  $\phi'$  represents the phase difference between the applied force and the resulting steady-state vibration of the damped system.

The *magnification factor* MF has been defined in Sec. 22.3 as the ratio of the amplitude of deflection caused by the forced vibration to the deflection caused by a static force  $F_0$ . Thus,

$$\text{MF} = \frac{X'}{F_0/k} = \frac{1}{\sqrt{[1 - (\omega_0/\omega_n)^2]^2 + [2(c/c_c)(\omega_0/\omega_n)]^2}} \quad (22-40)$$

The MF is plotted in Fig. 22-17 versus the frequency ratio  $\omega_0/\omega_n$  for various values of the damping factor  $c/c_c$ . It can be seen from this graph that the magnification of the amplitude increases as the damping factor decreases. Resonance obviously occurs only when the damping factor is zero and the frequency ratio equals 1.

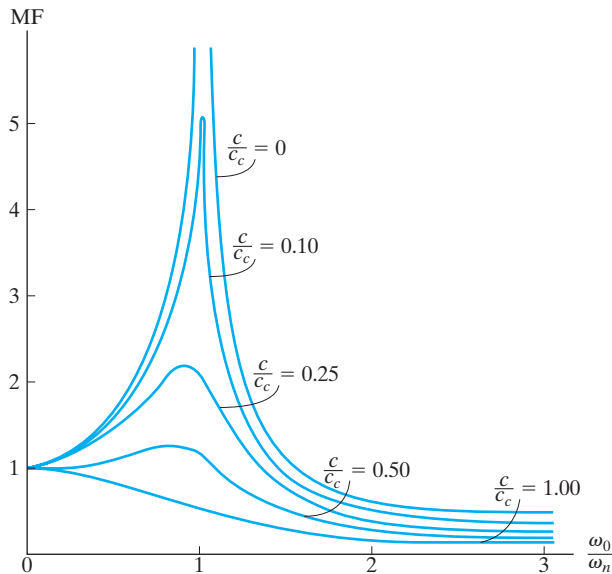
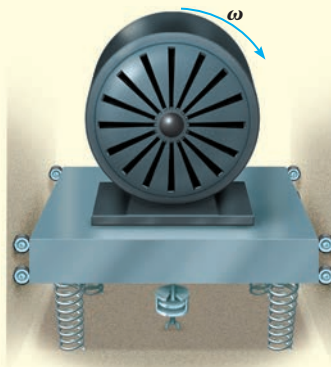


Fig. 22-17

**EXAMPLE 22.8**

The 30-kg electric motor shown in Fig. 22–18 is confined to move vertically, and is supported by *four* springs, each spring having a stiffness of 200 N/m. If the rotor is unbalanced such that its effect is equivalent to a 4-kg mass located 60 mm from the axis of rotation, determine the amplitude of vibration when the rotor is turning at  $\omega_0 = 10$  rad/s. The damping factor is  $c/c_c = 0.15$ .

**Fig. 22–18****SOLUTION**

The periodic force which causes the motor to vibrate is the centrifugal force due to the unbalanced rotor. This force has a constant magnitude of

$$F_0 = ma_n = m r \omega_0^2 = 4 \text{ kg}(0.06 \text{ m})(10 \text{ rad/s})^2 = 24 \text{ N}$$

The stiffness of the entire system of four springs is  $k = 4(200 \text{ N/m}) = 800 \text{ N/m}$ . Therefore, the natural frequency of vibration is

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{800 \text{ N/m}}{30 \text{ kg}}} = 5.164 \text{ rad/s}$$

Since the damping factor is known, the steady-state amplitude can be determined from the first of Eqs. 22–39, i.e.,

$$\begin{aligned} X' &= \frac{F_0/k}{\sqrt{[1 - (\omega_0/\omega_n)^2]^2 + [2(c/c_c)(\omega_0/\omega_n)]^2}} \\ &= \frac{24/800}{\sqrt{[1 - (10/5.164)^2]^2 + [2(0.15)(10/5.164)]^2}} \\ &= 0.0107 \text{ m} = 10.7 \text{ mm} \end{aligned}$$

*Ans.*

## \*22.6 Electrical Circuit Analogs

The characteristics of a vibrating mechanical system can be represented by an electric circuit. Consider the circuit shown in Fig. 22-19a, which consists of an inductor  $L$ , a resistor  $R$ , and a capacitor  $C$ . When a voltage  $E(t)$  is applied, it causes a current of magnitude  $i$  to flow through the circuit. As the current flows past the inductor the voltage drop is  $L(di/dt)$ , when it flows across the resistor the drop is  $Ri$ , and when it arrives at the capacitor the drop is  $(1/C) \int i dt$ . Since current cannot flow past a capacitor, it is only possible to measure the charge  $q$  acting on the capacitor. The charge can, however, be related to the current by the equation  $i = dq/dt$ . Thus, the voltage drops which occur across the inductor, resistor, and capacitor become  $L d^2q/dt^2$ ,  $R dq/dt$ , and  $q/C$ , respectively. According to Kirchhoff's voltage law, the applied voltage balances the sum of the voltage drops around the circuit. Therefore,

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C}q = E(t) \quad (22-41)$$

Consider now the model of a single-degree-of-freedom mechanical system, Fig. 22-19b, which is subjected to both a general forcing function  $F(t)$  and damping. The equation of motion for this system was established in the previous section and can be written as

$$m \frac{d^2x}{dt^2} + c \frac{dx}{dt} + kx = F(t) \quad (22-42)$$

By comparison, it is seen that Eqs. 22-41 and 22-42 have the same form, and hence mathematically the procedure of analyzing an electric circuit is the same as that of analyzing a vibrating mechanical system. The analogs between the two equations are given in Table 22-1.

This analogy has important application to experimental work, for it is much easier to simulate the vibration of a complex mechanical system using an electric circuit, which can be constructed on an analog computer, than to make an equivalent mechanical spring-and-dashpot model.

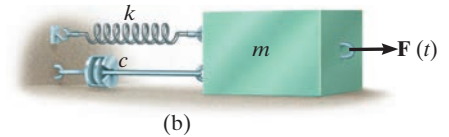
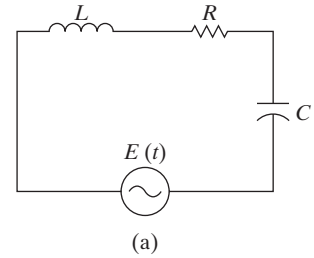


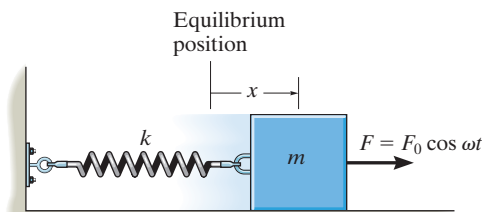
Fig. 22-19

TABLE 22-1  
Electrical-Mechanical Analogs

| Electrical                |        | Mechanical                  |         |
|---------------------------|--------|-----------------------------|---------|
| Electric charge           | $q$    | Displacement                | $x$     |
| Electric current          | $i$    | Velocity                    | $dx/dt$ |
| Voltage                   | $E(t)$ | Applied force               | $F(t)$  |
| Inductance                | $L$    | Mass                        | $m$     |
| Resistance                | $R$    | Viscous damping coefficient | $c$     |
| Reciprocal of capacitance | $1/C$  | Spring stiffness            | $k$     |

## PROBLEMS

**22–41.** If the block-and-spring model is subjected to the periodic force  $F = F_0 \cos \omega t$ , show that the differential equation of motion is  $\ddot{x} + (k/m)x = (F_0/m) \cos \omega t$ , where  $x$  is measured from the equilibrium position of the block. What is the general solution of this equation?



Prob. 22–41

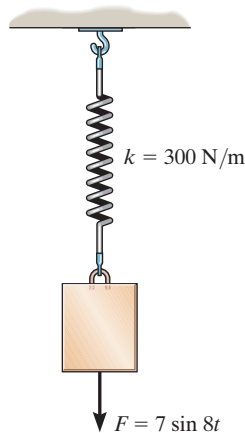
**22–42.** A block which has a mass  $m$  is suspended from a spring having a stiffness  $k$ . If an impressed downward vertical force  $F = F_0 \sin \omega t$  acts on the weight, determine the equation which describes the position of the block as a function of time.

**22–43.** The block shown in Fig. 22–15 has a mass of 20 kg, and the spring has a stiffness  $k = 600 \text{ N/m}$ . When the block is displaced and released, two successive amplitudes are measured as  $x_1 = 150 \text{ mm}$  and  $x_2 = 87 \text{ mm}$ . Determine the coefficient of viscous damping,  $c$ .

**\*22–44.** A 4-kg block is suspended from a spring that has a stiffness of  $k = 600 \text{ N/m}$ . The block is drawn downward 50 mm from the equilibrium position and released from rest when  $t = 0$ . If the support moves with an impressed displacement of  $\delta = (10 \sin 4t) \text{ mm}$ , where  $t$  is in seconds, determine the equation that describes the vertical motion of the block. Assume positive displacement is downward.

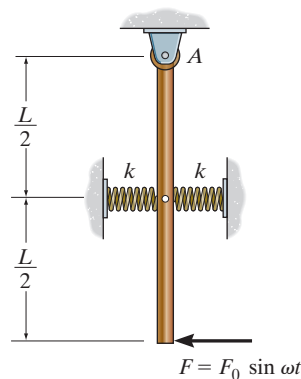
**22–45.** Use a block-and-spring model like that shown in Fig. 22–13a, but suspended from a vertical position and subjected to a periodic support displacement  $\delta = \delta_0 \sin \omega_0 t$ , determine the equation of motion for the system, and obtain its general solution. Define the displacement  $y$  measured from the static equilibrium position of the block when  $t = 0$ .

**22–46.** A 5-kg block is suspended from a spring having a stiffness of 300 N/m. If the block is acted upon by a vertical force  $F = (7 \sin 8t) \text{ N}$ , where  $t$  is in seconds, determine the equation which describes the motion of the block when it is pulled down 100 mm from the equilibrium position and released from rest at  $t = 0$ . Assume that positive displacement is downward.



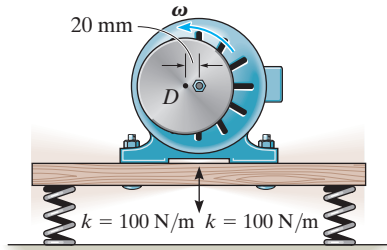
Prob. 22–46

**22–47.** The uniform rod has a mass of  $m$ . If it is acted upon by a periodic force of  $F = F_0 \sin \omega t$ , determine the amplitude of the steady-state vibration.



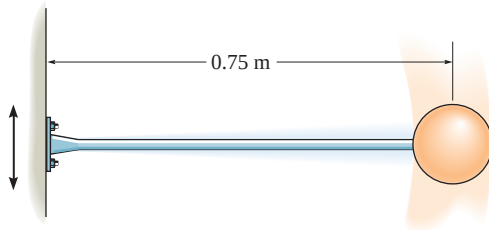
Prob. 22–47

**\*22–48.** The electric motor has a mass of 50 kg and is supported by *four springs*, each spring having a stiffness of 100 N/m. If the motor turns a disk  $D$  which is mounted eccentrically, 20 mm from the disk's center, determine the angular velocity  $\omega$  at which resonance occurs. Assume that the motor only vibrates in the vertical direction.



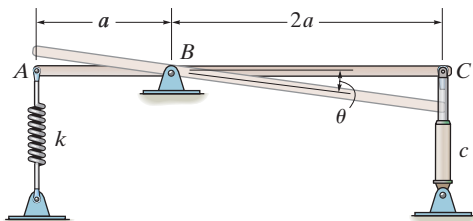
**Prob. 22–48**

**22–49.** The light elastic rod supports a 4-kg sphere. When an 18-N vertical force is applied to the sphere, the rod deflects 14 mm. If the wall oscillates with harmonic frequency of 2 Hz and has an amplitude of 15 mm, determine the amplitude of vibration for the sphere.



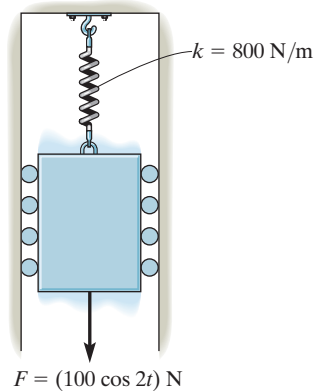
**Prob. 22–49**

**22–50.** Find the differential equation for small oscillations in terms of  $\theta$  for the uniform rod of mass  $m$ . Also show that if  $c < \sqrt{mk}/2$ , then the system remains underdamped. The rod is in a horizontal position when it is in equilibrium.



**Prob. 22–50**

**22–51.** The 40-kg block is attached to a spring having a stiffness of 800 N/m. A force  $F = (100 \cos 2t)$  N, where  $t$  is in seconds is applied to the block. Determine the maximum speed of the block for the steady-state vibration.

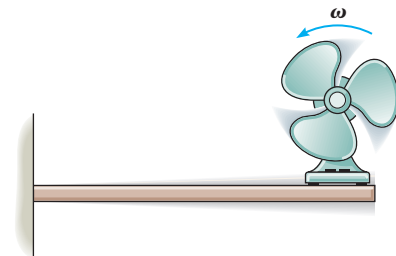


**Prob. 22–51**

**\*22–52.** The fan has a mass of 25 kg and is fixed to the end of a horizontal beam that has a negligible mass. The fan blade is mounted eccentrically on the shaft such that it is equivalent to an unbalanced 3.5-kg mass located 100 mm from the axis of rotation. If the static deflection of the beam is 50 mm as a result of the weight of the fan, determine the angular velocity of the fan blade at which resonance will occur. *Hint:* See the first part of Example 22.8.

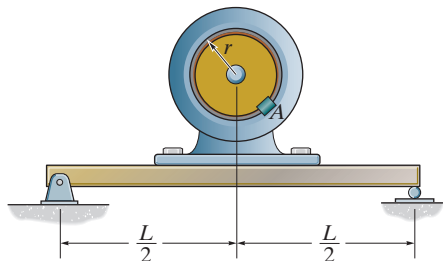
**22–53.** In Prob. 22–52, determine the amplitude of steady-state vibration of the fan if its angular velocity is 10 rad/s.

**22–54.** What will be the amplitude of steady-state vibration of the fan in Prob. 22–52 if the angular velocity of the fan blade is 18 rad/s? *Hint:* See the first part of Example 22.8.



**Probs. 22–52/53/54**

**22-55.** The motor of mass  $M$  is supported by a simply supported beam of negligible mass. If block  $A$  of mass  $m$  is clipped onto the rotor, which is turning at constant angular velocity of  $\omega$ , determine the amplitude of the steady-state vibration. *Hint:* When the beam is subjected to a concentrated force of  $P$  at its mid-span, it deflects  $\delta = PL^3/48EI$  at this point. Here  $E$  is Young's modulus of elasticity, a property of the material, and  $I$  is the moment of inertia of the beam's cross-sectional area.

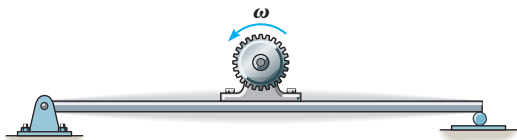


**Prob. 22-55**

**\*22-56.** The electric motor turns an eccentric flywheel which is equivalent to an unbalanced 0.125-kg mass located 250 mm from the axis of rotation. If the static deflection of the beam is 25 mm because of the mass of the motor, determine the angular velocity of the flywheel at which resonance will occur. The motor has a mass of 75 kg. Neglect the mass of the beam.

**22-57.** What will be the amplitude of steady-state vibration of the motor in Prob. 22-56 if the angular velocity of the flywheel is 20 rad/s?

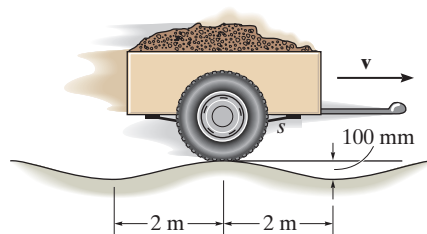
**22-58.** Determine the angular velocity of the flywheel in Prob. 22-56 which will produce an amplitude of vibration of 6 mm.



**Probs. 22-56/57/58**

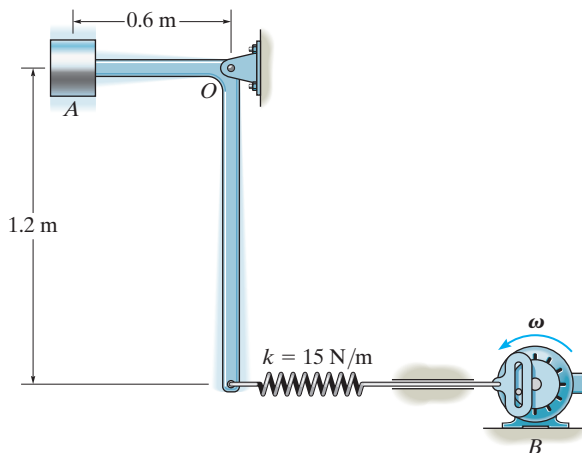
**22-59.** The 450-kg trailer is pulled with a constant speed over the surface of a bumpy road, which may be approximated by a cosine curve having an amplitude of 50 mm and wave length of 4 m. If the two springs  $s$  which support the trailer each have a stiffness of 800 N/m, determine the speed  $v$  which will cause the greatest vibration (resonance) of the trailer. Neglect the weight of the wheels.

**\*22-60.** Determine the amplitude of vibration of the trailer in Prob. 22-59 if the speed  $v = 15$  km/h.



**Probs. 22-59/60**

**22-61.** The small block at  $A$  has a mass of 4 kg and is mounted on the bent rod having negligible mass. If the rotor at  $B$  causes a harmonic movement  $\delta_B = (0.1 \cos 15t)$  m, where  $t$  is in seconds, determine the steady-state amplitude of vibration of the block.

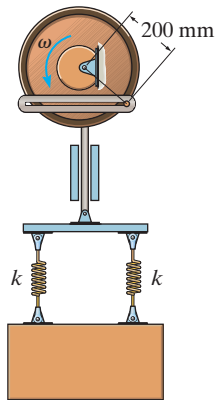


**Prob. 22-61**



**22–62.** The spring system is connected to a crosshead that oscillates vertically when the wheel rotates with a constant angular velocity of  $\omega$ . If the amplitude of the steady-state vibration is observed to be 400 mm, and the springs each have a stiffness of  $k = 2500$  N/m, determine the two possible values of  $\omega$  at which the wheel must rotate. The block has a mass of 50 kg.

**22–63.** The spring system is connected to a crosshead that oscillates vertically when the wheel rotates with a constant angular velocity of  $\omega = 5$  rad/s. If the amplitude of the steady-state vibration is observed to be 400 mm, determine the two possible values of the stiffness  $k$  of the springs. The block has a mass of 50 kg.



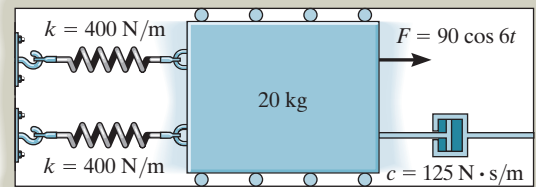
**Probs. 22–62/63**

**\*22–64.** A 3.5-kg block is suspended from a spring having a stiffness of  $k = 1250$  N/m. The support to which the spring is attached is given simple harmonic motion which can be expressed as  $\delta = (0.045 \sin 2t)$  m, where  $t$  is in seconds. If the damping factor is  $c/c_c = 0.8$ , determine the phase angle  $\phi$  of forced vibration.

**22–65.** Determine the magnification factor of the block, spring, and dashpot combination in Prob. 22–64.

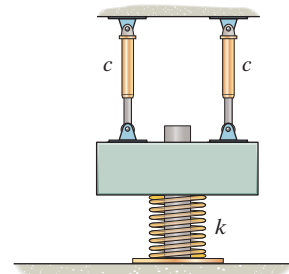
**22–66.** A block having a mass of 7 kg is suspended from a spring that has a stiffness  $k = 600$  N/m. If the block is given an upward velocity of 0.6 m/s from its equilibrium position at  $t = 0$ , determine its position as a function of time. Assume that positive displacement of the block is downward and that motion takes place in a medium which furnishes a damping force  $F = (50|v|)$  N, where  $v$  is in m/s.

**22–67.** The 20-kg block is subjected to the action of the harmonic force  $F = (90 \cos 6t)$  N, where  $t$  is in seconds. Write the equation which describes the steady-state motion.



**Prob. 22–67**

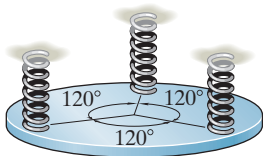
**\*22–68.** Two identical dashpots are arranged parallel to each other, as shown. Show that if the damping coefficient  $c < \sqrt{mk}$ , then the block of mass  $m$  will vibrate as an underdamped system.



**Prob. 22–68**

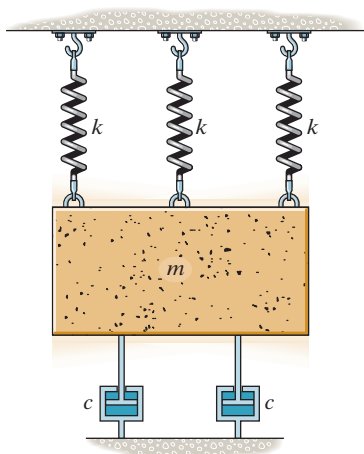
**22–69.** The damping factor,  $c/c_c$ , may be determined experimentally by measuring the successive amplitudes of vibrating motion of a system. If two of these maximum displacements can be approximated by  $x_1$  and  $x_2$ , as shown in Fig. 22–16, show that  $\ln(x_1/x_2) = 2\pi(c/c_c)/\sqrt{1-(c/c_c)^2}$ . The quantity  $\ln(x_1/x_2)$  is called the *logarithmic decrement*.

**22–70.** The 4-kg circular disk is attached to three springs, each spring having a stiffness  $k = 180 \text{ N/m}$ . If the disk is immersed in a fluid and given a downward velocity of  $0.3 \text{ m/s}$  at the equilibrium position, determine the equation which describes the motion. Consider positive displacement to be measured downward, and that fluid resistance acting on the disk furnishes a damping force having a magnitude  $F = (60|v|) \text{ N}$ , where  $v$  is the velocity of the block in  $\text{m/s}$ .



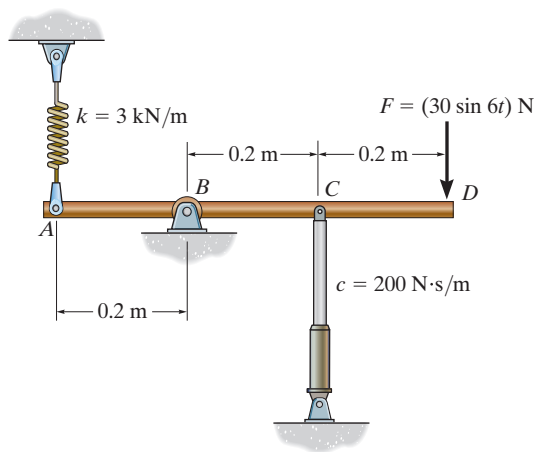
**Prob. 22–70**

**22–71.** Determine the differential equation of motion for the damped vibratory system shown. What type of motion occurs? Take  $k = 100 \text{ N/m}$ ,  $c = 200 \text{ N}\cdot\text{s/m}$ ,  $m = 25 \text{ kg}$ .



**Prob. 22–71**

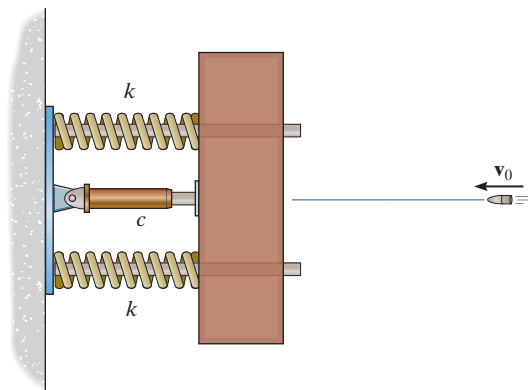
**\*22–72.** If the 12-kg rod is subjected to a periodic force of  $F = (30 \sin 6t) \text{ N}$ , where  $t$  is in seconds, determine the steady-state vibration amplitude  $\theta_{\max}$  of the rod about the pin  $B$ . Assume  $\theta$  is small.



**Prob. 22–72**

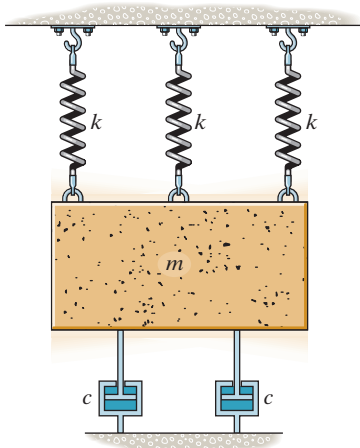
**22–73.** A bullet of mass  $m$  has a velocity of  $\mathbf{v}_0$  just before it strikes the target of mass  $M$ . If the bullet embeds in the target, and the vibration is to be critically damped, determine the dashpot's critical damping coefficient, and the springs' maximum compression. The target is free to move along the two horizontal guides that are "nested" in the springs.

**22–74.** A bullet of mass  $m$  has a velocity  $\mathbf{v}_0$  just before it strikes the target of mass  $M$ . If the bullet embeds in the target, and the dashpot's damping coefficient is  $0 < c < c_c$ , determine the springs' maximum compression. The target is free to move along the two horizontal guides that are "nested" in the springs.



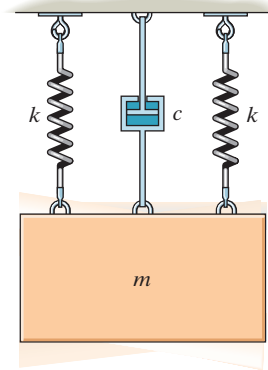
**Probs. 22–73/74**

**22-75.** Determine the differential equation of motion for the damped vibratory system shown. What type of motion occurs? Take  $k = 100 \text{ N/m}$ ,  $c = 200 \text{ N} \cdot \text{s/m}$ ,  $m = 25 \text{ kg}$ .



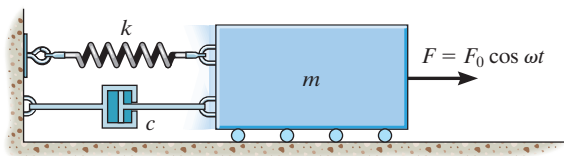
**Prob. 22-75**

**22-77.** Draw the electrical circuit that is equivalent to the mechanical system shown. What is the differential equation which describes the charge  $q$  in the circuit?



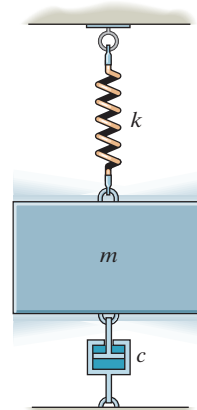
**Prob. 22-77**

**\*22-76.** Draw the electrical circuit that is equivalent to the mechanical system shown. Determine the differential equation which describes the charge  $q$  in the circuit.



**Prob. 22-76**

**22-78.** Draw the electrical circuit that is equivalent to the mechanical system shown. Determine the differential equation which describes the charge  $q$  in the circuit.



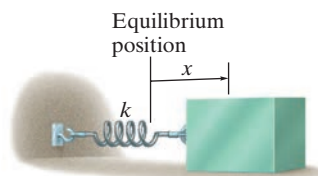
**Prob. 22-78**

## CHAPTER REVIEW

### Undamped Free Vibration

A body has free vibration when gravitational or elastic restoring forces cause the motion. This motion is undamped when friction forces are neglected. The periodic motion of an undamped, freely vibrating body can be studied by displacing the body from the equilibrium position and then applying the equation of motion along the path.

For a one-degree-of-freedom system, the resulting differential equation can be written in terms of its natural frequency  $\omega_n$ .



$$\ddot{x} + \omega_n^2 x = 0 \quad \tau = \frac{2\pi}{\omega_n} \quad f = \frac{1}{\tau} = \frac{\omega_n}{2\pi}$$

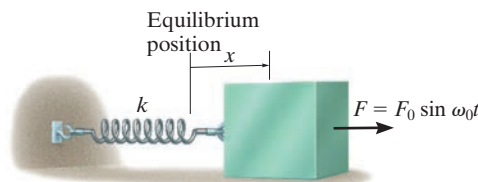
### Energy Methods

Provided the restoring forces acting on the body are gravitational and elastic, then conservation of energy can also be used to determine its simple harmonic motion. To do this, the body is displaced a small amount from its equilibrium position, and an expression for its kinetic and potential energy is written. The time derivative of this equation can then be rearranged in the standard form  $\ddot{x} + \omega_n^2 x = 0$ .

### Undamped Forced Vibration

When the equation of motion is applied to a body, which is subjected to a periodic force, or the support has a displacement with a frequency  $\omega_0$ , then the solution of the differential equation consists of a complementary solution and a particular solution. The complementary solution is caused by the free vibration and can be neglected. The particular solution is caused by the forced vibration.

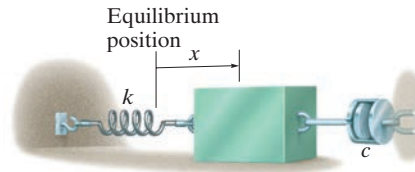
Resonance will occur if the natural frequency of vibration  $\omega_n$  is equal to the forcing frequency  $\omega_0$ . This should be avoided, since the motion will tend to become unbounded.



$$x_p = \frac{F_0/k}{1 - (\omega_0/\omega_n)^2} \sin \omega_0 t$$

### Viscous Damped Free Vibration

A viscous damping force is caused by fluid drag on the system as it vibrates. If the motion is slow, this drag force will be proportional to the velocity, that is,  $F = c\dot{x}$ . Here  $c$  is the coefficient of viscous damping. By comparing its value to the critical damping coefficient  $c_c = 2m\omega_n$ , we can specify the type of vibration that occurs. If  $c > c_c$ , it is an overdamped system; if  $c = c_c$ , it is a critically damped system; if  $c < c_c$ , it is an underdamped system.



### Viscous Damped Forced Vibration

The most general type of vibration for a one-degree-of-freedom system occurs when the system is damped and subjected to periodic forced motion. The solution provides insight as to how the damping factor,  $c/c_c$ , and the frequency ratio,  $\omega_0/\omega_n$ , influence the vibration.

Resonance is avoided provided  $c/c_c \neq 0$  and  $\omega_0/\omega_n \neq 1$ .

### Electrical Circuit Analogs

The vibrating motion of a complex mechanical system can be studied by modeling it as an electrical circuit. This is possible since the differential equations that govern the behavior of each system are the same.

## A

## Mathematical Expressions

## Quadratic Formula

$$\text{If } ax^2 + bx + c = 0, \text{ then } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

## Hyperbolic Functions

$$\sinh x = \frac{e^x - e^{-x}}{2}, \cosh x = \frac{e^x + e^{-x}}{2}, \tanh x = \frac{\sinh x}{\cosh x}$$

## Trigonometric Identities

$$\sin \theta = \frac{A}{C}, \csc \theta = \frac{C}{A}$$

$$\cos \theta = \frac{B}{C}, \sec \theta = \frac{C}{B}$$

$$\tan \theta = \frac{A}{B}, \cot \theta = \frac{B}{A}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin(\theta \pm \phi) = \sin \theta \cos \phi \pm \cos \theta \sin \phi$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

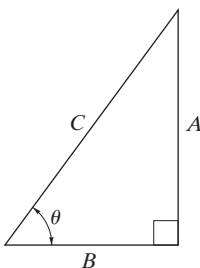
$$\cos(\theta \pm \phi) = \cos \theta \cos \phi \mp \sin \theta \sin \phi$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos \theta = \pm \sqrt{\frac{1 + \cos 2\theta}{2}}, \sin \theta = \pm \sqrt{\frac{1 - \cos 2\theta}{2}}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$1 + \tan^2 \theta = \sec^2 \theta \quad 1 + \cot^2 \theta = \csc^2 \theta$$



## Derivatives

$$\frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}$$

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\frac{d}{dx}(\cot u) = -\csc^2 u \frac{du}{dx}$$

$$\frac{d}{dx}(\sec u) = \tan u \sec u \frac{du}{dx}$$

$$\frac{d}{dx}(\csc u) = -\csc u \cot u \frac{du}{dx}$$

$$\frac{d}{dx}(\sin u) = \cos u \frac{du}{dx}$$

$$\frac{d}{dx}(\cos u) = -\sin u \frac{du}{dx}$$

$$\frac{d}{dx}(\tan u) = \sec^2 u \frac{du}{dx}$$

$$\frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$$

$$\frac{d}{dx}(\cosh u) = \sinh u \frac{du}{dx}$$

## Power-Series Expansions

$$\sin x = x - \frac{x^3}{3!} + \cdots \quad \sinh x = x + \frac{x^3}{3!} + \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \cdots \quad \cosh x = 1 + \frac{x^2}{2!} + \cdots$$

## Integrals

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$

$$\int \frac{dx}{a+bx} = \frac{1}{b} \ln(a+bx) + C$$

$$\int \frac{dx}{a+bx^2} = \frac{1}{2\sqrt{-ba}} \ln \left[ \frac{a+x\sqrt{-ab}}{a-x\sqrt{-ab}} \right] + C, ab < 0$$

$$\int \frac{x dx}{a+bx^2} = \frac{1}{2b} \ln(bx^2+a) + C$$

$$\int \frac{x^2 dx}{a+bx^2} = \frac{x}{b} - \frac{a}{b\sqrt{ab}} \tan^{-1} \frac{x\sqrt{ab}}{a} + C, ab > 0$$

$$\int \frac{dx}{a^2-x^2} = \frac{1}{2a} \ln \left[ \frac{a+x}{a-x} \right] + C, a^2 > x^2$$

$$\int \sqrt{a+bx} dx = \frac{2}{3b} \sqrt{(a+bx)^3} + C$$

$$\int x\sqrt{a+bx} dx = \frac{-2(2a-3bx)\sqrt{(a+bx)^3}}{15b^2} + C$$

$$\int x^2\sqrt{a+bx} dx = \frac{2(8a^2-12abx+15b^2x^2)\sqrt{(a+bx)^3}}{105b^3} + C$$

$$\int \sqrt{a^2-x^2} dx = \frac{1}{2} \left[ x\sqrt{a^2-x^2} + a^2 \sin^{-1} \frac{x}{a} \right] + C, a > 0$$

$$\int x\sqrt{x^2 \pm a^2} dx = \frac{1}{3} \sqrt{(x^2 \pm a^2)^3} + C$$

$$\int x^2\sqrt{a^2-x^2} dx = -\frac{x}{4}\sqrt{(a^2-x^2)^3} + \frac{a^2}{8} \left( x\sqrt{a^2-x^2} + a^2 \sin^{-1} \frac{x}{a} \right) + C, a > 0$$

$$\int \sqrt{x^2 \pm a^2} dx = \frac{1}{2} [x\sqrt{x^2 \pm a^2} \pm a^2 \ln(x + \sqrt{x^2 \pm a^2})] + C$$

$$\int x\sqrt{a^2-x^2} dx = -\frac{1}{3}\sqrt{(a^2-x^2)^3} + C$$

$$\int x^2\sqrt{x^2 \pm a^2} dx = \frac{x}{4}\sqrt{(x^2 \pm a^2)^3} \mp \frac{a^2}{8}x\sqrt{x^2 \pm a^2} - \frac{a^4}{8} \ln(x + \sqrt{x^2 \pm a^2}) + C$$

$$\int \frac{dx}{\sqrt{a+bx}} = \frac{2\sqrt{a+bx}}{b} + C$$

$$\int \frac{x dx}{\sqrt{x^2 \pm a^2}} = \sqrt{x^2 \pm a^2} + C$$

$$\int \frac{dx}{\sqrt{a+bx+cx^2}} = \frac{1}{\sqrt{c}} \ln \left[ \sqrt{a+bx+cx^2} + x\sqrt{c} + \frac{b}{2\sqrt{c}} \right] + C, c > 0$$

$$= \frac{1}{\sqrt{-c}} \sin^{-1} \left( \frac{-2cx-b}{\sqrt{b^2-4ac}} \right) + C, c < 0$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int x \cos(ax) dx = \frac{1}{a^2} \cos(ax) + \frac{x}{a} \sin(ax) + C$$

$$\int x^2 \cos(ax) dx = \frac{2x}{a^2} \cos(ax) + \frac{a^2x^2-2}{a^3} \sin(ax) + C$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C$$

$$\int xe^{ax} dx = \frac{e^{ax}}{a^2} (ax-1) + C$$

$$\int \sinh x dx = \cosh x + C$$

$$\int \cosh x dx = \sinh x + C$$

# Vector Analysis

The following discussion provides a brief review of vector analysis. A more detailed treatment of these topics is given in *Engineering Mechanics: Statics*.

**Vector.** A vector,  $\mathbf{A}$ , is a quantity which has magnitude and direction, and adds according to the parallelogram law. As shown in Fig. B-1,  $\mathbf{A} = \mathbf{B} + \mathbf{C}$ , where  $\mathbf{A}$  is the *resultant vector* and  $\mathbf{B}$  and  $\mathbf{C}$  are *component vectors*.

**Unit Vector.** A unit vector,  $\mathbf{u}_A$ , has a magnitude of one “dimensionless” unit and acts in the same direction as  $\mathbf{A}$ . It is determined by dividing  $\mathbf{A}$  by its magnitude  $A$ , i.e.,

$$\mathbf{u}_A = \frac{\mathbf{A}}{A}$$

(B-1)

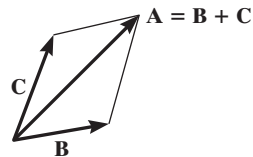


Fig. B-1



**Cartesian Vector Notation.** The directions of the positive  $x$ ,  $y$ ,  $z$  axes are defined by the Cartesian unit vectors  $\mathbf{i}$ ,  $\mathbf{j}$ ,  $\mathbf{k}$ , respectively.

As shown in Fig. B-2, vector  $\mathbf{A}$  is formulated by the addition of its  $x$ ,  $y$ ,  $z$  components as

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k} \quad (\text{B-2})$$

The *magnitude* of  $\mathbf{A}$  is determined from

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} \quad (\text{B-3})$$

The *direction* of  $\mathbf{A}$  is defined in terms of its *coordinate direction angles*,  $\alpha$ ,  $\beta$ ,  $\gamma$ , measured from the *tail* of  $\mathbf{A}$  to the *positive*  $x$ ,  $y$ ,  $z$  axes, Fig. B-3. These angles are determined from the *direction cosines* which represent the  $\mathbf{i}$ ,  $\mathbf{j}$ ,  $\mathbf{k}$  components of the unit vector  $\mathbf{u}_A$ ; i.e., from Eqs. B-1 and B-2

$$\mathbf{u}_A = \frac{A_x}{A} \mathbf{i} + \frac{A_y}{A} \mathbf{j} + \frac{A_z}{A} \mathbf{k} \quad (\text{B-4})$$

so that the direction cosines are

$$\cos \alpha = \frac{A_x}{A} \quad \cos \beta = \frac{A_y}{A} \quad \cos \gamma = \frac{A_z}{A} \quad (\text{B-5})$$

Hence,  $\mathbf{u}_A = \cos \alpha \mathbf{i} + \cos \beta \mathbf{j} + \cos \gamma \mathbf{k}$ , and using Eq. B-3, it is seen that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1 \quad (\text{B-6})$$

**The Cross Product.** The cross product of two vectors  $\mathbf{A}$  and  $\mathbf{B}$ , which yields the resultant vector  $\mathbf{C}$ , is written as

$$\mathbf{C} = \mathbf{A} \times \mathbf{B} \quad (\text{B-7})$$

and reads  $\mathbf{C}$  equals  $\mathbf{A}$  “cross”  $\mathbf{B}$ . The *magnitude* of  $\mathbf{C}$  is

$$C = AB \sin \theta \quad (\text{B-8})$$

where  $\theta$  is the angle made between the *tails* of  $\mathbf{A}$  and  $\mathbf{B}$  ( $0^\circ \leq \theta \leq 180^\circ$ ). The *direction* of  $\mathbf{C}$  is determined by the right-hand rule, whereby the fingers of the right hand are curled *from*  $\mathbf{A}$  *to*  $\mathbf{B}$  and the thumb points in the direction of  $\mathbf{C}$ , Fig. B-4. This vector is perpendicular to the plane containing vectors  $\mathbf{A}$  and  $\mathbf{B}$ .

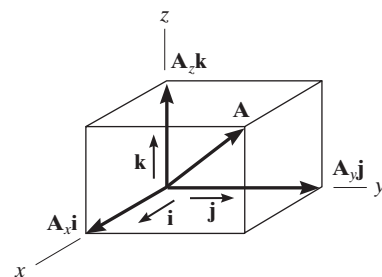


Fig. B-2

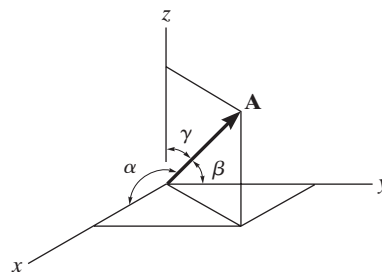


Fig. B-3

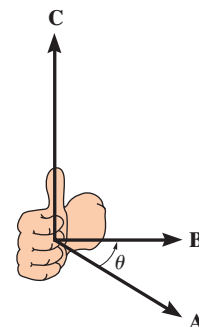


Fig. B-4



**The Dot Product.** The dot product of two vectors **A** and **B**, which yields a scalar, is defined as

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta \quad (\text{B-14})$$

and reads **A** “dot” **B**. The angle  $\theta$  is formed between the *tails* of **A** and **B** ( $0^\circ \leq \theta \leq 180^\circ$ ).

The dot product is commutative; i.e.,

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A} \quad (\text{B-15})$$

The distributive law is valid; i.e.,

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{D}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{D} \quad (\text{B-16})$$

And scalar multiplication can be performed in any manner, i.e.,

$$m(\mathbf{A} \cdot \mathbf{B}) = (m\mathbf{A}) \cdot \mathbf{B} = \mathbf{A} \cdot (m\mathbf{B}) = (\mathbf{A} \cdot \mathbf{B})m \quad (\text{B-17})$$

Using Eq. B-14, the dot product between any two Cartesian vectors can be determined. For example,  $\mathbf{i} \cdot \mathbf{i} = (1)(1) \cos 0^\circ = 1$  and  $\mathbf{i} \cdot \mathbf{j} = (1)(1) \cos 90^\circ = 0$ .

If **A** and **B** are expressed in Cartesian component form, then the dot product, Eq. C-14, can be determined from

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z \quad (\text{B-18})$$

The dot product may be used to determine the *angle  $\theta$  formed between two vectors*. From Eq. B-14,

$$\theta = \cos^{-1} \left( \frac{\mathbf{A} \cdot \mathbf{B}}{AB} \right) \quad (\text{B-19})$$

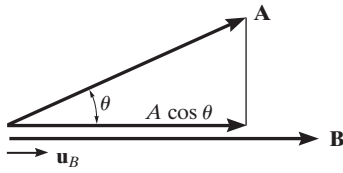


Fig. B-6

It is also possible to find the *component of a vector in a given direction* using the dot product. For example, the magnitude of the component (or projection) of vector  $\mathbf{A}$  in the direction of  $\mathbf{B}$ , Fig. B-6, is defined by  $A \cos \theta$ . From Eq. B-14, this magnitude is

$$A \cos \theta = \mathbf{A} \cdot \frac{\mathbf{B}}{B} = \mathbf{A} \cdot \mathbf{u}_B \quad (\text{B-20})$$

where  $\mathbf{u}_B$  represents a unit vector acting in the direction of  $\mathbf{B}$ , Fig. B-6.

**Differentiation and Integration of Vector Functions.** The rules for differentiation and integration of the sums and products of scalar functions also apply to vector functions. Consider, for example, the two vector functions  $\mathbf{A}(s)$  and  $\mathbf{B}(s)$ . Provided these functions are smooth and continuous for all  $s$ , then

$$\frac{d}{ds}(\mathbf{A} + \mathbf{B}) = \frac{d\mathbf{A}}{ds} + \frac{d\mathbf{B}}{ds} \quad (\text{B-21})$$

$$\int (\mathbf{A} + \mathbf{B}) ds = \int \mathbf{A} ds + \int \mathbf{B} ds \quad (\text{B-22})$$

For the cross product,

$$\frac{d}{ds}(\mathbf{A} \times \mathbf{B}) = \left( \frac{d\mathbf{A}}{ds} \times \mathbf{B} \right) + \left( \mathbf{A} \times \frac{d\mathbf{B}}{ds} \right) \quad (\text{B-23})$$

Similarly, for the dot product,

$$\frac{d}{ds}(\mathbf{A} \cdot \mathbf{B}) = \frac{d\mathbf{A}}{ds} \cdot \mathbf{B} + \mathbf{A} \cdot \frac{d\mathbf{B}}{ds} \quad (\text{B-24})$$

# The Chain Rule

The chain rule of calculus can be used to determine the time derivative of a composite function. For example, if  $y$  is a function of  $x$  and  $x$  is a function of  $t$ , then we can find the derivative of  $y$  with respect to  $t$  as follows

$$\dot{y} = \frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} \quad (\text{C-1})$$

In other words, to find  $\dot{y}$  we take the ordinary derivative  $(dy/dx)$  and multiply it by the time derivative  $(dx/dt)$ .

If several variables are functions of time and they are multiplied together, then the product rule  $d(uv) = du v + u dv$  must be used along with the chain rule when taking the time derivatives. Here are some examples.

**EXAMPLE C-1**

If  $y = x^3$  and  $x = t^4$ , find  $\ddot{y}$ , the second derivative of  $y$  with respect to time.

**SOLUTION**

Using the chain rule, Eq. C-1,

$$\dot{y} = 3x^2\dot{x}$$

To obtain the second time derivative we must use the product rule since  $x$  and  $\dot{x}$  are both functions of time, and also, for  $3x^2$  the chain rule must be applied. Thus, with  $u = 3x^2$  and  $v = \dot{x}$ , we have

$$\begin{aligned}\ddot{y} &= [6x\dot{x}]\dot{x} + 3x^2[\ddot{x}] \\ &= 3x[2\dot{x}^2 + x\ddot{x}]\end{aligned}$$

Since  $x = t^4$ , then  $\dot{x} = 4t^3$  and  $\ddot{x} = 12t^2$  so that

$$\begin{aligned}\ddot{y} &= 3(t^4)[2(4t^3)^2 + t^4(12t^2)] \\ &= 132t^{10}\end{aligned}$$

Note that this result can also be obtained by combining the functions, then taking the time derivatives, that is,

$$\begin{aligned}y &= x^3 = (t^4)^3 = t^{12} \\ \dot{y} &= 12t^{11} \\ \ddot{y} &= 132t^{10}\end{aligned}$$

**EXAMPLE C-2**

If  $y = xe^x$ , find  $\ddot{y}$ .

**SOLUTION**

Since  $x$  and  $e^x$  are both functions of time the product and chain rules must be applied. Have  $u = x$  and  $v = e^x$ .

$$\dot{y} = [\dot{x}]e^x + x[e^x\dot{x}]$$

The second time derivative also requires application of the product and chain rules. Note that the product rule applies to the three time variables in the last term, i.e.,  $x$ ,  $e^x$ , and  $\dot{x}$ .

$$\begin{aligned}\ddot{y} &= \{[\ddot{x}]e^x + \dot{x}[e^x\dot{x}]\} + \{[\dot{x}]e^x\dot{x} + x[e^x\dot{x}]\dot{x} + xe^x[\ddot{x}]\} \\ &= e^x[\ddot{x}(1 + x) + \dot{x}^2(2 + x)]\end{aligned}$$

If  $x = t^2$  then  $\dot{x} = 2t$ ,  $\ddot{x} = 2$  so that in terms in  $t$ , we have

$$\ddot{y} = e^{t^2}[2(1 + t^2) + 4t^2(2 + t^2)]$$

**EXAMPLE C-3**

If the path in radial coordinates is given as  $r = 5\theta^2$ , where  $\theta$  is a known function of time, find  $\ddot{r}$ .

**SOLUTION**

First, using the chain rule then the chain and product rules where  $u = 10\theta$  and  $v = \dot{\theta}$ , we have

$$\begin{aligned} r &= 5\theta^2 \\ \dot{r} &= 10\theta\dot{\theta} \\ \ddot{r} &= 10[(\dot{\theta})\dot{\theta} + \theta(\ddot{\theta})] \\ &= 10\dot{\theta}^2 + 10\theta\ddot{\theta} \end{aligned}$$

**EXAMPLE C-4**

If  $r^2 = 6\theta^3$ , find  $\ddot{r}$ .

**SOLUTION**

Here the chain and product rules are applied as follows.

$$\begin{aligned} r^2 &= 6\theta^3 \\ 2r\dot{r} &= 18\theta^2\dot{\theta} \\ 2[(\dot{r})\dot{r} + r(\ddot{r})] &= 18[(2\theta\dot{\theta})\dot{\theta} + \theta^2(\ddot{\theta})] \\ \dot{r}^2 + r\ddot{r} &= 9(2\theta\dot{\theta}^2 + \theta^2\ddot{\theta}) \end{aligned}$$

To find  $\ddot{r}$  at a specified value of  $\theta$  which is a known function of time, we can first find  $\dot{\theta}$  and  $\ddot{\theta}$ . Then using these values, evaluate  $r$  from the first equation,  $\dot{r}$  from the second equation and  $\ddot{r}$  using the last equation.

# Fundamental Problems

## Partial Solutions and Answers

### Chapter 12

**F12-1.**  $v = v_0 + a_c t$   
 $10 = 35 + a_c(15)$   
 $a_c = -1.67 \text{ m/s}^2 = 1.67 \text{ m/s}^2 \leftarrow$

Ans.

**F12-2.**  $s = s_0 + v_0 t + \frac{1}{2} a_c t^2$   
 $0 = 0 + 15t + \frac{1}{2}(-9.81)t^2$   
 $t = 3.06 \text{ s}$

Ans.

**F12-3.**  $ds = v dt$   
 $\int_0^s ds = \int_0^t (4t - 3t^2) dt$   
 $s = (2t^2 - t^3) \text{ m}$   
 $s = 2(4^2) - 4^3$   
 $= -32 \text{ m} = 32 \text{ m} \leftarrow$

Ans.

**F12-4.**  $a = \frac{dv}{dt} = \frac{d}{dt}(0.5t^3 - 8t)$   
 $a = (1.5t^2 - 8) \text{ m/s}^2$

When  $t = 2 \text{ s}$ ,  
 $a = 1.5(2^2) - 8 = -2 \text{ m/s}^2 = 2 \text{ m/s}^2 \leftarrow$

Ans.

**F12-5.**  $v = \frac{ds}{dt} = \frac{d}{dt}(2t^2 - 8t + 6) = (4t - 8) \text{ m/s}$   
 $v = 0 = (4t - 8)$   
 $t = 2 \text{ s}$   
 $s|_{t=0} = 2(0^2) - 8(0) + 6 = 6 \text{ m}$   
 $s|_{t=2} = 2(2^2) - 8(2) + 6 = -2 \text{ m}$   
 $s|_{t=3} = 2(3^2) - 8(3) + 6 = 0 \text{ m}$   
 $(\Delta s)_{\text{Tot}} = 8 \text{ m} + 2 \text{ m} = 10 \text{ m}$

Ans.

**F12-6.**  $\int v dv = \int a ds$   
 $\int_{5 \text{ m/s}}^v v dv = \int_0^s (10 - 0.2s) ds$   
 $v = (\sqrt{20s - 0.2s^2} + 25) \text{ m/s}$   
 At  $s = 10 \text{ m}$ ,  
 $v = \sqrt{20(10) - 0.2(10^2)} + 25$   
 $= 14.3 \text{ m/s} \rightarrow$

Ans.

**F12-7.**  $v = \int (4t^2 - 2) dt$   
 $v = \frac{4}{3} t^3 - 2t + C_1$   
 $s = \int (\frac{4}{3} t^3 - 2t + C_1) dt$   
 $s = \frac{1}{3} t^4 - t^2 + C_1 t + C_2$   
 $t = 0, s = -2, C_2 = -2$   
 $t = 2, s = -20, C_1 = -9.67$   
 $t = 4, s = 28.7 \text{ m}$

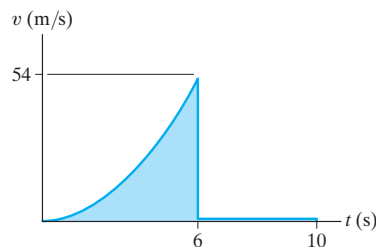
Ans.

**F12-8.**  $a = v \frac{dv}{ds}$   
 $= (20 - 0.05s^2)(-0.1s)$   
 At  $s = 15 \text{ m}$ ,  
 $a = -13.1 \text{ m/s}^2 = 13.1 \text{ m/s}^2 \leftarrow$

Ans.

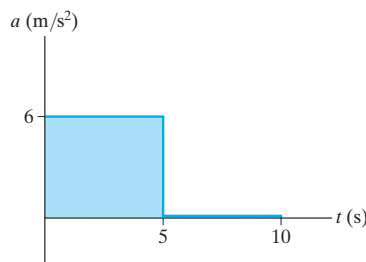
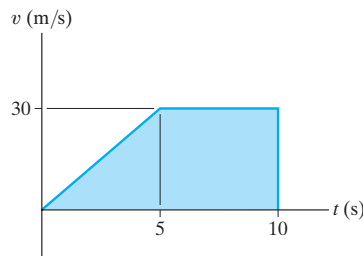
**F12-9.**  $v = \frac{ds}{dt} = \frac{d}{dt}(0.5t^3) = 1.5t^2$   
 $v = \frac{ds}{dt} = \frac{d}{dt}(108) = 0$

Ans.



**F12-10.**  $0 \leq t < 5 \text{ s}$ ,  
 $v = \frac{ds}{dt} = \frac{d}{dt}(3t^2) = (6t) \text{ m/s}$   
 $5 \text{ s} < t \leq 10 \text{ s}$ ,  
 $v = \frac{ds}{dt} = \frac{d}{dt}(30t - 75) = 30 \text{ m/s}$   
 $v = \frac{\Delta s}{\Delta t} = \frac{225 \text{ m} - 75 \text{ m}}{10 \text{ m} - 5 \text{ m}} = 30 \text{ m/s}$

$0 \leq t < 5 \text{ s}$ ,  
 $a = \frac{dv}{dt} = \frac{d}{dt}(6t) = 6 \text{ m/s}^2$   
 $5 \text{ s} < t \leq 10 \text{ s}$ ,  
 $a = \frac{dv}{dt} = \frac{d}{dt}(30) = 0$   
 $0 \leq t < 5 \text{ s}, a = \Delta v / \Delta t = 6 \text{ m/s}^2$   
 $5 \text{ s} < t \leq 10 \text{ s}, a = \Delta v / \Delta t = 0$

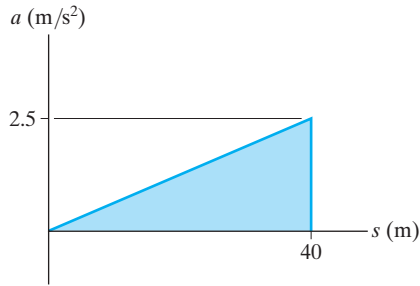




**F12-11.**  $a \, ds = v \, dv$ 

$$a = v \frac{dv}{ds} = 0.25s \frac{d}{ds} (0.25s) = 0.0625s$$

$$a|_{s=40 \text{ m}} = 0.0625(40 \text{ m}) = 2.5 \text{ m/s}^2 \rightarrow$$

**F12-12.** For  $0 \leq s \leq 10 \text{ m}$ 

$$a = s$$

$$\int_0^v v \, dv = \int_0^s s \, ds$$

$$v = s$$

$$\text{at } s = 10 \text{ m}, v = 10 \text{ m/s}$$

$$\text{For } 10 \text{ m} \leq s \leq 15$$

$$a = 10$$

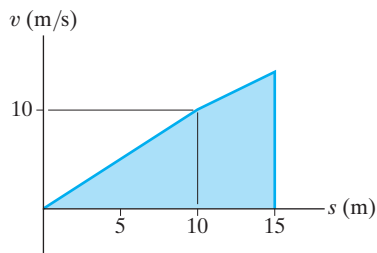
$$\int_{10}^v v \, dv = \int_{10}^s 10 \, ds$$

$$\frac{1}{2}v^2 - 50 = 10s - 100$$

$$v = \sqrt{20s - 100}$$

$$\text{at } s = 15 \text{ m}$$

$$v = 14.1 \text{ m/s}$$

**F12-13.**  $0 \leq t < 5 \text{ s}$ ,

$$dv = a \, dt \quad \int_0^v dv = \int_0^t 20 \, dt$$

$$v = (20t) \text{ m/s}$$

$$5 \text{ s} < t \leq t',$$

$$(\rightarrow) \quad dv = a \, dt \quad \int_{100 \text{ m/s}}^v dv = \int_{5 \text{ s}}^t -10 \, dt$$

$$v - 100 = (50 - 10t) \text{ m/s},$$

$$0 = 150 - 10t'$$

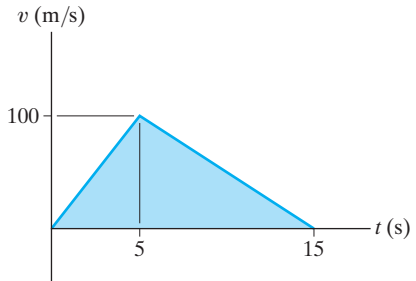
$$t' = 15 \text{ s}$$

Also,

$$\Delta v = 0 = \text{Area under the } a-t \text{ graph}$$

$$0 = (20 \text{ m/s}^2)(5 \text{ s}) + [-(10 \text{ m/s})(t' - 5) \text{ s}]$$

$$t' = 15 \text{ s}$$

**F12-14.**  $0 \leq t \leq 5 \text{ s}$ ,

$$ds = v \, dt \quad \int_0^s ds = \int_0^t 30t \, dt$$

$$s|_0^s = 15t^2|_0^t$$

$$s = (15t^2) \text{ m}$$

$$5 \text{ s} < t \leq 15 \text{ s},$$

$$(\rightarrow) \, ds = v \, dt; \quad \int_{375 \text{ m}}^s ds = \int_{5 \text{ s}}^t (-15t + 225) dt$$

$$s = (-7.5t^2 + 225t - 562.5) \text{ m}$$

$$s = (-7.5)(15)^2 + 225(15) - 562.5 \text{ m}$$

$$= 1125 \text{ m}$$

Ans.

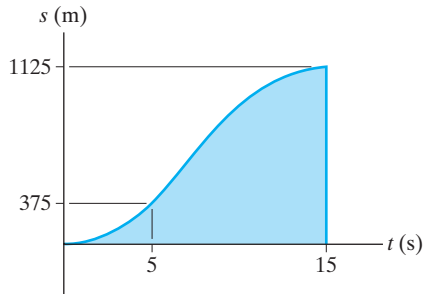
Also,

$$\Delta s = \text{Area under the } v-t \text{ graph}$$

$$= \frac{1}{2} (150 \text{ m/s})(15 \text{ s})$$

$$= 1125 \text{ m}$$

Ans.



$$\text{F12-15.} \quad \int_0^x dx = \int_0^t 32t \, dt$$

$$x = (16t^2) \text{ m}$$

(1)

$$\int_0^y dy = \int_0^t 8 \, dt$$

$$t = \frac{y}{8}$$

(2)

Substituting Eq. (2) into Eq. (1), get

$$y = 2\sqrt{x}$$

*Ans.*

**F12-16.**  $y = 0.75(8t) = 6t$

$$v_x = \dot{x} = \frac{dx}{dt} = \frac{d}{dt}(8t) = 8 \text{ m/s} \rightarrow$$

$$v_y = \dot{y} = \frac{dy}{dt} = \frac{d}{dt}(6t) = 6 \text{ m/s} \uparrow$$

The magnitude of the particle's velocity is

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(8 \text{ m/s})^2 + (6 \text{ m/s})^2}$$

$$= 10 \text{ m/s}$$

*Ans.*

**F12-17.**  $y = (4t^2) \text{ m}$

$$v_x = \dot{x} = \frac{d}{dt}(4t^4) = (16t^3) \text{ m/s} \rightarrow$$

$$v_y = \dot{y} = \frac{d}{dt}(4t^2) = (8t) \text{ m/s} \uparrow$$

When  $t = 0.5 \text{ s}$ ,

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(2 \text{ m/s})^2 + (4 \text{ m/s})^2}$$

$$= 4.47 \text{ m/s}$$

*Ans.*

$$a_x = \dot{v}_x = \frac{d}{dt}(16t^3) = (48t^2) \text{ m/s}^2$$

$$a_y = \dot{v}_y = \frac{d}{dt}(8t) = 8 \text{ m/s}^2$$

When  $t = 0.5 \text{ s}$ ,

$$a = \sqrt{a_x^2 + a_y^2} = \sqrt{(12 \text{ m/s}^2)^2 + (8 \text{ m/s}^2)^2}$$

$$= 14.4 \text{ m/s}^2$$

*Ans.*

**F12-18.**  $y = 0.5x$

$$\dot{y} = 0.5\dot{x}$$

$$v_y = t^2$$

When  $t = 4 \text{ s}$ ,

$$v_x = 32 \text{ m/s} \quad v_y = 16 \text{ m/s}$$

$$v = \sqrt{v_x^2 + v_y^2} = 35.8 \text{ m/s}$$

*Ans.*

$$a_x = \dot{v}_x = 4t$$

$$a_y = \dot{v}_y = 2t$$

When  $t = 4 \text{ s}$ ,

$$a_x = 16 \text{ m/s}^2 \quad a_y = 8 \text{ m/s}^2$$

$$a = \sqrt{a_x^2 + a_y^2} = \sqrt{16^2 + 8^2} = 17.9 \text{ m/s}^2$$

*Ans.*

**F12-19.**  $v_y = \dot{y} = 0.5x\dot{x} = 0.5(8)(8) = 32 \text{ m/s}$

Thus,

$$v = \sqrt{v_x^2 + v_y^2} = 33.0 \text{ m/s}$$

*Ans.*

$$a_y = \dot{v}_y = 0.5\dot{x}^2 + 0.5x\ddot{x}$$

$$= 0.5(8)^2 + 0.5(8)(4)$$

$$= 48 \text{ m/s}^2$$

$$a_x = 4 \text{ m/s}^2$$

Thus,

$$a = \sqrt{a_x^2 + a_y^2} = \sqrt{4^2 + 48^2} = 48.2 \text{ m/s}^2 \quad \text{Ans.}$$

**F12-20.**  $\dot{y} = 0.1x\dot{x}$

$$v_y = 0.1(5)(-3) = -1.5 \text{ m/s} = 1.5 \text{ m/s} \downarrow \quad \text{Ans.}$$

$$\ddot{y} = 0.1[\dot{x}\dot{x} + x\ddot{x}]$$

$$a_y = 0.1[(-3)^2 + 5(-1.5)] = 0.15 \text{ m/s}^2 \uparrow \quad \text{Ans.}$$

**F12-21.**  $(v_B)_y^2 = (v_A)_y^2 + 2a_y(v_B - v_A)$

$$0^2 = (5 \text{ m/s})^2 + 2(-9.81 \text{ m/s}^2)(h - 0)$$

$$h = 1.27 \text{ m}$$

*Ans.*

**F12-22.**  $y_C = y_A + (v_A)_y t_{AC} + \frac{1}{2} a_y t_{AC}^2$

$$0 = 0 + (5 \text{ m/s}) t_{AC} + \frac{1}{2} (-9.81 \text{ m/s}^2) t_{AC}^2$$

$$t_{AC} = 1.0194 \text{ s}$$

$$(v_C)_y = (v_A)_y + a_y t_{AC}$$

$$(v_C)_y = 5 \text{ m/s} + (-9.81 \text{ m/s}^2)(1.0194 \text{ s})$$

$$= -5 \text{ m/s} = 5 \text{ m/s} \downarrow$$

$$v_C = \sqrt{(v_C)_x^2 + (v_C)_y^2}$$

$$= \sqrt{(8.660 \text{ m/s})^2 + (5 \text{ m/s})^2} = 10 \text{ m/s} \quad \text{Ans.}$$

$$R = x_A + (v_A)_x t_{AC} = 0 + (8.660 \text{ m/s})(1.0194 \text{ s})$$

$$= 8.83 \text{ m}$$

*Ans.*

**F12-23.**  $s = s_0 + v_0 t$

$$10 = 0 + v_A \cos 30^\circ t$$

$$s = s_0 + v_0 t + \frac{1}{2} a_c t^2$$

$$3 = 1.5 + v_A \sin 30^\circ t + \frac{1}{2} (-9.81) t^2$$

$$t = 0.9334 \text{ s}, \quad v_A = 12.4 \text{ m/s}$$

*Ans.*

**F12-24.**  $s = s_0 + v_0 t$

$$R\left(\frac{4}{5}\right) = 0 + 20\left(\frac{3}{5}\right)t$$

$$s = s_0 + v_0 t + \frac{1}{2} a_c t^2$$

$$-R\left(\frac{3}{5}\right) = 0 + 20\left(\frac{4}{5}\right)t + \frac{1}{2} (-9.81)t^2$$

$$t = 5.10 \text{ s}$$

$$R = 76.5 \text{ m}$$

*Ans.*

**F12-25.**  $x_B = x_A + (v_A)_x t_{AB}$

$$3.6 \text{ m} = 0 + (0.8660 v_A) t_{AB}$$

$$v_A t_{AB} = 4.157$$

(1)

$$y_B = y_A + (v_A)_y t_{AB} + \frac{1}{2} a_y t_{AB}^2$$

$$(2.4 - 0.9) \text{ m} = 0 + 0.5v_{AtAB} + \frac{1}{2}(-9.81 \text{ m/s}^2)t_{AB}^2$$

Using Eq. (1),

$$1.5 = 0.5(4.157) - 4.905 t_{AB}^2$$

$$t_{AB} = 0.3434 \text{ s}$$

$$v_A = 12.1 \text{ m/s}$$

*Ans.*

**F12-26.**  $y_B = y_A + (v_A)_y t_{AB} + \frac{1}{2} a_y t_{AB}^2$

$$-150 \text{ m} = 0 + (90 \text{ m/s})t_{AB} + \frac{1}{2}(-9.81 \text{ m/s}^2)t_{AB}^2$$

$$t_{AB} = 19.89 \text{ s}$$

$$x_B = x_A + (v_A)_x t_{AB}$$

$$R = 0 + 120 \text{ m/s}(19.89 \text{ s}) = 2386.37 \text{ m}$$

$$= 2.39 \text{ km}$$

*Ans.*

**F12-27.**  $a_t = \dot{v} = \frac{dv}{dt} = \frac{d}{dt}(0.0625t^2) = (0.125t) \text{ m/s}^2 \Big|_{t=10 \text{ s}}$

$$= 1.25 \text{ m/s}^2$$

$$a_n = \frac{v^2}{\rho} = \frac{(0.0625t^2)^2}{40 \text{ m}} = [97.656(10^{-6})t^4] \text{ m/s}^2 \Big|_{t=10 \text{ s}}$$

$$= 0.9766 \text{ m/s}^2$$

$$a = \sqrt{a_t^2 + a_n^2} = \sqrt{(1.25 \text{ m/s}^2)^2 + (0.9766 \text{ m/s}^2)^2}$$

$$= 1.59 \text{ m/s}^2$$

*Ans.*

**F12-28.**  $v = 2s \Big|_{s=10} = 20 \text{ m/s}$

$$a_n = \frac{v^2}{\rho} = \frac{(20 \text{ m/s})^2}{50 \text{ m}} = 8 \text{ m/s}^2$$

$$a_t = v \frac{dv}{ds} = 4s \Big|_{s=10} = 40 \text{ m/s}^2$$

$$a = \sqrt{a_t^2 + a_n^2} = \sqrt{(40 \text{ m/s}^2)^2 + (8 \text{ m/s}^2)^2}$$

$$= 40.8 \text{ m/s}^2$$

*Ans.*

**F12-29.**  $v_C^2 = v_A^2 + 2a_t(s_C - s_A)$

$$(15 \text{ m/s})^2 = (25 \text{ m/s})^2 + 2a_t(300 \text{ m} - 0)$$

$$a_t = -0.6667 \text{ m/s}^2$$

$$v_B^2 = v_A^2 + 2a_t(s_B - s_A)$$

$$v_B^2 = (25 \text{ m/s})^2 + 2(-0.6667 \text{ m/s}^2)(250 \text{ m} - 0)$$

$$v_B = 17.08 \text{ m/s}$$

$$(a_B)_n = \frac{v_B^2}{\rho} = \frac{(17.08 \text{ m/s})^2}{300 \text{ m}} = 0.9722 \text{ m/s}^2$$

$$\begin{aligned} a_B &= \sqrt{(a_B)_t^2 + (a_B)_n^2} \\ &= \sqrt{(-0.6667 \text{ m/s}^2)^2 + (0.9722 \text{ m/s}^2)^2} \\ &= 1.18 \text{ m/s}^2 \end{aligned}$$

*Ans.*

**F12-30.**  $\tan \theta = \frac{dy}{dx} = \frac{d}{dx} \left( \frac{1}{8} x^2 \right) = \frac{1}{4} x$

$$\theta = \tan^{-1} \left( \frac{1}{4} x \right) \Big|_{x=3 \text{ m}}$$

$$= \tan^{-1} \left( \frac{3}{4} \right) = 36.87^\circ = 36.9^\circ \quad \text{Ans.}$$

$$\begin{aligned} \rho &= \frac{[1 + (dy/dx)^2]^{3/2}}{|d^2 y/dx^2|} = \frac{[1 + (\frac{1}{4} x)^2]^{3/2}}{|\frac{1}{4}|} \Big|_{x=3 \text{ m}} \\ &= 7.8125 \text{ m} \end{aligned}$$

$$a_n = \frac{v^2}{\rho} = \frac{(6 \text{ m/s})^2}{7.8125 \text{ m}} = 4.608 \text{ m/s}^2$$

$$\begin{aligned} a &= \sqrt{(a_t)^2 + (a_n)^2} = \sqrt{(2 \text{ m/s}^2)^2 + (4.608 \text{ m/s}^2)^2} \\ &= 5.02 \text{ m/s}^2 \end{aligned}$$

*Ans.*

**F12-31.**  $(a_B)_t = -0.001s = (-0.001)(300 \text{ m}) \left( \frac{\pi}{2} \text{ rad} \right) \text{ m/s}^2$

$$= -0.4712 \text{ m/s}^2$$

$$v dv = a_t ds$$

$$\int_{25 \text{ m/s}}^{v_B} v dv = \int_0^{150\pi \text{ m}} -0.001s ds$$

$$v_B = 20.07 \text{ m/s}$$

$$(a_B)_n = \frac{v_B^2}{\rho} = \frac{(20.07 \text{ m/s})^2}{300 \text{ m}} = 1.343 \text{ m/s}^2$$

$$\begin{aligned} a_B &= \sqrt{(a_B)_t^2 + (a_B)_n^2} \\ &= \sqrt{(-0.4712 \text{ m/s}^2)^2 + (1.343 \text{ m/s}^2)^2} \\ &= 1.42 \text{ m/s}^2 \end{aligned}$$

*Ans.*

**F12-32.**  $a_t ds = v dv$

$$a_t = v \frac{dv}{ds} = (0.2s)(0.2) = (0.04s) \text{ m/s}^2$$

$$a_t = 0.04(50 \text{ m}) = 2 \text{ m/s}^2$$

$$v = 0.2(50 \text{ m}) = 10 \text{ m/s}$$

$$a_n = \frac{v^2}{\rho} = \frac{(10 \text{ m/s})^2}{500 \text{ m}} = 0.2 \text{ m/s}^2$$

$$a = \sqrt{a_t^2 + a_n^2} = \sqrt{(2 \text{ m/s}^2)^2 + (0.2 \text{ m/s}^2)^2}$$

$$= 2.01 \text{ m/s}^2 \quad \text{Ans.}$$

**F12-33.**  $v_r = \dot{r} = 0$

$$v_\theta = r\dot{\theta} = (400\dot{\theta}) \text{ m/s}$$

$$v = \sqrt{v_r^2 + v_\theta^2}$$

$$15 \text{ m/s} = \sqrt{0^2 + [(400\dot{\theta}) \text{ m/s}]^2}$$

$$\dot{\theta} = 0.0375 \text{ rad/s} \quad \text{Ans.}$$

**F12-34.**  $r = 0.1t^3|_{t=1.5 \text{ s}} = 0.3375 \text{ m}$

$$\dot{r} = 0.3t^2|_{t=1.5 \text{ s}} = 0.675 \text{ m/s}$$

$$\dot{r} = 0.6t|_{t=1.5 \text{ s}} = 0.900 \text{ m/s}^2$$

$$\theta = 4t^{3/2}|_{t=1.5 \text{ s}} = 7.348 \text{ rad}$$

$$\dot{\theta} = 6t^{1/2}|_{t=1.5 \text{ s}} = 7.348 \text{ rad/s}$$

$$\ddot{\theta} = 3t^{-1/2}|_{t=1.5 \text{ s}} = 2.449 \text{ rad/s}^2$$

$$v_r = \dot{r} = 0.675 \text{ m/s}$$

$$v_\theta = r\dot{\theta} = (0.3375 \text{ m})(7.348 \text{ rad/s}) = 2.480 \text{ m/s}$$

$$a_r = \ddot{r} - r\dot{\theta}^2$$

$$= (0.900 \text{ m/s}^2) - (0.3375 \text{ m})(7.348 \text{ rad/s})^2$$

$$= -17.325 \text{ m/s}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = (0.3375 \text{ m})(2.449 \text{ rad/s}^2)$$

$$+ 2(0.675 \text{ m/s})(7.348 \text{ rad/s}) = 10.747 \text{ m/s}^2$$

$$v = \sqrt{v_r^2 + v_\theta^2}$$

$$= \sqrt{(0.675 \text{ m/s})^2 + (2.480 \text{ m/s})^2}$$

$$= 2.57 \text{ m/s} \quad \text{Ans.}$$

$$a = \sqrt{a_r^2 + a_\theta^2}$$

$$= \sqrt{(-17.325 \text{ m/s}^2)^2 + (10.747 \text{ m/s}^2)^2}$$

$$= 20.4 \text{ m/s}^2 \quad \text{Ans.}$$

**F12-35.**  $r = 2\theta$

$$\dot{r} = 2\dot{\theta}$$

$$\ddot{r} = 2\ddot{\theta}$$

At  $\theta = \pi/4 \text{ rad}$ ,

$$r = 2\left(\frac{\pi}{4}\right) = \frac{\pi}{2} \text{ m}$$

$$\dot{r} = 2(3 \text{ rad/s}) = 6 \text{ m/s}$$

$$\ddot{r} = 2(1 \text{ rad/s}) = 2 \text{ m/s}^2$$

$$a_r = \ddot{r} - r\dot{\theta}^2 = 2 \text{ m/s}^2 - \left(\frac{\pi}{2} \text{ m}\right)(3 \text{ rad/s})^2$$

$$= -12.14 \text{ m/s}^2$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

$$= \left(\frac{\pi}{2} \text{ m}\right)(1 \text{ rad/s}^2) + 2(6 \text{ m/s})(3 \text{ rad/s})$$

$$= 37.57 \text{ m/s}^2$$

$$a = \sqrt{a_r^2 + a_\theta^2}$$

$$= \sqrt{(-12.14 \text{ m/s}^2)^2 + (37.57 \text{ m/s}^2)^2}$$

$$= 39.5 \text{ m/s}^2 \quad \text{Ans.}$$

**F12-36.**  $r = e^\theta$

$$\dot{r} = e^\theta \dot{\theta}$$

$$\ddot{r} = e^\theta \ddot{\theta} + e^\theta \dot{\theta}^2$$

$$a_r = \ddot{r} - r\dot{\theta}^2 = (e^\theta \ddot{\theta} + e^\theta \dot{\theta}^2) - e^\theta \dot{\theta}^2 = e^\theta \ddot{\theta} = e^{\pi/4}(4)$$

$$= 8.77 \text{ m/s}^2 \quad \text{Ans.}$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = (e^\theta \ddot{\theta}) + (2(e^\theta \dot{\theta})\dot{\theta}) = e^\theta(\ddot{\theta} + 2\dot{\theta}^2)$$

$$= e^{\pi/4}(4 + 2(2)^2)$$

$$= 26.3 \text{ m/s}^2 \quad \text{Ans.}$$

**F12-37.**  $r = [0.2(1 + \cos \theta)] \text{ m}|_{\theta=30^\circ} = 0.3732 \text{ m}$

$$\dot{r} = [-0.2(\sin \theta)\dot{\theta}] \text{ m/s}|_{\theta=30^\circ}$$

$$= -0.2 \sin 30^\circ(3 \text{ rad/s})$$

$$= -0.3 \text{ m/s}$$

$$v_r = \dot{r} = -0.3 \text{ m/s}$$

$$v_\theta = r\dot{\theta} = (0.3732 \text{ m})(3 \text{ rad/s}) = 1.120 \text{ m/s}$$

$$v = \sqrt{v_r^2 + v_\theta^2} = \sqrt{(-0.3 \text{ m/s})^2 + (1.120 \text{ m/s})^2}$$

$$= 1.16 \text{ m/s} \quad \text{Ans.}$$

**F12-38.**  $30 \text{ m} = r \sin \theta$

$$r = \left(\frac{30 \text{ m}}{\sin \theta}\right) = (30 \csc \theta) \text{ m}$$

$$r = (30 \csc \theta)|_{\theta=45^\circ} = 42.426 \text{ m}$$

$$\dot{r} = -30 \csc \theta \cot \theta \dot{\theta}|_{\theta=45^\circ} = -(42.426\dot{\theta}) \text{ m/s}$$

$$v_r = \dot{r} = -(42.426\dot{\theta}) \text{ m/s}$$

$$v_\theta = r\dot{\theta} = (42.426\dot{\theta}) \text{ m/s}$$

$$v = \sqrt{v_r^2 + v_\theta^2}$$

$$2 = \sqrt{(-42.426\dot{\theta})^2 + (42.426\dot{\theta})^2}$$

$$\dot{\theta} = 0.0333 \text{ rad/s} \quad \text{Ans.}$$

**F12-39.**  $l_T = 3s_D + s_A$

$$0 = 3v_D + v_A$$

$$0 = 3v_D + 3 \text{ m/s}$$

$$v_D = -1 \text{ m/s} = 1 \text{ m/s} \uparrow \quad \text{Ans.}$$

$$\begin{aligned}\text{F12-40. } s_B + 2s_A + 2h &= l \\ v_B + 2v_A &= 0 \\ 6 + 2v_A &= 0 \quad v_A = -3 \text{ m/s} = 3 \text{ m/s} \uparrow \quad \text{Ans.}\end{aligned}$$

$$\begin{aligned}\text{F12-41. } 3s_A + s_B &= l \\ 3v_A + v_B &= 0 \\ 3v_A + 1.5 &= 0 \quad v_A = -0.5 \text{ m/s} = 0.5 \text{ m/s} \uparrow \quad \text{Ans.}\end{aligned}$$

$$\begin{aligned}\text{F12-42. } l_T &= 4s_A + s_F \\ 0 &= 4v_A + v_F \\ 0 &= 4v_A + 3 \text{ m/s} \\ v_A &= -0.75 \text{ m/s} = 0.75 \text{ m/s} \uparrow \quad \text{Ans.}\end{aligned}$$

$$\begin{aligned}\text{F12-43. } s_A + 2(s_A - a) + (s_A - s_P) &= l \\ 4s_A - s_P &= l + 2a \\ 4v_A - v_P &= 0 \\ 4v_A - (-4) &= 0 \\ 4v_A + 4 &= 0 \quad v_A = -1 \text{ m/s} = 1 \text{ m/s} \nearrow \quad \text{Ans.}\end{aligned}$$

$$\begin{aligned}\text{F12-44. } s_C + s_B &= l_{CED} \quad (1) \\ (s_A - s_C) + (s_B - s_C) + s_B &= l_{ACDF} \\ s_A + 2s_B - 2s_C &= l_{ACDF} \quad (2)\end{aligned}$$

Thus,

$$\begin{aligned}v_C + v_B &= 0 \\ v_A + 2v_B - 2v_C &= 0\end{aligned}$$

Eliminating  $v_C$ ,

$$v_A + 4v_B = 0$$

Thus,

$$4 \text{ m/s} + 4v_B = 0$$

$$v_B = -1 \text{ m/s} = 1 \text{ m/s} \uparrow \quad \text{Ans.}$$

$$\begin{aligned}\text{F12-45. } \mathbf{v}_B &= \mathbf{v}_A + \mathbf{v}_{B/A} \\ 100\mathbf{i} &= 80\mathbf{j} + \mathbf{v}_{B/A} \\ \mathbf{v}_{B/A} &= 100\mathbf{i} - 80\mathbf{j} \\ v_{B/A} &= \sqrt{(v_{B/A})_x^2 + (v_{B/A})_y^2} \\ &= \sqrt{(100 \text{ km/h})^2 + (-80 \text{ km/h})^2} \\ &= 128 \text{ km/h} \quad \text{Ans.}\end{aligned}$$

$$\theta = \tan^{-1} \left[ \frac{(v_{B/A})_y}{(v_{B/A})_x} \right] = \tan^{-1} \left( \frac{80 \text{ km/h}}{100 \text{ km/h}} \right) = 38.7^\circ \swarrow \quad \text{Ans.}$$

$$\begin{aligned}\text{F12-46. } \mathbf{v}_B &= \mathbf{v}_A + \mathbf{v}_{B/A} \\ (-400\mathbf{i} - 692.82\mathbf{j}) &= (650\mathbf{i}) + \mathbf{v}_{B/A} \\ \mathbf{v}_{B/A} &= [-1050\mathbf{i} - 692.82\mathbf{j}] \text{ km/h} \\ v_{B/A} &= \sqrt{(v_{B/A})_x^2 + (v_{B/A})_y^2} \\ &= \sqrt{(1050 \text{ km/h})^2 + (692.82 \text{ km/h})^2} \\ &= 1258 \text{ km/h} \quad \text{Ans.}\end{aligned}$$

$$\theta = \tan^{-1} \left[ \frac{(v_{B/A})_y}{(v_{B/A})_x} \right] = \tan^{-1} \left( \frac{692.82 \text{ km/h}}{1050 \text{ km/h}} \right) = 33.4^\circ \nearrow \quad \text{Ans.}$$

$$\begin{aligned}\text{F12-47. } \mathbf{v}_B &= \mathbf{v}_A + \mathbf{v}_{B/A} \\ (5\mathbf{i} + 8.660\mathbf{j}) &= (12.99\mathbf{i} + 7.5\mathbf{j}) + \mathbf{v}_{B/A} \\ \mathbf{v}_{B/A} &= [-7.990\mathbf{i} + 1.160\mathbf{j}] \text{ m/s} \\ v_{B/A} &= \sqrt{(-7.990 \text{ m/s})^2 + (1.160 \text{ m/s})^2} \\ &= 8.074 \text{ m/s} \\ d_{AB} &= v_{B/A} t = (8.074 \text{ m/s})(4 \text{ s}) = 32.3 \text{ m} \quad \text{Ans.}\end{aligned}$$

$$\begin{aligned}\text{F12-48. } \mathbf{v}_A &= \mathbf{v}_B + \mathbf{v}_{A/B} \\ -20 \cos 45^\circ \mathbf{i} + 20 \sin 45^\circ \mathbf{j} &= 65\mathbf{i} + \mathbf{v}_{A/B} \\ \mathbf{v}_{A/B} &= -79.14\mathbf{i} + 14.14\mathbf{j} \\ v_{A/B} &= \sqrt{(-79.14)^2 + (14.14)^2} \\ &= 80.4 \text{ km/h} \quad \text{Ans.}\end{aligned}$$

$$\begin{aligned}\mathbf{a}_A &= \mathbf{a}_B + \mathbf{a}_{A/B} \\ \frac{(20)^2}{0.1} \cos 45^\circ \mathbf{i} + \frac{(20)^2}{0.1} \sin 45^\circ \mathbf{j} &= 1200\mathbf{i} + \mathbf{a}_{A/B} \\ \mathbf{a}_{A/B} &= 1628\mathbf{i} + 2828\mathbf{j} \\ a_{A/B} &= \sqrt{(1628)^2 + (2828)^2} \\ &= 3.26(10^3) \text{ km/h}^2 \quad \text{Ans.}\end{aligned}$$

## Chapter 13

$$\begin{aligned}\text{F13-1. } s &= s_0 + v_0 t + \frac{1}{2} a_c t^2 \\ 6 \text{ m} &= 0 + 0 + \frac{1}{2} a(3 \text{ s})^2 \\ a &= 1.333 \text{ m/s}^2 \\ \Sigma F_y &= ma_y; \quad N_A - 20(9.81) \text{ N} \cos 30^\circ = 0 \\ N_A &= 169.91 \text{ N} \\ \Sigma F_x &= ma_x; \quad T - 20(9.81) \text{ N} \sin 30^\circ \\ &\quad - 0.3(169.91 \text{ N}) = (20 \text{ kg})(1.333 \text{ m/s}^2) \\ T &= 176 \text{ N} \quad \text{Ans.}\end{aligned}$$

$$\begin{aligned}\text{F13-2. } (F_f)_{\max} &= \mu_s N_A = 0.3(245.25 \text{ N}) = 73.575 \text{ N.} \\ \text{Since } F &= 100 \text{ N} > (F_f)_{\max} \text{ when } t = 0, \text{ the crate} \\ &\text{will start to move immediately after } \mathbf{F} \text{ is applied.} \\ +\uparrow \Sigma F_y &= ma_y; \quad N_A - 25(9.81) \text{ N} = 0 \\ N_A &= 245.25 \text{ N} \\ +\rightarrow \Sigma F_x &= ma_x; \\ 10t^2 + 100 - 0.25(245.25 \text{ N}) &= (25 \text{ kg})a \\ a &= (0.4t^2 + 1.5475) \text{ m/s}^2 \\ dv &= a \, dt \\ \int_0^v dv &= \int_0^{4 \text{ s}} (0.4t^2 + 1.5475) \, dt \\ v &= 14.7 \text{ m/s} \rightarrow \quad \text{Ans.}\end{aligned}$$

**F13-3.**  $\rightarrow \Sigma F_x = ma_x;$   
 $\left(\frac{4}{5}\right)500 \text{ N} - (500s) \text{ N} = (10 \text{ kg})a$   
 $a = (40 - 50s) \text{ m/s}^2$   
 $v dv = a ds$   
 $\int_0^v v dv = \int_0^{0.5 \text{ m}} (40 - 50s) ds$   
 $\frac{v^2}{2} \Big|_0^v = (40s - 25s^2) \Big|_0^{0.5 \text{ m}}$   
 $v = 5.24 \text{ m/s}$

Ans.

**F13-4.**  $\rightarrow \Sigma F_x = ma_x$   $100(s + 1) \text{ N} = (2000 \text{ kg})a$   
 $a = (0.05(s + 1)) \text{ m/s}^2$   
 $v dv = a ds$   
 $\int_0^v v dv = \int_0^{10 \text{ m}} 0.05(s + 1) ds$   
 $v = 2.45 \text{ m/s}$

**F13-5.**  $F_{sp} = k(l - l_0) = (200 \text{ N/m})(0.5 \text{ m} - 0.3 \text{ m})$   
 $= 40 \text{ N}$   
 $\theta = \tan^{-1}\left(\frac{0.3 \text{ m}}{0.4 \text{ m}}\right) = 36.86^\circ$   
 $\rightarrow \Sigma F_x = ma_x;$   
 $100 \text{ N} - (40 \text{ N})\cos 36.86^\circ = (25 \text{ kg})a$   
 $a = 2.72 \text{ m/s}^2$

**F13-6.** Blocks A and B:  
 $\rightarrow \Sigma F_x = ma_x;$   $30 = 35a;$   $a = 0.857 \text{ m/s}^2$   
 Check if slipping occurs between A and B.  
 $\rightarrow \Sigma F_x = ma_x;$   $30 - F = 10(0.857);$   
 $F = 21.43 \text{ N} < 0.4(10)(9.81) = 39.24 \text{ N}$   
 $a_A = a_B = 0.857 \text{ m/s}^2$

Ans.

**F13-7.**  $\Sigma F_n = m \frac{v^2}{\rho};$   $(0.3)m(9.81) = m \frac{v^2}{2}$   
 $v = 2.43 \text{ m/s}$

Ans.

**F13-8.**  $+\downarrow \Sigma F_n = ma_n;$   $m(9.81) = m\left(\frac{v^2}{250}\right)$   
 $v = 27.1 \text{ m/s}$

Ans.

**F13-9.**  $+\downarrow \Sigma F_n = ma_n;$   $70(9.81) + N_p = 70\left(\frac{(36)^2}{120}\right)$   
 $N_p = 69.3 \text{ N}$

Ans.

**F13-10.**  $\leftarrow \Sigma F_n = ma_n;$   
 $N_c \sin 30^\circ + 0.2 N_c \cos 30^\circ = m \frac{v^2}{500}$   
 $+\uparrow \Sigma F_b = 0;$   
 $N_c \cos 30^\circ - 0.2 N_c \sin 30^\circ - m(9.81) = 0$   
 $v = 35.96 \text{ m/s}$

Ans.

**F13-11.**  $\Sigma F_t = ma_t;$   $10(9.81) \text{ N} \cos 45^\circ = (10 \text{ kg})a_t$   
 $a_t = 6.94 \text{ m/s}^2$

Ans.

$\Sigma F_n = ma_n;$   
 $T - 10(9.81) \text{ N} \sin 45^\circ = (10 \text{ kg}) \frac{(3 \text{ m/s})^2}{2 \text{ m}}$   
 $T = 114 \text{ N}$

Ans.

**F13-12.**  $\Sigma F_n = ma_n;$   
 $F_n = (500 \text{ kg}) \frac{(15 \text{ m/s})^2}{200 \text{ m}} = 562.5 \text{ N}$   
 $\Sigma F_t = ma_t;$   
 $F_t = (500 \text{ kg})(1.5 \text{ m/s}^2) = 750 \text{ N}$   
 $F = \sqrt{F_n^2 + F_t^2} = \sqrt{(562.5 \text{ N})^2 + (750 \text{ N})^2}$   
 $= 938 \text{ N}$

Ans.

**F13-13.**  $a_r = \ddot{r} - r\dot{\theta}^2 = 0 - (1.5 \text{ m} + (8 \text{ m})\sin 45^\circ)\dot{\theta}^2$   
 $= (-7.157 \dot{\theta}^2) \text{ m/s}^2$   
 $\Sigma F_z = ma_z;$   
 $T \cos 45^\circ - m(9.81) = m(0)$   $T = 13.87 \text{ m}$   
 $\Sigma F_r = ma_r;$   
 $-(13.87 \text{ m}) \sin 45^\circ = m(-7.157 \dot{\theta}^2)$   
 $\dot{\theta} = 1.17 \text{ rad/s}$

Ans.

**F13-14.**  $\theta = \pi t^2 \Big|_{t=0.5 \text{ s}} = (\pi/4) \text{ rad}$   
 $\dot{\theta} = 2\pi t \Big|_{t=0.5 \text{ s}} = \pi \text{ rad/s}$   
 $\ddot{\theta} = 2\pi \text{ rad/s}^2$   
 $r = 0.6 \sin \theta \Big|_{\theta=\pi/4 \text{ rad}} = 0.4243 \text{ m}$   
 $\dot{r} = 0.6 (\cos \theta)\dot{\theta} \Big|_{\theta=\pi/4 \text{ rad}} = 1.3329 \text{ m/s}$   
 $\ddot{r} = 0.6 [(\cos \theta)\ddot{\theta} - (\sin \theta)\dot{\theta}^2] \Big|_{\theta=\pi/4 \text{ rad}} = -1.5216 \text{ m/s}^2$   
 $a_r = \ddot{r} - r\dot{\theta}^2 = -1.5216 \text{ m/s}^2 - (0.4243 \text{ m})(\pi \text{ rad/s})^2$   
 $= -5.7089 \text{ m/s}^2$   
 $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0.4243 \text{ m}(2\pi \text{ rad/s}^2)$   
 $+ 2(1.3329 \text{ m/s})(\pi \text{ rad/s})$   
 $= 11.0404 \text{ m/s}^2$   
 $\Sigma F_r = ma_r;$   
 $F \cos 45^\circ - N \cos 45^\circ - 0.2(9.81) \cos 45^\circ$   
 $= 0.2(-5.7089)$   
 $\Sigma F_\theta = ma_\theta;$   
 $F \sin 45^\circ + N \sin 45^\circ - 0.2(9.81) \sin 45^\circ$   
 $= 0.2(11.0404)$   
 $N = 2.37 \text{ N}$   $F = 2.72 \text{ N}$

Ans.

**F13-15.**  $r = 50e^{2\theta} \Big|_{\theta=\pi/6 \text{ rad}} = [50e^{2(\pi/6)}] \text{ m} = 142.48 \text{ m}$   
 $\dot{r} = 50(2e^{2\theta} \dot{\theta}) = 100e^{2\theta} \dot{\theta} \Big|_{\theta=\pi/6 \text{ rad}}$   
 $= [100e^{2(\pi/6)}(0.05)] = 14.248 \text{ m/s}$   
 $\ddot{r} = 100[(2e^{2\theta} \ddot{\theta}) + e^{2\theta}(\dot{\theta}^2)] \Big|_{\theta=\pi/6 \text{ rad}}$   
 $= 100[2e^{2(\pi/6)}(0.05^2) + e^{2(\pi/6)}(0.01)]$   
 $= 4.274 \text{ m/s}^2$   
 $a_r = \ddot{r} - r\dot{\theta}^2 = 4.274 \text{ m/s}^2 - 142.48 \text{ m}(0.05 \text{ rad/s})^2$   
 $= 3.918 \text{ m/s}^2$   
 $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 142.48 \text{ m}(0.01 \text{ rad/s}^2)$   
 $+ 2(14.248 \text{ m/s})(0.05 \text{ rad/s})$   
 $= 2.850 \text{ m/s}^2$   
 $\Sigma F_r = ma_r;$   
 $F_r = (2000 \text{ kg})(3.918 \text{ m/s}^2) = 7836.55 \text{ N}$   
 $\Sigma F_\theta = ma_\theta;$   
 $F_\theta = (2000 \text{ kg})(2.850 \text{ m/s}^2) = 5699.31 \text{ N}$   
 $F = \sqrt{F_r^2 + F_\theta^2}$   
 $= \sqrt{(7836.55 \text{ N})^2 + (5699.31 \text{ N})^2}$   
 $= 9689.87 \text{ N} = 9.69 \text{ kN}$

**F13-16.**  $r = (0.6 \cos 2\theta) \text{ m} \Big|_{\theta=0^\circ} = [0.6 \cos 2(0^\circ)] \text{ m} = 0.6 \text{ m}$   
 $\dot{r} = (-1.2 \sin 2\theta \dot{\theta}) \text{ m/s} \Big|_{\theta=0^\circ}$   
 $= [-1.2 \sin 2(0^\circ)(-3)] \text{ m/s} = 0$   
 $\ddot{r} = -1.2(\sin 2\theta \ddot{\theta} + 2\cos 2\theta \dot{\theta}^2) \text{ m/s}^2 \Big|_{\theta=0^\circ}$   
 $= -21.6 \text{ m/s}^2$   
 Thus,  
 $a_r = \ddot{r} - r\dot{\theta}^2 = -21.6 \text{ m/s}^2 - 0.6 \text{ m}(-3 \text{ rad/s})^2$   
 $= -27 \text{ m/s}^2$   
 $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0.6 \text{ m}(0) + 2(0)(-3 \text{ rad/s}) = 0$   
 $\Sigma F_\theta = ma_\theta; F - 0.2(9.81) \text{ N} = 0.2 \text{ kg}(0)$   
 $F = 1.96 \text{ N} \uparrow$  *Ans.*

## Chapter 14

**F14-1.**  $T_1 + \Sigma U_{1-2} = T_2$   
 $0 + \left(\frac{4}{5}\right)(500 \text{ N})(0.5 \text{ m}) - \frac{1}{2}(500 \text{ N/m})(0.5 \text{ m})^2$   
 $= \frac{1}{2}(10 \text{ kg})v^2$   
 $v = 5.24 \text{ m/s}$  *Ans.*

**F14-2.**  $\Sigma F_y = ma_y; N_A - 20(9.81) \text{ N} \cos 30^\circ = 0$   
 $N_A = 169.91 \text{ N}$   
 $T_1 + \Sigma U_{1-2} = T_2$

$$0 + 300 \text{ N}(10 \text{ m}) - 0.3(169.91 \text{ N})(10 \text{ m})$$

$$- 20(9.81) \text{ N}(10 \text{ m}) \sin 30^\circ$$

$$= \frac{1}{2}(20 \text{ kg})v^2$$

$$v = 12.3 \text{ m/s}$$
 *Ans.*

**F14-3.**  $T_1 + \Sigma U_{1-2} = T_2$   
 $0 + 2 \left[ \int_0^{15 \text{ m}} (600 + 2s^2) \text{ N } ds \right] - 100(9.81) \text{ N}(15 \text{ m})$   
 $= \frac{1}{2}(100 \text{ kg})v^2$   
 $v = 12.5 \text{ m/s}$  *Ans.*

**F14-4.**  $T_1 + \Sigma U_{1-2} = T_2$   
 $\frac{1}{2}(1800 \text{ kg})(125 \text{ m/s})^2 - \left[ \frac{(50\,000 \text{ N} + 20\,000 \text{ N})}{2}(400 \text{ m}) \right]$   
 $= \frac{1}{2}(1800 \text{ kg})v^2$   
 $v = 8.33 \text{ m/s}$  *Ans.*

**F14-5.**  $T_1 + \Sigma U_{1-2} = T_2$   
 $\frac{1}{2}(10 \text{ kg})(5 \text{ m/s})^2 + 100 \text{ N}s' + [10(9.81) \text{ N}] s' \sin 30^\circ$   
 $- \frac{1}{2}(200 \text{ N/m})(s')^2 = 0$   
 $s' = 2.09 \text{ m}$   
 $s = 0.6 \text{ m} + 2.09 \text{ m} = 2.69 \text{ m}$  *Ans.*

**F14-6.**  $T_A + \Sigma U_{A-B} = T_B$   
 Consider difference in cord length  $AC - BC$ ,  
 which is distance  $F$  moves.  
 $0 + 50 \text{ N}(\sqrt{(1.5 \text{ m})^2 + (2 \text{ m})^2} - (1.5 \text{ m}))$   
 $= \frac{1}{2}(2.5 \text{ kg})v_B^2$   
 $v_B = 6.32 \text{ m/s}$  *Ans.*

**F14-7.**  $\rightarrow \Sigma F_x = ma_x;$   
 $30\left(\frac{4}{5}\right) = 20a \quad a = 1.2 \text{ m/s}^2 \rightarrow$   
 $v = v_0 + a_c t$   
 $v = 0 + 1.2(4) = 4.8 \text{ m/s}$   
 $P = \mathbf{F} \cdot \mathbf{v} = F(\cos \theta)v$   
 $= 30\left(\frac{4}{5}\right)(4.8)$   
 $= 115 \text{ W}$  *Ans.*

**F14-8.**  $\rightarrow \Sigma F_x = ma_x;$   
 $10s = 20a \quad a = 0.5s \text{ m/s}^2 \rightarrow$   
 $v dv = a ds$   
 $\int_1^v v dv = \int_0^{5 \text{ m}} 0.5 s ds$   
 $v = 3.674 \text{ m/s}$   
 $P = \mathbf{F} \cdot \mathbf{v} = [10(5)](3.674) = 184 \text{ W}$  *Ans.*

**F14-9.**  $(+\uparrow)\Sigma F_y = 0;$   
 $T_1 - 100 \text{ N} = 0 \quad T_1 = 100 \text{ N}$   
 $(+\uparrow)\Sigma F_y = 0;$   
 $100 \text{ N} + 100 \text{ N} - T_2 = 0 \quad T_2 = 200 \text{ N}$   
 $P_{\text{out}} = \mathbf{T}_B \cdot \mathbf{v}_B = (200 \text{ N})(3 \text{ m/s}) = 600 \text{ W}$   
 $P_{\text{in}} = \frac{P_{\text{out}}}{\varepsilon} = \frac{600 \text{ W}}{0.8} = 750 \text{ W}$  *Ans.*

**F14-10.**  $\Sigma F_{y'} = ma_{y'}; \quad N - 20(9.81) \cos 30^\circ = 20(0)$   
 $N = 169.91 \text{ N}$

$\Sigma F_{x'} = ma_{x'};$   
 $F - 20(9.81) \sin 30^\circ - 0.2(169.91) = 0$   
 $F = 132.08 \text{ N}$   
 $P = \mathbf{F} \cdot \mathbf{v} = 132.08(5) = 660 \text{ W}$  *Ans.*

**F14-11.**  $+\uparrow \Sigma F_y = ma_y;$   
 $T - 50(9.81) = 50(0) \quad T = 490.5 \text{ N}$   
 $P_{\text{out}} = \mathbf{T} \cdot \mathbf{v} = 490.5(1.5) = 735.75 \text{ W}$   
 Also, for a point on the other cable  
 $P_{\text{out}} = \left(\frac{490.5}{2}\right)(1.5)(2) = 735.75 \text{ W}$   
 $P_{\text{in}} = \frac{P_{\text{out}}}{\varepsilon} = \frac{735.75}{0.8} = 920 \text{ W}$  *Ans.*

**F14-12.**  $2s_A + s_P = l$   
 $2a_A + a_P = 0$   
 $2a_A + 6 = 0$   
 $a_A = -3 \text{ m/s}^2 = 3 \text{ m/s}^2 \uparrow$   
 $\Sigma F_y = ma_y; \quad T_A - 490.5 \text{ N} = (50 \text{ kg})(3 \text{ m/s}^2)$   
 $T_A = 640.5 \text{ N}$   
 $P_{\text{out}} = \mathbf{T} \cdot \mathbf{v} = (640.5 \text{ N}/2)(12) = 3843 \text{ W}$   
 $P_{\text{in}} = \frac{P_{\text{out}}}{\varepsilon} = \frac{3843}{0.8} = 4803.75 \text{ W} = 4.80 \text{ kW}$  *Ans.*

**F14-13.**  $T_A + V_A = T_B + V_B$   
 $0 + 2(9.81)(1.5) = \frac{1}{2}(2)(v_B)^2 + 0$   
 $v_B = 5.42 \text{ m/s}$  *Ans.*  
 $+\uparrow \Sigma F_n = ma_n; \quad T - 2(9.81) = 2\left(\frac{(5.42)^2}{1.5}\right)$   
 $T = 58.9 \text{ N}$  *Ans.*

**F14-14.**  $T_A + V_A = T_B + V_B$   
 $\frac{1}{2}m_A v_A^2 + mgh_A = \frac{1}{2}m_B v_B^2 + mgh_B$   
 $\left[\frac{1}{2}(2 \text{ kg})(1 \text{ m/s})^2\right] + [2(9.81) \text{ N}(4 \text{ m})]$   
 $= \left[\frac{1}{2}(2 \text{ kg})v_B^2\right] + [0]$   
 $v_B = 8.915 \text{ m/s} = 8.92 \text{ m/s}$  *Ans.*  
 $+\uparrow \Sigma F_n = ma_n; \quad N_B - 2(9.81) \text{ N}$   
 $= (2 \text{ kg})\left(\frac{(8.915 \text{ m/s})^2}{2 \text{ m}}\right)$   
 $N_B = 99.1 \text{ N}$  *Ans.*

**F14-15.**  $T_1 + V_1 = T_2 + V_2$   
 $\frac{1}{2}(2)(4)^2 + \frac{1}{2}(30)(2 - 1)^2$   
 $= \frac{1}{2}(2)(v)^2 - 2(9.81)(1) + \frac{1}{2}(30)(\sqrt{5} - 1)^2$   
 $v = 5.26 \text{ m/s}$  *Ans.*

**F14-16.**  $T_A + V_A = T_B + V_B$   
 $0 + \frac{1}{2}(4)(2.5 - 0.5)^2 + 5(9.81)(2.5)$   
 $= \frac{1}{2}(5)v_B^2 + \frac{1}{2}(4)(1 - 0.5)^2$   
 $v_B = 7.21 \text{ m/s}$  *Ans.*

**F14-17.**  $T_1 + V_1 = T_2 + V_2$   
 $\frac{1}{2}mv_1^2 + mgy_1 + \frac{1}{2}ks_1^2$   
 $= \frac{1}{2}mv_2^2 + mgy_2 + \frac{1}{2}ks_2^2$   
 $[0] + [0] + [0] = [0] +$   
 $[-35 \text{ kg}(9.81 \text{ m/s}^2)(1.5 \text{ m} + s)] + \left[2\left(\frac{1}{2}(16 \text{ 000 N/m})s^2\right)\right.$   
 $\left. + \frac{1}{2}(24 \text{ 000 N/m})(s - 0.075 \text{ m})^2\right]$   
 $s = s_A = s_C = 0.170 \text{ m}$  *Ans.*  
 Also,  
 $s_B = 0.170 \text{ m} - 0.075 \text{ m} = 0.095 \text{ m}$  *Ans.*

**F14-18.**  $T_A + V_A = T_B + V_B$   
 $\frac{1}{2}mv_A^2 + \left(\frac{1}{2}ks_A^2 + mgy_A\right)$   
 $= \frac{1}{2}mv_B^2 + \left(\frac{1}{2}ks_B^2 + mgy_B\right)$   
 $\frac{1}{2}(4 \text{ kg})(2 \text{ m/s})^2 + \frac{1}{2}(400 \text{ N/m})(0.1 \text{ m} - 0.2 \text{ m})^2 + 0$   
 $= \frac{1}{2}(4 \text{ kg})v_B^2 + \frac{1}{2}(400 \text{ N/m})(\sqrt{(0.4 \text{ m})^2 + (0.3 \text{ m})^2}$   
 $- 0.2 \text{ m})^2 + [4(9.81) \text{ N}](-0.1 \text{ m} + 0.3 \text{ m})$   
 $v_B = 1.962 \text{ m/s} = 1.96 \text{ m/s}$  *Ans.*

## Chapter 15

**F15-1.**  $(\rightarrow) \quad m(v_1)_x + \Sigma \int_{t_1}^{t_2} F_x dt = m(v_2)_x$   
 $(0.5 \text{ kg})(25 \text{ m/s}) \cos 45^\circ - \int F_x dt$   
 $= (0.5 \text{ kg})(10 \text{ m/s}) \cos 30^\circ$   
 $I_x = \int F_x dt = 4.509 \text{ N} \cdot \text{s}$   
 $(+\uparrow) \quad m(v_1)_y + \Sigma \int_{t_1}^{t_2} F_y dt = m(v_2)_y$   
 $-(0.5 \text{ kg})(25 \text{ m/s}) \sin 45^\circ + \int F_y dt$   
 $= (0.5 \text{ kg})(10 \text{ m/s}) \sin 30^\circ$   
 $I_y = \int F_y dt = 11.339 \text{ N} \cdot \text{s}$   
 $I = \int F dt = \sqrt{(4.509 \text{ N} \cdot \text{s})^2 + (11.339 \text{ N} \cdot \text{s})^2}$   
 $= 12.2 \text{ N} \cdot \text{s}$  *Ans.*



**F15-2.**  $(+\uparrow) \quad m(v_1)_y + \Sigma \int_{t_1}^{t_2} F_y dt = m(v_2)_y$   
 $0 + N(4 \text{ s}) + (500 \text{ N})(4 \text{ s}) \sin 30^\circ$   
 $- (75 \text{ kg})(9.81 \text{ m/s}^2)(4 \text{ s}) = 0$   
 $N = 485.75 \text{ N}$

$(\rightarrow) \quad m(v_1)_x + \Sigma \int_{t_1}^{t_2} F_x dt = m(v_2)_x$   
 $0 + (500 \text{ N})(4 \text{ s}) \cos 30^\circ - 0.2(485.75)(4 \text{ s})$   
 $= (75 \text{ kg})v$   
 $v = 17.9 \text{ m/s}$

Ans.

**F15-3.** Time to start motion,  
 $+\uparrow \Sigma F_y = 0; \quad N - 25(9.81) \text{ N} = 0 \quad N = 245.25 \text{ N}$   
 $\rightarrow \Sigma F_x = 0; \quad 20t^2 - 0.3(245.25 \text{ N}) = 0 \quad t = 1.918 \text{ s}$   
 $(\rightarrow) \quad m(v_1)_x + \Sigma \int_{t_1}^{t_2} F_x dt = m(v_2)_x$   
 $0 + \int_{1.918 \text{ s}}^{4 \text{ s}} 20t^2 dt - (0.25(245.25 \text{ N}))(4 \text{ s} - 1.918 \text{ s})$   
 $= (25 \text{ kg})v$   
 $v = 10.1 \text{ m/s}$

Ans.

**F15-4.**  $(\rightarrow) \quad m(v_1)_x + \Sigma \int_{t_1}^{t_2} F_x dt = m(v_2)_x$   
 $(1500 \text{ kg})(0) + \left[ \frac{1}{2}(6000 \text{ N})(2 \text{ s}) + (6000 \text{ N})(6 \text{ s} - 2 \text{ s}) \right]$   
 $= (1500 \text{ kg})v$   
 $v = 20 \text{ m/s}$

Ans.

**F15-5.** SUV and trailer,  
 $m(v_1)_x + \Sigma \int_{t_1}^{t_2} F_x dt = m(v_2)_x$   
 $0 + (9000 \text{ N})(20 \text{ s}) = (1500 \text{ kg} + 2500 \text{ kg})v$   
 $v = 45.0 \text{ m/s}$

Ans.

Trailer,  
 $m(v_1)_x + \Sigma \int_{t_1}^{t_2} F_x dt = m(v_2)_x$   
 $0 + T(20 \text{ s}) = (1500 \text{ kg})(45.0 \text{ m/s})$   
 $T = 3375 \text{ N} = 3.375 \text{ kN}$

Ans.

**F15-6.** Block B:  
 $(+\downarrow) \quad mv_1 + \int F dt = mv_2$   
 $0 + 8(9.81)(5) - T(5) = 8(1)$   
 $T = 76.88 \text{ N}$   
 Block A:  
 $(\rightarrow) \quad mv_1 + \int F dt = mv_2$   
 $0 + 76.88(5) - \mu_k(10)(9.81)(5) = 10(1)$   
 $\mu_k = 0.763$

Ans.

**F15-7.**  $(\rightarrow) \quad m_A(v_A)_1 + m_B(v_B)_1 = m_A(v_A)_2 + m_B(v_B)_2$   
 $(20(10^3) \text{ kg})(3 \text{ m/s}) + (15(10^3) \text{ kg})(-1.5 \text{ m/s})$   
 $= (20(10^3) \text{ kg})(v_A)_2 + (15(10^3) \text{ kg})(2 \text{ m/s})$   
 $(v_A)_2 = 0.375 \text{ m/s} \rightarrow$

Ans.

$(\rightarrow) \quad m(v_B)_1 + \Sigma \int_{t_1}^{t_2} F dt = m(v_B)_2$   
 $(15(10^3) \text{ kg})(-1.5 \text{ m/s}) + F_{\text{avg}}(0.5 \text{ s})$   
 $= (15(10^3) \text{ kg})(2 \text{ m/s})$

$F_{\text{avg}} = 105(10^3) \text{ N} = 105 \text{ kN}$

Ans.

**F15-8.**  $(\rightarrow) \quad m_p[(v_p)_1]_x + m_c[(v_c)_1]_x = (m_p + m_c)v_2$   
 $5\left[10\left(\frac{4}{5}\right)\right] + 0 = (5 + 20)v_2$   
 $v_2 = 1.6 \text{ m/s}$

Ans.

**F15-9.**  $T_1 + V_1 = T_2 + V_2$   
 $\frac{1}{2}m_A(v_A)_1^2 + (V_g)_1 = \frac{1}{2}m_A(v_A)_2^2 + (V_g)_2$   
 $\frac{1}{2}(5)(5)^2 + 5(9.81)(1.5) = \frac{1}{2}(5)(v_A)_2^2 + 0$   
 $(v_A)_2 = 7.378 \text{ m/s}$   
 $(\rightarrow) \quad m_A(v_A)_2 + m_B(v_B)_2 = (m_A + m_B)v$   
 $5(7.378) + 0 = (5 + 8)v$   
 $v = 2.84 \text{ m/s}$

Ans.

**F15-10.**  $(\rightarrow) \quad m_A(v_A)_1 + m_B(v_B)_1 = m_A(v_A)_2 + m_B(v_B)_2$   
 $0 + 0 = 10(v_A)_2 + 15(v_B)_2 \quad (1)$   
 $T_1 + V_1 = T_2 + V_2$   
 $\frac{1}{2}m_A(v_A)_1^2 + \frac{1}{2}m_B(v_B)_1^2 + (V_e)_1$   
 $= \frac{1}{2}m_A(v_A)_2^2 + \frac{1}{2}m_B(v_B)_2^2 + (V_e)_2$   
 $0 + 0 + \frac{1}{2}\left[5(10^3)\right](0.2^2)$   
 $= \frac{1}{2}(10)(v_A)_2^2 + \frac{1}{2}(15)(v_B)_2^2 + 0$   
 $5(v_A)_2^2 + 7.5(v_B)_2^2 = 100 \quad (2)$   
 Solving Eqs. (1) and (2),  
 $(v_B)_2 = 2.31 \text{ m/s} \rightarrow$   
 $(v_A)_2 = -3.464 \text{ m/s} = 3.46 \text{ m/s} \leftarrow$

Ans.

Ans.

**F15-11.**  $(\rightarrow) \quad m_A(v_A)_1 + m_B(v_B)_1 = (m_A + m_B)v_2$   
 $0 + 10(15) = (15 + 10)v_2$   
 $v_2 = 6 \text{ m/s}$   
 $T_1 + V_1 = T_2 + V_2$   
 $\frac{1}{2}(m_A + m_B)v_2^2 + (V_e)_2 = \frac{1}{2}(m_A + m_B)v_3^2 + (V_e)_3$   
 $\frac{1}{2}(15 + 10)(6^2) + 0 = 0 + \frac{1}{2}\left[10(10^3)\right]s_{\text{max}}^2$   
 $s_{\text{max}} = 0.3 \text{ m} = 300 \text{ mm}$

Ans.

**F15-12.**  $(\rightarrow) \quad 0 + 0 = m_p(v_p)_x - m_c v_c$   
 $0 = (20 \text{ kg})(v_p)_x - (250 \text{ kg})v_c$   
 $(v_p)_x = 12.5 v_c$  (1)  
 $\mathbf{v}_p = \mathbf{v}_c + \mathbf{v}_{p/c}$   
 $(v_p)_x \mathbf{i} + (v_p)_y \mathbf{j} = -v_c \mathbf{i} + [(400 \text{ m/s}) \cos 30^\circ \mathbf{i}$   
 $\quad + (400 \text{ m/s}) \sin 30^\circ \mathbf{j}]$   
 $(v_p)_x \mathbf{i} + (v_p)_y \mathbf{j} = (346.41 - v_c) \mathbf{i} + 200 \mathbf{j}$   
 $(v_p)_x = 346.41 - v_c$   
 $(v_p)_y = 200 \text{ m/s}$   
 $(v_p)_x = 320.75 \text{ m/s} \quad v_c = 25.66 \text{ m/s}$   
 $v_p = \sqrt{(v_p)_x^2 + (v_p)_y^2}$   
 $= \sqrt{(320.75 \text{ m/s})^2 + (200 \text{ m/s})^2}$   
 $= 378 \text{ m/s}$  *Ans.*

**F15-13.**  $(\rightarrow) \quad e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1}$   
 $= \frac{(9 \text{ m/s}) - (1 \text{ m/s})}{(8 \text{ m/s}) - (-2 \text{ m/s})} = 0.8$

**F15-14.**  $(\rightarrow) \quad m_A(v_A)_1 + m_B(v_B)_1 = m_A(v_A)_2 + m_B(v_B)_2$   
 $[15(10^3) \text{ kg}](5 \text{ m/s}) + [25(10^3)](-7 \text{ m/s})$   
 $= [15(10^3) \text{ kg}](v_A)_2 + [25(10^3)](v_B)_2$   
 $15(v_A)_2 + 25(v_B)_2 = -100$  (1)

Using the coefficient of restitution equation,

$(\rightarrow) \quad e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1}$   
 $0.6 = \frac{(v_B)_2 - (v_A)_2}{5 \text{ m/s} - (-7 \text{ m/s})}$   
 $(v_B)_2 - (v_A)_2 = 7.2$  (2)

Solving,

$(v_B)_2 = 0.2 \text{ m/s} \rightarrow$  *Ans.*

$(v_A)_2 = -7 \text{ m/s} = 7 \text{ m/s} \leftarrow$  *Ans.*

**F15-15.**  $T_1 + V_1 = T_2 + V_2$   
 $\frac{1}{2} m(v_A)_1^2 + mg(h_A)_1 = \frac{1}{2} m(v_A)_2^2 + mg(h_A)_2$   
 $\frac{1}{2}(15 \text{ kg})(1.5 \text{ m/s}^2) + 15 \text{ kg}(9.81 \text{ m/s}^2)(3 \text{ m})$   
 $= \frac{1}{2}(15 \text{ kg})(v_A)_2^2 + 0$   
 $(v_A)_2 = 7.817 \text{ m/s} \leftarrow$   
 $(\leftarrow) \quad m_A(v_A)_2 + m_B(v_B)_2 = m_A(v_A)_3 + m_B(v_B)_3$   
 $(15 \text{ kg})(7.817 \text{ m/s}) + 0$   
 $= (15 \text{ kg})(v_A)_3 + (40 \text{ kg})(v_B)_3$

$15(v_A)_3 + 40(v_B)_3 = 117.255$  (1)

$(\leftarrow) \quad e = \frac{(v_B)_3 - (v_A)_3}{(v_A)_2 - (v_B)_2}$   
 $0.6 = \frac{(v_B)_3 - (v_A)_3}{7.817 \text{ m/s} - 0}$   
 $(v_B)_3 - (v_A)_3 = 4.6902$  (2)

Solving Eqs. (1) and (2), yields

$(v_B)_3 = 3.411 \text{ m/s} \leftarrow$

$(v_A)_3 = -1.279 \text{ m/s} = 1.279 \text{ m/s} \rightarrow$  *Ans.*

**F15-16.**  $(+\uparrow) \quad m[(v_b)_1]_y = m[(v_b)_2]_y$   
 $[(v_b)_2]_y = [(v_b)_1]_y = (20 \text{ m/s}) \sin 30^\circ = 10 \text{ m/s} \uparrow$   
 $(\rightarrow) \quad e = \frac{(v_w)_2 - [(v_b)_2]_x}{[(v_b)_1]_x - (v_w)_1}$   
 $0.75 = \frac{0 - [(v_b)_2]_x}{(20 \text{ m/s}) \cos 30^\circ - 0}$   
 $[(v_b)_2]_x = -12.99 \text{ m/s} = 12.99 \text{ m/s} \leftarrow$   
 $(v_b)_2 = \sqrt{[(v_b)_2]_x^2 + [(v_b)_2]_y^2}$   
 $= \sqrt{(12.99 \text{ m/s})^2 + (10 \text{ m/s})^2}$   
 $= 16.4 \text{ m/s}$  *Ans.*

$\theta = \tan^{-1} \left( \frac{[(v_b)_2]_y}{[(v_b)_2]_x} \right) = \tan^{-1} \left( \frac{10 \text{ m/s}}{12.99 \text{ m/s}} \right)$   
 $= 37.6^\circ$  *Ans.*

**F15-17.**  $\Sigma m(v_x)_1 = \Sigma m(v_x)_2$   
 $0 + 0 = 2(1) + 11(v_{Bx})_2$   
 $(v_{Bx})_2 = -0.1818 \text{ m/s}$   
 $\Sigma m(v_y)_1 = \Sigma m(v_y)_2$   
 $2(3) + 0 = 0 + 11(v_{By})_2$   
 $(v_{By})_2 = 0.545 \text{ m/s}$   
 $(v_B)_2 = \sqrt{(-0.1818)^2 + (0.545)^2}$   
 $= 0.575 \text{ m/s}$  *Ans.*

**F15-18.**  $+\nearrow \quad 1(3)\left(\frac{3}{5}\right) - 1(4)\left(\frac{4}{5}\right)$   
 $= 1(v_B)_{2x} + 1(v_A)_{2x}$   
 $+\nearrow \quad 0.5 = [(v_A)_{2x} - (v_B)_{2x}] / [(3)\left(\frac{3}{5}\right) - (-4)\left(\frac{4}{5}\right)]$   
Solving,  
 $(v_A)_{2x} = 0.550 \text{ m/s}, (v_B)_{2x} = -1.95 \text{ m/s}$   
Disc A,  
 $+\nwarrow \quad -1(4)\left(\frac{3}{5}\right) = 1(v_A)_{2y}$   
 $(v_A)_{2y} = -2.40 \text{ m/s}$

Disc  $B$ ,

$$-1(3)\left(\frac{4}{5}\right) = 1(v_B)_{2y}$$

$$(v_B)_{2y} = -2.40 \text{ m/s}$$

$$(v_A)_2 = \sqrt{(0.550)^2 + (2.40)^2} = 2.46 \text{ m/s} \quad \text{Ans.}$$

$$(v_B)_2 = \sqrt{(1.95)^2 + (2.40)^2} = 3.09 \text{ m/s} \quad \text{Ans.}$$

$$\mathbf{F15-19.} \quad H_O = \Sigma mvd;$$

$$H_O = \left[ 2(10)\left(\frac{4}{5}\right) \right](4) - \left[ 2(10)\left(\frac{3}{5}\right) \right](3) \\ = 28 \text{ kg} \cdot \text{m}^2/\text{s}$$

$$\mathbf{F15-20.} \quad H_P = \Sigma mvd;$$

$$H_P = [2(15) \sin 30^\circ](2) - [2(15) \cos 30^\circ](5) \\ = -99.9 \text{ kg} \cdot \text{m}^2/\text{s} = 99.9 \text{ kg} \cdot \text{m}^2/\text{s} \curvearrowright$$

$$\mathbf{F15-21.} \quad (H_z)_1 + \Sigma \int M_z dt = (H_z)_2$$

$$5(2)(1.5) + 5(1.5)(3) = 5v(1.5)$$

$$v = 5 \text{ m/s} \quad \text{Ans.}$$

$$\mathbf{F15-22.} \quad (H_z)_1 + \Sigma \int M_z dt = (H_z)_2$$

$$0 + \int_0^{4\text{s}} (10t)\left(\frac{4}{5}\right)(1.5)dt = 5v(1.5)$$

$$v = 12.8 \text{ m/s} \quad \text{Ans.}$$

$$\mathbf{F15-23.} \quad (H_z)_1 + \Sigma \int M_z dt = (H_z)_2$$

$$0 + \int_0^{5\text{s}} 0.9t^2 dt = 2v(0.6)$$

$$v = 31.2 \text{ m/s} \quad \text{Ans.}$$

$$\mathbf{F15-24.} \quad (H_z)_1 + \Sigma \int M_z dt = (H_z)_2$$

$$0 + \int_0^{4\text{s}} 8tdt + 2(10)(0.5)(4) = 2[10v(0.5)]$$

$$v = 10.4 \text{ m/s} \quad \text{Ans.}$$

## Chapter 16

$$\mathbf{F16-1.} \quad \theta = (20 \text{ rev})\left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right) = 40\pi \text{ rad}$$

$$\omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0)$$

$$(30 \text{ rad/s})^2 = 0^2 + 2\alpha_c[(40\pi \text{ rad}) - 0]$$

$$\alpha_c = 3.581 \text{ rad/s}^2 = 3.58 \text{ rad/s}^2 \quad \text{Ans.}$$

$$\omega = \omega_0 + \alpha_c t$$

$$30 \text{ rad/s} = 0 + (3.581 \text{ rad/s}^2)t$$

$$t = 8.38 \text{ s} \quad \text{Ans.}$$

$$\mathbf{F16-2.} \quad \frac{d\omega}{d\theta} = 2(0.005\theta) = (0.01\theta)$$

$$\alpha = \omega \frac{d\omega}{d\theta} = (0.005\theta^2)(0.01\theta) = 50(10^{-6})\theta^3 \text{ rad/s}^2$$

$$\text{When } \theta = 20 \text{ rev}(2\pi \text{ rad/1 rev}) = 40\pi \text{ rad,}$$

$$\alpha = [50(10^{-6})(40\pi)^3] \text{ rad/s}^2$$

$$= 99.22 \text{ rad/s}^2 = 99.2 \text{ rad/s}^2 \quad \text{Ans.}$$

$$\mathbf{F16-3.} \quad \omega = 4\theta^{1/2}$$

$$150 \text{ rad/s} = 4\theta^{1/2}$$

$$\theta = 1406.25 \text{ rad}$$

$$dt = \frac{d\theta}{\omega}$$

$$\int_0^t dt = \int_{1 \text{ rad}}^{\theta} \frac{d\theta}{4\theta^{1/2}}$$

$$t \Big|_0^t = \frac{1}{2} \theta^{1/2} \Big|_{1 \text{ rad}}$$

$$t = \frac{1}{2} \theta^{1/2} - \frac{1}{2}$$

$$t = \frac{1}{2} (1406.25)^{1/2} - \frac{1}{2} = 18.25 \text{ s} \quad \text{Ans.}$$

$$\mathbf{F16-4.} \quad \omega = \frac{d\theta}{dt} = (1.5t^2 + 15) \text{ rad/s}$$

$$\alpha = \frac{d\omega}{dt} = (3t) \text{ rad/s}^2$$

$$\omega = [1.5(3^2) + 15] \text{ rad/s} = 28.5 \text{ rad/s}$$

$$\alpha = 3(3) \text{ rad/s}^2 = 9 \text{ rad/s}^2$$

$$v = \omega r = (28.5 \text{ rad/s})(0.2 \text{ m}) = 5.70 \text{ m/s} \quad \text{Ans.}$$

$$a = \alpha r = (9 \text{ rad/s}^2)(0.2 \text{ m}) = 1.8 \text{ m/s}^2 \quad \text{Ans.}$$

$$\mathbf{F16-5.} \quad \omega d\omega = \alpha d\theta$$

$$\int_{2 \text{ rad/s}}^{\omega} \omega d\omega = \int_0^{\theta} 0.5\theta d\theta$$

$$\frac{\omega^2}{2} \Big|_{2 \text{ rad/s}}^{\omega} = 0.25\theta^2 \Big|_0^{\theta}$$

$$\omega = (0.5\theta^2 + 4)^{1/2} \text{ rad/s}$$

$$\text{When } \theta = 2 \text{ rev} = 4\pi \text{ rad,}$$

$$\omega = [0.5(4\pi)^2 + 4]^{1/2} \text{ rad/s} = 9.108 \text{ rad/s}$$

$$v_P = \omega r = (9.108 \text{ rad/s})(0.2 \text{ m}) = 1.82 \text{ m/s} \quad \text{Ans.}$$

$$(a_P)_t = \alpha r = (0.5\theta \text{ rad/s}^2)(0.2 \text{ m}) \Big|_{\theta=4\pi \text{ rad}}$$

$$= 1.257 \text{ m/s}^2$$

$$(a_P)_n = \omega^2 r = (9.108 \text{ rad/s})^2(0.2 \text{ m}) = 16.59 \text{ m/s}^2$$

$$a_P = \sqrt{(a_P)_t^2 + (a_P)_n^2}$$

$$= \sqrt{(1.257 \text{ m/s}^2)^2 + (16.59 \text{ m/s}^2)^2}$$

$$= 16.6 \text{ m/s}^2 \quad \text{Ans.}$$

**F16-6.**  $\alpha_B = \alpha_A \left( \frac{r_A}{r_B} \right)$   
 $= (4.5 \text{ rad/s}^2) \left( \frac{0.075 \text{ m}}{0.225 \text{ m}} \right) = 1.5 \text{ rad/s}^2$   
 $\omega_B = (\omega_B)_0 + \alpha_B t$   
 $\omega_B = 0 + (1.5 \text{ rad/s}^2)(3 \text{ s}) = 4.5 \text{ rad/s}$   
 $\theta_B = (\theta_B)_0 + (\omega_B)_0 t + \frac{1}{2} \alpha_B t^2$   
 $\theta_B = 0 + 0 + \frac{1}{2} (1.5 \text{ rad/s}^2)(3 \text{ s})^2$   
 $\theta_B = 6.75 \text{ rad}$   
 $v_C = \omega_B r_D = (4.5 \text{ rad/s})(0.125 \text{ m})$   
 $= 0.5625 \text{ m/s}$  *Ans.*  
 $s_C = \theta_B r_D = (6.75 \text{ rad})(0.125 \text{ m}) = 0.84375 \text{ m}$   
 $= 844 \text{ mm}$  *Ans.*

**F16-7.** Vector Analysis  
 $\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$   
 $-v_B \mathbf{j} = (3\mathbf{i}) \text{ m/s}$   
 $+ (\boldsymbol{\omega} \mathbf{k}) \times (-1.5 \cos 30^\circ \mathbf{i} + 1.5 \sin 30^\circ \mathbf{j})$   
 $-v_B \mathbf{j} = [3 - \omega_{AB} (1.5 \sin 30^\circ)] \mathbf{i} - \omega (1.5 \cos 30^\circ) \mathbf{j}$   
 $0 = 3 - \omega (1.5 \sin 30^\circ)$  (1)  
 $-v_B = 0 - \omega (1.5 \cos 30^\circ)$  (2)  
 $\omega = 4 \text{ rad/s}$   $v_B = 5.20 \text{ m/s}$  *Ans.*

Scalar Solution

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$$

$$\left[ \downarrow v_B \right] = \left[ \begin{matrix} 3 \\ \rightarrow \end{matrix} \right] + \left[ \omega (1.5) \angle 30^\circ \right]$$

This yields Eqs. (1) and (2).

**F16-8.** Vector Analysis  
 $\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$   
 $(v_B)_x \mathbf{i} + (v_B)_y \mathbf{j} = \mathbf{0} + (-10\mathbf{k}) \times (-0.6\mathbf{i} + 0.6\mathbf{j})$   
 $(v_B)_x \mathbf{i} + (v_B)_y \mathbf{j} = 6\mathbf{i} + 6\mathbf{j}$   
 $(v_B)_x = 6 \text{ m/s}$  and  $(v_B)_y = 6 \text{ m/s}$   
 $v_B = \sqrt{(v_B)_x^2 + (v_B)_y^2}$   
 $= \sqrt{(6 \text{ m/s})^2 + (6 \text{ m/s})^2}$   
 $= 8.49 \text{ m/s}$  *Ans.*

Scalar Solution

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$$

$$\left[ \begin{matrix} (v_B)_x \\ \rightarrow \end{matrix} \right] + \left[ \begin{matrix} (v_B)_y \\ \uparrow \end{matrix} \right] = \left[ \begin{matrix} 0 \\ \rightarrow \end{matrix} \right] + \left[ \angle 45^\circ \quad 10 \left( \frac{0.6}{\cos 45^\circ} \right) \right]$$

$$\begin{aligned} \rightarrow (v_B)_x &= 0 + 10(0.6/\cos 45^\circ) \cos 45^\circ = 6 \text{ m/s} \rightarrow \\ \uparrow (v_B)_y &= 0 + 10(0.6/\cos 45^\circ) \sin 45^\circ = 6 \text{ m/s} \uparrow \end{aligned}$$

**F16-9.** Vector Analysis  
 $\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$   
 $(1.2 \text{ m/s}) \mathbf{i} = (-0.6 \text{ m/s}) \mathbf{i} + (-\boldsymbol{\omega} \mathbf{k}) \times (0.9 \text{ m}) \mathbf{j}$   
 $1.2 \mathbf{i} = (-0.6 + 0.9\omega) \mathbf{i}$   
 $\omega = 2 \text{ rad/s}$  *Ans.*

Scalar Solution

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$$

$$\left[ \begin{matrix} 1.2 \\ \rightarrow \end{matrix} \right] = \left[ \begin{matrix} 0.6 \\ \leftarrow \end{matrix} \right] + \left[ \begin{matrix} \omega(0.9) \\ \rightarrow \end{matrix} \right]$$

$$\rightarrow 1.2 = -0.6 + \omega(0.9); \quad \omega = 2 \text{ rad/s}$$

**F16-10.** Vector Analysis  
 $\mathbf{v}_A = \boldsymbol{\omega}_{OA} \times \mathbf{r}_A$   
 $= (12 \text{ rad/s}) \mathbf{k} \times (0.3 \text{ m}) \mathbf{j}$   
 $= [-3.6\mathbf{i}] \text{ m/s}$   
 $\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega}_{AB} \times \mathbf{r}_{B/A}$   
 $v_B \mathbf{j} = (-3.6 \text{ m/s}) \mathbf{i}$   
 $+ (\omega_{AB} \mathbf{k}) \times (0.6 \cos 30^\circ \mathbf{i} - 0.6 \sin 30^\circ \mathbf{j}) \text{ m}$   
 $v_B \mathbf{j} = [\omega_{AB} (0.6 \sin 30^\circ) - 3.6] \mathbf{i} + \omega_{AB} (0.6 \cos 30^\circ) \mathbf{j}$   
 $0 = \omega_{AB} (0.6 \sin 30^\circ) - 3.6$  (1)  
 $v_B = \omega_{AB} (0.6 \cos 30^\circ)$  (2)  
 $\omega_{AB} = 12 \text{ rad/s}$   $v_B = 6.24 \text{ m/s} \uparrow$  *Ans.*

Scalar Solution

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$$

$$\left[ \begin{matrix} v_B \\ \uparrow \end{matrix} \right] = \left[ \begin{matrix} \leftarrow \\ 12(0.3) \end{matrix} \right] + \left[ \begin{matrix} \nearrow 30^\circ \\ \omega(0.6) \end{matrix} \right]$$

This yields Eqs. (1) and (2).

**F16-11.** Vector Analysis  
 $\mathbf{v}_C = \mathbf{v}_B + \boldsymbol{\omega}_{BC} \times \mathbf{r}_{C/B}$   
 $v_C \mathbf{j} = (-18\mathbf{i}) \text{ m/s}$   
 $+ (-\omega_{BC} \mathbf{k}) \times (-0.75 \cos 30^\circ \mathbf{i} + 0.75 \sin 30^\circ \mathbf{j}) \text{ m}$   
 $v_C \mathbf{j} = (-18) \mathbf{i} + 0.6495 \omega_{BC} \mathbf{j} + 0.375 \omega_{BC} \mathbf{i}$   
 $0 = -18 + 0.375 \omega_{BC}$  (1)  
 $v_C = 0.6495 \omega_{BC}$  (2)  
 $\omega_{BC} = 48 \text{ rad/s}$  *Ans.*  
 $v_C = 31.2 \text{ m/s}$

Scalar Solution

$$\mathbf{v}_C = \mathbf{v}_B + \mathbf{v}_{C/B}$$

$$\left[ \begin{matrix} v_C \\ \uparrow \end{matrix} \right] = \left[ \begin{matrix} v_B \\ \leftarrow \end{matrix} \right] + \left[ \begin{matrix} \nearrow 30^\circ \\ \omega(0.75) \end{matrix} \right]$$

This yields Eqs. (1) and (2).

**F16-12. Vector Analysis**

$$\begin{aligned}
 \mathbf{v}_B &= \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A} \\
 -v_B \cos 30^\circ \mathbf{i} + v_B \sin 30^\circ \mathbf{j} &= (-3 \text{ m/s})\mathbf{j} + \\
 & \quad (-\omega \mathbf{k}) \times (-2 \sin 45^\circ \mathbf{i} - 2 \cos 45^\circ \mathbf{j}) \text{ m} \\
 -0.8660v_B \mathbf{i} + 0.5v_B \mathbf{j} & \\
 &= -1.4142\omega \mathbf{i} + (1.4142\omega - 3)\mathbf{j} \\
 -0.8660v_B &= -1.4142\omega \quad (1) \\
 0.5v_B &= 1.4142\omega - 3 \quad (2) \\
 \omega &= 5.02 \text{ rad/s} \quad v_B = 8.20 \text{ m/s} \quad \text{Ans.}
 \end{aligned}$$

Scalar Solution

$$\begin{aligned}
 \mathbf{v}_B &= \mathbf{v}_A + \mathbf{v}_{B/A} \\
 \left[ \begin{array}{c} \searrow 30^\circ \\ \downarrow 3 \end{array} v_B \right] &= \left[ \begin{array}{c} \downarrow 3 \end{array} \right] + \left[ \begin{array}{c} \searrow 45^\circ \\ \omega(2) \end{array} \right]
 \end{aligned}$$

This yields Eqs. (1) and (2).

$$\begin{aligned}
 \text{F16-13. } \omega_{AB} &= \frac{v_A}{r_{A/IC}} = \frac{6}{3} = 2 \text{ rad/s} \quad \text{Ans.} \\
 \phi &= \tan^{-1}\left(\frac{2}{1.5}\right) = 53.13^\circ \\
 r_{C/IC} &= \sqrt{(3)^2 + (2.5)^2 - 2(3)(2.5) \cos 53.13^\circ} = 2.5 \text{ m} \\
 v_C &= \omega_{AB} r_{C/IC} = 2(2.5) = 5 \text{ m/s} \quad \text{Ans.} \\
 \theta &= 90^\circ - \phi = 90^\circ - 53.13^\circ = 36.9^\circ \quad \text{Ans.}
 \end{aligned}$$

$$\begin{aligned}
 \text{F16-14. } v_B &= \omega_{AB} r_{B/A} = 12(0.6) = 7.2 \text{ m/s} \quad \downarrow \\
 v_C &= 0 \quad \text{Ans.} \\
 \omega_{BC} &= \frac{v_B}{r_{B/IC}} = \frac{7.2}{1.2} = 6 \text{ rad/s} \quad \text{Ans.}
 \end{aligned}$$

$$\begin{aligned}
 \text{F16-15. } \omega &= \frac{v_O}{r_{O/IC}} = \frac{6}{0.3} = 20 \text{ rad/s} \quad \text{Ans.} \\
 r_{A/IC} &= \sqrt{0.3^2 + 0.6^2} = 0.6708 \text{ m} \\
 \phi &= \tan^{-1}\left(\frac{0.3}{0.6}\right) = 26.57^\circ \\
 v_A &= \omega r_{A/IC} = 20(0.6708) = 13.4 \text{ m/s} \quad \text{Ans.} \\
 \theta &= 90^\circ - \phi = 90^\circ - 26.57^\circ = 63.4^\circ \quad \text{Ans.}
 \end{aligned}$$

**F16-16.** The location of *IC* can be determined using similar triangles.

$$\begin{aligned}
 \frac{0.5 - r_{C/IC}}{3} &= \frac{r_{C/IC}}{1.5} \quad r_{C/IC} = 0.1667 \text{ m} \\
 \omega &= \frac{v_C}{r_{C/IC}} = \frac{1.5}{0.1667} = 9 \text{ rad/s} \quad \text{Ans.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } r_{O/IC} &= 0.3 - r_{C/IC} = 0.3 - 0.1667 \\
 &= 0.1333 \text{ m.}
 \end{aligned}$$

$$v_O = \omega r_{O/IC} = 9(0.1333) = 1.20 \text{ m/s} \quad \text{Ans.}$$

$$\begin{aligned}
 \text{F16-17. } v_B &= \omega r_{B/A} = 6(0.2) = 1.2 \text{ m/s} \\
 r_{B/IC} &= 0.8 \tan 60^\circ = 1.3856 \text{ m} \\
 r_{C/IC} &= \frac{0.8}{\cos 60^\circ} = 1.6 \text{ m} \\
 \omega_{BC} &= \frac{v_B}{r_{B/IC}} = \frac{1.2}{1.3856} = 0.8660 \text{ rad/s} \\
 &= 0.866 \text{ rad/s} \quad \text{Ans.}
 \end{aligned}$$

Then,

$$v_C = \omega_{BC} r_{C/IC} = 0.8660(1.6) = 1.39 \text{ m/s} \quad \text{Ans.}$$

$$\begin{aligned}
 \text{F16-18. } v_B &= \omega_{AB} r_{B/A} = 10(0.2) = 2 \text{ m/s} \\
 v_C &= \omega_{CD} r_{C/D} = \omega_{CD} (0.2) \rightarrow \\
 r_{B/IC} &= \frac{0.4}{\cos 30^\circ} = 0.4619 \text{ m} \\
 r_{C/IC} &= 0.4 \tan 30^\circ = 0.2309 \text{ m} \\
 \omega_{BC} &= \frac{v_B}{r_{B/IC}} = \frac{2}{0.4619} = 4.330 \text{ rad/s} \\
 &= 4.33 \text{ rad/s} \quad \text{Ans.} \\
 v_C &= \omega_{BC} r_{C/IC} \\
 \omega_{CD} (0.2) &= 4.330(0.2309) \\
 \omega_{CD} &= 5 \text{ rad/s} \quad \text{Ans.}
 \end{aligned}$$

$$\text{F16-19. } \omega = \frac{v_A}{r_{A/IC}} = \frac{6}{3} = 2 \text{ rad/s}$$

Vector Analysis

$$\begin{aligned}
 \mathbf{a}_B &= \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A} \\
 a_B \mathbf{i} &= -5\mathbf{j} + (\alpha \mathbf{k}) \times (3\mathbf{i} - 4\mathbf{j}) - 2^2(3\mathbf{i} - 4\mathbf{j}) \\
 a_B \mathbf{i} &= (4\alpha - 12)\mathbf{i} + (3\alpha + 11)\mathbf{j} \\
 a_B &= 4\alpha - 12 \quad (1) \\
 0 &= 3\alpha + 11 \quad (2) \\
 \alpha &= -3.67 \text{ rad/s}^2 \quad \text{Ans.} \\
 a_B &= -26.7 \text{ m/s}^2 \quad \text{Ans.}
 \end{aligned}$$

Scalar Solution

$$\begin{aligned}
 \mathbf{a}_B &= \mathbf{a}_A + \mathbf{a}_{B/A} \\
 \left[ \begin{array}{c} a_B \\ \rightarrow \end{array} \right] &= \left[ \begin{array}{c} \downarrow 5 \end{array} \right] + \left[ \begin{array}{c} \alpha(5) \frac{5}{4} \end{array} \right] + \left[ \begin{array}{c} 4\frac{5}{3}(2)^2(5) \end{array} \right]
 \end{aligned}$$

This yields Eqs. (1) and (2).

**F16-20. Vector Analysis**

$$\begin{aligned}\mathbf{a}_A &= \mathbf{a}_O + \boldsymbol{\alpha} \times \mathbf{r}_{A/O} - \omega^2 \mathbf{r}_{A/O} \\ &= 1.8\mathbf{i} + (-6\mathbf{k}) \times (0.3\mathbf{j}) - 12^2(0.3\mathbf{j}) \\ &= \{3.6\mathbf{i} - 43.2\mathbf{j}\} \text{ m/s}^2\end{aligned}\quad \text{Ans.}$$

Scalar Analysis

$$\begin{aligned}\mathbf{a}_A &= \mathbf{a}_O + \mathbf{a}_{A/O} \\ \left[ \begin{matrix} (a_A)_x \\ \rightarrow \end{matrix} \right] + \left[ \begin{matrix} (a_A)_y \uparrow \end{matrix} \right] &= \left[ \begin{matrix} (6)(0.3) \\ \rightarrow \end{matrix} \right] + \left[ \begin{matrix} (6)(0.3) \\ \rightarrow \end{matrix} \right] \\ &\quad + \left[ \begin{matrix} \downarrow (12)^2(0.3) \end{matrix} \right]\end{aligned}$$

$$\begin{aligned}\pm \quad (a_A)_x &= 1.8 + 1.8 = 3.6 \text{ m/s}^2 \rightarrow \\ + \uparrow \quad (a_A)_y &= -43.2 \text{ m/s}^2\end{aligned}$$

**F16-21. Using**

$$\begin{aligned}v_O &= \omega r; \quad 6 = \omega(0.3) \\ \omega &= 20 \text{ rad/s} \\ a_O &= \alpha r; \quad 3 = \alpha(0.3) \\ \alpha &= 10 \text{ rad/s}^2\end{aligned}\quad \text{Ans.}$$

Vector Analysis

$$\begin{aligned}\mathbf{a}_A &= \mathbf{a}_O + \boldsymbol{\alpha} \times \mathbf{r}_{A/O} - \omega^2 \mathbf{r}_{A/O} \\ &= 3\mathbf{i} + (-10\mathbf{k}) \times (-0.6\mathbf{i}) - 20^2(-0.6\mathbf{i}) \\ &= \{243\mathbf{i} + 6\mathbf{j}\} \text{ m/s}^2\end{aligned}\quad \text{Ans.}$$

Scalar Analysis

$$\begin{aligned}\mathbf{a}_A &= \mathbf{a}_O + \mathbf{a}_{A/O} \\ \left[ \begin{matrix} (a_A)_x \\ \rightarrow \end{matrix} \right] + \left[ \begin{matrix} (a_A)_y \uparrow \end{matrix} \right] &= \left[ \begin{matrix} 3 \\ \rightarrow \end{matrix} \right] + \left[ \begin{matrix} 10(0.6) \\ \uparrow \end{matrix} \right] + \left[ \begin{matrix} (20)^2(0.6) \\ \rightarrow \end{matrix} \right] \\ \pm \quad (a_A)_x &= 3 + 240 = 243 \text{ m/s}^2 \\ + \uparrow \quad (a_A)_y &= 10(0.6) = 6 \text{ m/s}^2 \uparrow\end{aligned}$$

$$\text{F16-22. } \frac{r_{A/IC}}{3} = \frac{0.5 - r_{A/IC}}{1.5}; \quad r_{A/IC} = 0.3333 \text{ m}$$

$$\omega = \frac{v_A}{r_{A/IC}} = \frac{3}{0.3333} = 9 \text{ rad/s}$$

Vector Analysis

$$\begin{aligned}\mathbf{a}_A &= \mathbf{a}_C + \boldsymbol{\alpha} \times \mathbf{r}_{A/C} - \omega^2 \mathbf{r}_{A/C} \\ 1.5\mathbf{i} - (a_A)_n \mathbf{j} &= -0.75\mathbf{i} + (a_C)_n \mathbf{j} \\ &\quad + (-\alpha \mathbf{k}) \times 0.5\mathbf{j} - 9^2(0.5\mathbf{j}) \\ 1.5\mathbf{i} - (a_A)_n \mathbf{j} &= (0.5\alpha - 0.75)\mathbf{i} + [(a_C)_n - 40.5]\mathbf{j} \\ 1.5 &= 0.5\alpha - 0.75 \\ \alpha &= 4.5 \text{ rad/s}^2\end{aligned}\quad \text{Ans.}$$

Scalar Analysis

$$\begin{aligned}\mathbf{a}_A &= \mathbf{a}_C + \mathbf{a}_{A/C} \\ \left[ \begin{matrix} 1.5 \\ \rightarrow \end{matrix} \right] + \left[ \begin{matrix} (a_A)_n \downarrow \end{matrix} \right] &= \left[ \begin{matrix} 0.75 \\ \leftarrow \end{matrix} \right] + \left[ \begin{matrix} (a_C)_n \uparrow \end{matrix} \right] + \left[ \begin{matrix} \alpha(0.5) \\ \rightarrow \end{matrix} \right] \\ &\quad + \left[ \begin{matrix} (9)^2(0.5) \\ \downarrow \end{matrix} \right] \\ \pm \quad 1.5 &= -0.75 + \alpha(0.5) \\ \alpha &= 4.5 \text{ rad/s}^2\end{aligned}$$

$$\text{F16-23. } v_B = \omega r_{B/A} = 12(0.3) = 3.6 \text{ m/s}$$

$$\omega_{BC} = \frac{v_B}{r_{B/IC}} = \frac{3.6}{1.2} = 3 \text{ rad/s}$$

Vector Analysis

$$\begin{aligned}\mathbf{a}_B &= \boldsymbol{\alpha} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A} \\ &= (-6\mathbf{k}) \times (0.3\mathbf{i}) - 12^2(0.3\mathbf{i}) \\ &= \{-43.2\mathbf{i} - 1.8\mathbf{j}\} \text{ m/s}^2 \\ \mathbf{a}_C &= \mathbf{a}_B + \boldsymbol{\alpha}_{BC} \times \mathbf{r}_{C/B} - \omega_{BC}^2 \mathbf{r}_{C/B} \\ a_C \mathbf{i} &= (-43.2\mathbf{i} - 1.8\mathbf{j}) \\ &\quad + (\alpha_{BC} \mathbf{k}) \times (1.2\mathbf{i}) - 3^2(1.2\mathbf{i}) \\ a_C \mathbf{i} &= -54\mathbf{i} + (1.2\alpha_{BC} - 1.8)\mathbf{j} \\ a_C &= -54 \text{ m/s}^2 = 54 \text{ m/s}^2 \leftarrow \quad \text{Ans.} \\ 0 &= 1.2\alpha_{BC} - 1.8 \quad \alpha_{BC} = 1.5 \text{ rad/s}^2 \quad \text{Ans.}\end{aligned}$$

Scalar Analysis

$$\begin{aligned}\mathbf{a}_C &= \mathbf{a}_B + \mathbf{a}_{C/B} \\ \left[ \begin{matrix} a_C \\ \leftarrow \end{matrix} \right] &= \left[ \begin{matrix} 6(0.3) \\ \downarrow \end{matrix} \right] + \left[ \begin{matrix} (12)^2(0.3) \\ \leftarrow \end{matrix} \right] + \left[ \begin{matrix} \alpha_{BC}(1.2) \\ \uparrow \end{matrix} \right] + \left[ \begin{matrix} (3)^2(1.2) \\ \leftarrow \end{matrix} \right] \\ \leftarrow \quad a_C &= 43.2 + 10.8 = 54 \text{ m/s}^2 \leftarrow \\ + \uparrow \quad 0 &= -6(0.3) + 1.2\alpha_{BC} \\ \alpha_{BC} &= 1.5 \text{ rad/s}^2\end{aligned}$$

$$\text{F16-24. } v_B = \omega r_{B/A} = 6(0.2) = 1.2 \text{ m/s} \rightarrow$$

$$r_{B/IC} = 0.8 \tan 60^\circ = 1.3856 \text{ m}$$

$$\omega_{BC} = \frac{v_B}{r_{B/IC}} = \frac{1.2}{1.3856} = 0.8660 \text{ rad/s}$$

Vector Analysis

$$\begin{aligned}\mathbf{a}_B &= \boldsymbol{\alpha} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A} \\ &= (-3\mathbf{k}) \times (0.2\mathbf{j}) - 6^2(0.2\mathbf{j}) \\ &= [0.6\mathbf{i} - 7.2\mathbf{j}] \text{ m/s}^2 \\ \mathbf{a}_C &= \mathbf{a}_B + \boldsymbol{\alpha}_{BC} \times \mathbf{r}_{C/B} - \omega_{BC}^2 \mathbf{r}_{C/B} \\ a_C \cos 30^\circ \mathbf{i} + a_C \sin 30^\circ \mathbf{j} &= (0.6\mathbf{i} - 7.2\mathbf{j}) + (\alpha_{BC} \mathbf{k} \times 0.8\mathbf{i}) - 0.8660^2(0.8\mathbf{i})\end{aligned}$$

$$0.8660a_C \mathbf{i} + 0.5a_C \mathbf{j} = (0.8\alpha_{BC} - 7.2)\mathbf{j}$$

$$0.8660a_C = 0 \quad (1)$$

$$0.5a_C = 0.8\alpha_{BC} - 7.2 \quad (2)$$

$$a_C = 0 \quad \alpha_{BC} = 9 \text{ rad/s}^2 \quad \text{Ans.}$$

Scalar Analysis

$$\mathbf{a}_C = \mathbf{a}_B + \mathbf{a}_{C/B}$$

$$\begin{bmatrix} a_C \\ \nearrow 30^\circ \end{bmatrix} = \begin{bmatrix} 3(0.2) \\ \rightarrow \end{bmatrix} + \begin{bmatrix} (6)^2(0.2) \\ \downarrow \end{bmatrix} + \begin{bmatrix} \alpha_{BC}(0.8) \\ \uparrow \end{bmatrix} + \begin{bmatrix} (0.8660)^2(0.8) \\ \leftarrow \end{bmatrix}$$

This yields Eqs. (1) and (2).

## Chapter 17

**F17-1.**  $\pm \Sigma F_x = m(a_G)_x$ ;  $100\left(\frac{4}{5}\right) = 100a$

$$a = 0.8 \text{ m/s}^2 \rightarrow \quad \text{Ans.}$$

$$+\uparrow \Sigma F_y = m(a_G)_y;$$

$$N_A + N_B - 100\left(\frac{3}{5}\right) - 100(9.81) = 0 \quad (1)$$

$$\zeta + \Sigma M_G = 0;$$

$$N_A(0.6) + 100\left(\frac{3}{5}\right)(0.7) - N_B(0.4) - 100\left(\frac{4}{5}\right)(0.7) = 0 \quad (2)$$

$$N_A = 430.4 \text{ N} = 430 \text{ N} \quad \text{Ans.}$$

$$N_B = 610.6 \text{ N} = 611 \text{ N} \quad \text{Ans.}$$

**F17-2.**  $\Sigma F_{x'} = m(a_G)_{x'}$ ;  $80(9.81) \sin 15^\circ = 80a$

$$a = 2.54 \text{ m/s}^2 \quad \text{Ans.}$$

$$\Sigma F_{y'} = m(a_G)_{y'};$$

$$N_A + N_B - 80(9.81) \cos 15^\circ = 0 \quad (1)$$

$$\zeta + \Sigma M_G = 0;$$

$$N_A(0.5) - N_B(0.5) = 0 \quad (2)$$

$$N_A = N_B = 379 \text{ N} \quad \text{Ans.}$$

**F17-3.**  $\zeta + \Sigma M_A = \Sigma (\mathcal{M}_k)_A$ ;  $50\left(\frac{3}{5}\right)(2.1) = 10a(1.05)$

$$a = 6 \text{ m/s}^2 \quad \text{Ans.}$$

$$\pm \Sigma F_x = m(a_G)_x$$
;  $A_x + 50\left(\frac{3}{5}\right) = 10(6)$

$$A_x = 30 \text{ N} \quad \text{Ans.}$$

$$+\uparrow \Sigma F_y = m(a_G)_y$$
;  $A_y - 10(9.81) + 50\left(\frac{4}{5}\right) = 0$

$$A_y = 58.1 \text{ N} \quad \text{Ans.}$$

**F17-4.**  $F_A = \mu_s N_A = 0.2N_A$   $F_B = \mu_s N_B = 0.2N_B$

$$\pm \Sigma F_x = m(a_G)_x;$$

$$0.2N_A + 0.2N_B = 100a \quad (1)$$

$$+\uparrow \Sigma F_y = m(a_G)_y;$$

$$N_A + N_B - 100(9.81) = 0 \quad (2)$$

$$\zeta + \Sigma M_G = 0;$$

$$0.2N_A(0.75) + N_A(0.9) + 0.2N_B(0.75) - N_B(0.6) = 0 \quad (3)$$

Solving Eqs. (1), (2), and (3),

$$N_A = 294.3 \text{ N} = 294 \text{ N}$$

$$N_B = 686.7 \text{ N} = 687 \text{ N}$$

$$a = 1.96 \text{ m/s}^2 \quad \text{Ans.}$$

Since  $N_A$  is positive, the table will indeed slide before it tips.

**F17-5.**  $(a_G)_t = \alpha r = \alpha(1.5 \text{ m})$

$$(a_G)_n = \omega^2 r = (5 \text{ rad/s})^2(1.5 \text{ m}) = 37.5 \text{ m/s}^2$$

$$\Sigma F_t = m(a_G)_t; \quad 100 \text{ N} = 50 \text{ kg}[\alpha(1.5 \text{ m})] \\ \alpha = 1.33 \text{ rad/s}^2 \quad \text{Ans.}$$

$$\Sigma F_n = m(a_G)_n; \quad T_{AB} + T_{CD} - 50(9.81) \text{ N} \\ = 50 \text{ kg}(37.5 \text{ m/s}^2)$$

$$T_{AB} + T_{CD} = 2365.5$$

$$\zeta + \Sigma M_G = 0; \quad T_{CD}(1 \text{ m}) - T_{AB}(1 \text{ m}) = 0$$

$$T_{AB} = T_{CD} = 1182.75 \text{ N} = 1.18 \text{ kN} \quad \text{Ans.}$$

**F17-6.**  $\zeta + \Sigma M_C = 0;$

$$\mathbf{a}_G = \mathbf{a}_D = \mathbf{a}_B$$

$$D_y(0.6) - 450 = 0 \quad D_y = 750 \text{ N} \quad \text{Ans.}$$

$$(a_G)_n = \omega^2 r = 6^2(0.6) = 21.6 \text{ m/s}^2$$

$$(a_G)_t = \alpha r = \alpha(0.6)$$

$$+\uparrow \Sigma F_t = m(a_G)_t;$$

$$750 - 50(9.81) = 50[\alpha(0.6)]$$

$$\alpha = 8.65 \text{ rad/s}^2 \quad \text{Ans.}$$

$$\pm \Sigma F_n = m(a_G)_n;$$

$$F_{AB} + D_x = 50(21.6) \quad (1)$$

$$\zeta + \Sigma M_G = 0;$$

$$D_x(0.4) + 750(0.1) - F_{AB}(0.4) = 0 \quad (2)$$

$$D_x = 446.25 \text{ N} = 446 \text{ N} \quad \text{Ans.}$$

$$F_{AB} = 633.75 \text{ N} = 634 \text{ N} \quad \text{Ans.}$$

**F17-7.**  $I_O = mk_O^2 = 100(0.5^2) = 25 \text{ kg} \cdot \text{m}^2$   
 $\zeta + \Sigma M_O = I_O \alpha; \quad -100(0.6) = -25\alpha$   
 $\alpha = 2.4 \text{ rad/s}^2$   
 $\omega = \omega_0 + \alpha_c t$   
 $\omega = 0 + 2.4(3) = 7.2 \text{ rad/s}$

Ans.

**F17-8.**  $I_O = \frac{1}{2}mr^2 = \frac{1}{2}(50)(0.3^2) = 2.25 \text{ kg} \cdot \text{m}^2$   
 $\zeta + \Sigma M_O = I_O \alpha;$   
 $-9t = -2.25\alpha \quad \alpha = (4t) \text{ rad/s}^2$   
 $d\omega = \alpha dt$   
 $\int_0^\omega d\omega = \int_0^t 4t dt$   
 $\omega = (2t^2) \text{ rad/s}$   
 $\omega = 2(4^2) = 32 \text{ rad/s}$

Ans.

**F17-9.**  $(a_G)_t = \alpha r_G = \alpha(0.15)$   
 $(a_G)_n = \omega^2 r_G = 6^2(0.15) = 5.4 \text{ m/s}^2$   
 $I_O = I_G + md^2 = \frac{1}{12}(30)(0.9^2) + 30(0.15^2)$   
 $= 2.7 \text{ kg} \cdot \text{m}^2$   
 $\zeta + \Sigma M_O = I_O \alpha; \quad 60 - 30(9.81)(0.15) = 2.7\alpha$   
 $\alpha = 5.872 \text{ rad/s}^2 = 5.87 \text{ rad/s}^2$   
 $\leftarrow \Sigma F_n = m(a_G)_n; \quad O_n = 30(5.4) = 162 \text{ N}$   
 $+\uparrow \Sigma F_t = m(a_G)_t;$   
 $O_t - 30(9.81) = 30[5.872(0.15)]$   
 $O_t = 320.725 \text{ N} = 321 \text{ N}$

Ans.

Ans.

Ans.

**F17-10.**  $(a_G)_t = \alpha r_G = \alpha(0.3)$   
 $(a_G)_n = \omega^2 r_G = 10^2(0.3) = 30 \text{ m/s}^2$   
 $I_O = I_G + md^2 = \frac{1}{2}(30)(0.3^2) + 30(0.3^2)$   
 $= 4.05 \text{ kg} \cdot \text{m}^2$   
 $\zeta + \Sigma M_O = I_O \alpha;$   
 $50\left(\frac{3}{5}\right)(0.3) + 50\left(\frac{4}{5}\right)(0.3) = 4.05\alpha$   
 $\alpha = 5.185 \text{ rad/s}^2 = 5.19 \text{ rad/s}^2$   
 $+\uparrow \Sigma F_n = m(a_G)_n;$   
 $O_n + 50\left(\frac{3}{5}\right) - 30(9.81) = 30(30)$   
 $O_n = 1164.3 \text{ N} = 1.16 \text{ kN}$   
 $\rightarrow \Sigma F_t = m(a_G)_t;$   
 $O_t + 50\left(\frac{4}{5}\right) = 30[5.185(0.3)]$   
 $O_t = 6.67 \text{ N}$

Ans.

Ans.

Ans.

**F17-11.**  $I_G = \frac{1}{12}ml^2 = \frac{1}{12}(15 \text{ kg})(0.9 \text{ m})^2 = 1.0125 \text{ kg} \cdot \text{m}^2$   
 $(a_G)_n = \omega^2 r_G = 0$   
 $(a_G)_t = \alpha(0.15 \text{ m})$   
 $I_O = I_G + md_{OG}^2$   
 $= 1.0125 \text{ kg} \cdot \text{m}^2 + 15 \text{ kg}(0.15 \text{ m})^2$   
 $= 1.35 \text{ kg} \cdot \text{m}^2$   
 $\zeta + \Sigma M_O = I_O \alpha;$   
 $[15(9.81) \text{ N}](0.15 \text{ m}) = (1.35 \text{ kg} \cdot \text{m}^2)\alpha$   
 $\alpha = 16.35 \text{ rad/s}^2$   
 $+\downarrow \Sigma F_t = m(a_G)_t; \quad -O_t + 15(9.81) \text{ N}$   
 $= (15 \text{ kg})[16.35 \text{ rad/s}^2(0.15 \text{ m})]$   
 $O_t = 110.36 \text{ N} = 110 \text{ N}$   
 $\rightarrow \Sigma F_n = m(a_G)_n; \quad O_n = 0$

Ans.

Ans.

Ans.

**F17-12.**  $(a_G)_t = \alpha r_G = \alpha(0.45)$   
 $(a_G)_n = \omega^2 r_G = 6^2(0.45) = 16.2 \text{ m/s}^2$   
 $I_O = \frac{1}{3}ml^2 = \frac{1}{3}(30)(0.9^2) = 8.1 \text{ kg} \cdot \text{m}^2$   
 $\zeta + \Sigma M_O = I_O \alpha;$   
 $300\left(\frac{4}{5}\right)(0.6) - 30(9.81)(0.45) = 8.1\alpha$   
 $\alpha = 1.428 \text{ rad/s}^2 = 1.43 \text{ rad/s}^2$   
 $\leftarrow \Sigma F_n = m(a_G)_n; \quad O_n + 300\left(\frac{3}{5}\right) = 30(16.2)$   
 $O_n = 306 \text{ N}$   
 $+\uparrow \Sigma F_t = m(a_G)_t; \quad O_t + 300\left(\frac{4}{5}\right) - 30(9.81)$   
 $= 30[1.428(0.45)]$   
 $O_t = 73.58 \text{ N} = 73.6 \text{ N}$

Ans.

Ans.

Ans.

**F17-13.**  $I_G = \frac{1}{12}ml^2 = \frac{1}{12}(60)(3^2) = 45 \text{ kg} \cdot \text{m}^2$   
 $+\uparrow \Sigma F_y = m(a_G)_y;$   
 $80 - 20 = 60a_G \quad a_G = 1 \text{ m/s}^2 \uparrow$   
 $\zeta + \Sigma M_G = I_G \alpha; \quad 80(1) + 20(0.75) = 45\alpha$   
 $\alpha = 2.11 \text{ rad/s}^2$

Ans.

**F17-14.**  $\zeta + \Sigma M_A = (\mathcal{M}_k)_A;$   
 $-200(0.3) = -100a_G(0.3) - 4.5\alpha$   
 $30a_G + 4.5\alpha = 60 \quad (1)$   
 $a_G = \alpha r = \alpha(0.3) \quad (2)$   
 $\alpha = 4.44 \text{ rad/s}^2 \quad a_G = 1.33 \text{ m/s}^2 \rightarrow$

Ans.

**F17-15.**  $+\uparrow \Sigma F_y = m(a_G)_y;$   
 $N - 20(9.81) = 0 \quad N = 196.2 \text{ N}$   
 $\rightarrow \Sigma F_x = m(a_G)_x; \quad 0.5(196.2) = 20a_O$   
 $a_O = 4.905 \text{ m/s}^2 \rightarrow$

Ans.



$$\zeta + \Sigma M_O = I_O \alpha;$$

$$0.5(196.2)(0.4) - 100 = -1.8\alpha$$

$$\alpha = 33.8 \text{ rad/s}^2$$

Ans.

**F17-16.** Sphere  $I_G = \frac{2}{5}(20)(0.15)^2 = 0.18 \text{ kg} \cdot \text{m}^2$

$$\zeta + \Sigma M_{IC} = (\mathcal{M}_k)_{IC};$$

$$20(9.81)\sin 30^\circ(0.15) = 0.18\alpha + (20a_G)(0.15)$$

$$0.18\alpha + 3a_G = 14.715$$

$$a_G = \alpha r = \alpha(0.15)$$

$$\alpha = 23.36 \text{ rad/s}^2 = 23.4 \text{ rad/s}^2$$

Ans.

$$a_G = 3.504 \text{ m/s}^2 = 3.50 \text{ m/s}^2$$

Ans.

**F17-17.**  $+\uparrow \Sigma F_y = m(a_G)_y;$

$$N - 200(9.81) = 0 \quad N = 1962 \text{ N}$$

$$\rightarrow \Sigma F_x = m(a_G)_x;$$

$$T - 0.2(1962) = 200a_G \quad (1)$$

$$\begin{aligned} \zeta + \Sigma M_A = (\mathcal{M}_k)_A; \quad 450 - 0.2(1962)(1) \\ = 18\alpha + 200a_G(0.4) \end{aligned} \quad (2)$$

$$(a_A)_t = 0 \quad a_A = (a_A)_n$$

$$\mathbf{a}_G = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{G/A} - \omega^2 \mathbf{r}_{G/A}$$

$$a_G \mathbf{i} = -a_A \mathbf{j} + \alpha \mathbf{k} \times (-0.4 \mathbf{j}) - \omega^2(-0.4 \mathbf{j})$$

$$a_G \mathbf{i} = 0.4\alpha \mathbf{i} + (0.4\omega^2 - a_A) \mathbf{j}$$

$$a_G = 0.4\alpha \quad (3)$$

Solving Eqs. (1), (2), and (3),

$$\alpha = 1.15 \text{ rad/s}^2 \quad a_G = 0.461 \text{ m/s}^2$$

$$T = 485 \text{ N}$$

Ans.

**F17-18.**  $\rightarrow \Sigma F_x = m(a_G)_x; \quad 0 = 12(a_G)_x \quad (a_G)_x = 0$

$$\zeta + \Sigma M_A = (\mathcal{M}_k)_A$$

$$\begin{aligned} -12(9.81)(0.3) = 12(a_G)_y(0.3) - \frac{1}{12}(12)(0.6)^2\alpha \\ 0.36\alpha - 3.6(a_G)_y = 35.316 \end{aligned} \quad (1)$$

$$\omega = 0$$

$$\mathbf{a}_G = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{G/A} - \omega^2 \mathbf{r}_{G/A}$$

$$(a_G)_y \mathbf{j} = a_A \mathbf{i} + (-\alpha \mathbf{k}) \times (0.3 \mathbf{i}) - 0$$

$$(a_G)_y \mathbf{j} = (a_A) \mathbf{i} - 0.3 \mathbf{j}$$

$$a_A = 0$$

$$(a_G)_y = -0.3\alpha \quad (2)$$

Solving Eqs. (1) and (2)

$$\alpha = 24.5 \text{ rad/s}^2$$

$$(a_G)_y = -7.36 \text{ m/s}^2 = 7.36 \text{ m/s}^2 \downarrow \quad \text{Ans.}$$

## Chapter 18

**F18-1.**  $I_O = mk_O^2 = 80(0.4^2) = 12.8 \text{ kg} \cdot \text{m}^2$

$$T_1 = 0$$

$$T_2 = \frac{1}{2}I_O\omega^2 = \frac{1}{2}(12.8)\omega^2 = 6.4\omega^2$$

$$s = \theta r = 20(2\pi)(0.6) = 24\pi \text{ m}$$

$$T_1 + \Sigma U_{1-2} = T_2$$

$$0 + 50(24\pi) = 6.4\omega^2$$

$$\omega = 24.3 \text{ rad/s}$$

Ans.

**F18-2.**  $T_1 = 0$

$$\begin{aligned} T_2 &= \frac{1}{2}m(v_G)_2^2 + \frac{1}{2}I_G\omega_2^2 \\ &= \frac{1}{2}(25 \text{ kg})(0.75\omega_2)^2 \\ &\quad + \frac{1}{2}\left[\frac{1}{12}(25 \text{ kg})(1.5 \text{ m})^2\right]\omega_2^2 \end{aligned}$$

$$T_2 = 11.719\omega_2^2$$

Or,

$$\begin{aligned} I_O &= \frac{1}{3}ml^2 = \frac{1}{3}(25 \text{ kg})(1.5 \text{ m})^2 \\ &= 18.75 \text{ kg} \cdot \text{m}^2 \end{aligned}$$

So that

$$\begin{aligned} T_2 &= \frac{1}{2}I_O\omega_2^2 = \frac{1}{2}(18.75 \text{ kg} \cdot \text{m}^2)\omega_2^2 \\ &= 9.375\omega_2^2 \end{aligned}$$

$$T_1 + \Sigma U_{1-2} = T_2$$

$$T_1 + [-W_{yG} + M\theta] = T_2$$

$$\begin{aligned} 0 + [-(25(9.81 \text{ N})(0.75 \text{ m})) + (150 \text{ N} \cdot \text{m})(\frac{\pi}{2})] \\ = 9.375\omega_2^2 \end{aligned}$$

$$\omega_2 = 2.35 \text{ rad/s}$$

Ans.

**F18-3.**  $(v_G)_2 = \omega_2 r_{G/IC} = \omega_2(2.5)$

$$I_G = \frac{1}{12}ml^2 = \frac{1}{12}(50)(5^2) = 104.17 \text{ kg} \cdot \text{m}^2$$

$$T_1 = 0$$

$$\begin{aligned} T_2 &= \frac{1}{2}m(v_G)_2^2 + \frac{1}{2}I_G\omega_2^2 \\ &= \frac{1}{2}(50)[\omega_2(2.5)]^2 + \frac{1}{2}(104.17)\omega_2^2 = 208.33\omega_2^2 \end{aligned}$$

$$U_P = P s_P = 600(3) = 1800 \text{ J}$$

$$U_W = -Wh = -50(9.81)(2.5 - 2) = -245.25 \text{ J}$$

$$T_1 + \Sigma U_{1-2} = T_2$$

$$0 + 1800 + (-245.25) = 208.33\omega_2^2$$

$$\omega_2 = 2.732 \text{ rad/s} = 2.73 \text{ rad/s}$$

Ans.

$$\begin{aligned}\text{F18-4. } T &= \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2 \\ &= \frac{1}{2}(50 \text{ kg})(0.4\omega)^2 + \frac{1}{2}[50 \text{ kg}(0.3 \text{ m})^2]\omega^2 \\ &= 6.25\omega^2 \text{ J}\end{aligned}$$

Or,

$$\begin{aligned}T &= \frac{1}{2}I_{IC}\omega^2 \\ &= \frac{1}{2}[50 \text{ kg}(0.3 \text{ m})^2 + 50 \text{ kg}(0.4 \text{ m})^2]\omega^2 \\ &= 6.25\omega^2 \text{ J}\end{aligned}$$

$$s_G = \theta r = 10(2\pi \text{ rad})(0.4 \text{ m}) = 8\pi \text{ m}$$

$$T_1 + \Sigma U_{1-2} = T_2$$

$$T_1 + P \cos 30^\circ s_G = T_2$$

$$0 + (50 \text{ N})\cos 30^\circ(8\pi \text{ m}) = 6.25\omega^2 \text{ J}$$

$$\omega = 13.2 \text{ rad/s}$$

*Ans.*

$$\text{F18-5. } I_G = \frac{1}{12}ml^2 = \frac{1}{12}(30)(3^2) = 22.5 \text{ kg} \cdot \text{m}^2$$

$$T_1 = 0$$

$$\begin{aligned}T_2 &= \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2 \\ &= \frac{1}{2}(30)[\omega(0.5)]^2 + \frac{1}{2}(22.5)\omega^2 = 15\omega^2\end{aligned}$$

Or,

$$\begin{aligned}I_O &= I_G + md^2 = \frac{1}{12}(30)(3^2) + 30(0.5^2) \\ &= 30 \text{ kg} \cdot \text{m}^2\end{aligned}$$

$$T_2 = \frac{1}{2}I_O\omega^2 = \frac{1}{2}(30)\omega^2 = 15\omega^2$$

$$s_1 = \theta r_1 = 8\pi(0.5) = 4\pi \text{ m}$$

$$s_2 = \theta r_2 = 8\pi(1.5) = 12\pi \text{ m}$$

$$U_{P_1} = P_1 s_1 = 30(4\pi) = 120\pi \text{ J}$$

$$U_{P_2} = P_2 s_2 = 20(12\pi) = 240\pi \text{ J}$$

$$U_M = M\theta = 20[4(2\pi)] = 160\pi \text{ J}$$

$$U_W = (0 \text{ bar returns to same position})$$

$$T_1 + \Sigma U_{1-2} = T_2$$

$$0 + 120\pi + 240\pi + 160\pi = 15\omega^2$$

$$\omega = 10.44 \text{ rad/s} = 10.4 \text{ rad/s}$$

*Ans.*

$$\text{F18-6. } v_G = \omega r = \omega(0.4)$$

$$I_G = mk_G^2 = 20(0.3^2) = 1.8 \text{ kg} \cdot \text{m}^2$$

$$T_1 = 0$$

$$\begin{aligned}T_2 &= \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2 \\ &= \frac{1}{2}(20)[\omega(0.4)]^2 + \frac{1}{2}(1.8)\omega^2 \\ &= 2.5\omega^2\end{aligned}$$

$$U_M = M\theta = M\left(\frac{s_0}{r}\right) = 50\left(\frac{20}{0.4}\right) = 2500 \text{ J}$$

$$T_1 + \Sigma U_{1-2} = T_2$$

$$0 + 2500 = 2.5\omega^2$$

$$\omega = 31.62 \text{ rad/s} = 31.6 \text{ rad/s}$$

*Ans.*

$$\text{F18-7. } v_G = \omega r = \omega(0.3)$$

$$I_G = \frac{1}{2}mr^2 = \frac{1}{2}(30)(0.3^2) = 1.35 \text{ kg} \cdot \text{m}^2$$

$$T_1 = 0$$

$$\begin{aligned}T_2 &= \frac{1}{2}m(v_G)_2^2 + \frac{1}{2}I_G\omega_2^2 \\ &= \frac{1}{2}(30)[\omega_2(0.3)]^2 + \frac{1}{2}(1.35)\omega_2^2 = 2.025\omega_2^2\end{aligned}$$

$$(V_g)_1 = Wy_1 = 0$$

$$(V_g)_2 = -Wy_2 = -30(9.81)(0.3) = -88.29 \text{ J}$$

$$T_1 + V_1 = T_2 + V_2$$

$$0 + 0 = 2.025\omega_2^2 + (-88.29)$$

$$\omega_2 = 6.603 \text{ rad/s} = 6.60 \text{ rad/s}$$

*Ans.*

$$\text{F18-8. } v_O = \omega r_{O/IC} = \omega(0.2)$$

$$I_O = mk_O^2 = 50(0.3^2) = 4.5 \text{ kg} \cdot \text{m}^2$$

$$T_1 = 0$$

$$\begin{aligned}T_2 &= \frac{1}{2}m(v_O)_2^2 + \frac{1}{2}I_O\omega_2^2 \\ &= \frac{1}{2}(50)[\omega_2(0.2)]^2 + \frac{1}{2}(4.5)\omega_2^2 \\ &= 3.25\omega_2^2\end{aligned}$$

$$(V_g)_1 = Wy_1 = 0$$

$$\begin{aligned}(V_g)_2 &= -Wy_2 = -50(9.81)(6 \sin 30^\circ) \\ &= -1471.5 \text{ J}\end{aligned}$$

$$T_1 + V_1 = T_2 + V_2$$

$$0 + 0 = 3.25\omega_2^2 + (-1471.5)$$

$$\omega_2 = 21.28 \text{ rad/s} = 21.3 \text{ rad/s}$$

*Ans.*

$$\text{F18-9. } v_G = \omega r_G = \omega(1.5)$$

$$I_G = \frac{1}{12}(60)(3^2) = 45 \text{ kg} \cdot \text{m}^2$$

$$T_1 = 0$$

$$\begin{aligned}T_2 &= \frac{1}{2}m(v_G)_2^2 + \frac{1}{2}I_G\omega_2^2 \\ &= \frac{1}{2}(60)[\omega_2(1.5)]^2 + \frac{1}{2}(45)\omega_2^2 \\ &= 90\omega_2^2\end{aligned}$$

Or,

$$T_2 = \frac{1}{2}I_O\omega_2^2 = \frac{1}{2}[45 + 60(1.5^2)]\omega_2^2 = 90\omega_2^2$$

$$(V_g)_1 = Wy_1 = 0$$

$$\begin{aligned}(V_g)_2 &= -Wy_2 = -60(9.81)(1.5 \sin 45^\circ) \\ &= -624.30 \text{ J}\end{aligned}$$

$$(V_e)_1 = \frac{1}{2}ks_1^2 = 0$$

$$(V_e)_2 = \frac{1}{2}ks_2^2 = \frac{1}{2}(150)(3 \sin 45^\circ)^2 = 337.5 \text{ J}$$

$$T_1 + V_1 = T_2 + V_2$$

$$0 + 0 = 90\omega_2^2 + [-624.30 + 337.5]$$

$$\omega_2 = 1.785 \text{ rad/s} = 1.79 \text{ rad/s}$$

*Ans.*

$$\text{F18-10. } v_G = \omega r_G = \omega(0.75)$$

$$I_G = \frac{1}{12}(30)(1.5^2) = 5.625 \text{ kg} \cdot \text{m}^2$$

$$T_1 = 0$$

$$T_2 = \frac{1}{2} m(v_G)_2^2 + \frac{1}{2} I_G \omega_2^2 \\ = \frac{1}{2} (30)[\omega(0.75)]^2 + \frac{1}{2} (5.625)\omega_2^2 = 11.25\omega_2^2$$

Or,

$$T_2 = \frac{1}{2} I_O \omega_2^2 = \frac{1}{2} [5.625 + 30(0.75^2)] \omega_2^2 \\ = 11.25\omega_2^2$$

$$(V_g)_1 = Wy_1 = 0$$

$$(V_g)_2 = -Wy_2 = -30(9.81)(0.75) \\ = -220.725 \text{ J}$$

$$(V_e)_1 = \frac{1}{2} ks_1^2 = 0$$

$$(V_e)_2 = \frac{1}{2} ks_2^2 = \frac{1}{2} (80)(\sqrt{2^2 + 1.5^2} - 0.5)^2 = 160 \text{ J}$$

$$T_1 + V_1 = T_2 + V_2$$

$$0 + 0 = 11.25\omega_2^2 + (-220.725 + 160)$$

$$\omega_2 = 2.323 \text{ rad/s} = 2.32 \text{ rad/s} \quad \text{Ans.}$$

**F18-11.**  $(v_G)_2 = \omega_2 r_{G/IC} = \omega_2(0.75)$

$$I_G = \frac{1}{12} (30)(1.5^2) = 5.625 \text{ kg} \cdot \text{m}^2$$

$$T_1 = 0$$

$$T_2 = \frac{1}{2} m(v_G)_2^2 + \frac{1}{2} I_G \omega_2^2 \\ = \frac{1}{2} (30)[\omega_2(0.75)]^2 + \frac{1}{2} (5.625)\omega_2^2 = 11.25\omega_2^2$$

$$(V_g)_1 = Wy_1 = 30(9.81)(0.75 \sin 45^\circ) = 156.08 \text{ J}$$

$$(V_g)_2 = -Wy_2 = 0$$

$$(V_e)_1 = \frac{1}{2} ks_1^2 = 0$$

$$(V_e)_2 = \frac{1}{2} ks_2^2 = \frac{1}{2} (300)(1.5 - 1.5 \cos 45^\circ)^2 \\ = 28.95 \text{ J}$$

$$T_1 + V_1 = T_2 + V_2$$

$$0 + (156.08 + 0) = 11.25\omega_2^2 + (0 + 28.95)$$

$$\omega_2 = 3.362 \text{ rad/s} = 3.36 \text{ rad/s} \quad \text{Ans.}$$

**F18-12.**  $(V_g)_1 = -Wy_1 = -[20(9.81) \text{ N}](1 \text{ m}) = -196.2 \text{ J}$

$$(V_g)_2 = 0$$

$$(V_e)_1 = \frac{1}{2} ks_1^2 \\ = \frac{1}{2} (100 \text{ N/m}) \left( \sqrt{(3 \text{ m})^2 + (2 \text{ m})^2} - 0.5 \text{ m} \right)^2 \\ = 482.22 \text{ J}$$

$$(V_e)_2 = \frac{1}{2} ks_2^2 = \frac{1}{2} (100 \text{ N/m})(1 \text{ m} - 0.5 \text{ m})^2 \\ = 12.5 \text{ J}$$

$$T_1 = 0$$

$$T_2 = \frac{1}{2} I_A \omega^2 = \frac{1}{2} \left[ \frac{1}{3} (20 \text{ kg})(2 \text{ m})^2 \right] \omega^2 \\ = 13.3333\omega^2$$

$$T_1 + V_1 = T_2 + V_2$$

$$0 + [-196.2 \text{ J} + 482.22 \text{ J}] \\ = 13.3333\omega_2^2 + [0 + 12.5 \text{ J}]$$

$$\omega_2 = 4.53 \text{ rad/s} \quad \text{Ans.}$$

## Chapter 19

**F19-1.**  $\zeta + I_O \omega_1 + \sum \int_{t_1}^{t_2} M_O dt = I_O \omega_2$

$$0 + \int_0^{4 \text{ s}} 3t^2 dt = [60(0.3)^2] \omega_2$$

$$\omega_2 = 11.85 \text{ rad/s} = 11.9 \text{ rad/s} \quad \text{Ans.}$$

**F19-2.**  $\zeta + (H_A)_1 + \sum \int_{t_1}^{t_2} M_A dt = (H_A)_2$

$$0 + 300(6) = 300(0.4^2) \omega_2 + 300[\omega(0.6)](0.6)$$

$$\omega_2 = 11.54 \text{ rad/s} = 11.5 \text{ rad/s} \quad \text{Ans.}$$

$$\pm \quad m(v_1)_x + \sum \int_{t_1}^{t_2} F_x dt = m(v_2)_x$$

$$0 + F_f(6) = 300[11.54(0.6)]$$

$$F_f = 346 \text{ N} \quad \text{Ans.}$$

**F19-3.**  $v_A = \omega_A r_{A/IC} = \omega_A(0.15)$

$$\zeta + \sum M_O = 0; \quad 9 - A_t(0.45) = 0 \quad A_t = 20 \text{ N}$$

$$\zeta + (H_C)_1 + \sum \int_{t_1}^{t_2} M_C dt = (H_C)_2$$

$$0 + [20(5)](0.15) \\ = 10[\omega_A(0.15)](0.15) \\ + [10(0.1^2)] \omega_A$$

$$\omega_A = 46.2 \text{ rad/s} \quad \text{Ans.}$$

**F19-4.**  $I_A = mk_A^2 = 10(0.08^2) = 0.064 \text{ kg} \cdot \text{m}^2$

$$I_B = mk_B^2 = 50(0.15^2) = 1.125 \text{ kg} \cdot \text{m}^2$$

$$\omega_A = \left( \frac{r_B}{r_A} \right) \omega_B = \left( \frac{0.2}{0.1} \right) \omega_B = 2\omega_B$$

$$\zeta + I_A(\omega_A)_1 + \sum \int_{t_1}^{t_2} M_A dt = I_A(\omega_A)_2$$

$$0 + 10(5) - \int_0^{5 \text{ s}} F(0.1) dt = 0.064[2(\omega_B)_2]$$

$$\int_0^{5 \text{ s}} F dt = 500 - 1.28(\omega_B)_2 \quad (1)$$

$$\zeta + I_B(\omega_B)_1 + \sum \int_{t_1}^{t_2} M_B dt = I_B(\omega_B)_2$$

$$0 + \int_0^{5 \text{ s}} F(0.2) dt = 1.125(\omega_B)_2$$

$$\int_0^{5 \text{ s}} F dt = 5.625(\omega_B)_2 \quad (2)$$

Equating Eqs. (1) and (2),

$$500 - 1.28(\omega_B)_2 = 5.625(\omega_B)_2$$

$$(\omega_B)_2 = 72.41 \text{ rad/s} = 72.4 \text{ rad/s} \quad \text{Ans.}$$

**F19-5.**  $(\rightarrow) \quad m[(v_O)_x]_1 + \Sigma \int F_x dt = m[(v_O)_x]_2$

$$0 + (150 \text{ N})(3 \text{ s}) + F_A(3 \text{ s})$$

$$= (50 \text{ kg})(0.3\omega_2)$$

$$\zeta + I_G\omega_1 + \Sigma \int M_G dt = I_G\omega_2$$

$$0 + (150 \text{ N})(0.2 \text{ m})(3 \text{ s}) - F_A(0.3 \text{ m})(3 \text{ s})$$

$$= [(50 \text{ kg})(0.175 \text{ m})^2] \omega_2$$

$$\omega_2 = 37.3 \text{ rad/s}$$

$$F_A = 36.53 \text{ N}$$

Also,

$$I_{IC}\omega_1 + \Sigma \int M_{IC} dt = I_{IC}\omega_2$$

$$0 + [(150 \text{ N})(0.2 + 0.3) \text{ m}](3 \text{ s})$$

$$= [(50 \text{ kg})(0.175 \text{ m})^2 + (50 \text{ kg})(0.3 \text{ m})^2] \omega_2$$

$$\omega_2 = 37.3 \text{ rad/s}$$

*Ans.*

**F19-6.**  $(+\uparrow) \quad m[(v_G)_1]_y + \Sigma \int F_y dt = m[(v_G)_2]_y$

$$0 + N_A(3 \text{ s}) - [150(9.81 \text{ N})](3 \text{ s}) = 0$$

$$N_A = 1471.5 \text{ N}$$

$$\zeta + (H_{IC})_1 + \Sigma \int M_{IC} dt = (H_{IC})_2$$

$$0 + (250 \text{ N} \cdot \text{m})(3 \text{ s}) - [0.15(1471.5 \text{ N})(3 \text{ s})](0.5 \text{ m})$$

$$= [150 \text{ kg}(1.25 \text{ m})^2] \omega_2 + (150 \text{ kg})[\omega_2(1 \text{ m})](1 \text{ m})$$

$$\omega_2 = 1.09 \text{ rad/s}$$

*Ans.*

*Ans.*

# Preliminary Problems Dynamics Solutions

## Chapter 12

**P12-1.** a)  $v = \frac{ds}{dt} = \frac{d}{dt}(2t^3) = 6t^2 \Big|_{t=2\text{ s}} = 24 \text{ m/s}$

b)  $a ds = v dv, \quad v = 5s, \quad dv = 5 ds$   
 $a ds = (5s) 5 ds$

$a = 25s \Big|_{s=1\text{ m}} = 25 \text{ m/s}^2$

c)  $a = \frac{dv}{dt} = \frac{d}{dt}(4t + 5) = 4 \text{ m/s}^2$

d)  $v = v_0 + a_c t$   
 $v = 0 + 2(2) = 4 \text{ m/s}$

e)  $v^2 = v_0^2 + 2a_c(s - s_0)$   
 $v^2 = (3)^2 + 2(2)(4 - 0)$   
 $v = 5 \text{ m/s}$

f)  $a ds = v dv$   
 $\int_{s_1}^{s_2} s ds = \int_0^v v dv$   
 $s^2 \Big|_4^5 = v^2 \Big|_0^v$   
 $25 - 16 = v^2$   
 $v = 3 \text{ m/s}$

g)  $s = s_0 + v_0 t + \frac{1}{2} a_c t^2$   
 $s = 2 + 2(3) + \frac{1}{2}(4)(3)^2 = 26 \text{ m}$

h)  $dv = a dt$   
 $\int_0^v dv = \int_0^1 (8t^2) dt$   
 $v = 2.67t^3 \Big|_0^1 = 2.67 \text{ m/s}$

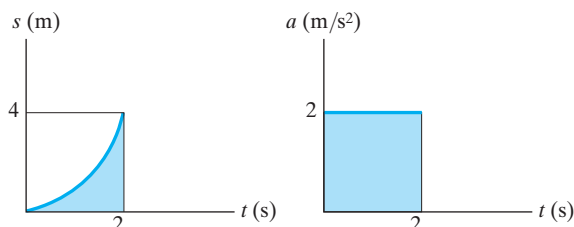
i)  $v = \frac{ds}{dt} = \frac{d}{dt}(3t^2 + 2) = 6t \Big|_{t=2\text{ s}} = 12 \text{ m/s}$

j)  $v_{\text{avg}} = \frac{\Delta s}{\Delta t} = \frac{6 \text{ m} - (-1 \text{ m})}{10 \text{ s} - 0} = 0.7 \text{ m/s} \rightarrow$   
 $(v_{\text{sp}})_{\text{avg}} = \frac{s_T}{\Delta t} = \frac{7 \text{ m} + 14 \text{ m}}{10 \text{ s} - 0} = 2.1 \text{ m/s}$

**P12-2.** a)  $v = 2t$

$s = t^2$

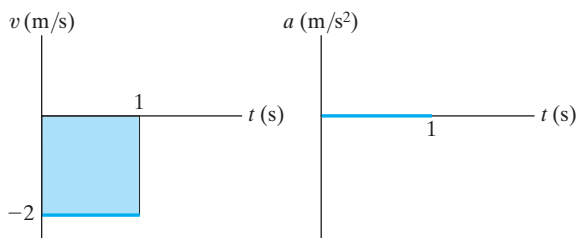
$a = 2$



b)  $s = -2t + 2$

$v = -2$

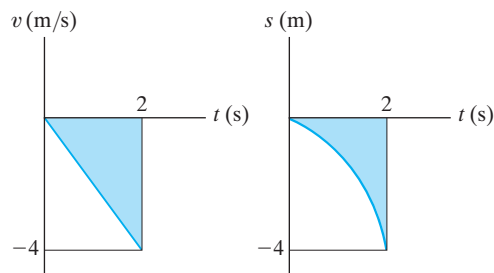
$a = 0$



c)  $a = -2$

$v = -2t$

$s = -t^2$



d)

$\Delta s = \int_0^3 v dt = \text{Area} = \frac{1}{2}(2)(2) + 2(3 - 2) = 4 \text{ m}$

$s - 0 = 4 \text{ m},$

$s = 4 \text{ m}$

$a = \frac{dv}{dt} = \text{slope at } t = 3 \text{ s}, a = 0$

- e) For  $a = 2$ ,  
 $v = 2t$

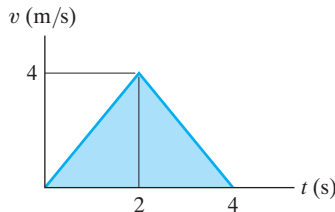
When  $t = 2$  s,  $v = 4$  m/s.

For  $a = -2$ ,

$$\int_4^v dv = \int_2^t -2 dt$$

$$v - 4 = -2t + 4$$

$$v = -2t + 8$$



f)  $\int_1^v v dv = \int_0^2 a ds = \text{Area}$

$$\frac{1}{2}v^2 - \frac{1}{2}(1)^2 = \frac{1}{2}(2)(4)$$

$$v = 3 \text{ m/s}$$

- g)  $v dv = a ds$  At  $s = 1$  m,  $v = 2$  m/s.

$$a = v \frac{dv}{ds} = v(\text{slope}) = 2(-2) = -4 \text{ m/s}^2$$

**P12-3.** a)  $y = 4x^2$   
 $\dot{y} = 8x\dot{x}$

$$\ddot{y} = (8\dot{x})\dot{x} + 8x(\ddot{x})$$

b)  $y = 3e^x$   
 $\dot{y} = 3e^x\dot{x}$

$$\ddot{y} = (3e^x\dot{x})\dot{x} + 3e^x(\ddot{x})$$

c)  $y = 6 \sin x$

$$\dot{y} = (6 \cos x)\dot{x}$$

$$\ddot{y} = [(-6 \sin x)\dot{x}]\dot{x} + (6 \cos x)(\ddot{x})$$

**P12-4.**  $y_A, t_{AB}, (v_B)_y$

$$20 = 0 + 40t_{AB}$$

$$0 = y_A + 0 + \frac{1}{2}(-9.81)(t_{AB})^2$$

$$(v_B)_y^2 = 0^2 + 2(-9.81)(0 - y_A)$$

**P12-5.**  $x_B, t_{AB}, (v_B)_y$

$$x_B = 0 + (10 \cos 30^\circ)(t_{AB})$$

$$0 = 8 + (10 \sin 30^\circ)t_{AB} + \frac{1}{2}(-9.81)t_{AB}^2$$

$$(v_B)_y^2 = 0^2 + 2(-9.81)(0 - 8)$$

**P12-6.**

$$x_B, y_B, (v_B)_y$$

$$x_B = 0 + (60 \cos 20^\circ)(5)$$

$$y_B = 0 + (60 \sin 20^\circ)(5) + \frac{1}{2}(-9.81)(5)^2$$

$$(v_B)_y = 60 \sin 20^\circ + (-9.81)(5)$$

**P12-7.** a)  $a_t = \dot{v} = 3 \text{ m/s}^2$

$$a_n = \frac{v^2}{\rho} = \frac{(2)^2}{1} = 4 \text{ m/s}^2$$

$$a = \sqrt{(3)^2 + (4)^2} = 5 \text{ m/s}^2$$

b)  $a_t = \dot{v} = 4 \text{ m/s}^2$

$$v^2 = v_0^2 + 2a_t(s - s_0)$$

$$v^2 = 0 + 2(4)(2 - 0)$$

$$v = 4 \text{ m/s}$$

$$a_n = \frac{v^2}{\rho} = \frac{(4)^2}{2} = 8 \text{ m/s}^2$$

c)  $a_t = 0$

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}{\frac{d^2y}{dx^2}} \bigg|_{x=0} = \frac{1 + 0}{4} = \frac{1}{4}$$

$$a_n = \frac{v^2}{\rho} = \frac{(2)^2}{\frac{1}{4}} = 16 \text{ m/s}^2$$

$$a = \sqrt{(0)^2 + (16)^2} = 16 \text{ m/s}^2$$

d)  $a_t ds = v dv$

$$a_t ds = (4s + 1)(4 ds)$$

$$a_t = (16s + 4)|_{s=0} = 4 \text{ m/s}^2$$

$$a_n = \frac{v^2}{\rho} = \frac{(4(0) + 1)^2}{2} = 0.5 \text{ m/s}^2$$

e)  $a_t ds = v dv$

$$\int_0^s 2s ds = \int_1^v v dv$$

$$s^2 = \frac{1}{2}(v^2 - 1)$$

$$v = \sqrt{2s^2 + 1} \bigg|_{s=2 \text{ m}} = 3 \text{ m/s}$$

$$a_t = \dot{v} = 2(2) = 4 \text{ m/s}^2$$

$$a_n = \frac{v^2}{\rho} = \frac{(3)^2}{3} = 3 \text{ m/s}^2$$

$$a = \sqrt{(4)^2 + (3)^2} = 5 \text{ m/s}^2$$

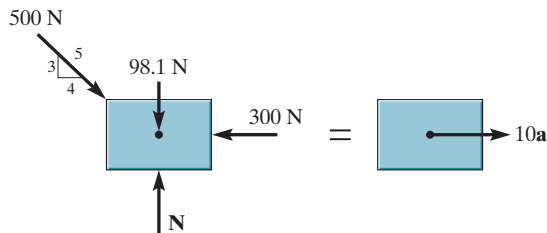
f)  $a_t = \dot{v} = 8t \bigg|_{t=1} = 8 \text{ m/s}^2$

$$a_n = \frac{v^2}{\rho} = \frac{(4(1)^2 + 2)^2}{6} = 6 \text{ m/s}^2$$

$$a = \sqrt{(8)^2 + (6)^2} = 10 \text{ m/s}^2$$

## Chapter 13

P13-1. a)

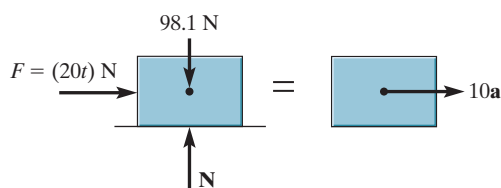


$$\pm \Sigma F_x = ma_x; \quad \left(\frac{4}{5}\right)(500 \text{ N}) - 300 \text{ N} = 10a$$

$$a = 10 \text{ m/s}^2$$

$$\pm v = v_0 + a_c t; \quad v = 0 + 10(2) = 20 \text{ m/s}$$

b)



$$\pm \Sigma F_x = ma_x; \quad 20t = 10a$$

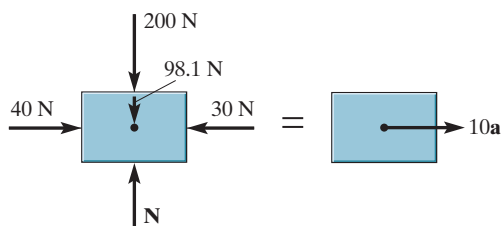
$$a = 2t$$

$$dv = a dt; \quad \int_0^v dv = \int_0^2 2t dt$$

$$v = 4 \text{ m/s}$$

P13-2.

a)



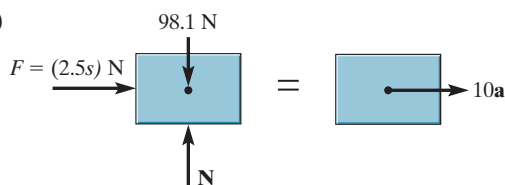
$$\pm \Sigma F_x = ma_x; \quad 40 \text{ N} - 30 \text{ N} = 10a$$

$$a = 1 \text{ m/s}^2$$

$$\pm v^2 = v_0^2 + 2a_c(s - s_0); \quad v^2 = (3)^2 + 2(1)(8 - 0)$$

$$v = 5 \text{ m/s}$$

b)



$$\pm \Sigma F_x = ma_x; \quad 2.5s = 10a$$

$$a = 2.5s$$

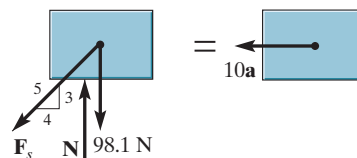
$$v dv = a ds$$

$$\int_3^v v dv = \int_0^8 2.5s ds$$

$$v^2 - (3)^2 = 2.5(8 - 0)^2$$

$$v = 13 \text{ m/s}$$

P13-3.

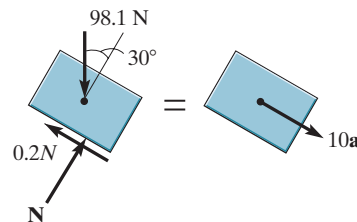


$$F_s = kx = (10 \text{ N/m})(5 \text{ m} - 1 \text{ m}) = 40 \text{ N}$$

$$\leftarrow \Sigma F_x = ma_x; \quad \frac{4}{5}(40 \text{ N}) = 10a$$

$$a = 3.2 \text{ m/s}^2$$

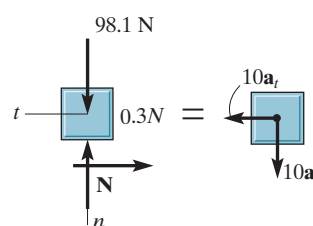
P13-4.



$$\searrow \Sigma F_x = ma_x; \quad 98.1 \sin 30^\circ - 0.2N = 10a$$

$$\nearrow \Sigma F_y = ma_y; \quad N - 98.1 \cos 30^\circ = 0$$

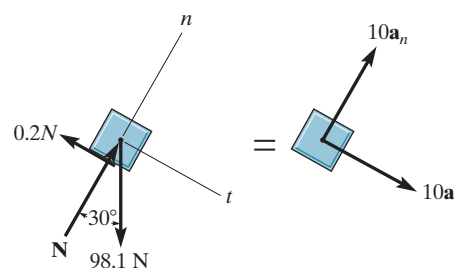
P13-5. a)



$$\leftarrow \Sigma F_t = ma_t; \quad -0.3N = 10a_t$$

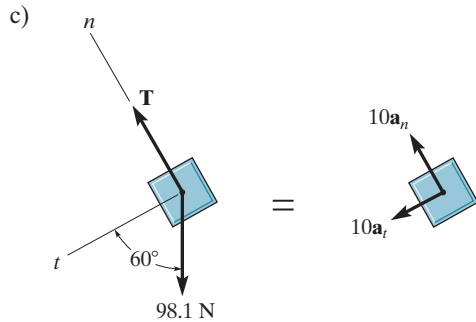
$$+\downarrow \Sigma F_n = ma_n; \quad 98.1 - N = 10\left(\frac{(6)^2}{10}\right)$$

b)



$$\swarrow \Sigma F_t = ma_t; 98.1 \sin 30^\circ - 0.2N = 10a_t$$

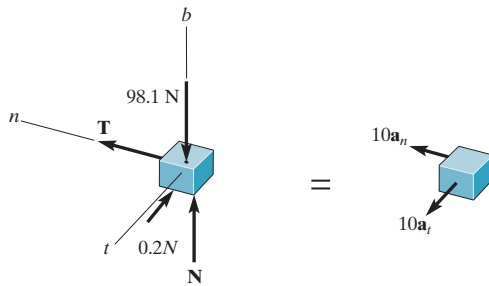
$$\nearrow \Sigma F_n = ma_n; N - 98.1 \cos 30^\circ = 10\left(\frac{4^2}{5}\right)$$



$$\nwarrow \Sigma F_t = ma_t; 98.1 \cos 60^\circ = 10a_t$$

$$\nwarrow \Sigma F_n = ma_n; T - 98.1 \sin 60^\circ = 10\left(\frac{8^2}{6}\right)$$

**P13-6.** a)

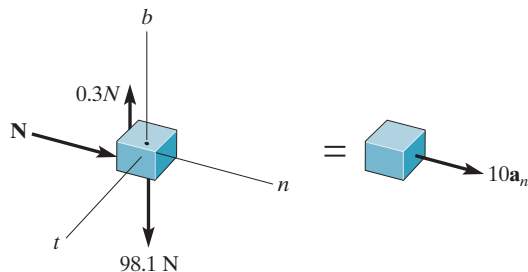


$$\Sigma F_b = 0; N - 98.1 = 0$$

$$\Sigma F_t = ma_t; -0.2N = 10a_t$$

$$\Sigma F_n = ma_n; T = 10\frac{(8)^2}{4}$$

b)



$$\Sigma F_b = 0; 0.3N - 98.1 = 0$$

$$\Sigma F_t = ma_t; 0 = 0$$

$$\Sigma F_n = ma_n; N = 10\frac{v^2}{2}$$

## Chapter 14

**P14-1.** a)  $U = \frac{3}{5}(500 \text{ N})(2 \text{ m}) = 600 \text{ J}$

b)  $U = 0$

c)  $U = \int_0^2 6s^2 ds = 2(2)^3 = 16 \text{ J}$

d)  $U = 100 \text{ N} \left( \frac{3}{5}(2 \text{ m}) \right) = \frac{3}{5}(100 \text{ N})(2 \text{ m}) = 120 \text{ J}$

e)  $U = \frac{4}{5}(\text{Area}) = \frac{4}{5} \left[ \frac{1}{2}(1)(20) + (1)(20) \right] = 24 \text{ J}$

f)  $U = \frac{1}{2}(10 \text{ N/m})((3 \text{ m})^2 - (1 \text{ m})^2) = 40 \text{ J}$

g)  $U = -\left(\frac{4}{5}\right)(100 \text{ N})(2 \text{ m}) = -160 \text{ J}$

**P14-2.** a)  $T = \frac{1}{2}(10 \text{ kg})(2 \text{ m/s})^2 = 20 \text{ J}$

b)  $T = \frac{1}{2}(10 \text{ kg})(6 \text{ m/s})^2 = 180 \text{ J}$

**P14-3.** a)  $V = (100 \text{ N})(2 \text{ m}) = 200 \text{ J}$

b)  $V = (100 \text{ N})(3 \text{ m}) = 300 \text{ J}$

c)  $V = 0$

**P14-4.** a)  $V = \frac{1}{2}(10 \text{ N/m})(5 \text{ m} - 4 \text{ m})^2 = 5 \text{ J}$

b)  $V = \frac{1}{2}(10 \text{ N/m})(10 \text{ m} - 4 \text{ m})^2 = 180 \text{ J}$

c)  $V = \frac{1}{2}(10 \text{ N/m})(5 \text{ m} - 4 \text{ m})^2 = 5 \text{ J}$

## Chapter 15

**P15-1.** a)  $I = (100 \text{ N})(2 \text{ s}) = 200 \text{ N} \cdot \text{s} \swarrow$

b)  $I = (200 \text{ N})(2 \text{ s}) = 400 \text{ N} \cdot \text{s} \downarrow$

c)  $I = \int_0^2 6t dt = 3(2)^2 = 12 \text{ N} \cdot \text{s} \searrow$

d)  $I = \text{Area} = \frac{1}{2}(1)(20) + (2)(20) = 50 \text{ N} \cdot \text{s} \nearrow$

e)  $I = (80 \text{ N})(2 \text{ s}) = 160 \text{ N} \cdot \text{s} \rightarrow$

f)  $I = (60 \text{ N})(2 \text{ s}) = 120 \text{ N} \cdot \text{s} \nearrow$

**P15-2.** a)  $L = (10 \text{ kg})(10 \text{ m/s}) = 100 \text{ kg} \cdot \text{m/s} \searrow$

b)  $L = (10 \text{ kg})(2 \text{ m/s}) = 20 \text{ kg} \cdot \text{m/s} \swarrow$

c)  $L = (10 \text{ kg})(3 \text{ m/s}) = 30 \text{ kg} \cdot \text{m/s} \rightarrow$



## Chapter 16

**P16-1.** a)  $\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \text{ (pin)}$

$$v_B = 18 \text{ m/s} + 2\omega$$


Also,

$$-v_B \mathbf{j} = -18 \mathbf{j} + (-\omega \mathbf{k}) \times (-2 \cos 60^\circ \mathbf{i} - 2 \sin 60^\circ \mathbf{j})$$

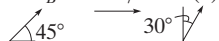
b)  $\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \text{ (pin)}$

$$(v_B)_x + (v_B)_y = 4(0.5) \text{ m/s} + 4(0.5) \text{ m/s}$$


Also,

$$(v_B)_x \mathbf{i} + (v_B)_y \mathbf{j} = 2 \mathbf{i} + (-4 \mathbf{k}) \times (-0.5 \cos 30^\circ \mathbf{i} + 0.5 \sin 30^\circ \mathbf{j})$$

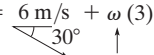
c)  $\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \text{ (pin)}$

$$v_B = 6 \text{ m/s} + \omega(5)$$


Also,

$$v_B \cos 45^\circ \mathbf{i} + v_B \sin 45^\circ \mathbf{j} = 6 \mathbf{i} + (\omega \mathbf{k}) \times (4 \mathbf{i} - 3 \mathbf{j})$$

d)  $\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \text{ (pin)}$

$$v_B = 6 \text{ m/s} + \omega(3)$$


Also,

$$v_B \mathbf{i} = 6 \cos 30^\circ \mathbf{i} + 6 \sin 30^\circ \mathbf{j} + (\omega \mathbf{k}) \times (3 \mathbf{i})$$

e)  $v_A = 12 \text{ m/s} = \omega(0.5 \text{ m}) \quad \omega = 24 \text{ rad/s}$

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \text{ (pin)}$$

$$(v_B)_x + (v_B)_y = 12 \text{ m/s} + (24)(0.5)$$


Also,

$$(v_B)_x \mathbf{i} + (v_B)_y \mathbf{j} = 12 \mathbf{j} + (24 \mathbf{k}) \times (0.5 \mathbf{j})$$

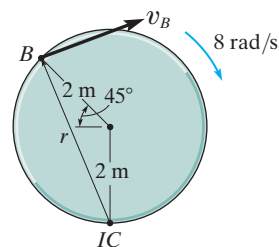
f)  $\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \text{ (pin)}$

$$v_B = 6 \text{ m/s} + \omega(5)$$

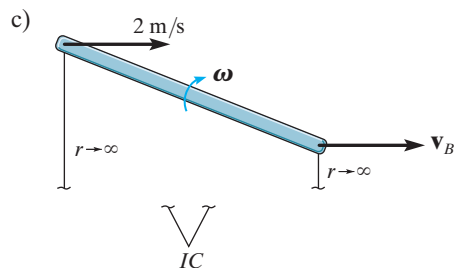
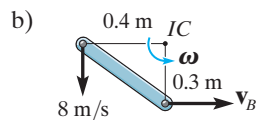

Also,

$$v_B \mathbf{i} = 6 \mathbf{i} + (\omega \mathbf{k}) \times (4 \mathbf{i} + 3 \mathbf{j})$$

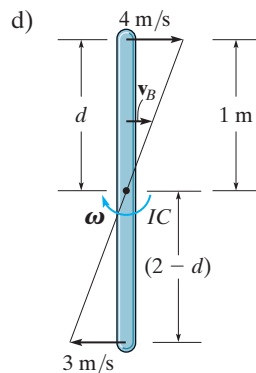
**P16-2.** a)

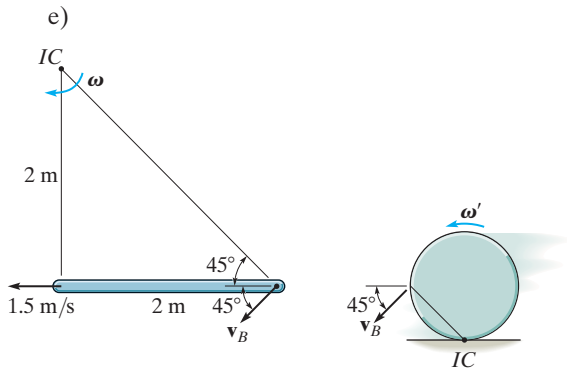


$$r = \sqrt{(2 \cos 45^\circ)^2 + (2 + 2 \sin 45^\circ)^2}$$

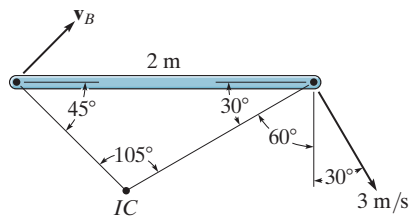


$$v_B = 2 \text{ m/s}, \omega = 0$$





f)



P16-3. a)

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A \text{ (pin)}}$$

$$a_B = 2 \text{ m/s}^2 + \frac{(3)^2}{3} + 2\alpha + (2.12)^2(2)$$

Also,

$$-a_B \mathbf{j} = -2\mathbf{i} + 3\mathbf{j} + (-\alpha \mathbf{k}) \times (2 \sin 45^\circ \mathbf{i} + 2 \cos 45^\circ \mathbf{j}) - (2.12)^2(2 \sin 45^\circ \mathbf{i} + 2 \cos 45^\circ \mathbf{j})$$

b)

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A \text{ (pin)}}$$

$$(a_B)_x + (a_B)_y = (2)(2) \text{ m/s}^2 + \alpha(2) + (4)^2(2)$$

Also,

$$(a_B)_x \mathbf{i} + (a_B)_y \mathbf{j} = 4\mathbf{i} + (-\alpha \mathbf{k}) \times (-2 \cos 45^\circ \mathbf{i} + 2 \sin 45^\circ \mathbf{j}) - (4)^2(-2 \cos 45^\circ \mathbf{i} + 2 \sin 45^\circ \mathbf{j})$$

c)

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A \text{ (pin)}}$$

$$(a_B)_x + (6)^2(1) = 2(2) + (3)^2(2) + \alpha(4)$$

Also,

$$(a_B)_x \mathbf{i} - 36\mathbf{j} = 4\mathbf{i} - 18\mathbf{j} + (-\alpha \mathbf{k}) \times (4\mathbf{i})$$

d)

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A \text{ (pin)}}$$

$$a_B = 6 + \alpha(2) + (3)^2(2)$$

Also,

$$a_B \mathbf{i} = -6 \cos 60^\circ \mathbf{i} - 6 \sin 60^\circ \mathbf{j} + (-\alpha \mathbf{k}) \times (-2\mathbf{i}) - (3)^2(-2\mathbf{i})$$

e)

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A \text{ (pin)}}$$

$$a_B = 8(0.5) + (4)^2(0.5) + \alpha(2) + (1.15)^2(2)$$

Also,

$$-a_B \mathbf{i} = -4\mathbf{j} + 8\mathbf{i} + (-\alpha \mathbf{k}) \times (-2 \cos 30^\circ \mathbf{i} - 2 \sin 30^\circ \mathbf{j}) - (1.15)^2(-2 \cos 30^\circ \mathbf{i} - 2 \sin 30^\circ \mathbf{j})$$

f)

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A \text{ (pin)}}$$

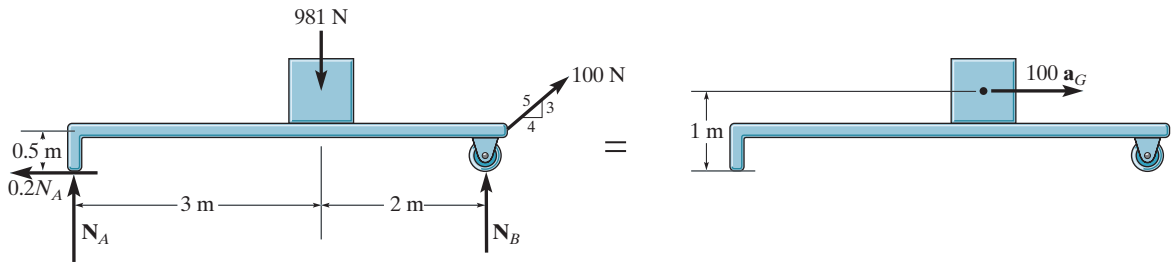
$$(a_B)_x + (a_B)_y = 2(0.5) + 2(0.5) + (4)^2(0.5)$$

Also,

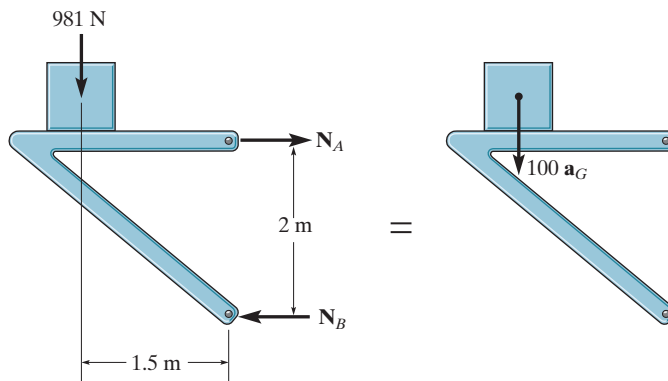
$$(a_B)_x \mathbf{i} + (a_B)_y \mathbf{j} = -1\mathbf{j} + (-2\mathbf{k}) \times (0.5\mathbf{j}) - (4)^2(0.5\mathbf{j})$$

## Chapter 17

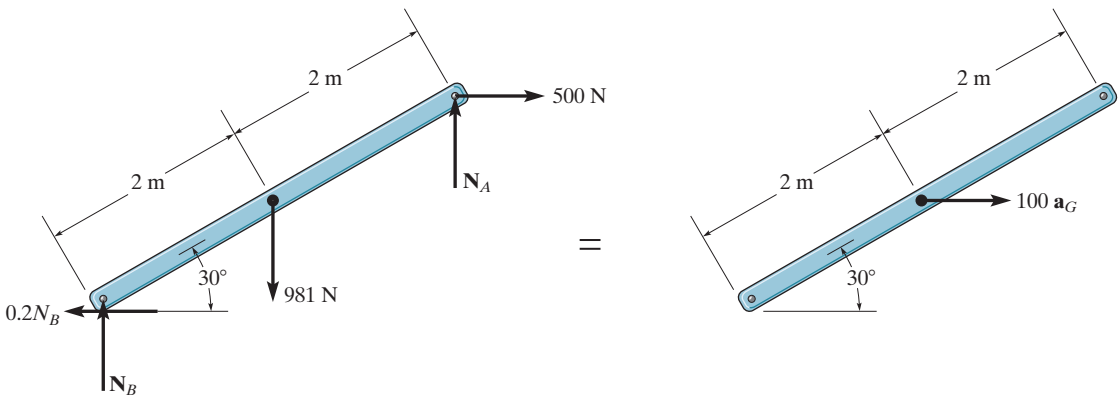
P17-1. a)



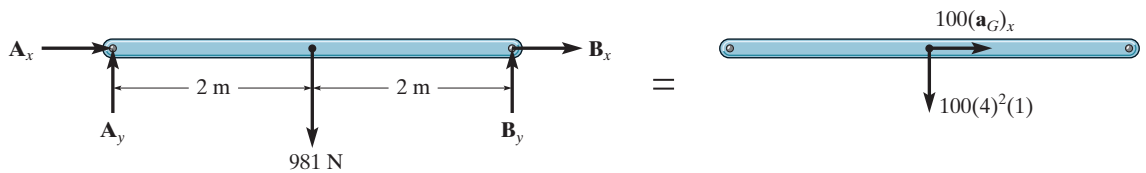
b)



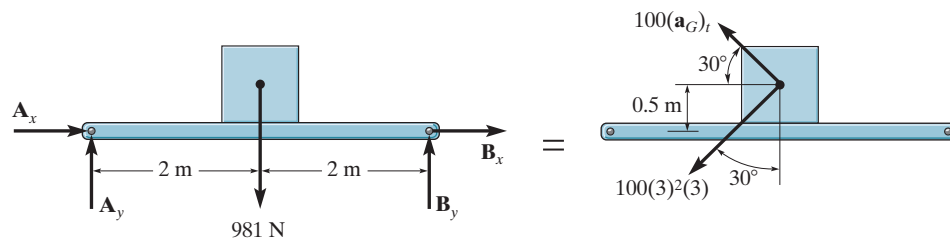
c)



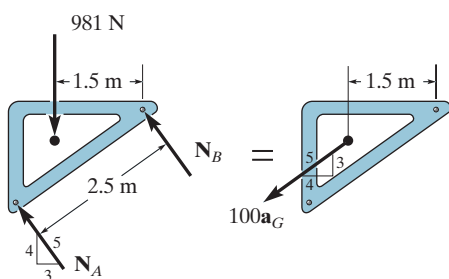
d)



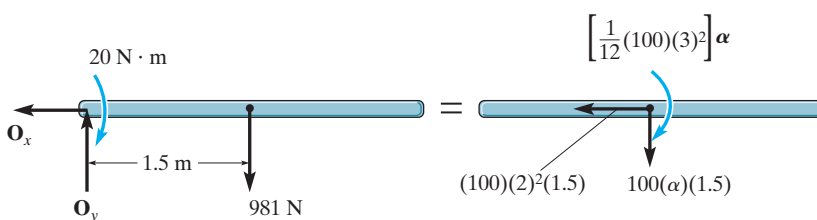
e)



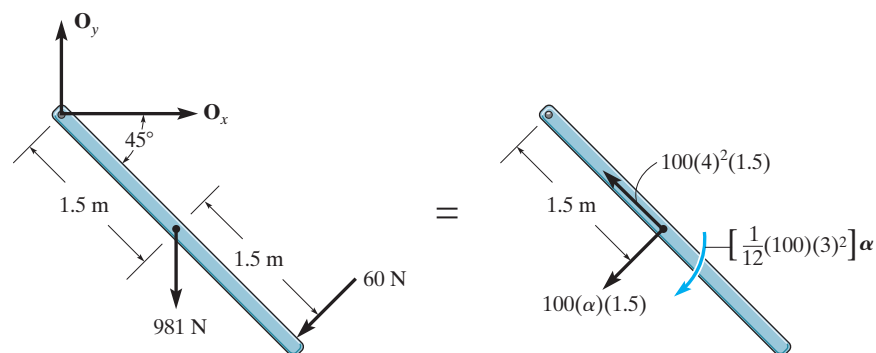
f)



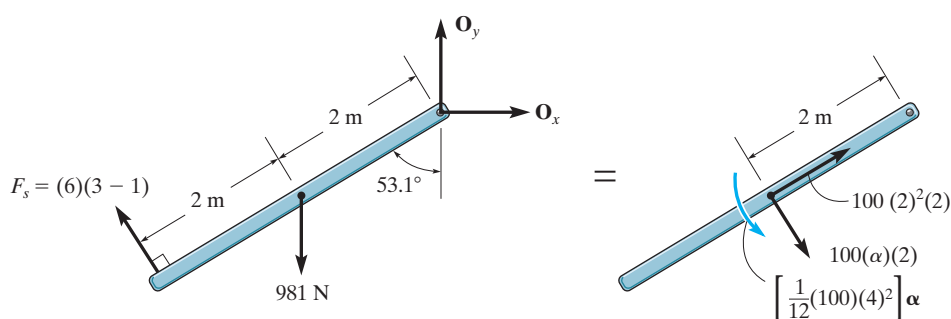
P17-2. a)

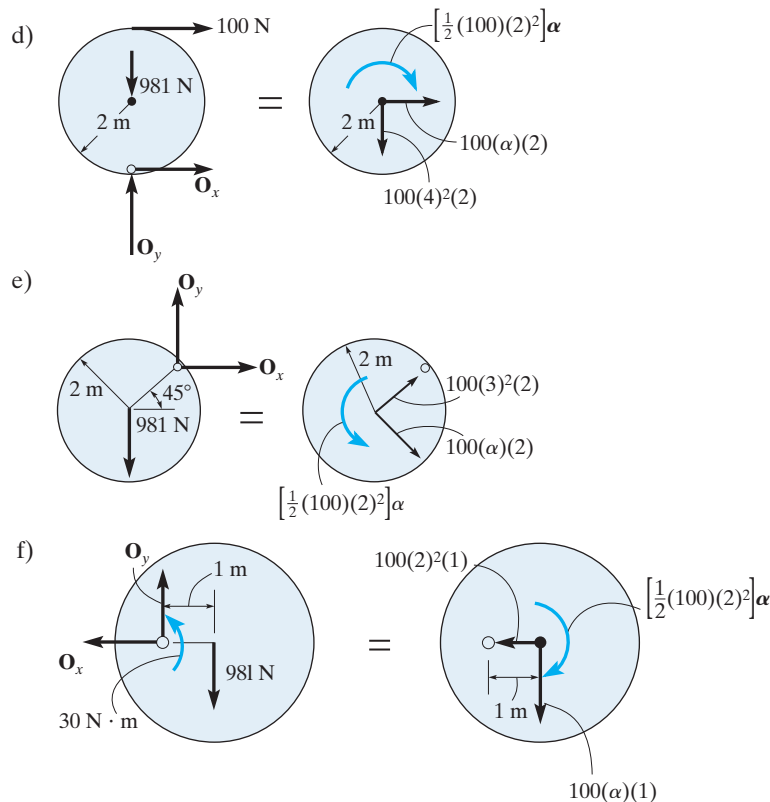


b)



c)





## Chapter 18

**P18-1.** a)  $T = \frac{1}{2} \left[ \frac{100(2)^2}{2} \right] (3)^2 = 900 \text{ J}$

b)  $T = \frac{1}{2} (100) [2(1)]^2 + \frac{1}{2} \left[ \frac{1}{12} (100)(6)^2 \right] (2)^2$   
 $= 800 \text{ J}$

Also,

$$T = \frac{1}{2} \left[ \frac{1}{12} (100)(6)^2 + 100(1)^2 \right] (2)^2 = 800 \text{ J}$$

c)  $T = \frac{1}{2} (100) [2(2)]^2 + \frac{1}{2} \left[ \frac{1}{2} (100)(2)^2 \right] (2)^2$   
 $= 1200 \text{ J}$

Also,

$$T = \frac{1}{2} \left[ \frac{1}{2} (100)(2)^2 + 100(2)^2 \right] (2)^2$$
  
 $= 1200 \text{ J}$

d)  $T = \frac{1}{2} (100) [2(1.5)]^2 + \frac{1}{2} \left[ \frac{1}{12} (100)(3)^2 \right] (2)^2$   
 $= 600 \text{ J}$

Also,

$$T = \frac{1}{2} \left[ \frac{1}{12} (100)(3)^2 + 100(1.5)^2 \right] (2)^2$$
  
 $= 600 \text{ J}$

e)  $T = \frac{1}{2} (100) [4(2)]^2 + \frac{1}{2} \left[ \frac{1}{2} (100)(2)^2 \right] (4)^2$   
 $= 4800 \text{ J}$

Also,

$$T = \frac{1}{2} \left[ \frac{1}{2} (100)(2)^2 + 100(2)^2 \right] (4)^2$$
  
 $= 4800 \text{ J}$

f)  $T = \frac{1}{2} (100) [(4)(2)]^2 = 3200 \text{ J}$

## Chapter 19

**P19-1.** a)  $H_G = \left[ \frac{1}{2} (100)(2)^2 \right] (3) = 600 \text{ kg} \cdot \text{m}^2/\text{s}^2$

$$H_O = \left[ \frac{1}{2} (100)(2)^2 + 100(2)^2 \right] (3)$$
  
 $= 1800 \text{ kg} \cdot \text{m}^2/\text{s}^2$

$$\text{b) } H_G = \left[ \frac{1}{12}(100)(3)^2 \right](4) = 300 \text{ kg} \cdot \text{m}^2/\text{s} \curvearrowright$$

$$\begin{aligned} H_O &= \left[ \frac{1}{12}(100)(3)^2 + (100)(1.5)^2 \right](4) \\ &= 1200 \text{ kg} \cdot \text{m}^2/\text{s} \curvearrowright \end{aligned}$$

$$\text{c) } H_G = \left[ \frac{1}{2}(100)(2)^2 \right](4) = 800 \text{ kg} \cdot \text{m}^2/\text{s} \curvearrowright$$

$$\begin{aligned} H_O &= \left[ \frac{1}{2}(100)(2)^2 + (100)(2)^2 \right](4) \\ &= 2400 \text{ kg} \cdot \text{m}^2/\text{s} \curvearrowright \end{aligned}$$

$$\text{d) } H_G = \left[ \frac{1}{12}(100)(4)^2 \right]3 = 400 \text{ kg} \cdot \text{m}^2/\text{s} \curvearrowright$$

$$\begin{aligned} H_O &= \left[ \frac{1}{12}(100)(4)^2 + (100)(1)^2 \right]3 \\ &= 700 \text{ kg} \cdot \text{m}^2/\text{s} \curvearrowright \end{aligned}$$

$$\mathbf{P19-2.} \quad \text{a) } \int M_O dt = \left( \frac{4}{5} \right)(500)(2)(3) = 2400 \text{ N} \cdot \text{s} \cdot \text{m} \curvearrowright$$

$$\begin{aligned} \text{b) } \int M_O dt &= \left[ 2(20) + \frac{1}{2}(3 - 2)(20) \right]4 \\ &= 200 \text{ N} \cdot \text{s} \cdot \text{m} \curvearrowright \end{aligned}$$

$$\text{c) } \int M_O dt = \frac{3}{5} \int_0^3 4(2t + 2)dt = 36 \text{ N} \cdot \text{s} \cdot \text{m} \curvearrowright$$

$$\text{d) } \int M_O dt = \int_0^3 (30t^2)dt = 270 \curvearrowright$$

# Review Problem Solutions

## Chapter 12

**R12-1.**  $s = t^3 - 9t^2 + 15t$

$$v = \frac{ds}{dt} = 3t^2 - 18t + 15$$

$$a = \frac{dv}{dt} = 6t - 18$$

$a_{\max}$  occurs at  $t = 10$  s.

$$a_{\max} = 6(10) - 18 = 42 \text{ m/s}^2 \quad \text{Ans.}$$

$v_{\max}$  occurs when  $t = 10$  s

$$v_{\max} = 3(10)^2 - 18(10) + 15 = 135 \text{ m/s} \quad \text{Ans.}$$

**R12-2.**  $(\rightarrow) \quad s = s_0 + v_0 t + \frac{1}{2} a_c t^2$

$$s = 0 + 12(10) + \frac{1}{2}(-2)(10)^2$$

$$s = 20.0 \text{ m} \quad \text{Ans.}$$

**R12-3.**  $v = \frac{ds}{dt} = 1800(1 - e^{-0.3t})$

$$\int_0^x ds = \int_0^t 1800(1 - e^{-0.3t}) dt$$

$$s = 1800 \left( t + \frac{1}{0.3} e^{-0.3t} \right) - 6000$$

Thus, in  $t = 3$  s

$$s = 1800 \left( 3 + \frac{1}{0.3} e^{-0.3(3)} \right) - 6000$$

$$s = 1839.4 \text{ mm} = 1.84 \text{ m} \quad \text{Ans.}$$

**R12-4.**  $0 \leq t \leq 5 \quad a = \frac{\Delta v}{\Delta t} = \frac{20}{5} = 4 \text{ m/s}^2 \quad \text{Ans.}$

$$5 \leq t \leq 20 \quad a = \frac{\Delta v}{\Delta t} = \frac{20 - 20}{20 - 5} = 0 \text{ m/s}^2 \quad \text{Ans.}$$

$$20 \leq t \leq 30 \quad a = \frac{\Delta v}{\Delta t} = \frac{0 - 20}{30 - 20} = -2 \text{ m/s}^2 \quad \text{Ans.}$$

At  $t_1 = 5$  s,  $t_2 = 20$  s, and  $t_3 = 30$  s,

$$s_1 = A_1 = \frac{1}{2}(5)(20) = 50 \text{ m} \quad \text{Ans.}$$

$$s_2 = A_1 + A_2 = 50 + 20(20 - 5) = 350 \text{ m} \quad \text{Ans.}$$

$$s_3 = A_1 + A_2 + A_3 = 350$$

$$+ \frac{1}{2}(30 - 20)(20) = 450 \text{ m} \quad \text{Ans.}$$

**R12-5.**  $v_A = 20\mathbf{i}$

$$v_B = 21.21\mathbf{i} + 21.21\mathbf{j}$$

$$v_C = 40\mathbf{i}$$

$$\mathbf{a}_{AB} = \frac{\Delta v}{\Delta t} = \frac{21.21\mathbf{i} + 21.21\mathbf{j} - 20\mathbf{i}}{3}$$

$$\mathbf{a}_{AB} = \{0.404\mathbf{i} + 7.07\mathbf{j}\} \text{ m/s}^2 \quad \text{Ans.}$$

$$\mathbf{a}_{AC} = \frac{\Delta v}{\Delta t} = \frac{40\mathbf{i} - 20\mathbf{i}}{8}$$

$$\mathbf{a}_{AC} = \{2.50\mathbf{i}\} \text{ m/s}^2 \quad \text{Ans.}$$

**R12-6.**  $(\rightarrow) \quad s = s_0 + v_0 t$

$$40 = 0 + (v_0)_x (3.6)$$

$$(v_0)_x = 11.11 \text{ m/s}$$

$$(+\uparrow) \quad s = s_0 + v_0 t + \frac{1}{2} a_c t^2$$

$$0 = 0 + (v_0)_y (3.6) + \frac{1}{2}(-9.81)(3.6)^2$$

$$(v_0)_y = 17.658 \text{ m/s}$$

$$v_0 = \sqrt{(11.11)^2 + (17.658)^2} = 20.9 \text{ m/s} \quad \text{Ans.}$$

$$\theta = \tan^{-1} \left( \frac{17.658}{11.11} \right) = 57.8^\circ \quad \text{Ans.}$$

**R12-7.**  $v dv = a_t ds$

$$\int_4^v v dv = \int_0^{10} 0.05s ds$$

$$0.5v^2 - 8 = \frac{0.05}{2}(10)^2$$

$$v = 4.583 = 4.58 \text{ m/s} \quad \text{Ans.}$$

$$a_n = \frac{v^2}{\rho} = \frac{(4.583)^2}{50} = 0.420 \text{ m/s}^2$$

$$a_t = 0.05(10) = 0.5 \text{ m/s}^2$$

$$a = \sqrt{(0.420)^2 + (0.5)^2} = 0.653 \text{ m/s}^2 \quad \text{Ans.}$$

**R12-8.**  $dv = a dt$

$$\int_0^v dv = \int_0^t 0.5e^t dt$$

$$v = 0.5(e^t - 1)$$

When  $t = 2$  s,  $v = 0.5(e^2 - 1) = 3.195 \text{ m/s}$

$$= 3.19 \text{ m/s} \quad \text{Ans.}$$

When  $t = 2 \text{ s}$   $a_t = 0.5e^2 = 3.695 \text{ m/s}^2$

$$a_n = \frac{v^2}{\rho} = \frac{3.195^2}{5} = 2.041 \text{ m/s}^2$$

$$a = \sqrt{a_t^2 + a_n^2} = \sqrt{3.695^2 + 2.041^2} = 4.22 \text{ m/s}^2$$

*Ans.*

**R12-9.**  $r = 2 \text{ m}$        $\theta = 5t^2$   
 $\dot{r} = 0$        $\dot{\theta} = 10t$   
 $\ddot{r} = 0$        $\ddot{\theta} = 10$   
 $a = (\ddot{r} - r\dot{\theta}^2)\mathbf{u}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{u}_\theta$   
 $= [0 - 2(10t)^2]\mathbf{u}_r + [2(10) + 0]\mathbf{u}_\theta$   
 $= \{-200t^2\mathbf{u}_r + 20\mathbf{u}_\theta\} \text{ m/s}^2$

When  $\theta = 30^\circ = 30\left(\frac{\pi}{180}\right) = 0.524 \text{ rad}$   
 $0.524 = 5t^2$   
 $t = 0.324 \text{ s}$

$a = [-200(0.324)^2]\mathbf{u}_r + 20\mathbf{u}_\theta$   
 $= \{-20.9\mathbf{u}_r + 20\mathbf{u}_\theta\} \text{ m/s}^2$   
 $a = \sqrt{(-20.9)^2 + (20)^2} = 29.0 \text{ m/s}^2$

*Ans.*

**R12-10.**  $4s_B + s_A = l$   
 $4v_B = -v_A$   
 $4a_B = -a_A$   
 $4a_B = -0.2$   
 $a_B = -0.05 \text{ m/s}^2$   
 $(+\downarrow) \quad v_B = (v_B)_0 + a_B t$   
 $-8 = 0 - (0.05)(t)$   
 $t = 160 \text{ s}$

*Ans.*

**R12-11.**  $\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$   
 $[500 \leftarrow] = [600 \swarrow_{75^\circ} \theta] + v_{B/A}$   
 $(\leftarrow) \quad 500 = -600 \cos 75^\circ + (v_{B/A})_x$   
 $(v_{B/A})_x = 655.29 \leftarrow$   
 $(+\uparrow) \quad 0 = -600 \sin 75^\circ + (v_{B/A})_y$   
 $(v_{B/A})_y = 579.56 \uparrow$   
 $(v_{B/A}) = \sqrt{(655.29)^2 + (579.56)^2}$   
 $v_{B/A} = 875 \text{ km/h}$   
 $\theta = \tan^{-1}\left(\frac{579.56}{655.29}\right) = 41.5^\circ \swarrow$

*Ans.*

*Ans.*

## Chapter 13

**R13-1.**  $20 \text{ km/h} = \frac{20(10)^3}{3600} = 5.556 \text{ m/s}$   
 $(\leftarrow) \quad v^2 = v_0^2 + 2a_c(s - s_0)$

$$a = -0.3429 \text{ m/s}^2 = 0.3429 \text{ m/s}^2 \rightarrow$$

$$\rightarrow \Sigma F_x = ma_x; \quad F = 250(0.3429) = 85.7 \text{ N} \quad \text{Ans.}$$

**R13-2.**  $\nwarrow \Sigma F_y = ma_y; \quad N_C - 50(9.81) \cos 30^\circ = 0$   
 $N_C = 424.79$

$$\nearrow \Sigma F_x = ma_x; \quad 3T - 0.3(424.79) - 50(9.81) \sin 30^\circ = 50a_C \quad (1)$$

Kinematics,  $2s_C + (s_C - s_p) = l$

Taking two time derivatives, yields

$$3a_C = a_p$$

Thus,  $a_C = \frac{6}{3} = 2$

Substituting into Eq. (1) and solving,

$$T = 158 \text{ N} \quad \text{Ans.}$$

**R13-3.** Suppose the two blocks move together.  
 Then

$$250 \text{ N} = (10 + 25) a$$

$$a = 7.143 \text{ m/s}^2$$

Then the friction force on block B is

$$F_B = 25(7.143) = 178.57 \text{ N}$$

The maximum friction force between blocks A and B is

$$F_{\max} = 0.4[10(9.81)] = 39.24 \text{ N} < 178.57 \text{ N}$$

The blocks have different accelerations.

Block A:

$$\rightarrow \Sigma F_x = ma_x; \quad 250 - 0.3[10(9.81)] = 10a_A$$

$$a_A = 22.1 \text{ m/s}^2 \quad \text{Ans.}$$

Block B:

$$\rightarrow \Sigma F_x = ma_x; \quad 0.3[10(9.81)] = 25a_B$$

$$a_B = 1.18 \text{ m/s}^2 \quad \text{Ans.}$$

**R13-4. Kinematics:** Since the motion of the crate is known, its acceleration  $\mathbf{a}$  will be determined first.

$$a = v \frac{dv}{ds} = (0.05s^{3/2}) \left[ (0.05) \left( \frac{3}{2} \right) s^{1/2} \right]$$

$$= 0.00375s^2 \text{ m/s}^2$$

When  $s = 10 \text{ m}$ ,

$$a = 0.00375(10^2) = 0.375 \text{ m/s}^2 \rightarrow$$



**Free-Body Diagram:** The kinetic friction  $F_f = \mu_k N = 0.2N$  must act to the left to oppose the motion of the crate which is to the right.

**Equations of Motion:** Here,  $a_y = 0$ . Thus,

$$+\uparrow \Sigma F_y = ma_y; \quad N - 20(9.81) = 20(0) \\ N = 196.2 \text{ N}$$

Using the results of **N** and **a**,

$$+\rightarrow \Sigma F_x = ma_x; \quad T - 0.2(196.2) = 20(0.375) \\ T = 46.7 \text{ N} \quad \text{Ans.}$$

**R13-5.**  $+\nwarrow \Sigma F_n = ma_n; \quad T - 30(9.81) \cos \theta = 30\left(\frac{v^2}{4}\right)$

$$+\nearrow \Sigma F_t = ma_t; \quad -30(9.81) \sin \theta = 30a_t \\ a_t = -9.81 \sin \theta$$

$a_t ds = v dv$  Since  $ds = 4 d\theta$ , then

$$-9.81 \int_0^\theta \sin \theta (4 d\theta) = \int_4^v v dv$$

$$9.81(4) \cos \theta \Big|_0^\theta = \frac{1}{2}(v)^2 - \frac{1}{2}(4)^2$$

$$39.24(\cos \theta - 1) + 8 = \frac{1}{2}v^2$$

At  $\theta = 20^\circ$

$$v = 3.357 \text{ m/s}$$

$$a_t = -3.36 \text{ m/s}^2 = 3.36 \text{ m/s}^2 \quad \checkmark \quad \text{Ans.}$$

$$T = 361 \text{ N} \quad \text{Ans.}$$

**R13-6.**  $\Sigma F_z = ma_z; \quad N_z - mg = 0 \quad N_z = mg$

$$\Sigma F_x = ma_n; \quad 0.3(mg) = m\left(\frac{v^2}{r}\right)$$

$$v = \sqrt{0.3gr} = \sqrt{0.3(9.81)(1.5)} = 2.10 \text{ m/s} \quad \text{Ans.}$$

**R13-7.**  $v = \frac{1}{8}x^2$

$$\frac{dy}{dx} = \tan \theta = \frac{1}{4}x \Big|_{x=-3 \text{ m}} = -0.75 \quad \theta = -36.87^\circ$$

$$\frac{d^2y}{dx^2} = \frac{1}{4}$$

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}{\left|\frac{d^2y}{dx^2}\right|} = \frac{\left[1 + (-0.75)^2\right]^{\frac{3}{2}}}{\left|\frac{1}{4}\right|} = 7.8125 \text{ m}$$

$$+\nearrow \Sigma F_n = ma_n;$$

$$N - [5(9.81)] \cos 36.87^\circ = 5\left(\frac{2^2}{7.8125}\right) \\ N = 41.8 \text{ N} \quad \text{Ans.}$$

$$+\searrow \Sigma F_t = ma_t; \quad -0.2(41.8) + [5(9.81)] \sin 36.87^\circ \\ = 5a_t$$

$$a_t = 4.214 \text{ m/s}^2 = 4.21 \text{ m/s}^2 \quad \text{Ans.}$$

**R13-8.**  $r = 0.5 \text{ m}$

$$\dot{r} = 3 \text{ m/s} \quad \dot{\theta} = 6 \text{ rad/s}$$

$$\ddot{r} = 1 \text{ m/s}^2 \quad \ddot{\theta} = 2 \text{ rad/s}^2$$

$$a_r = \ddot{r} - r\dot{\theta}^2 = 1 - 0.5(6)^2 = -17$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0.5(2) + 2(3)(6) = 37$$

$$\Sigma F_r = ma_r; \quad F_r = 4(-17) = -68 \text{ N}$$

$$\Sigma F_\theta = ma_\theta; \quad N_\theta = 4(37) = 148 \text{ N}$$

$$\Sigma F_z = ma_z; \quad N_z = 4(9.81) = 0$$

$$N_z = 39.24 \text{ N}$$

$$F_r = -68 \text{ N} \quad \text{Ans.}$$

$$N = \sqrt{(148)^2 + (39.24)^2} = 153 \text{ N} \quad \text{Ans.}$$

## Chapter 14

**R14-1.**  $+\nwarrow \Sigma F_y = 0; \quad N_C - 50(9.81) \cos 30^\circ = 0$

$$N_C = 424.79 \text{ N}$$

$$T_1 + \Sigma U_{1-2} = T_2$$

$$0 + [50(9.81) \sin 30^\circ](10) - [0.3(424.79)](10) = \frac{1}{2}(50)v_2^2$$

$$v_2 = 6.86 \text{ m/s} \quad \text{Ans.}$$

**R14-2.**  $T_1 + \Sigma \int F ds = T_2$

$$0 + 1(9.81)(5 - 0.5) + \int_{2 \text{ m}}^0 50 dx + \int_0^{4 \text{ m}} 30y dy$$

$$+ \int_{5 \text{ m}}^{0.5 \text{ m}} 10z dz = \frac{1}{2}(1)v_B^2$$

$$v_B = 11.0 \text{ m/s} \quad \text{Ans.}$$

**R14-3.**  $T_1 + V_1 = T_2 + V_2$

$$0 + [0.75(9.81)](5) = \frac{1}{2}(0.75)v_B^2$$

$$v_B = 9.90 \text{ m/s} \quad \text{Ans.}$$

**R14-4.** The work done by  $F$  depends upon the difference in the cord length  $AC-BC$ .

$$T_A + \Sigma U_{A-B} = T_B$$

$$0 + F[\sqrt{(0.3)^2 + (0.3)^2} - \sqrt{(0.3)^2 + (0.3 - 0.15)^2}] - 0.5(9.81)(0.15)$$

$$-\frac{1}{2}(100)(0.15)^2 = \frac{1}{2}(0.5)(2.5)^2$$

$$F(0.0889) = 3.423$$

$$F = 38.5 \text{ N}$$

*Ans.*

**R14-5.**  $(+\uparrow) \quad v^2 = v_0^2 + 2a_c(s - s_0)$   
 $(6)^2 = 0 + 2a_c(3 - 0)$   
 $a_c = 6 \text{ m/s}^2$

$$+\uparrow \Sigma F_y = ma_y; \quad 2T - 25(9.81) = 25(6)$$

$$T = 197.625 \text{ N}$$

$$s_C + (s_C - s_M) = l$$

$$v_M = 2v_C$$

$$v_M = 2(6) = 12 \text{ m/s}$$

$$P_0 = \mathbf{T} \cdot \mathbf{v} = 197.625(12) = 2371.5 \text{ W}$$

$$P_i = \frac{2371.5}{0.74} = 3204.73 \text{ W} = 3.20 \text{ kW} \quad \text{Ans.}$$

**R14-6.**  $+\uparrow \Sigma F_y = ma_y; \quad 2(150) - 25(9.81) = 25a_B$   
 $a_B = 2.19 \text{ m/s}^2$

$$(+\uparrow) \quad v^2 = v_0^2 + 2a_c(s - s_0)$$

$$v_B^2 = 0 + 2(2.19)(3 - 0)$$

$$v_B = 3.625 \text{ m/s}$$

$$2s_B + s_M = l$$

$$v_M = -2v_B$$

$$v_M = -2(3.625) = 7.250 \text{ m/s}$$

$$P_o = \mathbf{F} \cdot \mathbf{v} = 150(7.250) = 1087.47 \text{ W}$$

$$P_i = \frac{1087.47}{0.76} = 1430.89 \text{ W} = 1.43 \text{ kW} \quad \text{Ans.}$$

**R14-7.**  $T_A + V_A = T_B + V_B$

$$0 + (0.25)(9.81)(0.6) + \frac{1}{2}(150)(0.6 - 0.1)^2$$

$$= \frac{1}{2}(0.25)(v_B)^2 + \frac{1}{2}(150)(0.4 - 0.1)^2$$

$$v_B = 10.4 \text{ m/s}$$

*Ans.*

**R14-8.**  $\frac{z}{5} = \frac{2}{\sqrt{0.8^2 + 5^2}}$

$$z = 1.9749 \text{ m}$$

$$T_1 + V_1 = T_2 + V_2$$

$$0 + 0 = \frac{1}{2}(5)v_2^2 + \frac{1}{2}(15)v_2^2$$

$$+ 5(9.81)(1.9749) - 15(9.81)(1.9749)$$

$$v_2 = 4.40 \text{ m/s}$$

*Ans.*

## Chapter 15

**R15-1.**  $(+\uparrow) \quad m(v_1)_y + \Sigma \int F_y dt = m(v_2)_y$

$$0 + N_p(t) - 58.86(t) = 0$$

$$N_p = 58.86 \text{ N}$$

$$(\rightarrow) \quad m(v_1)_x + \Sigma \int F_x dt = m(v_2)_x$$

$$6(3) - 0.2(58.86)(t) = 6(1)$$

$$t = 1.02 \text{ s}$$

*Ans.*

**R15-2.**  $+\searrow \Sigma F_x = 0; \quad N_B - 50(9.81) \cos 30^\circ = 0$   
 $N_B = 424.79 \text{ N}$

$$(+\nearrow) \quad m(v_x)_1 + \Sigma \int F_x dt = m(v_x)_2$$

$$50(2) + \int_0^2 (300 + 120\sqrt{t}) dt - 0.4(424.79)(2)$$

$$- 50(9.81) \sin 30^\circ(2) = 50v_2$$

$$v_2 = 1.92 \text{ m/s}$$

*Ans.*

**R15-3.** The crate starts moving when

$$F = F_r = 0.6(196.2) = 117.72 \text{ N}$$

From the graph since

$$F = \frac{200}{5}t, \quad 0 \leq t \leq 5 \text{ s}$$

The time needed for the crate to start moving is

$$t = \frac{5}{200}(117.72) = 2.943 \text{ s}$$

Hence, the impulse due to  $F$  is equal to the area under the curve from  $2.943 \text{ s} \leq t \leq 10 \text{ s}$

$$\rightarrow \quad m(v_x)_1 + \Sigma \int F_x dt = m(v_x)_2$$

$$0 + \int_{2.943}^5 \frac{200}{5}t dt + \int_5^{10} 200 dt$$

$$- (0.5)196.2(10 - 2.943) = 20v_2$$

$$40\left(\frac{1}{2}t^2\right)\Big|_{2.943}^5 + 200(10 - 5) - 692.292 = 20v_2$$

$$634.483 = 20v_2$$

$$v_2 = 31.7 \text{ m/s} \quad \text{Ans.}$$

**R15-4.**  $(v_A)_1 = \left[20(10^3)\frac{\text{m}}{\text{h}}\right]\left(\frac{1\text{h}}{3600\text{s}}\right) = 5.556 \text{ m/s}$

$$(v_B)_1 = \left[5(10^3)\frac{\text{m}}{\text{h}}\right]\left(\frac{1\text{h}}{3600\text{s}}\right) = 1.389 \text{ m/s},$$

$$\text{and } (v_C)_1 = \left[25(10^3)\frac{\text{m}}{\text{h}}\right]\left(\frac{1\text{h}}{3600\text{s}}\right) = 6.944 \text{ m/s}$$

For the first case,

$$(\rightarrow) \quad m_A(v_A)_1 + m_B(v_B)_1 = (m_A + m_B)v_2$$

$$10000(5.556) + 5000(1.389) = (10000 + 5000)v_{AB}$$

$$v_{AB} = 4.167 \text{ m/s} \rightarrow$$

Using the result of  $v_{AB}$  and considering the second case,

$$(\rightarrow) \quad (m_A + m_B)v_{AB} + m_C(v_C)_1$$

$$= (m_A + m_B + m_C)v_{ABC}$$

$$(10000 + 5000)(4.167) + [-20000(6.944)]$$

$$= (10000 + 5000 + 20000)v_{ABC}$$

$$v_{ABC} = -2.183 \text{ m/s} = 2.18 \text{ m/s} \leftarrow \text{Ans.}$$

**R15-5.**  $(\rightarrow) \quad m_P(v_P)_1 + m_B(v_B)_1 = m_P(v_P)_2 + m_B(v_B)_2$

$$0.2(900) + 15(0) = 0.2(300) + 15(v_B)_2$$

$$(v_B)_2 = 8 \text{ m/s} \rightarrow \quad \text{Ans.}$$

$$(+\uparrow) \quad m(v_1)_y + \Sigma \int_{t_1}^{t_2} F_y dt = m(v_2)_y$$

$$15(0) + N(t) - 15(9.81)(t) = 15(0)$$

$$N = 147.15 \text{ N}$$

$$(\rightarrow) \quad m(v_1)_x + \Sigma \int_{t_1}^{t_2} F_x dt = m(v_2)_x$$

$$15(8) + [-0.2(147.15)(t)] = 15(0)$$

$$t = 4.077 \text{ s} = 4.08 \text{ s} \quad \text{Ans.}$$

**R15-6.**  $(\rightarrow) \quad \Sigma mv_1 = \Sigma mv_2$

$$3(2) + 0 = 3(v_A)_2 + 2(v_B)_2$$

$$(\rightarrow) \quad e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1}$$

$$1 = \frac{(v_B)_2 - (v_A)_2}{2 - 0}$$

Solving

$$(v_A)_2 = 0.400 \text{ m/s} \rightarrow \quad \text{Ans.}$$

$$(v_B)_2 = 2.40 \text{ m/s} \rightarrow \quad \text{Ans.}$$

Block A:

$$T_1 + \Sigma U_{1-2} = T_2$$

$$\frac{1}{2}(3)(0.400)^2 - 3(9.81)(0.3)d_A = 0$$

$$d_A = 0.0272 \text{ m}$$

Block B:

$$T_1 + \Sigma U_{1-2} = T_2$$

$$\frac{1}{2}(2)(2.40)^2 - 2(9.81)(0.3)d_B = 0$$

$$d_B = 0.9786 \text{ m}$$

$$d = d_B - d_A = 0.951 \text{ m} \quad \text{Ans.}$$

**R15-7.**  $(v_A)_{x_1} = -2 \cos 40^\circ = -1.532 \text{ m/s}$

$$(v_A)_{y_1} = -2 \sin 40^\circ = -1.285 \text{ m/s}$$

$$(\rightarrow) \quad m_A(v_A)_{x_1} + m_B(v_B)_{x_1} = m_A(v_A)_{x_2}$$

$$+ m_B(v_B)_{x_2}$$

$$-2(1.532) + 0 = 0.2(v_A)_{x_2}$$

$$+ 0.2(v_B)_{x_2} \quad (1)$$

$$(\rightarrow) \quad e = \frac{(v_{ref})_2}{(v_{ref})_1}$$

$$0.75 = \frac{(v_A)_{x_2} - (v_B)_{x_1}}{1.532} \quad (2)$$

Solving Eqs. (1) and (2)

$$(v_A)_{x_2} = -0.1915 \text{ m/s}$$

$$(v_B)_{x_2} = -1.3405 \text{ m/s}$$

For A:

$$(+\downarrow) \quad m_A(v_A)_{y_1} = m_A(v_A)_{y_2}$$

$$(v_A)_{y_2} = 1.285 \text{ m/s}$$

For B:

$$(+\uparrow) \quad m_B(v_B)_{y_1} = m_B(v_B)_{y_2}$$

$$(v_B)_{y_2} = 0$$

Hence  $(v_B)_2 = (v_B)_{x_2} = 1.34 \text{ m/s} \leftarrow \text{Ans.}$

$$(v_A)_2 = \sqrt{(-0.1915)^2 + (1.285)^2} = 1.30 \text{ m/s} \quad \text{Ans.}$$

$$(\theta_A)_2 = \tan^{-1}\left(\frac{0.1915}{1.285}\right) = 8.47^\circ \quad \text{Ans.}$$

**R15-8.**  $(H_z)_1 + \Sigma \int M_z dt = (H_z)_2$

$$(10)(2)(0.75) + 60(2)\left(\frac{3}{5}\right)(0.75) +$$

$$\int_0^2 (8t^2 + 5)dt = 10v(0.75)$$

$$69 + \left[\frac{8}{3}t^3 + 5t\right]_0^2 = 7.5v$$

$$v = 13.4 \text{ m/s} \quad \text{Ans.}$$

## Chapter 16

**R16-1.**  $(\omega_A)_O = 60 \text{ rad/s}$

$$\alpha_A = -1 \text{ rad/s}^2$$

$$\omega_A = (\omega_A)_O + \alpha_A t$$

$$\omega_A = 60 + (-1)(3) = 57 \text{ rad/s}$$

$$v_p = \omega_A r_A = 57(0.3) = 17.1 \text{ m/s}$$

$$\omega_B = \frac{v_p}{r_B} = \frac{17.1}{0.6} = 28.5 \text{ rad/s}$$

$$v_W = \omega_B r_h = (28.5)(0.15) = 4.275 \text{ m/s} \quad \text{Ans.}$$

$$\alpha_A = 1 \text{ rad/s}^2$$

$$(a_p)_t = \alpha_A r_A = (1)(0.3) = 0.3 \text{ m/s}^2$$

$$\alpha_B = \frac{(a_p)_t}{r_B} = \frac{0.3}{0.6} = 0.5 \text{ rad/s}^2$$

$$a_W = \alpha_B r_h = (0.5)(0.15) = 0.075 \text{ m/s}^2 \downarrow \quad \text{Ans.}$$

**R16-2.**  $\alpha_a = 0.6\theta_A$

$$\theta_C = \frac{0.5}{0.075} = 6.667 \text{ rad}$$

$$\theta_A(0.05) = (6.667)(0.15)$$

$$\theta_A = 20 \text{ rad}$$

$$\alpha d\theta = \omega d\omega$$

$$\int_0^{20} 0.6\theta_A d\theta_A = \int_3^{\omega_A} \omega_A d\omega_A$$

$$0.3\theta_A^2 \Big|_0^{20} = \frac{1}{2}\omega_A^2 \Big|_3^{\omega_A}$$

$$120 = \frac{1}{2}\omega_A^2 - 4.5$$

$$\omega_A = 15.780 \text{ rad/s}$$

$$15.780(0.05) = \omega_C(0.15)$$

$$\omega_C = 5.260 \text{ rad/s}$$

$$v_B = 5.260(0.075) = 0.394 \text{ m/s} \quad \text{Ans.}$$

**R16-3.** A point on the drum which is in contact with the board has a tangential acceleration of

$$a_t = 0.5 \text{ m/s}^2$$

$$a^2 = a_t^2 + a_n^2$$

$$(3)^2 = (0.5)^2 + a_n^2$$

$$a_n = 2.96 \text{ m/s}^2$$

$$a_n = \omega^2 r, \quad \omega = \sqrt{\frac{2.96}{0.25}} = 3.44 \text{ rad/s}$$

$$v_B = \omega r = 3.44(0.25) = 0.860 \text{ m/s} \quad \text{Ans.}$$

**R16-4.**  $\mathbf{v}_B = \omega_{AB} \times \mathbf{r}_{B/A}$

$$\mathbf{v}_C = \mathbf{v}_B + \omega \times \mathbf{r}_{C/B}$$

$$v_C \mathbf{i} = (6\mathbf{k}) \times (0.2 \cos 45^\circ \mathbf{i} + 0.2 \sin 45^\circ \mathbf{j}) +$$

$$(\omega \mathbf{k}) \times (0.5 \cos 30^\circ \mathbf{i} - 0.5 \sin 30^\circ \mathbf{j})$$

$$v_C = -0.8485 + \omega(0.25)$$

$$0 = 0.8485 + 0.433 \omega$$

Solving

$$\omega = 1.96 \text{ rad/s} \curvearrowright$$

$$v_C = 1.34 \text{ m/s} \quad \text{Ans.}$$

**R16-5.**  $\omega = \frac{2}{0.08} = 25 \text{ rad/s}$

$$\alpha = \frac{4}{0.08} = 50 \text{ rad/s}^2$$

$$\mathbf{a}_C = \mathbf{a}_A + (\mathbf{a}_{C/A})_n + (\mathbf{a}_{C/A})_t$$

$$\mathbf{a}_C = 4\mathbf{j} + (25)^2(0.08)\mathbf{i} + 50(0.08)\mathbf{j}$$

$$\rightarrow a_C \cos \theta = 0 + 50$$

$$+\uparrow a_C \sin \theta = 4 + 0 + 4$$

$$\text{Solving, } a_C = 50.6 \text{ m/s}^2$$

Ans.

$$\theta = 9.09^\circ \swarrow \theta$$

Ans.

The cylinder moves up with an acceleration

$$a_B = (a_C)_t = 50.6 \sin 9.09^\circ = 8.00 \text{ m/s}^2 \uparrow \quad \text{Ans.}$$

**R16-6.**  $\mathbf{a}_C = \mathbf{a}_B + \mathbf{a}_{C/B}$

$$2.057 + (a_C)_t = 1.8 + 1.2 + \alpha_{CB}(0.5)$$

$$\rightarrow \downarrow \quad \downarrow \quad \leftarrow \quad \searrow \theta 30^\circ$$

$$(\pm) \quad 2.057 = -1.2 + \alpha_{CB}(0.5) \cos 30^\circ$$

$$(+\downarrow) \quad (a_C)_t = 1.8 + \alpha_{CB}(0.5) \sin 30^\circ$$

$$\alpha_{CB} = 7.52 \text{ rad/s}^2$$

Ans.

$$(a_C)_t = 3.68 \text{ m/s}^2$$

$$a_C = \sqrt{(3.68)^2 + (2.057)^2} = 4.22 \text{ m/s}^2 \quad \text{Ans.}$$

$$\theta = \tan^{-1}\left(\frac{3.68}{2.057}\right) = 60.8^\circ \swarrow \quad \text{Ans.}$$

Also,

$$\mathbf{a}_C = \mathbf{a}_B + \alpha_{CB} \times \mathbf{r}_{C/B} - \omega^2 \mathbf{r}_{C/B}$$

$$-(a_C)_t \mathbf{j} + \frac{(0.6)^2}{0.175} \mathbf{i} = -(2)^2(0.3)\mathbf{i} - 6(0.3)\mathbf{j}$$

$$+ (\alpha_{CB} \mathbf{k}) \times (-0.5 \cos 60^\circ \mathbf{i} - 0.5 \sin 60^\circ \mathbf{j}) - 0$$

$$2.057 = -1.20 + \alpha_{CB}(0.433)$$

$$-(a_C)_t = -1.8 - \alpha_{CB}(0.250)$$

$$\alpha_{CB} = 7.52 \text{ rad/s}^2$$

Ans.

$$a_t = 3.68 \text{ m/s}^2$$

$$a_C = \sqrt{(3.68)^2 + (2.057)^2} = 4.22 \text{ m/s}^2 \quad \text{Ans.}$$

$$\theta = \tan^{-1}\left(\frac{3.68}{2.057}\right) = 60.8^\circ \swarrow \theta \quad \text{Ans.}$$

**R16-7.**  $a_C = 0.5(8) = 4 \text{ m/s}^2$

$$\mathbf{a}_B = \mathbf{a}_C + \mathbf{a}_{B/C}$$

$$\mathbf{a}_B = \left[ \begin{array}{c} 4 \\ \leftarrow \end{array} \right] + \left[ \begin{array}{c} (3)^2(0.5) \\ \nearrow 30^\circ \end{array} \right] + \left[ \begin{array}{c} (0.5)(8) \\ \searrow 30^\circ \end{array} \right]$$

$$(\pm \rightarrow) \quad (a_B)_x = -4 + 4.5 \cos 30^\circ + 4 \sin 30^\circ$$

$$= 1.897 \text{ m/s}^2$$

$$(+\uparrow) \quad (a_B)_y = 0 + 4.5 \sin 30^\circ - 4 \cos 30^\circ$$

$$= -1.214 \text{ m/s}^2$$

$$a_B = \sqrt{(1.897)^2 + (-1.214)^2}$$

$$= 2.25 \text{ m/s}^2 \quad \text{Ans.}$$

$$\theta = \tan^{-1}\left(\frac{1.214}{1.897}\right) = 32.6^\circ \quad \text{Ans.}$$

Also,

$$\mathbf{a}_B = \mathbf{a}_C + \alpha \times \mathbf{r}_{B/C} - \omega^2 \mathbf{r}_{B/C}$$

$$(a_B)_x \mathbf{i} + (a_B)_y \mathbf{j} = -4\mathbf{i} + (8\mathbf{k}) \times (-0.5 \cos 30^\circ \mathbf{i} - 0.5 \sin 30^\circ \mathbf{j}) - (3)^2(-0.5 \cos 30^\circ \mathbf{i} - 0.5 \sin 30^\circ \mathbf{j})$$

$$(\pm \rightarrow) \quad (a_B)_x = -4 + 8(0.5 \sin 30^\circ) + (3)^2(0.5 \cos 30^\circ)$$

$$= 1.897 \text{ m/s}^2$$

$$(+\uparrow) \quad (a_B)_y = 0 - 8(0.5 \cos 30^\circ) + (3)^2(0.5 \sin 30^\circ)$$

$$= -1.214 \text{ m/s}^2$$

$$\theta = \tan^{-1}\left(\frac{1.214}{1.897}\right) = 32.6^\circ \quad \text{Ans.}$$

$$a_B = \sqrt{(1.897)^2 + (-1.214)^2} = 2.25 \text{ m/s}^2 \quad \text{Ans.}$$

**R16-8.**  $v_B = 3(0.7) = 2.10 \text{ m/s} \leftarrow$

$$\mathbf{v}_C = \mathbf{v}_B + \omega \times \mathbf{r}_{C/B}$$

$$-v_C\left(\frac{4}{5}\right)\mathbf{i} - v_C\left(\frac{3}{5}\right)\mathbf{j} = -2.10\mathbf{i} + \omega \mathbf{k} \times (-0.5\mathbf{i} - 1.2\mathbf{j})$$

$$(\pm \rightarrow) \quad -0.8v_C = -2.10 + 1.2\omega$$

$$(+\uparrow) \quad -0.6v_C = -0.5\omega$$

Solving:

$$\omega = 1.125 \text{ rad/s}$$

$$v_C = 0.9375 \text{ m/s} \quad \text{Ans.}$$

$$(a_B)_n = (3)^2(0.7) = 6.30 \text{ m/s}^2 \downarrow$$

$$(a_B)_t = (2)(0.7) = 1.40 \text{ m/s}^2 \leftarrow$$

$$\mathbf{a}_C = \mathbf{a}_B + \alpha \times \mathbf{r}_{C/B} - \omega^2 \mathbf{r}_{C/B}$$

$$-a_C\left(\frac{4}{5}\right)\mathbf{i} - a_C\left(\frac{3}{5}\right)\mathbf{j} = -1.40\mathbf{i} - 6.30\mathbf{j} + (\alpha \mathbf{k})$$

$$\times (-0.5\mathbf{i} - 1.2\mathbf{j}) - (1.125)^2(-0.5\mathbf{i} - 1.2\mathbf{j})$$

$$(\pm \rightarrow) \quad -0.8a_C = -1.40 + 1.2\alpha + 0.6328$$

$$(+\uparrow) \quad -0.6a_C = -6.30 - 0.5\alpha + 1.51875$$

$$a_C = 5.47 \text{ m/s}^2 \quad \text{Ans.}$$

$$\alpha = -3.00 \text{ rad/s}^2$$

## Chapter 17

**R17-1.**  $\leftarrow \Sigma F_x = ma_x; \quad 50 \cos 60^\circ = 200a_G \quad (1)$

$$+\uparrow \Sigma F_y = ma_y; \quad N_A + N_B - 200(9.81)$$

$$-50 \sin 60^\circ = 0 \quad (2)$$

$$\zeta + \Sigma M_G = 0; \quad -N_A(0.3) + N_B(0.2) +$$

$$50 \cos 60^\circ(0.3)$$

$$-50 \sin 60^\circ(0.6) = 0 \quad (3)$$

Solving,

$$a_G = 0.125 \text{ m/s}^2$$

$$N_A = 765.2 \text{ N}$$

$$N_B = 1240 \text{ N}$$

At each wheel

$$N'_A = \frac{N_A}{2} = 383 \text{ N} \quad \text{Ans.}$$

$$N'_B = \frac{N_B}{2} = 620 \text{ N} \quad \text{Ans.}$$

### R17-2. Curvilinear Translation:

$$(a_G)_t = 8(1.5) = 12 \text{ m/s}^2$$

$$(a_G)_n = (5)^2(1.5) = 37.5 \text{ m/s}^2$$

$$\bar{x} = \frac{\Sigma \bar{x}m}{\Sigma m} = \frac{0.5(1.5) + 1(1.5)}{2(1.5)} = 0.75 \text{ m}$$

$$+\downarrow \Sigma F_y = m(a_G)_y;$$

$$E_y + 2(1.5)(9.81) = [2(1.5)](12 \cos 30^\circ + 37.5 \sin 30^\circ)$$

$$E_y = 58.00 \text{ N} = 58.0 \text{ N} \quad \text{Ans.}$$

$$\pm \Sigma F_x = m(a_G)_x;$$

$$E_x = [2(1.5)](37.5 \cos 30^\circ - 12 \sin 30^\circ)$$

$$E_x = 79.43 \text{ N} = 79.4 \text{ N} \quad \text{Ans.}$$

$$\zeta + \Sigma M_G = 0; \quad M_E - 58.00(0.75) = 0$$

$$M_E = 43.50 \text{ N} \cdot \text{m} = 43.5 \text{ N} \cdot \text{m} \quad \text{Ans.}$$

### R17-3. (a) Rear wheel drive

**Equations of motion:**

$$\pm \Sigma F_x = m(a_G)_x; \quad 0.3N_B = 1.5(10)^3 a_G \quad (1)$$

$$\zeta + \Sigma M_A = \Sigma (M_k)_A; \quad 1.5(10)^3(9.81)(1.3)$$

$$-N_B(2.9) = -1.5(10)^3 a_G(0.4) \quad (2)$$

Solving Eqs. (1) and (2) yields:

$$N_B = 6881 \text{ N} = 6.88 \text{ kN}$$

$$a_G = 1.38 \text{ m/s}^2 \quad \text{Ans.}$$

$$\text{R17-4. } \pm \Sigma F_x = m(a_G)_x; 40 \sin 60^\circ + N_C - \left(\frac{5}{13}\right)T = 0$$

$$+ \uparrow \Sigma F_y = m(a_G)_y; -40 \cos 60^\circ + 0.3N_C - 20(9.81) + \frac{12}{13}T = 0$$

$$\zeta + \Sigma M_A = I_A \alpha; 40(0.120) - 0.3N_C(0.120) = \left[\frac{1}{2}(20)(0.120)^2\right]\alpha$$

Solving,

$$T = 218 \text{ N} \quad \text{Ans.}$$

$$N_C = 49.28 \text{ N}$$

$$\alpha = 21.0 \text{ rad/s}^2 \quad \text{Ans.}$$

$$\text{R17-5. } (a_G)_t = \alpha(27)$$

$$\leftarrow \Sigma F_t = m(a_G)_x; F + 100 - 25 = 15[\alpha(2)]$$

$$\zeta + \Sigma M_O = I_O \alpha; 100(1.5) + F(3) = \left[\frac{1}{3}(15)(4^2)\right]\alpha$$

Solving,

$$\alpha = 7.50 \text{ rad/s}^2 \quad \text{Ans.}$$

$$F = 150 \text{ N} \quad \text{Ans.}$$

$$\text{R17-6. } I_O = \frac{2}{5}(15)(0.3^2) + 15(0.9^2)$$

$$+ \frac{1}{3}(5)(0.6^2) = 13.29 \text{ kg} \cdot \text{m}^2$$

$$\bar{x} = \frac{0.9(15) + 0.3(5)}{15 + 5} = 0.75 \text{ m}$$

$$\pm \Sigma F_n = ma_n; O_x = 0 \quad \text{Ans.}$$

$$+ \downarrow \Sigma F_t = m(a_G); 20(981) - O_y = 20a_G$$

$$\zeta + \Sigma M_O = I_O \alpha; [20(9.81)](0.75) = 13.29\alpha$$

$$\alpha = 11.07 \text{ rad/s}$$

Kinematic:

$$a_G = \alpha r_G = 11.07(0.75) = 8.304 \text{ m/s}^2$$

Then

$$O_y = 30.12 \text{ N} = 30.1 \text{ N} \quad \text{Ans.}$$

$$\text{R17-7. } + \uparrow \Sigma F_y = m(a_G)_y; N_B - 20(9.81) = 0$$

$$N_B = 196.2 \text{ N}$$

$$F_B = 0.1(196.2) = 19.62 \text{ N}$$

$$\zeta + \Sigma M_{IC} = \Sigma (M_k)_{IC}; 30 - 19.62(0.6) = 20(0.2\alpha)(0.2) + [20(0.25)^2]\alpha$$

$$\alpha = 8.89 \text{ rad/s}^2 \quad \text{Ans.}$$

$$\text{R17-8. } \leftarrow \Sigma F_x = m(a_G)_x; 0.3N_A = 9a_G$$

$$+ \uparrow \Sigma F_y = m(a_G)_y; N_A - 9(9.81) = 0$$

$$\zeta + \Sigma M_G = I_G \alpha; 0.3N_A(0.15) = \left[\frac{2}{5}(9)(0.15)^2\right]\alpha$$

Solving,

$$N_A = 88.29 \text{ N}$$

$$a_G = 2.943 \text{ m/s}^2 \leftarrow$$

$$\alpha = 49.05 \text{ rad/s}^2 \searrow$$

$$(\zeta +) \quad \omega = \omega_0 + \alpha_c t$$

$$0 = \omega_0 - 49.05t$$

$$\omega_0 = 49.05t$$

$$(\pm) \quad v = v_0 + a_c t$$

$$0 = 6 - 2.943\left(\frac{\omega_0}{49.05}\right)$$

$$\omega_0 = 100 \text{ rad/s} \quad \text{Ans.}$$

## Chapter 18

$$\text{R18-1. } T_1 + \Sigma U_{1-2} = T_2$$

$$0 + (50)(9.81)(1.25) = \frac{1}{2}\left[(50)(1.75)^2\right]\omega_2^2$$

$$\omega_2 = 2.83 \text{ rad/s} \quad \text{Ans.}$$

**R18-2. Kinetic Energy and Work:** The mass moment inertia of the flywheel about its mass center is  $I_O = mk_O^2 = 50(0.2^2) = 2 \text{ kg} \cdot \text{m}^2$ . Thus,

$$T = \frac{1}{2}I_O\omega^2 = \frac{1}{2}(2)\omega^2 = \omega^2$$

Since the wheel is initially at rest,  $T_1 = 0$ .  $\mathbf{W}$ ,  $\mathbf{O}_x$ , and  $\mathbf{O}_y$  do no work while  $\mathbf{M}$  does positive work. When the wheel rotates

$$\theta = (5 \text{ rev})\left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right) = 10\pi, \text{ the work done by } M \text{ is}$$

$$\begin{aligned}
 U_M &= \int M d\theta = \int_0^{10\pi} (9\theta^{1/2} + 1) d\theta \\
 &= (6\theta^{3/2} + \theta) \Big|_0^{10\pi} \\
 &= 1087.93 \text{ J}
 \end{aligned}$$

**Principle of Work and Energy:**

$$\begin{aligned}
 T_1 + \Sigma U_{1-2} &= T_2 \\
 0 + 1087.93 &= \omega^2 \\
 \omega &= 33.0 \text{ rad/s} \quad \text{Ans.}
 \end{aligned}$$

**R18-3.** Before braking:

$$\begin{aligned}
 T_1 + \Sigma U_{1-2} &= T_2 \\
 0 + 15(9.81)(3) &= \frac{1}{2}(15)v_B^2 + \frac{1}{2}[50(0.23)^2] \left( \frac{v_B}{0.15} \right)^2 \\
 v_B &= 2.58 \text{ m/s} \quad \text{Ans.} \\
 \frac{s_B}{0.15} &= \frac{s_C}{0.25}
 \end{aligned}$$

Set  $s_B = 3 \text{ m}$ , then  $s_C = 5 \text{ m}$ .

$$\begin{aligned}
 T_1 + \Sigma U_{1-2} &= T_2 \\
 0 - F(5) + 15(9.81)(6) &= 0 \\
 F &= 176.6 \text{ N} \\
 N &= \frac{176.6}{0.5} = 353.2 \text{ N}
 \end{aligned}$$

Brake arm:

$$\begin{aligned}
 \zeta + \Sigma M_A &= 0; \quad -353.2(0.5) + P(1.25) = 0 \\
 P &= 141 \text{ N} \quad \text{Ans.}
 \end{aligned}$$

**R18-4.**

$$\begin{aligned}
 \frac{s_G}{0.3} &= \frac{s_A}{(0.5 - 0.3)} \\
 s_A &= 0.6667s_G \\
 +\curvearrowright \Sigma F_y &= 0; \quad N_A - 60(9.81) \cos 30^\circ = 0 \\
 N_A &= 509.7 \text{ N} \\
 T_1 + \Sigma U_{1-2} &= T_2 \\
 0 + 60(9.81) \sin 30^\circ (s_G) - 0.2(509.7)(0.6667s_G) \\
 &= \frac{1}{2}[60(0.3)^2](6)^2 \\
 &\quad + \frac{1}{2}(60)[(0.3)(6)]^2 \\
 s_G &= 0.859 \text{ m} \quad \text{Ans.}
 \end{aligned}$$

**R18-5. Conservation of Energy:** Originally, both gears are rotating with an angular velocity of  $\omega_1 = \frac{2}{0.05} = 40 \text{ rad/s}$ . After the rack has traveled

$s = 600 \text{ mm}$ , both gears rotate with an angular velocity of  $\omega_2 = \frac{v_2}{0.05}$ , where  $v_2$  is the speed of the rack at that moment.

$$\begin{aligned}
 T_1 + V_1 &= T_2 + V_2 \\
 \frac{1}{2}(6)(2)^2 + 2 \left\{ \frac{1}{2}[4(0.03)^2](40)^2 \right\} + 0 \\
 &= \left\{ \frac{1}{2}[4(0.03)^2] \left( \frac{v_2}{0.05} \right)^2 \right\} - 6(9.81)(0.6) \\
 v_2 &= 3.46 \text{ m/s} \quad \text{Ans.}
 \end{aligned}$$

**R18-6.** Datum through A.

$$\begin{aligned}
 T_1 + V_1 &= T_2 + V_2 \\
 \frac{1}{2} \left[ \frac{1}{3}(25)(3)^2 \right] (2)^2 + \frac{1}{2}(90)(2 - 1)^2 \\
 &= \frac{1}{2} \left[ \frac{1}{3}(25)(3)^2 \right] (\omega^2) + \frac{1}{2}(90)(3.5 - 1)^2 \\
 &\quad - 25(9.81)(1.5 \sin 30^\circ) \\
 \omega &= 1.61 \text{ rad/s} \quad \text{Ans.}
 \end{aligned}$$

**R18-7.**

$$\begin{aligned}
 T_1 + V_1 &= T_2 + V_2 \\
 \text{At } \theta = 0^\circ, v_B &= 0 \text{ and } \omega_{BC} = v_C \\
 0 + [2(9.81)(0.5 \sin 45^\circ) + [0.5(9.81)](1 \sin 45^\circ) \\
 &\quad + \frac{1}{2} \left[ \frac{1}{3}(2)(1^2) \right] v_C^2 + \frac{1}{2}(0.5)v_C^2 + 0 \\
 v_C &= 4.22 \text{ m/s} \quad \text{Ans.}
 \end{aligned}$$

**R18-8.** Datum at lowest point.

$$\begin{aligned}
 T_1 + V_1 &= T_2 + V_2 \\
 \frac{1}{2} \left[ \frac{1}{2}(40)(0.3)^2 \right] \left( \frac{4}{0.3} \right)^2 + \frac{1}{2}(40)(4)^2 \\
 &\quad + 40(9.81)d \sin 30^\circ = 0 + \frac{1}{2}(200)d^2 \\
 100d^2 - 196.2d - 480 &= 0 \\
 \text{Solving for the positive root,} \\
 d &= 3.38 \text{ m} \quad \text{Ans.}
 \end{aligned}$$

**Chapter 19****R19-1.**  $I_O = mk_O^2 = 75(0.375^2) = 10.547 \text{ kg} \cdot \text{m}^2$ 

$$\begin{aligned}
 I_O \omega_1 + \Sigma \int_{t_1}^{t_2} M_O dt &= I_O \omega_2 \\
 0 + \int_0^{3 \text{ s}} (50t^2)(0.3) dt &= 10.547 \omega_2 \\
 3t^3 \Big|_0^{3 \text{ s}} &= 10.547 \omega_2 \\
 \omega_2 &= 7.68 \text{ rad/s} \quad \text{Ans.}
 \end{aligned}$$

$$\begin{aligned} \mathbf{R19-3.} \quad +\swarrow m(v_G)_1 + \Sigma \int F dt &= m(v_G)_2 \\ 0 + 9(9.81)(\sin 30^\circ)(3) - \int_0^3 F dt &= 9(v_G)_2 \quad (1) \end{aligned}$$

$$\begin{aligned} \zeta + (H_G)_1 + \Sigma \int M_G dt &= (H_G)_2 \\ 0 + \left( \int_0^3 F dt \right)(0.3) &= [9(0.225)^2] \omega_2 \quad (2) \end{aligned}$$

Since  $(v_G)_2 = 0.3\omega_2$ ,

Eliminating  $\int_0^3 F dt$  from Eqs. (1) and (2) and solving for  $(v_G)_2$  yields.

$$(v_G)_2 = 9.42 \text{ m/s} \quad \text{Ans.}$$

Also,

$$\begin{aligned} \zeta + (H_A)_1 + \Sigma \int M_A dt &= (H_A)_2 \\ 0 + 9(9.81) \sin 30^\circ(3)(0.3) &= [9(0.225)^2 + 9(0.3)^2] \omega \\ \omega &= 31.39 \text{ rad/s} \\ v &= 0.3(31.39) = 9.42 \text{ m/s} \quad \text{Ans.} \end{aligned}$$

$$\begin{aligned} \mathbf{R19-4.} \quad \leftarrow \quad m(v_x)_1 + \Sigma \int_{t_1}^{t_2} F_x dt &= m(v_x)_2 \\ 0 + 200(3) &= 100(v_O)_2 \\ (v_O)_2 &= 6 \text{ m/s} \quad \text{Ans.} \end{aligned}$$

and

$$\begin{aligned} I_x \omega_1 + \Sigma \int_{t_1}^{t_2} M_z dt &= I_x \omega_2 \\ 0 - [200(0.4)(3)] &= -9\omega_2 \\ \omega_2 &= 26.7 \text{ rad/s} \quad \text{Ans.} \end{aligned}$$

$$\begin{aligned} \mathbf{R19-5.} \quad (+\uparrow) \quad mv_1 + \Sigma \int F dt &= mv_2 \\ 0 + T(3) - 15(9.81)(3) + 200(3) &= 15v_o \end{aligned}$$

$$\begin{aligned} (\zeta+) \quad (H_O)_1 + \Sigma \int M_O dt &= (H_O)_2 \\ 0 - T(0.15)3 + 200(0.3)(3) &= [15(0.2)^2] \omega \end{aligned}$$

Kinematics,

$$v_o = 0.15\omega$$

Solving,

$$\begin{aligned} T &= 110.18 \text{ N} \\ \omega &= 217 \text{ rad/s} \\ v_O &= 32.61 \text{ m/s} \quad \text{Ans.} \end{aligned}$$

Also,

$$\begin{aligned} (\zeta+) \quad (H_{IC})_1 + \Sigma \int M_{IC} dt &= (H_{IC})_2 \\ 0 + 200(0.45)(3) - 15(9.81)(0.15)(3) &= [15(0.2)^2 + 15(0.15)^2] \omega \\ \omega &= 217 \text{ rad/s} \quad \text{Ans.} \end{aligned}$$

$$\begin{aligned} \mathbf{R19-6.} \quad \zeta + (H_A)_1 + \Sigma \int M_A dt &= (H_A)_2 \\ [15(0.25)^2](6) - \int T(0.4) dt &= [15(0.25)^2] \omega_A \end{aligned}$$

$$\begin{aligned} \zeta + (H_B)_1 + \Sigma \int M_B dt &= (H_B)_2 \\ 0 + \int T(0.15) dt &= [7.5(0.18)^2] \omega_B \end{aligned}$$

Kinematics:

$$\begin{aligned} 0.4\omega_A &= 0.15\omega_B \\ \omega_B &= 2.6667\omega_A \end{aligned}$$

Thus,

$$\begin{aligned} \omega_A &= 2.11 \text{ rad/s} \quad \text{Ans.} \\ \omega_B &= 5.63 \text{ rad/s} \quad \text{Ans.} \end{aligned}$$

$$\begin{aligned} \mathbf{R19-7.} \quad H_1 &= H_2 \\ \left( \frac{1}{2}mr^2 \right) \omega_1 &= \left[ \frac{1}{2}mr^2 + mr^2 \right] \omega_2 \\ \omega_2 &= \frac{1}{3}\omega_1 \quad \text{Ans.} \end{aligned}$$

$$\begin{aligned} \mathbf{R19-8.} \quad H_1 &= H_2 \\ (0.940)(0.5) + (4) \left[ \frac{1}{12}(20) \left( (0.75)^2 + (0.2)^2 \right) \right. &+ \left. (20)(0.375 + 0.2)^2 \right] (0.5) \\ &= (0.940)(\omega) + 4 \left[ \frac{1}{12}(20)(0.2)^2 + (20)(0.2)^2 \right] \omega \\ \omega &= 3.56 \text{ rad/s} \quad \text{Ans.} \end{aligned}$$



# Answers to Selected Problems

## Chapter 12

- 12-1.**  $v = 0$   
 $s = 80.7 \text{ m}$
- 12-2.**  $v = 625 \text{ m/s}$   
 $s = 639.5 \text{ m}$   
 $d = 637.5 \text{ m}$
- 12-3.**  $a = -12 \text{ m/s}^2, s = 0$   
 $s_T = 4 + 4 = 8 \text{ m}, (v_{sp})_{\text{avg}} = 2.67 \text{ m/s}$
- 12-5.**  $v_{\text{avg}} = 0, (v_{sp})_{\text{avg}} = 3 \text{ m/s}, a \Big|_{t=6} = 2 \text{ m/s}^2$
- 12-6.**  $\Delta s = 14.71 \text{ m}$
- 12-7.**  $t = 25 \text{ s}$   
 $s = 312.5 \text{ m}$
- 12-9.**  $S_{BA} = \left| \frac{2v_A v_B - v_A^2}{2a_A} \right|$
- 12-10.**  $v_{\text{avg}} = 0.222 \text{ m/s}, (v_{sp})_{\text{avg}} = 2.22 \text{ m/s}$
- 12-11.**  $t = 30 \text{ s}; s = 792 \text{ m}$
- 12-13.**  $v = 32 \text{ m/s}, s = 67 \text{ m}, d = 66 \text{ m}$
- 12-14.**  $s = 28.4 \text{ km}$
- 12-15.**  $v = \frac{16}{(8t + 1)^2} \text{ m/s}$   
 $a = \frac{256}{(8t + 1)^3} \text{ m/s}^2$
- 12-17.**  $v = \left( 2kt + \frac{1}{v_0^2} \right)^{-1/2},$   
 $s = \frac{1}{k} \left[ \left( 2kt + \left( \frac{1}{v_0^2} \right) \right)^{1/2} - \frac{1}{v_0} \right]$
- 12-18.**  $s = 6.53 \text{ m}, t = 3.27 \text{ s}$
- 12-19.**  $v_{pI} = 322.49 \frac{\text{m}}{\text{s}}, t_I = 19.27 \text{ s}$
- 12-21.**  $v = 3.93 \text{ m/s}, s = 9.98 \text{ m}$
- 12-22.** (a)  $s = -30.5 \text{ m},$   
 (b)  $s_{To t} = 56.0 \text{ m},$   
 (c)  $v = 10 \text{ m/s}$
- 12-23.** (a)  $v = 45.5 \text{ m/s},$  (b)  $v_{\text{max}} = 100 \text{ m/s}$
- 12-25.**  $v = (20e^{-2t}) \text{ m/s}, a = (-40e^{-2t}) \text{ m/s}^2,$   
 $s = 10(1 - e^{-2t}) \text{ m}$
- 12-26.**  $v = 0.781 \text{ m/s}$
- 12-27.**  $t = 0.549 \left( \frac{v_f}{g} \right)$
- 12-29.**  $d = (30 - 9.62) = 20.4 \text{ m}, t = 2 \text{ s}$   
 $s_B = 20.4 \text{ m}$
- 12-30.**  $v_1 = 3.68 \text{ m/s}, t_2 = 1.98 \text{ s}, v_2 = 15.8 \text{ m/s} \downarrow$
- 12-31.**  $s = \frac{v_0}{k}(1 - e^{-kt}), a = -kv_0 e^{-kt}$
- 12-33.**  $v = 11.2 \text{ km/s}$
- 12-34.**  $v = -R \sqrt{\frac{2g_0(y_0 - y)}{(R + y)(R + y_0)}}, v = 3.02 \text{ km/s}$
- 12-35.**  $v_{\text{max}} = 16.7 \text{ m/s}$
- 12-37.**  $a \Big|_{t=0} = -4 \text{ m/s}^2, a \Big|_{t=2 \text{ s}} = 0,$   
 $a \Big|_{t=4 \text{ s}} = 4 \text{ m/s}^2, v \Big|_{t=0} = 3 \text{ m/s},$   
 $v \Big|_{t=2 \text{ s}} = -1 \text{ m/s}, v \Big|_{t=4 \text{ s}} = 3 \text{ m/s}$
- 12-38.**  $\Delta s = 1.11 \text{ km}$
- 12-39.**  $s = 2 \sin \left( \frac{\pi}{5} t \right) + 4$   
 $v = \frac{2\pi}{5} \cos \left( \frac{\pi}{5} t \right)$   
 $a = -\frac{2\pi^2}{25} \sin \left( \frac{\pi}{5} t \right)$
- 12-41.**  $s = 600 \text{ m. For } 0 \leq t < 40 \text{ s}, a = 0.$   
 For  $40 \text{ s} < t \leq 80 \text{ s}, a = -0.250 \text{ m/s}^2.$
- 12-43.**  $v_{\text{max}} = 0.894 \text{ m/s}, t' = 0.447 \text{ s}, s = 0.2 \text{ m}$
- 12-45.**  $v = 150 \text{ m/s}$   
 $v = \left\{ \frac{1}{4}t^2 - 15t + 275 \right\} \text{ m/s}$   
 $v = \{-5t + 175\} \text{ m/s}$   
 $t' = 35.0 \text{ s}$   
 $s = \{150t\} \text{ m}$   
 $s = \left\{ \frac{1}{12}t^3 - \frac{15}{2}t^2 + 275t - 583 \right\} \text{ m}$   
 $s = \left\{ \frac{-5}{2}t^2 + 175t + 83.3 \right\} \text{ m}$
- 12-46.** For  $0 \leq t < 30 \text{ s}, v = \left\{ \frac{1}{5}t^2 \right\} \text{ m/s}, s = \left\{ \frac{1}{15}t^3 \right\} \text{ m}$   
 For  $30 \leq t \leq 60 \text{ s}, v = \{24t - 540\} \text{ m/s},$
- 12-47.**  $t' = 133 \text{ s}, s \Big|_{t=133.09 \text{ s}} = 8857 \text{ m}$
- 12-49.**  $a = 0.32 \text{ m/s}^2, a = -0.32 \text{ m/s}^2$
- 12-50.**  $v_{\text{max}} = 100 \text{ m/s}, t' = 40 \text{ s}$
- 12-51.**  $a_{\text{max}} = 4.00 \text{ m/s}^2$
- 12-53.**  $a_3 = 8.00 \text{ m/s}^2, a_4 = 4.50 \text{ m/s}^2$
- 12-55.**  $s = \{0.2t^3\} \text{ m}$   
 $s = \left\{ \frac{1}{4}(90t - 3t^2 - 275) \right\} \text{ m}$   
 $s = \frac{1}{4}[90(15) - 3(15^2) - 275] = 100 \text{ m}$   
 $v_{\text{avg}} = 6.67 \text{ m/s}$

- 12-57.** At  $t = 8$  s,  $a = 0$  and  $s = 30$  m.  
At  $t = 12$  s,  $a = -1$  m/s<sup>2</sup> and  $s = 48$  m.
- 12-58.** When  $t = 5$  s,  $s_B = 62.5$  m.  
When  $t = 10$  s,  $v_A = (v_A)_{\max} = 40$  m/s and  $s_A = 200$  m.  
When  $t = 15$  s,  $s_A = 400$  m and  $s_B = 312.5$  m.  
 $\Delta s = s_A - s_B = 87.5$  m
- 12-61.** For  $0 \leq t < 15$  s,  
$$v = \left\{ \frac{1}{2} t^2 \right\} \text{ m/s}$$
$$s = \left\{ \frac{1}{6} t^3 \right\} \text{ m}$$
For  $15 \leq t \leq 40$  s,  
 $v = \{20t - 187.5 \text{ m/s}\}$   
 $s = \{10t^2 - 187.5t + 115\} \text{ m}$
- 12-62.**  $v = \left\{ \frac{1}{10} s \right\} \text{ m/s}$
- 12-63.**  $v = \{5 - 6t\} \text{ m/s}$ ,  $a = -6 \text{ m/s}^2$
- 12-65.** At  $s = 100$  s,  $a = 11.1 \text{ m/s}^2$ ,  
At  $s = 175$  m,  $a = -25 \text{ m/s}^2$
- 12-66.**  $t' = 22.5$  s
- 12-67.** When  $s = 100$  m,  $t = 10$  s.  
When  $s = 400$  m,  $t = 16.9$  s.  
 $a|_{s=100 \text{ m}} = 4 \text{ m/s}^2$ ,  
 $a|_{s=400 \text{ m}} = 16 \text{ m/s}^2$
- 12-69.**  $a = 80.2 \text{ m/s}^2$ , (42.7, 16.0, 14.0) m
- 12-70.**  $v = 9.68 \text{ m/s}$ ,  $a = 169.8 \text{ m/s}^2$
- 12-71.**  $\Delta \mathbf{r} = \{6\mathbf{i} + 4\mathbf{j}\} \text{ m}$
- 12-73.**  $v = 201 \text{ m/s}$ ,  $a = 405 \text{ m/s}^2$
- 12-74.**  $d_1 = 8.34 \text{ m}$ ,  $a_2 = 0.541 \text{ m/s}^2$
- 12-75.**  $v_{\text{sp}} = 4.28 \text{ m/s}$
- 12-77.**  $v = 8.55 \text{ m/s}$ ,  $a = 5.82 \text{ m/s}^2$
- 12-78.**  $v = 1003 \text{ m/s}$ ,  $a = 103 \text{ m/s}^2$
- 12-79.**  $d = 4.00 \text{ m}$ ,  $a = 37.8 \text{ m/s}^2$
- 12-81.**  $(\mathbf{v}_{BC})_{\text{avg}} = \{3.88\mathbf{i} + 6.72\mathbf{j}\} \text{ m/s}$
- 12-82.**  $v = \sqrt{c^2 k^2 + b^2}$ ,  $a = ck^2$
- 12-83.**  $d = 204 \text{ m}$ ,  $v = 41.8 \text{ m/s}$   
 $a = 4.66 \text{ m/s}^2$
- 12-85.**  $v_A = 23.2 \text{ m/s}$
- 12-86.**  $v_A = 11.7 \text{ m/s}$ ,  $v_B = 10.8 \text{ m/s}$ ,  $\theta = 20.7^\circ \swarrow$
- 12-87.**  $v_A = 28.0 \text{ m/s}$
- 12-89.**  $d = \frac{v_0^2}{g \cos \theta} (\sin 2\phi - 2 \tan \theta \cos^2 \phi)$
- 12-90.**  $\phi = \frac{1}{2} \tan^{-1}(-\cot \theta)$
- 12-91.**  $R_{\max} = 10.2 \text{ m}$ ,  $\theta = 45^\circ$
- 12-93.**  $\theta_A = 51.38^\circ = 51.4^\circ$   
 $d = 7.18 \text{ m}$
- 12-94.**  $t = 1.195 \text{ s}$ ,  $d = 12.7 \text{ m}$
- 12-95.**  $v_A = 11.1 \text{ m/s}$ ,  $h = 3.45 \text{ m}$
- 12-97.**  $v_A = 19.4 \text{ m/s}$ ,  $t_{AB} = 4.54 \text{ s}$
- 12-98.**  $v_A = 19.4 \text{ m/s}$ ,  $v_B = 40.4 \text{ m/s}$
- 12-99.**  $R = \frac{v_0}{g} \sin 2\theta$ ,  $t = \frac{2v_0}{g} \sin \theta$ ,  $h = \frac{v_0^2}{2g} \sin^2 \theta$
- 12-101.**  $t_A = 0.553 \text{ s}$ ,  $x = 3.46 \text{ m}$
- 12-102.**  $\theta_D = 14.7^\circ$ ,  $\theta_C = 75.3^\circ$ ,  
 $\Delta t = 1.45 \text{ s}$
- 12-103.**  $\theta_A = 11.6^\circ$ ,  $t = 0.408 \text{ s}$ ,  $\theta_B = 11.6^\circ \nwarrow$
- 12-105.**  $v_{\min} = 0.838 \text{ m/s}$ ,  $v_{\max} = 1.76 \text{ m/s}$
- 12-106.**  $t = 3.55 \text{ s}$ ,  $v = 32.0 \frac{\text{m}}{2}$
- 12-107.**  $R = 19.0 \text{ m}$ ,  $t = 2.48 \text{ s}$
- 12-109.**  $R_{\min} = 0.189 \text{ m}$ ,  $R_{\max} = 1.19 \text{ m}$
- 12-110.**  $v_n = 0$  and  $v_t = 7.21 \text{ m/s}$ ,  $a_n = 0.555 \text{ m/s}^2$ ,  
 $a_t = 2.77 \text{ m/s}^2$
- 12-111.**  $v = 38.7 \text{ m/s}$
- 12-113.**  $v = 4.40 \text{ m/s}$ ,  $a_t = 5.04 \text{ m/s}^2$ ,  $a_n = 1.39 \text{ m/s}^2$
- 12-114.**  $a = 4.22 \text{ m/s}^2$
- 12-115.**  $a = 1.30 \text{ m/s}^2$
- 12-117.**  $\rho = 100 \text{ m}$
- 12-118.**  $v = 3.68 \text{ m/s}$ ,  $a = 4.98 \text{ m/s}^2$
- 12-119.**  $v = 1.5 \text{ m/s}$ ,  $a = 0.117 \text{ m/s}^2$
- 12-121.**  $a = 1.68 \text{ m/s}^2$
- 12-122.**  $v = 19.9 \text{ m/s}$ ,  $a = 24.2 \text{ m/s}^2$
- 12-123.**  $h = 5.99 \text{ Mm}$
- 12-125.**  $t = 7.00 \text{ s}$ ,  $s = 98.0 \text{ m}$
- 12-126.**  $a_t = 3.62 \text{ m/s}^2$ ,  $\rho = 29.6 \text{ m}$
- 12-127.**  $\theta = 38.2^\circ \swarrow$
- 12-129.**  $a = 8.61 \text{ m/s}^2$
- 12-130.**  $a = 32.2 \text{ m/s}^2$
- 12-131.**  $a = 3.05 \text{ m/s}^2$
- 12-133.**  $a_t = 17.1 \text{ m/s}^2$ ,  $a_n = 46.98 \text{ m/s}^2$ ,  
 $\rho = 6.44 \text{ km}$
- 12-134.**  $a = 2.36 \text{ m/s}^2$
- 12-135.**  $v = 1.96 \text{ m/s}$ ,  $a = 0.930 \text{ m/s}^2$
- 12-137.**  $a = 0.952 \text{ m/s}^2$
- 12-138.**  $y = -0.0766x^2$ ,  $v = 8.37 \text{ m/s}$ ,  
 $a_n = 9.38 \text{ m/s}^2$ ,  $a_t = 2.88 \text{ m/s}^2$
- 12-139.**  $a_{\max} = \frac{v^2 a}{b^2}$
- 12-141.**  $t = 1.21 \text{ s}$
- 12-142.**  $y = \{0.839x - 0.131x^2\} \text{ m}$   
 $a_t = -3.94 \text{ m/s}^2$   
 $a_n = 8.98 \text{ m/s}^2$
- 12-143.**  $a_{\min} = 3.09 \text{ m/s}^2$
- 12-145.**  $a = 322 \text{ mm/s}^2$ ,  $\theta = 26.6^\circ \nearrow$
- 12-146.**  $a = 0.511 \text{ m/s}^2$
- 12-147.**  $a = 0.309 \text{ m/s}^2$
- 12-149.**  $a = 26.9 \text{ m/s}^2$
- 12-150.**  $d = 11.0 \text{ m}$ ,  $a_A = 19.0 \text{ m/s}^2$ ,  
 $a_B = 12.8 \text{ m/s}^2$

- 12-151.**  $t = 2.51 \text{ s}$ ,  $a_A = 22.2 \text{ m/s}^2$ ,  
 $a_B = 65.1 \text{ m/s}^2$   
**12-153.**  $a = 0.525 \text{ m/s}^2$   
**12-154.**  $\alpha = 52.5^\circ$ ,  $\beta = 142^\circ$ ,  $\gamma = 85.1^\circ$   
 $\alpha = 128^\circ$ ,  $\beta = 37.9^\circ$ ,  $\gamma = 94.9^\circ$   
**12-155.**  $v_r = -1.66 \text{ m/s}$   
 $v_\theta = -2.07 \text{ m/s}$   
 $a_r = -3.6372 - 7.637(-0.27067)^2$   
 $-4.20 \text{ m/s}$   
 $a_\theta = 2.97 \text{ m/s}^2$   
**12-157.**  $v = 2.4 \text{ m/s}$   
 $a = 19.3 \text{ m/s}^2$   
**12-158.**  $v_r = 691 \text{ m/s}$ ,  $v_\theta = 402 \text{ m/s}$   
**12-159.**  $a = 7.26 \text{ m/s}^2$   
**12-161.**  $v_r = -2.33 \text{ m/s}$   
 $v_\theta = 7.91 \text{ m/s}$   
 $a_r = -158 \text{ m/s}^2$   
 $a_\theta = -18.6 \text{ m/s}^2$   
**12-162.**  $v_r = ae^{at}$ ,  $v_\theta = e^{at}$ ,  
 $a_r = e^{at}(a^2 - 1)$ ,  $a_\theta = 2ae^{at}$   
**12-163.**  $v = 20 \text{ m/s}$ ,  $a = 5.39 \text{ m/s}^2$   
**12-165.**  $\dot{\mathbf{a}} = (\ddot{r} - 3\dot{r}\dot{\theta}^2 - 3r\dot{\theta}\ddot{\theta})\mathbf{u}_r$   
 $+ (3\dot{r}\ddot{\theta} + \dot{r}\ddot{\theta} + 3\ddot{r}\dot{\theta} - r\dot{\theta}^3)\mathbf{u}_\theta + (\ddot{z})\mathbf{u}_z$   
**12-166.**  $a = 2.32 \text{ m/s}^2$   
**12-167.**  $v_r = 1.20 \text{ m/s}$ ,  $v_\theta = 1.26 \text{ m/s}$ ,  
 $a_r = -3.77 \text{ m/s}^2$ ,  $a_\theta = 7.20 \text{ m/s}^2$   
**12-169.**  $v = 4.24 \text{ m/s}$ ,  $a = 17.6 \text{ m/s}^2$   
**12-170.**  $v_r = 8.21 \text{ mm/s}$   
 $a = -665 \text{ mm/s}^2$   
**12-171.**  $v = 8.21 \text{ mm/s}$   
 $a = -659 \text{ mm/s}^2$   
**12-173.**  $v_r = -14.1 \text{ m/s}$ ,  $v_\theta = 14.1 \text{ m/s}$ ,  
 $a_r = -1.41 \text{ m/s}^2$ ,  $a_\theta = -1.41 \text{ m/s}^2$   
**12-174.**  $v_r = -1.84 \text{ m/s}$ ,  $v_\theta = 19.1 \text{ m/s}$ ,  
 $a_r = -2.29 \text{ m/s}^2$ ,  $a_\theta = 4.60 \text{ m/s}^2$   
**12-175.**  $v = 12.6 \text{ m/s}$ ,  $a = 83.2 \text{ m/s}^2$   
**12-177.**  $v = 3.49 \text{ m/s}$   
**12-178.**  $a = 27.8 \text{ m/s}^2$   
**12-179.**  $v = 1.32 \text{ m/s}$   
**12-181.**  $v_r = a\dot{\theta}$   
 $v_\theta = a\theta\dot{\theta}$   
 $a_r = -a\theta\dot{\theta}^2$   
 $a_\theta = 2a\dot{\theta}^2$   
**12-182.**  $v_r = 6.00 \text{ m/s}$   
 $v_\theta = 18.3 \text{ m/s}$   
 $a_r = -67.1 \text{ m/s}^2$   
 $a_\theta = 66.3 \text{ m/s}^2$   
**12-183.**  $v_r = -250 \text{ mm/s}$   
 $a_r = -9330 \text{ mm/s}^2$   
**12-185.**  $\dot{\theta} = 0.333 \text{ rad/s}$ ,  $a = 6.67 \text{ m/s}^2$   
**12-186.**  $a_r = -6.67 \text{ m/s}^2$ ,  $a_\theta = 3 \text{ m/s}^2$   
**12-187.**  $v_r = 0.242 \text{ m/s}$ ,  $v_\theta = 0.943 \text{ m/s}$ ,  
 $a_r = -2.33 \text{ m/s}^2$ ,  $a_\theta = 1.74 \text{ m/s}^2$   
**12-189.**  $v_r = -2.80 \text{ m/s}$   
 $v_\theta = 19.8 \text{ m/s}$   
**12-190.**  $v_r = -4.00 \text{ m/s}$   
 $v_\theta = 28.3 \text{ m/s}$   
 $a_r = -5.43 \text{ m/s}^2$   
 $a_\theta = -1.60 \text{ m/s}^2$   
**12-191.**  $\dot{\theta} = 0.302 \text{ rad/s}$   
**12-193.**  $v_r = 25.9 \text{ mm/s}$   
 $a_r = -195 \text{ mm/s}^2$   
**12-194.**  $v = \dot{r} = 25.9 \text{ mm/s}$   
 $a = \ddot{r} = 42.5 \text{ mm/s}^2$   
**12-195.**  $v_B = 1.67 \text{ m/s}$   
**12-197.**  $v_B = -2 \text{ m/s}$  Positive means down,  
Negative means up.  
**12-198.**  $v_A = 32 \text{ m/s}$  ↓  
**12-199.**  $t = 3.83 \text{ s}$   
**12-201.**  $t = 1.07 \text{ s}$   
 $v_{A/B} = 5.93 \text{ m/s} \rightarrow$   
**12-202.**  $v_B = 1.50 \text{ m/s}$   
**12-203.**  $2 + 2v_B - 1 = 0$   
**12-205.**  $v_B = 0.75 \text{ m/s}$   
**12-206.**  $v_{A/B} = 2.90 \text{ m/s} \uparrow$   
**12-207.**  $t = 5.00 \text{ s}$   
**12-209.**  $v_A = 2 \text{ m/s} \leftarrow$   
**12-210.**  $v_C = (1.8 \sec \theta) \text{ m/s} \rightarrow$   
**12-211.**  $v_C = 1.2 \text{ m/s} \uparrow$   
 $a_C = 0.512 \text{ m/s}^2 \uparrow$   
**12-213.**  $v_B = 0.671 \text{ m/s} \uparrow$   
**12-214.**  $v_B = 1.41 \text{ m/s} \uparrow$   
**12-215.**  $v_A = 10.0 \text{ m/s} \leftarrow$ ,  $a_A = 46.0 \text{ m/s}^2 \leftarrow$   
**12-217.**  $v_{A/B} = 13.4 \text{ m/s}$ ,  $\theta_v = 31.7^\circ \nearrow$   
 $a_{A/B} = 4.32 \text{ m/s}^2$ ,  $\theta_a = 79.0^\circ \searrow$   
**12-218.**  $\theta = 15.1^\circ$   
**12-219.**  $v_{r/c} = 31.2 \text{ m/s}$   
 $\theta = 9.58^\circ$   
 $v_{r/c} = 19.9 \text{ m/s}$   
 $\theta = 9.58^\circ$   
**12-221.**  $v_w = 58.3 \text{ km/h}$ ,  $\theta = 59.0^\circ \searrow$   
**12-222.**  $v_{A/B} = 15.7 \text{ m/s}$ ,  $\theta = 7.11^\circ \nearrow$ ,  $t = 38.1 \text{ s}$   
**12-223.**  $v_b = 6.21 \text{ m/s}$   
 $t = 11.4 \text{ s}$   
**12-225.**  $v_{A/B} = 120 \text{ km/h} \downarrow$   
 $a_{A/B} = 4000 \text{ km/h}^2$ ,  $\theta = 0.716^\circ \nearrow$   
**12-226.**  $v_b = 5.56 \text{ m/s}$   
 $\theta = 84.4^\circ$   
**12-227.**  $v_r = 34.6 \text{ km/h} \downarrow$   
**12-229.**  $v_{B/A} = 20.5 \text{ m/s}$ ,  $\theta_v = 43.1^\circ \nearrow$   
 $a_{B/A} = 4.92 \text{ m/s}^2$ ,  $\theta_a = 6.04^\circ \nearrow$

**12-230.**  $v_{r/m} = 16.6 \text{ km/h}$ ,  $\theta = 25.0^\circ \swarrow$

**12-231.**  $v_{B/A} = 11.2 \text{ m/s}$ ,  $\theta = 50.3^\circ$

**12-233.**  $v_{A/C} = 21.5 \text{ m/s}$

$\theta_v = 34.9^\circ \swarrow$

$a_{A/C} = 4.20 \text{ m/s}^2$

$\theta_a = 75.4^\circ \swarrow$

**12-234.**  $v_{B/C} = 18.6 \text{ m/s}$

$\theta_v = 66.2^\circ \swarrow$

$a_{B/C} = 0.959 \text{ m/s}^2$

$\theta_a = 8.57^\circ \swarrow$

**12-235.**  $v_{w/s} = 19.9 \text{ m/s}$ ,  $\theta = 74.0^\circ \swarrow$

## Chapter 13

**13-1.**  $a = 1.66 \text{ m/s}^2$

**13-2.**  $a = 1.75 \text{ m/s}^2$

**13-3.**  $F = 6.37 \text{ N}$

**13-5.**  $P = 392 \text{ N}$

**13-6.**  $v = 3.36 \text{ m/s}$ ,  $s = 5.04 \text{ m}$

**13-7.**  $P = 224 \text{ N}$

**13-9.**  $s = 8.49 \text{ m}$

**13-10.**  $T = 4.92 \text{ kN}$

**13-11.**  $F = \frac{m(a_B + g)\sqrt{4y^2 + d^2}}{4y}$

**13-13.**  $v = 14.1 \text{ m/s}$

**13-14.**  $s = 5.43 \text{ m}$

**13-15.**  $a = 1.96 \text{ m/s}^2$

**13-17.**  $a = \frac{1}{2}(1 - \mu_k)g$

**13-18.**  $A_x = 685 \text{ N}$ ,  $A_y = 1.19 \text{ kN}$ ,

$M_A = 4.74 \text{ kN} \cdot \text{m}$

**13-19.**  $\mu_k = 0.559$

**13-21.**  $v = 0.301 \text{ m/s}$

**13-22.**  $v = \left(\frac{F_0 t_0}{\pi m}\right) \left[1 - \cos\left(\frac{\pi t}{t_0}\right)\right]$

$v_{\max} = \frac{2F_0 t_0}{\pi m}$

$s = \left(\frac{F_0 t_0}{\pi m}\right) \left[t - \frac{t_0}{\pi} \sin\left(\frac{\pi t}{t_0}\right)\right]$

**13-23.**  $v_B = 5.70 \text{ m/s} \uparrow$

**13-25.**  $(\pm) s_x = 2.68 \text{ m}$ ,

$t_{AB} = 1.87 \text{ s}$

**13-26.**  $R = 2.45 \text{ m}$ ,  $t_{AB} = 1.72 \text{ s}$

**13-27.**  $v_B = 4.52 \text{ m/s}$

**13-29.**  $t = 2.11 \text{ s}$

**13-30.**  $R = \{150t\} \text{ N}$

**13-31.**  $T = \left(\frac{mg}{2}\right) \sin(2\theta)$

**13-33.**  $t = 5.66 \text{ s}$

**13-34.**  $s = 16.7 \text{ m}$

**13-35.**  $v = \frac{1}{m} \sqrt{1.09 F_0^2 t^2 + 2 F_0 t m v_0 + m^2 v_0^2}$

$x = \frac{y}{0.3} + v_0 \left(\sqrt{\frac{2m}{0.3 F_0}}\right) y^{1/2}$

**13-37.**  $t = 0.519 \text{ s}$

**13-38.**  $P = 2mg \left(\frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta}\right)$ ,

$a = \left(\frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta}\right) g$

**13-39.**  $T = 1.63 \text{ kN}$

**13-41.**  $v = 5.13 \text{ m/s}$

**13-42.**  $B_y = 1.92 \text{ kN}$

$A_y = 2.11 \text{ kN}$

$A_x = 0$

**13-43.**  $v = 32.2 \text{ m/s}$

**13-45.**  $a_B = 0$

$a_c = 4.11 \text{ m/s}^2 \rightarrow$

$a_D = 0.162 \text{ m/s}^2 \rightarrow$

**13-46.**  $P = 2mg \tan \theta$

**13-47.**  $P = 2mg \left(\frac{\sin \theta + \mu_s \cos \theta}{\cos \theta - \mu_s \sin \theta}\right)$

**13-49.**  $x = d$ ,  $v = \sqrt{\frac{kd^2}{m_A + m_B}}$

**13-50.**  $x = d$  for separation.

**13-51.**  $d = \frac{(m_A + m_B)g}{k}$

**13-53.**  $v = 3.13 \text{ m/s}$

**13-54.**  $v = 1.63 \text{ m/s}$ ,  $N = 7.36 \text{ N}$

**13-55.**  $T = 51.5 \text{ kN}$

**13-57.**  $r = 1.36 \text{ m}$

**13-58.**  $v = 10.5 \text{ m/s}$

**13-59.**  $\rho = 9.32 \text{ m}$

**13-60.**  $N = 6.18 \text{ kN}$

**13-61.**  $v = 0.969 \text{ m/s}$

**13-62.**  $v = 1.48 \text{ m/s}$

**13-63.**  $F_N = 15.86 \text{ kN}$ ,  $\rho = 282 \text{ m}$

**13-65.**  $\theta = 26.7^\circ$

**13-66.**  $v = 6.30 \text{ m/s}$ ,  $F_n = 283 \text{ N}$ ,

$F_t = 0$ ,  $F_b = 490 \text{ N}$

**13-67.**  $\theta = \cos^{-1}\left(\frac{m_B}{m_A}\right)$

$v_B = \sqrt{\frac{g(l-h)(m_A^2 - m_B^2)}{m_A m_B}}$

**13-69.**  $N = 1.02 \text{ kN}$

**13-70.**  $N = 7.69 \text{ kN}$

**13-71.**  $N_A = -38.6 \text{ N}$

$N_B = -96.6 \text{ N}$

**13-73.**  $F_f = 1.11 \text{ kN}$ ,  $N = 6.73 \text{ kN}$

**13-74.**  $T = 1.82 \text{ N}$ ,  $N_B = 0.844 \text{ N}$

**13-75.**  $v = \sqrt{gr}$ ,  $N = 2mg$

- 13-77.  $a_t = g \left( \frac{x}{\sqrt{1+x^2}} \right)$ ,  $v = \sqrt{v_0^2 + gx^2}$ ,  
 $N = \frac{m}{\sqrt{1+x^2}} \left[ g - \frac{v_0^2 + gx^2}{1+x^2} \right]$
- 13-78.  $v = 49.5 \text{ m/s}$
- 13-79.  $d = 0.581 \text{ m}$
- 13-81.  $T = 64.0 \text{ N}$ ,  
 $T = 34.6 \text{ N}$
- 13-82.  $\theta = 37.7^\circ$
- 13-83.  $N_P = 2.65 \text{ kN}$ ,  $\rho = 68.3 \text{ m}$
- 13-85.  $\psi = 84.29^\circ$ ,  $v' = 12.00 \text{ m/s}^2$
- 13-86.  $\psi = 84.29^\circ$ ,  $v' = 11.02 \text{ m/s}^2$
- 13-87.  $F = 2.271 \text{ N}$
- 13-89.  $F = -0.898 \text{ N}$
- 13-90.  $F_r = -29.4 \text{ N}$ ,  $F_\theta = 0$ ,  $F_z = 392 \text{ N}$
- 13-91.  $F_r = 102 \text{ N}$ ,  $F_z = 375 \text{ N}$ ,  $F_\theta = 79.7 \text{ N}$
- 13-93.  $F_{z\max} = 547 \text{ N}$ ,  $F_{z\min} = 434 \text{ N}$
- 13-94.  $N = 0.883 \text{ N}$ ,  $F = 3.92 \text{ N}$
- 13-95.  $F_r = -900 \text{ N}$ ,  $F_\theta = -200 \text{ N}$ ,  $F_z = 1.96 \text{ kN}$
- 13-97.  $F = 17.0 \text{ N}$
- 13-98.  $N = 12.0 \text{ N}$
- 13-99.  $(N)_{\max} = 36.0 \text{ N}$ ,  $(N)_{\min} = 4.00 \text{ N}$
- 13-101.  $F_{OA} = 3.20 \text{ N}$
- 13-102.  $\theta = \tan^{-1} \left( \frac{4r_c \dot{\theta}_0^2}{g} \right)$
- 13-103.  $N_C = 25.6 \text{ N}$ ,  $F_{OA} = 0$
- 13-105.  $F = -6.483 \text{ N}$ ,  $F_N = 5.76 \text{ N}$
- 13-106.  $N = 24.8 \text{ N}$ ,  $F = 24.8 \text{ N}$
- 13-107.  $N = 2.95 \text{ kN}$
- 13-109.  $F_r = 1.78 \text{ N}$ ,  $N_s = 5.79 \text{ N}$
- 13-110.  $F_r = 2.93 \text{ N}$ ,  $N_s = 6.37 \text{ N}$
- 13-111.  $r = 0.198 \text{ m}$
- 13-113.  $v_0 = 30.4 \text{ km/s}$ ,  
 $1/r = 0.348(10^{-12}) \cos \theta + 6.74(10^{-12})$
- 13-114.  $h = 35.9 \text{ mm}$ ,  $v_s = 3.07 \text{ km/s}$
- 13-115.  $v_o = 7.45 \text{ km/s}$
- 13-117.  $T^2 = \left( \frac{4\pi^2}{GM_s} \right) a^3$
- 13-118.  $v_B = 7.71 \text{ km/s}$ ,  
 $v_A = 4.63 \text{ km/s}$
- 13-119.  $v_A = 6.67(10^3) \text{ m/s}$ ,  
 $v_B = 2.77(10^3) \text{ m/s}$
- 13-121.  $v_A = 7.47 \text{ km/s}$
- 13-122.  $r_0 = 11.1 \text{ Mm}$ ,  $\Delta v_A = 814 \text{ m/s}$
- 13-123.  $(v_A)_C = 5.27(10^3) \text{ m/s}$ ,  $\Delta v = 684 \text{ m/s}$
- 13-125.  $\Delta v = \sqrt{\frac{GM_e}{r_0}} (\sqrt{2} - \sqrt{1+e})$
- 13-126.  $v_A = 4.89(10^3) \text{ m/s}$ ,  $v_B = 3.26(10^3) \text{ m/s}$
- 13-127.  $v_A = 11.5 \text{ Mm/h}$ ,  $d = 27.3 \text{ Mm}$
- 13-129.  $v_A = 2.01(10^3) \text{ m/s}$

13-130.  $v_{A'} = 521 \text{ m/s}$ ,  $t = 21.8 \text{ h}$

13-131.  $v_A = 7.01(10^3) \text{ m/s}$

## Chapter 14

- 14-1.  $v = 10.7 \text{ m/s}$
- 14-2.  $s = 1.35 \text{ m}$
- 14-3.  $s = 5.99 \text{ m}$
- 14-5.  $x^{\max} = 0.825 \text{ m}$
- 14-6.  $v = 4.08 \text{ m/s}$
- 14-7.  $v = 2.84 \text{ m/s}$
- 14-9.  $d = 12 \text{ m}$
- 14-10.  $x_{\max} = 0.173 \text{ m}$
- 14-11.  $s = 20.5 \text{ m}$
- 14-13.  $k = 15.0 \text{ MN/m}^2$
- 14-14.  $v_A = 1.98 \text{ m/s} \downarrow$ ,  $v_B = 3.96 \text{ m/s} \uparrow$
- 14-15.  $v_A = 3.82 \text{ m/s}$
- 14-17.  $\theta = 41.4^\circ$
- 14-18.  $v_B = 3.34 \text{ m/s}$
- 14-19.  $v = 1.07 \text{ m/s}$
- 14-21.  $k_B = 11.1 \text{ kN/m}$
- 14-22.  $v = 3.58 \text{ m/s}$
- 14-23.  $F_{\text{avg}} = 44.1 \text{ kN}$ ,  $x = 38.2 \text{ mm}$
- 14-25.  $s_{\text{Tot}} = 2.04 \text{ m}$
- 14-26.  $x = 0.688 \text{ m}$
- 14-27.  $R = 2.83 \text{ m}$ ,  $v_C = 7.67 \text{ m/s}$
- 14-29.  $s_1 = 0.628 \text{ m}$
- 14-30.  $s = 179 \text{ mm}$
- 14-31.  $v_B = 18.0 \text{ m/s}$ ,  $N_B = 12.5 \text{ kN}$
- 14-33.  $y = 0.815 \text{ m}$ ,  
 $N_b = 568 \text{ N}$ ,  
 $a = a_t = 6.23 \text{ m/s}^2$
- 14-34.  $v_B = 4.52 \frac{\text{m}}{\text{s}}$
- 14-35.  $F = 43.9 \text{ N}$
- 14-37.  $v_A = 10.5 \text{ m/s}$
- 14-38.  $h_A = 22.5 \text{ m}$ ,  $h_C = 12.5 \text{ m}$
- 14-39.  $v_B = 30.0 \text{ m/s}$ ,  $d = 130.2 \text{ m}$
- 14-41.  $F = 367 \text{ N}$
- 14-42.  $t = 4.44 \text{ s}$
- 14-43.  $t = 51.4 \text{ min}$
- 14-45.  $v = 18.7 \text{ m/s}$
- 14-46.  $P_{\max} = 113 \text{ kW}$ ,  $P_{\text{avg}} = 56.5 \text{ kW}$
- 14-47.  $s = \frac{Mv^3}{3P}$
- 14-49.  $P = 8.31 \text{ t MW}$
- 14-50.  $P_i = 483 \text{ kW}$
- 14-51.  $P_i = 622 \text{ kW}$
- 14-53.  $P_{in} = 1.60 \text{ kW}$
- 14-54.  $P_0 = 35.4 \text{ kW}$
- 14-55.  $P = 12.6 \text{ kW}$
- 14-57.  $P_i = 22.2 \text{ kW}$

- 14-58.**  $P = 1.12 \text{ kW}$   
**14-59.**  $P_{in} = 19.5(10^3) \text{ kW}$ ,  $W = 19.5 \text{ kJ}$   
**14-61.**  $P = \{160 t - 533 t^2\} \text{ kW}$ ,  $U = 1.69 \text{ kJ}$   
**14-62.**  $P_{\max} = 10.7 \text{ kW}$   
**14-63.**  $P = \{400(10^3)t\} \text{ W}$   
**14-65.**  $P = 58.1 \text{ kW}$   
**14-66.**  $s_B = 5.70 \text{ m}$   
**14-67.**  $v_A = 1.54 \text{ m/s}$ ,  $v_B = 4.62 \text{ m/s}$   
**14-69.**  $h = 1154 \text{ mm}$   
**14-70.**  $k_B = 287 \text{ N/m}$   
**14-71.**  $v_B = 5.33 \text{ m/s}$   
 $N = 694 \text{ N}$   
**14-73.**  $N_B = 0$   
 $h = 18.3 \text{ m}$   
 $N_C = 17.2 \text{ kN}$   
**14-74.**  $N_B = 0$ ,  $h = 18.75 \text{ m}$ ,  $N_C = 17.2 \text{ kN}$   
**14-75.**  $d = 8.53 \text{ m}$   
 $v_D = 10 \text{ m/s}$   
**14-77.**  $v_B = 15.5 \text{ m/s}$   
**14-78.**  $h = 23.75 \text{ m}$ ,  $v_C = 21.6 \text{ m/s}$   
**14-79.**  $y = 213 \text{ mm}$   
**14-81.**  $F = \frac{-G M_e m}{r^2}$   
**14-82.**  $F = G M_e m \left( \frac{1}{r_1} - \frac{1}{r_2} \right)$   
**14-83.**  $v = 5.94 \text{ m/s}$   
 $N = 8.53 \text{ N}$   
**14-85.**  $N = 78.6 \text{ N}$   
**14-86.**  $y = 5.10 \text{ m}$ ,  $N = 15.3 \text{ N}$ ,  $a = 9.32 \text{ m/s}^2 \searrow$   
**14-87.**  $\theta = 22.3^\circ$ ,  $s = 0.587 \text{ m}$   
**14-89.**  $v_B = 34.8 \text{ Mm/h}$   
**14-90.**  $s_B = 0.638 \text{ m}$ ,  $s_A = 1.02 \text{ m}$   
**14-91.**  $v_B = 10.6 \text{ m/s}$ ,  $T_B = 142 \text{ N}$   
 $v_C = 9.47 \text{ m/s}$ ,  $T_C = 48.7 \text{ N}$   
**14-93.**  $k = 773 \text{ N/m}$   
**14-94.**  $v = 6.97 \text{ m/s}$   
**14-95.**  $v_2 = \sqrt{\frac{2}{\pi}(\pi - 2)gr}$   
**14-97.**  $d = 1.34 \text{ m}$

## Chapter 15

- 15-1.**  $v = 1.75 \text{ N} \cdot \text{s}$   
**15-2.**  $t = 0.432 \text{ s}$   
**15-3.**  $I = 5.68 \text{ N} \cdot \text{s}$   
**15-5.**  $v_2 = 21.8 \text{ m/s}$   
**15-6.**  $v_2 = 16.6 \text{ m/s}$   
**15-7.**  $P = 205 \text{ N}$   
**15-9.**  $v = 14.29 \text{ m/s}$   
 $F_D = 15.7 \text{ kN}$   
**15-10.**  $v = 6.62 \text{ m/s}$

- 15-11.**  $\mu_k = 0.340$   
**15-13.**  $v_2 = 15.6 \text{ m/s}$   
**15-14.**  $v = 4.05 \text{ m/s}$   
**15-15.**  $I = 15 \text{ kN} \cdot \text{s}$  in both cases  
**15-17.**  $v_{\max} = 108 \text{ m/s}$ ,  $s = 1.83 \text{ km}$   
**15-18.**  $t_1 = 1.242 \text{ s}$ ,  $t_2 = 1.929 \text{ s}$   
**15-19.**  $v = 8.81 \text{ m/s}$ ,  $s = 24.8 \text{ m}$   
**15-21.**  $T = 14.9 \text{ kN}$ ,  $F = 24.8 \text{ kN}$   
**15-22.**  $v_A = 8.41 \text{ m/s} \downarrow$   
**15-23.**  $v|_{t=3 \text{ s}} = 5.68 \text{ m/s} \downarrow$ ,  
 $v|_{t=6 \text{ s}} = 21.1 \text{ m/s} \uparrow$   
**15-25.**  $v = 7.21 \text{ m/s} \uparrow$   
**15-26.** Observer A:  $v = 7.40 \text{ m/s}$ ,  
Observer B:  $v = 5.40 \text{ m/s}$   
**15-27.**  $v = 5.07 \text{ m/s}$   
**15-29.**  $t = 1.02 \text{ s}$ ,  $I = 162 \text{ N} \cdot \text{s}$   
**15-30.**  $v = 16.1 \text{ m/s}$   
**15-31.**  $I = 77.3 \text{ N} \cdot \text{s}$   
**15-33.**  $v = 7.65 \text{ m/s}$   
**15-34.**  $F = 12.7 \text{ kN}$   
**15-35.**  $v = 18.6 \text{ m/s} \rightarrow$   
**15-37.**  $v = 5.21 \text{ m/s} \leftarrow$ ,  
 $\Delta T = -32.6 \text{ kJ}$   
**15-38.**  $v = 0.5 \text{ m/s}$ ,  $\Delta T = -16.9 \text{ kJ}$   
**15-39.**  $v = 733 \text{ m/s}$   
**15-41.**  $v_2 = \sqrt{v_1^2 + 2gh}$ ,  $\theta_2 = \sin^{-1} \left( \frac{v_1 \sin \theta}{\sqrt{v_1^2 + 2gh}} \right)$   
**15-42.**  $v_t = 8.62 \text{ m/s}$   
**15-43.**  $s = 4.00 \text{ m}$   
**15-45.**  $v_A = 3.09(10^3) \text{ m/s}$   
 $v_B = 2.62(10^3) \text{ m/s}$   
 $d_B = 104 \text{ m}$   
**15-46.**  $s_B = 6.67 \text{ m} \rightarrow$   
**15-47.**  $s_P = 0$   
 $t = 0.408 \text{ s}$   
**15-49.**  $d = 6.87 \text{ mm}$   
**15-50.**  $v_c = 5.04 \text{ m/s} \leftarrow$   
**15-51.**  $s_{\max} = 481 \text{ mm}$   
**15-53.**  $v_A = 3.29 \text{ m/s}$   
 $v_B = 2.19 \text{ m/s}$   
**15-54.** **a.**  $v_C = 0.390 \text{ m/s} \leftarrow$   
**b.**  $v_C = 0.390 \text{ m/s} \leftarrow$   
**15-55.**  $s_B = 71.4 \text{ mm} \rightarrow$   
**15-57.**  $v_{C/R} = 6.356 \text{ m/s}$   
 $\phi = 35.8^\circ \swarrow$   
**15-58.**  $(v_A)_2 = 0.353 \text{ m/s}$   
 $(v_B)_2 = 2.35 \text{ m/s}$   
**15-59.**  $e = 0.75$ ,  $\Delta T = -9.65 \text{ kJ}$   
**15-61.**  $v_B = 8.24 \text{ m/s} \rightarrow$   
 $v_A = 0.160 \text{ m/s} \leftarrow$   
**15-62.**  $x_{\max} = 0.839 \text{ m}$

- 15-63.  $v_C = 0, v_D = v$   
 $v_B = v, v_A = 0$   
 $v_C = v, v_B = 0$
- 15-65.  $(v_B)_2 = \frac{e(1+e)}{2}v_0$
- 15-66.  $(v_B)_1 = 8.81 \text{ m/s}$   
 $\theta_1 = 10.5^\circ \nearrow$   
 $(v_B)_2 = 4.62 \text{ m/s}$   
 $\theta_2 = 20.3^\circ \searrow$   
 $s = 3.96 \text{ m}$
- 15-67.  $v'_B = 22.2 \text{ m/s}, \theta = 13.0^\circ$
- 15-69.  $(v_C)_2 = \frac{v(1+e)^2}{4}$
- 15-70.  $(v_B)_2 = \frac{1}{3}\sqrt{2gh}(1+e)$
- 15-71.  $(v_B)_3 = 3.24 \text{ m/s}, \theta = 43.9^\circ$
- 15-73.  $h = 1.57 \text{ m}$
- 15-74.  $(v_P)_2 = 0.940 \text{ m/s}$
- 15-75.  $d = 34.8 \text{ mm}$
- 15-77.  $v_A = 1.35 \text{ m/s} \rightarrow, v_B = 5.89 \text{ m/s}, \theta = 32.9^\circ \searrow$
- 15-78.  $e = 0.0113$
- 15-79.  $v_c = \frac{v_0(1+e)m}{(m+M)}$   
 $v_b = v_0\left(\frac{m - eM}{m + M}\right)$   
 $t = \frac{d}{v_0}\left(1 + \frac{1}{e}\right)^2$
- 15-81.  $F = 2.62 \text{ kN}$
- 15-82.  $\phi = \arccos\left[1 - \left(\frac{1+e}{2}\right)^4(1 - \cos(\theta))\right]$
- 15-83. (a)  $(v_B)_1 = 8.81 \text{ m/s}, \theta = 10.5^\circ \nearrow$ ,  
 (b)  $(v_B)_2 = 4.62 \text{ m/s}, \phi = 20.3^\circ \searrow$ ,  
 (c)  $s = 3.96 \text{ m}$
- 15-85.  $(v_A)_2 = 8.19 \text{ m/s}, (v_B)_2 = 9.38 \text{ m/s}$
- 15-86.  $(v_B)_2 = 1.06 \text{ m/s} \leftarrow, (v_A)_2 = 0.968 \text{ m/s},$   
 $(\theta_A)_2 = 5.11^\circ \nearrow$
- 15-87.  $e = \frac{\sin \phi}{\sin \theta} \left( \frac{\cos \theta - \mu \sin \theta}{\mu \sin \phi + \cos \phi} \right)$
- 15-89.  $e = \frac{2}{3}$
- 15-90.  $v_{A2} = -4.533 \text{ m/s}, v_{B2} = -1.733 \text{ m/s}$
- 15-91.  $(v_B)_2 = \frac{\sqrt{3}}{4}(1+e)v,$   
 Negative  $x'$  axis.  
 $\Delta U_k = \frac{3mv^2}{16}(1 - e^2)$
- 15-93.  $(v_B)_3 = 1.50 \text{ m/s}$
- 15-94.  $\mathbf{H}_{AO} = 22.3 \text{ kg} \cdot \text{m}^2/\text{s}$   
 $\mathbf{H}_{BO} = -7.18 \text{ kg} \cdot \text{m}^2/\text{s}$   
 $\mathbf{H}_{CO} = -21.60 \text{ kg} \cdot \text{m}^2/\text{s}$
- 15-95.  $\mathbf{H}_{AP} = -57.6 \text{ kg} \cdot \text{m}^2/\text{s}$   
 $\mathbf{H}_{BP} = 94.4 \text{ kg} \cdot \text{m}^2/\text{s}$   
 $\mathbf{H}_{CP} = -41.2 \text{ kg} \cdot \text{m}^2/\text{s}$
- 15-97.  $(H_A)_P = \{-52.8\mathbf{k}\} \text{ kg} \cdot \text{m}^2/\text{s},$   
 $(H_B)_P = \{-118\mathbf{k}\} \text{ kg} \cdot \text{m}^2/\text{s}$
- 15-98.  $\{-21.5\mathbf{i} + 21.5\mathbf{j} + 37.6\} \text{ kg} \cdot \text{m}^2/\text{s}$
- 15-99.  $\{21.5\mathbf{i} + 21.5\mathbf{j} + 59.1\mathbf{k}\} \text{ kg} \cdot \text{m}^2/\text{s}$
- 15-101.  $v = 6.15 \text{ m/s}$
- 15-102.  $t = 1.34 \text{ s}$
- 15-103.  $v = 3.33 \text{ m/s}$
- 15-105.  $v_2 = 1.99 \text{ m/s}$   
 $U_F = 8.32 \text{ N} \cdot \text{m}$
- 15-106.  $\ddot{\theta} + \left(\frac{g}{R}\right)\sin \theta = 0$
- 15-107.  $v_2 = 6.738 \text{ m/s}, (v_2)_r = 2.52 \text{ m/s}$
- 15-109.  $v_1 = 3.47 \text{ m/s}$
- 15-110.  $v_2 = 4.03 \text{ m/s},$   
 $\Sigma U_{1-2} = 725 \text{ J}$
- 15-111.  $h = 196 \text{ mm}$   
 $\theta = 75.0^\circ$
- 15-113.  $v_B = 10.2 \text{ km/s}, r_B = 13.8 \text{ Mm}$
- 15-114.  $T = 40.1 \text{ kN}$
- 15-115.  $C_x = 4.97 \text{ kN}, D_x = 2.23 \text{ kN}, D_y = 7.20 \text{ kN}$
- 15-117.  $Q = 0.217 (10^{-3}) \text{ m}^3/\text{s}$
- 15-118.  $M_D = 14.45 \text{ kN} \cdot \text{m}$   
 $D_y = 25.88 \text{ kN}$   
 $D_x = 11.38 \text{ kN}$
- 15-119.  $F_x = 11.0 \text{ N}$
- 15-121.  $F = 6.24 \text{ N}, P = 3.12 \text{ N}$
- 15-122.  $F_x = 795 \text{ N}$   
 $F_y = 1.20 \text{ kN}$
- 15-123.  $T = 278 \text{ N}$
- 15-125.  $A_x = 3.98 \text{ kN}$   
 $A_y = 3.81 \text{ kN}$   
 $M = 1.99 \text{ kN} \cdot \text{m}$
- 15-126.  $T = 82.8 \text{ N}, N = 396 \text{ N}$
- 15-127.  $C_x = 95.8 \text{ N}$   
 $D_y = 38.9 \text{ N}$   
 $C_y = 118 \text{ N}$
- 15-129.  $v_{\max} = 625 \text{ m/s}$
- 15-130.  $F_B = 402 \text{ N}$
- 15-131.  $C_x = 4.26 \text{ kN}, C_y = 2.12 \text{ kN},$   
 $M_C = 5.16 \text{ kN} \cdot \text{m}$
- 15-133.  $v = \left\{ \frac{8000}{2000 + 50t} \right\} \text{ m/s}$
- 15-134.  $a = 0.125 \text{ m/s}^2, v = 4.05 \text{ m/s}$
- 15-135.  $F = 3.55 \text{ kN}$
- 15-137.  $v = 25.0 \text{ m/s}$
- 15-138.  $\frac{d}{dt}m = m_0 \left( \frac{a_0 + g}{v_{er}} \right) e^{\left( \frac{a_0 + g}{v_{er}} \right)t}$
- 15-139.  $F_D = 11.5 \text{ kN}$

$$15-141. v = \sqrt{\frac{2}{3}g\left(\frac{y^3 - h^3}{y^2}\right)}$$

$$15-142. m' = \frac{M}{s}$$

$$15-143. v_p = 594 \text{ km/h}$$

$$15-145. m' = \frac{3M}{s}$$

$$15-146. F = \{7.85t + 0.320\} \text{ N}$$

$$15-147. a = 0.0476 \text{ m/s}^2$$

$$15-149. F = m'v^2$$

## Chapter 16

$$16-1. v_A = 2.60 \text{ m/s}, a_A = 9.35 \text{ m/s}^2$$

$$16-2. v_A = 22.0 \text{ m/s}, (a_A)_t = 12.0 \text{ m/s}^2, (a_A)_n = 968 \text{ m/s}^2$$

$$16-3. v_A = 26.0 \text{ m/s}, (a_A)_t = 10.0 \text{ m/s}^2, (a_A)_n = 1352 \text{ m/s}^2$$

$$16-5. \theta = 5443 \text{ rev}, \omega = 740 \text{ rad/s}, \alpha = 8 \text{ rad/s}^2$$

$$16-6. \theta = 3.32 \text{ rev}, t = 1.67 \text{ s}$$

$$16-7. t = 6.98 \text{ s}, \theta_D = 34.9 \text{ rev}$$

$$16-9. a_B = 29.0 \text{ m/s}^2$$

$$16-10. a_B = 16.5 \text{ m/s}^2$$

$$16-11. \alpha = 60 \text{ rad/s}^2, \omega = 90.0 \text{ rad/s}, \theta = 90.0 \text{ rad}$$

$$16-13. \omega_B = 180 \text{ rad/s}, \omega_C = 360 \text{ rad/s}$$

$$16-14. \omega = 42.7 \text{ rad/s}, \theta = 42.7 \text{ rad}$$

$$16-15. a_t = 2.83 \text{ m/s}^2, a_n = 35.6 \text{ m/s}^2$$

$$16-17. \omega_B = 21.9 \text{ rad/s} \curvearrowright$$

$$16-18. \omega_B = 31.7 \text{ rad/s} \curvearrowright$$

$$16-19. \omega_E = 0.150 \text{ rad/s}$$

$$16-21. v_A = 8.10 \text{ m/s}, (a_A)_t = 4.95 \text{ m/s}^2, (a_A)_n = 437 \text{ m/s}^2$$

$$16-22. \omega_D = 4.00 \text{ rad/s}, \alpha_D = 0.400 \text{ rad/s}^2$$

$$16-23. \omega_D = 12.0 \text{ rad/s}, \alpha_D = 0.600 \text{ rad/s}^2$$

$$16-25. \omega_A = 225 \text{ rad/s}$$

$$16-26. \omega_C = 1.68 \text{ rad/s}, \theta_C = 1.68 \text{ rad}$$

$$16-27. r_A = 31.8 \text{ mm}, r_B = 31.8 \text{ mm}, n = 1.91 \text{ canisters per minute}$$

$$16-29. v_E = 3 \text{ m/s}, (a_E)_t = 2.70 \text{ m/s}^2, (a_E)_n = 600 \text{ m/s}^2$$

$$16-30. s_W = 2.89 \text{ m}$$

$$16-31. \omega_B = 312 \text{ rad/s}, \alpha_B = 176 \text{ rad/s}^2$$

$$16-33. a = \frac{\omega^2}{2\pi} \left( \frac{r_2 - r_1}{L} \right) d$$

$$16-34. a = \frac{s}{2\pi} \omega^2$$

$$16-35. v_C = \{-4.8\mathbf{i} - 3.6\mathbf{j} - 1.2\mathbf{k}\} \text{ m/s}, a_C = \{38.4\mathbf{i} - 64.8\mathbf{j} + 40.8\mathbf{k}\} \text{ m/s}^2$$

$$16-37. v_C = 2.50 \text{ m/s}, a_C = 13.1 \text{ m/s}^2$$

$$16-38. v_A = 40 \text{ mm/s}, v_W = 34.6 \text{ mm/s}$$

$$16-39. \omega = \frac{rv_A}{y\sqrt{y^2 - r^2}}, \alpha = \frac{rv_A^2(2y^2 - r^2)}{y^2(y^2 - r^2)^{3/2}}$$

$$16-41. \omega = 8.70 \text{ rad/s}, \alpha = -50.5 \text{ rad/s}^2$$

$$16-42. \omega = -19.2 \text{ rad/s}, \alpha = -183 \text{ rad/s}^2$$

$$16-43. \omega_{AB} = 0$$

$$16-45. v = -\left( \frac{r_1^2 \omega \sin \theta}{2\sqrt{r_1^2 \cos^2 \theta + r_2^2 + 2r_1 r_2}} + r_1 \omega \sin \theta \right)$$

$$16-46. v = \omega d \left( \sin \theta + \frac{d \sin 2\theta}{2\sqrt{(R+r)^2 - d^2 \sin^2 \theta}} \right)$$

$$16-47. v = -r\omega \sin \theta$$

$$16-49. v_C = L\omega \uparrow, a_C = 0.577 L\omega^2 \uparrow$$

$$16-50. \omega = \frac{2v_0}{r} \sin^2 \theta / 2, \alpha = \frac{2v_0^2}{r^2} (\sin \theta)(\sin^2 \theta / 2)$$

$$16-51. v_B = \left( \frac{h}{d} \right) v_A$$

$$16-53. \dot{\theta} = \frac{v \sin \phi}{L \cos(\phi - \theta)}$$

$$16-54. \omega = \frac{v}{2r}$$

$$16-55. \omega' = \frac{(R+r)\omega}{r}, \alpha' = \frac{(R+r)\alpha}{r}$$

$$16-57. v_A = 2.4 \text{ m/s}$$

$$16-58. \omega_{BC} = 2.83 \text{ rad/s} \curvearrowright, \omega_{AB} = 2.83 \text{ rad/s} \curvearrowright$$

$$16-59. v_G = -9.00 \text{ m/s}$$

$$16-61. v_C = 1.06 \text{ m/s} \leftarrow, \omega_{BC} = 0.707 \text{ rad/s} \curvearrowright$$

$$16-62. v_A = 2.45 \text{ m/s} \uparrow$$

$$16-63. v_C = 2.45 \text{ m/s}, \omega_{BC} = 7.81 \text{ rad/s}$$

$$16-65. \omega_{BC} = 10.6 \text{ rad/s} \curvearrowright, v_C = 29.0 \text{ m/s} \rightarrow$$

$$16-66. v_C = -1.64 \text{ m/s}$$

$$16-67. v_C = -1.70 \text{ m/s}$$

$$16-69. \omega_{BC} = 2.69 \text{ rad/s}, \omega_{AB} = 4.39 \text{ rad/s}$$

$$16-70. \mathbf{v}_O = \left( \frac{R}{R-r} \right) \mathbf{v} \rightarrow$$

$$16-71. v_A = \left( \frac{2R}{R-r} \right) \mathbf{v} \rightarrow$$

$$16-73. \omega_B = 90 \text{ rad/s} \curvearrowright, \omega_A = 180 \text{ rad/s} \curvearrowright$$

$$16-74. v_B = 3.00 \text{ m/s},$$

$$v_C = 0.587 \text{ m/s},$$

$$v_B = 3.00 \text{ m/s},$$

$$v_C = 0.587 \text{ m/s}$$

$$16-75. v_E = 4.00 \text{ m/s}, \theta = 52.7^\circ \curvearrowright$$



- 16-77.  $v_E = 312 \text{ mm/s}$ ,  $\phi = 8.13^\circ$   
 $v_E = 312 \text{ mm/s}$ ,  $\phi = 8.13^\circ$
- 16-78.  $v_P = 4.88 \text{ m/s} \leftarrow$
- 16-79.  $\omega_D = 105 \text{ rad/s} \curvearrowright$
- 16-82.  $v_O = 1.04 \text{ m/s} \rightarrow$
- 16-83.  $v_C = 8.69 \text{ m/s}$
- 16-85.  $\omega_{BC} = 6.79 \text{ rad/s}$
- 16-86.  $v_A = 0$ ,  $v_B = 1.2 \text{ m/s}$ ,  
 $v_C = 0.849 \text{ m/s} \searrow 45^\circ$
- 16-87.  $\omega_{BPD} = 3.00 \text{ rad/s} \curvearrowright$ ,  $v_D = 1.20 \text{ m/s} \swarrow$ ,  
 $\omega_{CD} = 6.00 \text{ rad/s} \curvearrowright$
- 16-89.  $\omega_{BC} = 8.66 \text{ rad/s} \curvearrowright$ ,  $\omega_{AB} = 4.00 \text{ rad/s} \curvearrowright$
- 16-90.  $v_C = 1.34 \text{ m/s} \leftarrow$
- 16-91.  $v_C = 0.897 \text{ m/s}$
- 16-93.  $v_A = \omega(r_2 - r_1)$
- 16-94.  $v_C = 2.40 \text{ m/s} \leftarrow$
- 16-95.  $\omega_B = 6.67 \text{ rad/s}$
- 16-97.  $\omega_S = 57.5 \text{ rad/s} \curvearrowright$ ,  $\omega_{OA} = 10.6 \text{ rad/s} \curvearrowright$
- 16-98.  $\omega_S = 15.0 \text{ rad/s}$ ,  $\omega_R = 3.00 \text{ rad/s}$
- 16-99.  $\omega_{CD} = 57.7 \text{ rad/s} \curvearrowright$
- 16-101.  $\omega_R = 4 \text{ rad/s}$
- 16-102.  $\omega_R = 4 \text{ rad/s}$
- 16-103.  $\alpha_B = 1.43 \text{ rad/s}^2$
- 16-105.  $a_C = 13.0 \text{ m/s}^2 \swarrow$ ,  $\alpha_{BC} = 12.4 \text{ rad/s}^2 \curvearrowright$
- 16-106.  $\omega = 6.67 \text{ rad/s} \curvearrowright$ ,  $v_B = 4.00 \text{ m/s} \searrow$   
 $\alpha = 15.7 \text{ rad/s}^2 \curvearrowright$ ,  $a_B = 24.8 \text{ m/s}^2 \nwarrow$
- 16-107.  $v_C = 3.86 \text{ m/s} \leftarrow$ ,  $a_C = 17.7 \text{ m/s}^2 \leftarrow$
- 16-109.  $v_B = 4v \rightarrow$ ,  
 $v_A = 2\sqrt{2}v$ ,  $\theta = 45^\circ \nearrow$ ,  
 $a_B = \frac{2v^2}{r} \downarrow$ ,  $a_A = \frac{2v^2}{r} \rightarrow$
- 16-110.  $v_A = 0.424 \text{ m/s}$ ,  $\theta_v = 45^\circ \searrow$ ,  
 $a_A = 0.806 \text{ m/s}^2$ ,  $\theta_a = 7.13^\circ \nearrow$
- 16-111.  $v_B = 0.6 \text{ m/s} \downarrow$ ,  
 $a_B = 1.84 \text{ m/s}^2$ ,  $\theta = 60.6^\circ \searrow$
- 16-113.  $a_C = 10.0 \text{ m/s}^2$ ,  $\theta = 2.02^\circ \swarrow$
- 16-114.  $\omega_{BC} = 0$ ,  $v_C = v_B$ ;  
 $\omega_{CD}(0.1) = 0.4$ ,  $\omega_{CD} = 4.00 \text{ rad/s} \curvearrowright$ ,  
 $\alpha_{BC} = 6.16 \text{ rad/s}^2 \curvearrowright$ ,  $\alpha_{CD} = 21.9 \text{ rad/s}^2 \curvearrowright$
- 16-115.  $\alpha_{CD} = 474 \text{ rad/s}^2 \curvearrowright$ ,
- 16-117.  $\omega_C = 20.0 \text{ rad/s} \curvearrowright$ ,  $\alpha_C = 127 \text{ rad/s} \curvearrowright$
- 16-118.  $a_A = 1.34 \text{ m/s}^2$ ,  $\theta = 26.6^\circ \searrow$
- 16-119.  $\omega_{CD} = 7.79 \text{ rad/s} \curvearrowright$ ,  $\alpha_{CD} = 136 \text{ rad/s}^2 \curvearrowright$
- 16-121.  $v_B = 1.58\omega a$ ,  $a_B = 1.58\alpha a - 1.77\omega^2 a$
- 16-122.  $\alpha = 40.0 \text{ rad/s}^2$ ,  $a_A = 2.00 \text{ m/s}^2 \leftarrow$
- 16-123.  $\omega_C = 4.40 \text{ rad/s} \curvearrowright$ ,  $\alpha_C = 8374 \text{ rad/s}^2 \curvearrowright$
- 16-125.  $\alpha = 60 \text{ rad/s}^2$   
 $a_B = 167 \text{ m/s}^2$
- 16-126.  $\mathbf{a}_A = \{-3.50\mathbf{i} - 4.80\mathbf{j}\} \text{ m/s}^2$ ,  $|\mathbf{a}_A| = 5.94 \text{ m/s}^2$
- 16-127.  $\alpha_{AB} = -36 \frac{\text{rad}}{\text{s}^2}$

- 16-129.  $\mathbf{v}_B = \{0.6\mathbf{i} + 2.4\mathbf{j}\} \text{ m/s}$ ,  
 $\mathbf{a}_B = \{-14.2\mathbf{i} + 8.40\mathbf{j}\} \text{ m/s}^2$
- 16-130.  $\mathbf{a}_A = \{-5.60\mathbf{i} - 16\mathbf{j}\} \text{ m/s}^2$
- 16-131.  $\mathbf{v}_C = [0.6\mathbf{i}] \text{ m/s}$ ,  $\mathbf{a}_C = [-1.2\mathbf{j}] \text{ m/s}^2$
- 16-133.  $v_C = 2.40 \text{ m/s}$ ,  $\theta = 60^\circ \searrow$
- 16-134.  $v_C = 2.40 \text{ m/s}$ ,  $a_C = \{-14.4\mathbf{j}\} \text{ m/s}^2$
- 16-135.  $\mathbf{a}_B = \{-1\mathbf{i} + 0.2\mathbf{j}\} \text{ m/s}^2$ ,  
 $\mathbf{a}_B = \{-1.69\mathbf{i} + 0.6\mathbf{j}\} \text{ m/s}^2$
- 16-137.  $\mathbf{v}_A = \{-17.2\mathbf{i} + 12.5\mathbf{j}\} \text{ m/s}$ ,  
 $\mathbf{a}_A = \{349\mathbf{i} + 597\mathbf{j}\} \text{ m/s}^2$
- 16-138.  $\omega_{AB} = 5 \text{ rad/s} \curvearrowright$ ,  $\alpha_{AB} = 2.5 \text{ rad/s}^2 \curvearrowright$
- 16-139.  $\omega_{CD} = 3.00 \text{ rad/s} \curvearrowright$ ,  $\alpha_{CD} = 12.0 \text{ rad/s}^2 \curvearrowright$
- 16-141.  $\omega_{CD} = 6.93 \text{ rad/s}$
- 16-142.  $v_B = 7.7 \text{ m/s}$ ,  $a_B = 201 \text{ m/s}^2$
- 16-143.  $\omega_{CB} = 1.33 \text{ rad/s} \curvearrowright$ ,  $\alpha_{CD} = 3.08 \text{ rad/s}^2 \curvearrowright$
- 16-145.  $\mathbf{v}_C = \{-3.5\mathbf{i} + 8.66\mathbf{j}\} \text{ m/s}^2$ ,  
 $\mathbf{a}_C = \{-19.42\mathbf{i} - 3.42\mathbf{j}\} \text{ m/s}^2$
- 16-146.  $\omega_{AB} = 0.667 \text{ rad/s} \curvearrowright$ ,  $\alpha_{AB} = 3.08 \text{ rad/s}^2 \curvearrowright$
- 16-147.  $(\mathbf{v}_{\text{rel}})_{xyz} = [27\mathbf{i} + 25\mathbf{j}] \text{ m/s}$ ,  
 $(\mathbf{a}_{\text{rel}})_{xyz} = [2.4\mathbf{i} + 0.38\mathbf{j}] \text{ m/s}$
- 16-149.  $\mathbf{v}_B = \{7.13\mathbf{j}\} \text{ m/s}$ ,  $|\mathbf{v}_B| = 7.13 \text{ m/s}$ ,  
 $\mathbf{a}_B = \{-15.94\mathbf{i}\} \text{ m/s}^2$ ,  $|\mathbf{a}_B| = 15.94 \text{ m/s}^2$
- 16-150.  $\omega_{DC} = 2.96 \text{ rad/s} \curvearrowright$
- 16-151.  $\omega_{AC} = 0$ ,  $\alpha_{AC} = 14.4 \text{ rad/s}^2 \curvearrowright$

## Chapter 17

- 17-1.  $I_y = \frac{1}{3} ml^2$
- 17-2.  $k_x = 57.7 \text{ mm}$
- 17-3.  $I_z = \frac{\pi h R^4}{2} \left[ k + \frac{2aR^2}{3} \right]$   
 $m = \pi h R^2 \left( k + \frac{aR^2}{2} \right)$
- 17-5.  $I_y = \frac{2}{5} m r^2$
- 17-6.  $I_x = \frac{93}{70} mb^2$
- 17-7.  $I_x = \frac{2}{5} m r^2$
- 17-9.  $I_z = m(R^2 + \frac{3}{4} a^2)$
- 17-10.  $I_G = 0.230 \text{ kg} \cdot \text{m}^2$
- 17-11.  $I_O = 0.560 \text{ kg} \cdot \text{m}^2$
- 17-13.  $I_y = \frac{m}{6} (a^2 + h^2)$
- 17-14.  $I_A = 7.67 \text{ kg} \cdot \text{m}^2$
- 17-15.  $I_O = 1.36 \text{ kg} \cdot \text{m}^2$
- 17-17.  $k_O = 2.17 \text{ m}$

- 17-18.  $I_O = 6.23 \text{ kg} \cdot \text{m}^2$   
 17-19.  $\bar{y} = 0.888 \text{ m}$ ,  $I_G = 5.61 \text{ kg} \cdot \text{m}^2$   
 17-21.  $\bar{y} = 1.78 \text{ m}$ ,  $I_G = 4.45 \text{ kg} \cdot \text{m}^2$   
 17-22.  $I_x = 3.25 \text{ g} \cdot \text{m}^2$   
 17-23.  $I_{x'} = 7.19 \text{ g} \cdot \text{m}^2$   
 17-25.  $T_{AB} = T_{CD} = T = 23.6 \text{ kN}$ ,  
 $T_{EF} = T_{GH} = T' = 27.6 \text{ kN}$   
 17-26.  $a_{\max} = 4.73 \text{ m/s}^2$   
 17-27.  $F_{CD} = 289 \text{ kN}$   
 17-29.  $t = 3.94 \text{ s}$   
 17-30.  $P = 314 \text{ N}$   
 17-31.  $P = 785 \text{ N}$   
 17-33.  $N = 29.6 \text{ kN}$ ,  $V = 0$ ,  $M = 51.2 \text{ kN} \cdot \text{m}$   
 17-34.  $a = 5.61 \text{ m/s}^2$   
 17-35.  $N_B = 237 \text{ N}$   
 $N_A = 744 \text{ N}$   
 17-37.  $a = 4.33 \text{ m/s}^2 \leftarrow$ ,  $N_A = 113 \text{ N}$ ,  $N_B = 325 \text{ N}$   
 17-38.  $P = 579 \text{ N}$   
 17-39.  $F_{AB} = 112 \text{ N}$ ,  $C_x = 26.2 \text{ N}$ ,  $C_y = 49.8 \text{ N}$   
 17-41.  $\theta = \tan^{-1} \left( \frac{a}{g} \right)$   
 17-42. Since the required friction  $F_f > (F_f)_{\max} = \mu_k N_B = 0.6(14715) = 8829 \text{ N}$ , **it is not possible to lift the front wheels off the ground.**  
 17-43.  $N_A = 1.78 \text{ kN}$ ,  
 $N_B = 5.58 \text{ kN}$ ,  
 $F_B = 4.50 \text{ kN}$   
 17-45.  $N_A = 568 \text{ N}$ ,  
 $N_B = 544 \text{ N}$ ,  
 17-46.  $a = 3.96 \text{ m/s}^2$   
 17-47.  $a = 2.01 \text{ m/s}^2$ .  
 The crate slips.  
 17-49.  $a = 9.81 \text{ m/s}^2$ ,  $C_x = 12.3 \text{ kN}$ ,  $C_y = 12.3 \text{ kN}$   
 17-50.  $N = 0.433wx$ ,  $V = 0.25wx$ ,  $M = 0.125wx^2$   
 17-51.  $T = 1.52 \text{ kN}$ ,  $\theta = 18.6^\circ$   
 17-53.  $\alpha = 2.62 \text{ rad/s}^2$   
 17-54.  $V_E = 43.7 \text{ N}$ ,  $N_E = 27.5 \text{ N}$ ,  $M_E = 32.8 \text{ N} \cdot \text{m}$   
 17-55.  $\alpha = 5.95 \text{ rad/s}^2$   
 17-57.  $\omega = 56.2 \text{ rad/s}$ ,  $A_x = 0$ ,  $A_y = 98.1 \text{ N}$   
 17-58.  $\alpha = 14.7 \text{ rad/s}^2$ ,  $A_x = 88.3 \text{ N}$ ,  $A_y = 147 \text{ N}$   
 17-59.  $F_A = \frac{3}{2} mg$   
 17-61.  $\alpha = 0.694 \text{ rad/s}^2$   
 17-62.  $\alpha = 7.28 \text{ rad/s}^2$   
 17-63.  $F = 22.1 \text{ N}$   
 17-65.  $\omega_A = 38.3 \text{ rad/s}$ ,  $\omega_B = 57.5 \text{ rad/s}$   
 17-66.  $\omega = 0.474 \text{ rad/s}$   
 17-67.  $\omega = 2.71 \text{ rad/s}$   
 17-69.  $\alpha = 8.68 \text{ rad/s}^2$ ,  $A_n = 0$ ,  $A_t = 106 \text{ N}$   
 17-70.  $I_A = 175 \text{ kg} \cdot \text{m}^2$ ,  $\alpha = 1.25 \text{ rad/s}^2$   
 17-71.  $\omega = 65.6 \text{ rad/s}$   
 17-73.  $a = 2.97 \text{ m/s}^2$   
 17-74.  $a = \frac{g(m_B - m_A)}{\left( \frac{1}{2} M + m_B + m_A \right)}$   
 17-75.  $A_x = 294 \text{ N}$ ,  $A_y = 294 \text{ N}$ ,  $t = 1.91 \text{ s}$   
 17-77.  $F_{CB} = 193 \text{ N}$ ,  $t = 3.11 \text{ s}$   
 17-78.  $\alpha = 14.2 \text{ rad/s}^2$   
 17-79.  $t = 6.71 \text{ s}$   
 17-81.  $v = 0.548 \text{ m/s}$   
 17-82.  $\alpha = 12.5 \text{ rad/s}^2 \curvearrowright$ ,  $a_G = 18.75 \text{ m/s}^2 \downarrow$   
 17-83.  $N = wx \left[ \frac{\omega^2}{g} \left( L - \frac{x}{2} \right) + \cos \theta \right]$ ,  
 $V = wx \sin \theta$ ,  $M = \frac{1}{2} wx^2 \sin \theta$   
 17-85.  $N_A = 177 \text{ kN}$ ,  $V_A = 5.86 \text{ kN}$ ,  $M_A = 50.7 \text{ kN} \cdot \text{m}$   
 17-86.  $M = 0.3gmI$   
 17-87.  $N_B = 2.89 \text{ kN}$ ,  
 $A_x = 0$ ,  $A_y = 2.89 \text{ kN}$   
 17-89.  $\omega = 800 \text{ rad/s}$   
 17-90.  $\Sigma M_{IC} = I_{IC} \alpha$   
 17-91.  $\alpha = 2.45 \text{ rad/s}^2 \curvearrowright$ ,  $N_B = 2.23 \text{ N}$ ,  $N_A = 33.3 \text{ N}$   
 17-93.  $\alpha = 1.25 \text{ rad/s}$ ,  $T = 2.32 \text{ N}$   
 17-94.  $T = 3.13 \text{ kN}$ ,  $\alpha = 1.684 \text{ rad/s}$ ,  $a_C = 1.35 \text{ m/s}^2$   
 17-95.  $\alpha = 5.62 \text{ rad/s}^2$ ,  $T = 196 \text{ N}$   
 17-97. Since  $F_C = 13.08 \text{ N} > F_{\max} = 10.19 \text{ N}$  then our no-slip assumption is wrong and we know that slipping does occur.  
 $\alpha = 5.66 \text{ rad/s}^2$   
 17-98.  $\alpha = 0.560 \text{ rad/s}^2 \curvearrowright$ ,  $a_G = 0.224 \text{ m/s}^2 \rightarrow$   
 17-99.  $F = 42.3 \text{ N}$   
 17-101.  $\alpha = 1.30 \text{ rad/s}^2$   
 17-102.  $\alpha = 0.500 \text{ rad/s}^2$   
 17-103.  $\alpha = 15.6 \text{ rad/s}^2$   
 17-105.  $\alpha = \frac{6(P - \mu_k mg)}{mL}$ ,  $a_B = \frac{2(P - \mu_k mg)}{m}$   
 17-106.  $a_{Gx} = -1.82 \text{ m/s}^2$ ,  $a_{Gy} = -1.69 \text{ m/s}^2$ ,  
 $\alpha = -0.283 \text{ rad/s}^2$   
 17-107.  $\alpha = 3 \text{ rad/s}^2$   
 17-109. Since  $F_f < (F_f)_{\max} = \mu_s N = 0.5(91.32) = 45.66 \text{ N}$  then the semicircular **disk does not slip.**  
 17-110.  $a_G = \mu_k g \leftarrow$ ,  $\alpha = \frac{2\mu_k g}{r} \curvearrowright$   
 17-111.  $\omega = \frac{1}{3} \omega_0$ ,  $t = \frac{\omega_0 r}{3\mu_k g}$   
 17-113.  $\alpha = 6.67 \text{ rad/s}^2$   
 17-114.  $\alpha = 3.33 \text{ rad/s}^2$   
 17-115.  $\alpha = \frac{10g}{13\sqrt{2}r}$   
 17-117.  $\alpha = \frac{2g}{3r} \sin \theta$ ,  $N = m \left( \frac{\omega^2 r^2}{R - r} + g \cos \theta \right)$

17-118.  $\alpha_A = 43.6 \text{ rad/s}^2$ ,  $\alpha_B = 43.6 \text{ rad/s}^2$ ,  $T = 15.7 \text{ N}$

17-119.  $T_A = \frac{4}{7}W$

## Chapter 18

18-2.  $\omega = 2.02 \text{ rad/s}$

18-3.  $\omega = 1.78 \text{ rad/s}$

18-5.  $\omega = 14.1 \text{ rad/s}$

18-6.  $\omega = 3.96 \text{ rad/s}$

18-7.  $v_C = 5.20 \text{ m/s}$   $\uparrow$

18-9.  $\omega = 3.16 \text{ rad/s}$

18-10.  $v = 2.10 \text{ m/s}$

18-11.  $\omega = 21.5 \text{ rad/s}$

18-13.  $\omega = 8.64 \text{ rad/s}$

18-14.  $s = 5.16 \text{ m}$ ,  $T = 78.5 \text{ N}$

18-15.  $\omega = 14.9 \text{ rad/s}$

18-17.  $\theta_0 = 1.66 \text{ rad}$

18-18.  $\omega = 3.13 \text{ rad/s}$

18-19.  $s_G = 1.60 \text{ m}$

18-21.  $\omega = \sqrt{\omega_0^2 + \frac{g}{r^2}s} \sin \theta$

18-22.  $U = 237 \text{ J}$

18-23.  $\omega = 6.92 \text{ rad/s}$

18-25.  $\omega_2 = 4.97 \text{ rad/s}$

18-26.  $\omega_2 = 2.06 \text{ rad/s}$

18-27.  $\omega = 5.40 \text{ rad/s}$

18-29.  $\omega = 2.50 \text{ rad/s}$

18-30.  $\omega = 44.6 \text{ rad/s}$

18-31.  $\omega = 3.62 \text{ rad/s}$

18-33.  $\omega = 4.60 \text{ rad/s}$

18-34.  $\omega = \sqrt{\frac{3g}{l}} \sin \theta$

18-35.  $\theta = 8.53^\circ$

18-37.  $v_b = 2.52 \text{ m/s}$

18-38.  $\omega = 19.8 \text{ rad/s}$

18-39.  $s = 0.301 \text{ m}$ ,  $T = 163 \text{ N}$

18-41.  $v_C = 1.52 \text{ m/s}$

18-42.  $s_C = 78.0 \text{ mm}$

18-43.  $\mathbf{v}_A = \{-2.48\mathbf{i} - 2.48\mathbf{j}\} \text{ m/s}$ ,  $|\mathbf{v}_A| = 3.50 \text{ m/s}$

18-45.  $v_D = 3.67 \text{ m/s}$

18-46.  $(\omega_{BC})_2 = 2.91 \text{ rad/s}$

$(\omega_{AB})_2 = 2.91 \text{ rad/s}$

18-47.  $(\omega_{AB})_2 = (\omega_{BC})_2 = 1.12 \text{ rad/s}$

18-49.  $\theta_0 = 8.94 \text{ rev}$

18-50.  $\omega = 3.28 \text{ rad/s}$

18-51.  $\omega = 3.78 \text{ rad/s}$

18-53.  $(\omega_{ABC})_2 = 7.24 \text{ rad/s}$

18-54.  $k = 18.4 \text{ N/m}$

18-55.  $(\omega_{BC})_2 = 0$

18-57.  $\omega_2 = 3.09 \text{ rad/s}$

18-58.  $v_A = 4.00 \text{ m/s}$

18-59.  $\omega = 3.44 \text{ rad/s}$

18-61.  $\omega_A = 19.7 \text{ rad/s}$

18-62.  $\omega = 3.92 \text{ rad/s}$

18-63.  $s = 0.301 \text{ m}$ ,  $T = 163 \text{ N}$

18-65.  $s = 0.708 \text{ m}$

18-66.  $\omega = 1.82 \text{ rad/s}$

18-67.  $(v_A)_2 = 7.24 \text{ m/s}$

## Chapter 19

19-1.  $r_{P/G} = \frac{k_G^2}{r_{G/O}}$

19-3. Since  $\omega$  is a free vector, so is  $\mathbf{H}_P$ .

19-5.  $\omega = 36.3 \text{ rad/s}$

19-6.  $t = 0.6125 \text{ s}$

19-7.  $\omega = 245 \text{ rad/s}$

19-9.  $h = \frac{7}{5}r$

19-10.  $\int M dt = 0.833 \text{ kg} \cdot \text{m}^2/\text{s}$

19-11.  $\omega_0 = 2.5 \left( \frac{v_0}{r} \right)$

19-13.  $y = \frac{\sqrt{2}}{3}a$

19-14.  $y = \frac{2}{3}l$

19-15.  $t = 9.74 \text{ s}$

19-17.  $\omega = 18.4 \text{ rad/s}$   $\curvearrowright$

$T = 313.59 \text{ N}$

19-18.  $v_A = 24.1 \text{ m/s}$

19-19. (a)  $\omega_{BC} = 68.7 \text{ rad/s}$ ,

(b)  $\omega_{BC} = 66.8 \text{ rad/s}$ ,

(c)  $\omega_{BC} = 68.7 \text{ rad/s}$

19-21.  $\omega = 27.9 \text{ rad/s}$

Since  $F_f = 51 \text{ N} < F_{\max} = 238 \text{ N}$  then our no-slip assumption is good.

19-22.  $v_0 = \sqrt{(2g \sin \theta d) \left( \frac{m_c}{m} \right)}$

19-23.  $F = 0.214 \text{ N}$ ,  $\omega_A = 63.3 \text{ rad/s}$ ,

$\omega_B = 127 \text{ rad/s}$

19-25.  $\omega = 16 \text{ rad/s}$ ,  $v = 4 \text{ m/s}$

19-26.  $v_B = 1.59 \text{ m/s}$

19-27.  $\omega = 21.0 \text{ rad/s}$

19-29.  $\omega_M = 0$ ,  $\omega_M = \frac{I}{I_z} \omega$ ,  $\omega_M = \frac{2I}{I_y} \omega$

19-30.  $(\omega_z)_2 = 5.10 \text{ rad/s}$

19-31.  $\omega_B = 10.9 \text{ rad/s}$

19-33.  $\omega_2 = 2.55 \text{ rev/s}$

19-34.  $\omega = 1.91 \text{ rad/s}$

**19-35.**  $\omega_2 = 0.656 \text{ rad/s}, \theta = 18.8^\circ$

**19-37.**  $y = \frac{1}{2}a$

**19-38.**  $\omega = 3.23 \text{ rad/s}, \theta = 32.8^\circ$

**19-39.**  $\omega_2 = 57 \text{ rad/s}, U_F = 367 \text{ J}$

**19-41.**  $\omega = \frac{1}{4}\omega_0, \omega = 1 \text{ rad/s}$

**19-42.**  $h = \frac{7}{5}r$

**19-43.**  $\omega = 6.45 \text{ rad/s}$

**19-45.**  $\omega_T = 1.19 \times 10^{-3} \text{ rad/s}$

**19-46.**  $\omega_2 = 1.01 \text{ rad/s}$

**19-47.**  $\omega = \sqrt{7.5 \frac{g}{L}}$

**19-49.**  $\omega_1 = 3.98 \text{ rad/s}$

**19-50.**  $\omega_3 = 2.73 \text{ rad/s}$

**19-51.**  $\omega_2 = 3.371 \text{ rad/s}, \theta = 47.4^\circ$

**19-53.**  $\omega_1 = 7.17 \text{ rad/s}$

**19-54.**  $\theta = 50.2^\circ$

**19-55.**  $\omega_2 = 2.41 \text{ rad/s}, \omega_3 = 1.86 \text{ rad/s}$

**19-57.**  $(v_G)_{y2} = e(v_G)_{y1} \uparrow,$

$$(v_G)_{x2} = \frac{5}{7} \left( (v_G)_{x1} - \frac{2}{5} \omega_1 r \right) \leftarrow$$

**19-58.**  $v_2 = 0.195 \text{ m/s}$

## Chapter 20

**20-1.**  $\omega = \{-8.24\mathbf{j}\} \text{ rad/s}, \alpha = \{24.7\mathbf{i} - 5.49\mathbf{j}\} \text{ rad/s}^2$

**20-2.**  $\mathbf{v}_A = \{-2.60\mathbf{i} - 0.750\mathbf{j} + 1.30\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_A = \{2.77\mathbf{i} - 11.7\mathbf{j} + 2.34\mathbf{k}\} \text{ m/s}^2$

**20-3.** (a)  $\alpha = \omega_s \omega_t \mathbf{j},$   
 (b)  $\alpha = -\omega_s \omega_t \mathbf{k}$

**20-5.**  $\mathbf{v}_B = \{-124\mathbf{i} - 15\mathbf{j} + 26.0\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_B = \{569\mathbf{i} - 2608\mathbf{j} - 75\mathbf{k}\} \text{ m/s}^2$

**20-6.**  $\mathbf{v}_B = \{4876\mathbf{i} - 15\mathbf{j} + 26.0\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_B = \{1069\mathbf{i} - 2608\mathbf{j} - 75\mathbf{k}\} \text{ m/s}^2$

**20-7.**  $v_B = 0, v_C = 0.283 \text{ m/s}, a_B = 1.13 \text{ m/s}^2,$   
 $a_C = 1.60 \text{ m/s}^2$

**20-9.**  $\omega = \{2\mathbf{i} + 42.4\mathbf{j} + 43.4\mathbf{k}\} \text{ rad/s},$   
 $\alpha = \{-42.4\mathbf{i} - 82.9\mathbf{j} + 84.9\mathbf{k}\} \text{ rad/s}^2$

**20-10.**  $(\omega_C)_{DE} = 40 \text{ rad/s}, (\omega_{DE})_y = 5 \text{ rad/s}$

**20-11.**  $\omega = \{-4.00\mathbf{j}\} \text{ rad/s}, \alpha = \{16.0\mathbf{i}\} \text{ rad/s}^2$   
 $\mathbf{v}_A = \{-0.283\mathbf{i}\} \text{ m/s}, \mathbf{a}_A = \{-1.13\mathbf{j} - 1.13\mathbf{k}\} \text{ m/s}^2$

**20-13.**  $\mathbf{v}_B = \{-0.4\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_B = \{-8.20\mathbf{i} + 40.6\mathbf{j} - \mathbf{k}\} \text{ rad/s}^2$

**20-14.**  $\mathbf{v}_A = \{0.083\mathbf{i} - 0.464\mathbf{j} + 0.124\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_A = \{-0.371\mathbf{i} - 0.108\mathbf{j} - 0.278\mathbf{k}\} \text{ m/s}^2$

**20-15.**  $\omega = \{30\mathbf{j} - 15\mathbf{k}\} \text{ rad/s}, \alpha = \{450\mathbf{i}\} \text{ rad/s}^2$

**20-17.**  $\omega = \{6\mathbf{j} + 15\mathbf{k}\} \text{ rad/s},$   
 $\mathbf{v}_B = \{-27\mathbf{i} - 4.5\mathbf{j} + 1.8\mathbf{k}\} \text{ m/s}$   
 $\mathbf{a}_B = \{72.9\mathbf{i} - 405.9\mathbf{j} + 0.45\mathbf{k}\} \text{ m/s}^2$

**20-18.**  $\mathbf{v}_B = \{-473\mathbf{i} - 4.5\mathbf{j} + 1.8\mathbf{k}\} \text{ m/s}$   
 $\mathbf{a}_B = \{122.9\mathbf{i} - 405.9\mathbf{j} + 0.45\mathbf{k}\} \text{ m/s}^2$

**20-19.**  $\mathbf{v}_P = \{-1.60\mathbf{i}\} \text{ m/s}$   
 $\mathbf{a}_P = \{-0.640\mathbf{i} - 12.0\mathbf{j} - 8.00\mathbf{k}\} \text{ m/s}^2$

**20-20.**  $\mathbf{v}_C = \{1.8\mathbf{j} - 1.5\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_C = \{-36.6\mathbf{i} + 0.45\mathbf{j} - 0.9\mathbf{k}\} \text{ m/s}^2$

**20-21.**  $\omega = \{4.35\mathbf{i} + 12.7\mathbf{j}\} \text{ rad/s},$   
 $\alpha = \{-26.1\mathbf{k}\} \text{ rad/s}^2$

**20-22.**  $\omega = 41.2 \text{ rad/s},$   
 $v_P = 4.00 \text{ m/s}, \alpha = \dot{\omega} = 400 \text{ rad/s}^2,$   
 $a_P = 100 \text{ m/s}^2$

**20-23.**  $\omega_A = 47.8 \text{ rad/s}, \omega_B = 7.78 \text{ rad/s}$

**20-25.**  $\mathbf{v}_B = \{-1.92\mathbf{j} + 2.56\mathbf{k}\} \text{ m/s}$

**20-26.**  $v_B = 5.00 \text{ m/s},$   
 $\omega_{AB} = \{-4.00\mathbf{i} - 0.600\mathbf{j} - 1.20\mathbf{k}\} \text{ rad/s}$

**20-27.**  $v_B = 5.00 \text{ m/s},$   
 $\omega_{AB} = \{-4.00\mathbf{i} - 0.600\mathbf{j} - 1.20\mathbf{k}\} \text{ rad/s}$

**20-29.**  $\mathbf{a}_A = \{-13.9\mathbf{k}\} \text{ m/s}^2$

**20-30.**  $\omega_{BC} = \{0.204\mathbf{i} - 0.612\mathbf{j} + 1.36\mathbf{k}\} \text{ rad/s},$   
 $\mathbf{v}_B = \{-0.333\mathbf{j}\} \text{ m/s}$

**20-31.**  $\omega_{AB} = \{-1.00\mathbf{i} - 0.500\mathbf{j} + 2.50\mathbf{k}\} \text{ rad/s},$   
 $\mathbf{v}_B = \{-2.50\mathbf{j} - 2.50\mathbf{k}\} \text{ m/s}$

**20-33.**  $\omega_{BD} = \{-1.20\mathbf{j}\} \text{ rad/s}$

**20-34.**  $\alpha_{BD} = \{-8.00\mathbf{j}\} \text{ rad/s}^2$

**20-35.**  $\omega_{BC} = \{0.769\mathbf{i} - 2.31\mathbf{j} + 0.513\mathbf{k}\} \text{ rad/s}$   
 $\mathbf{v}_B = \{-0.333\mathbf{j}\} \text{ m/s}$

**20-37.**  $\mathbf{v}_C = \{-1.00\mathbf{i} + 5.00\mathbf{j} + 0.800\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_C = \{-28.8\mathbf{i} - 5.45\mathbf{j} + 32.3\mathbf{k}\} \text{ m/s}^2$

**20-38.**  $\mathbf{v}_C = \{-1\mathbf{i} + 5\mathbf{j} + 0.8\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_C = \{-28.2\mathbf{i} - 5.45\mathbf{j} + 32.3\mathbf{k}\} \text{ m/s}^2$

**20-39.**  $\mathbf{v}_A = \{-5.70\mathbf{i} + 1.20\mathbf{j} - 1.60\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_A = \{-1.44\mathbf{i} - 3.74\mathbf{j} - 0.240\mathbf{k}\} \text{ m/s}^2$

**20-41.**  $\mathbf{v}_P = \{7.50\mathbf{i} + 1.50\mathbf{j} - 3.00\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_P = \{4.50\mathbf{i} - 40.5\mathbf{j} - 2.00\mathbf{k}\} \text{ m/s}^2$

**20-42.**  $\mathbf{v}_A = \{-8.66\mathbf{i} + 2.26\mathbf{j} + 2.26\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_A = \{-22.6\mathbf{i} - 47.8\mathbf{j} + 45.3\mathbf{k}\} \text{ m/s}^2$

**20-43.**  $\mathbf{v}_A = \{-8.66\mathbf{i} + 2.26\mathbf{j} + 2.26\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_A = \{-26.1\mathbf{i} - 44.4\mathbf{j} + 7.92\mathbf{k}\} \text{ m/s}^2$

**20-45.**  $\mathbf{v}_B = \{-17.8\mathbf{i} - 3\mathbf{j} + 5.20\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_B = \{3.05\mathbf{i} - 30.9\mathbf{j} + 1.10\mathbf{k}\} \text{ m/s}^2$

**20-46.**  $\mathbf{v}_C = \{3\mathbf{i} + 6\mathbf{j} - 3\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_C = \{-13.0\mathbf{i} + 28.5\mathbf{j} - 10.2\mathbf{k}\} \text{ m/s}^2$

**20-47.**  $\mathbf{v}_P = \{-0.849\mathbf{i} + 0.849\mathbf{j} + 0.566\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_P = \{-5.09\mathbf{i} - 7.35\mathbf{j} + 6.79\mathbf{k}\} \text{ m/s}^2$

**20-49.**  $\mathbf{v}_C = \{-6.75\mathbf{i} - 6.25\mathbf{j}\} \text{ m/s},$   
 $\mathbf{a}_C = \{28.75\mathbf{i} - 26.25\mathbf{j} - 4\mathbf{k}\} \text{ m/s}^2$

**20-50.**  $\mathbf{v}_B = \{-10.2\mathbf{i} - 30\mathbf{j} + 52.0\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_B = \{-31.0\mathbf{i} - 161\mathbf{j} - 90\mathbf{k}\} \text{ m/s}^2$

**20-51.**  $\mathbf{v}_B = \{-10.2\mathbf{i} - 28\mathbf{j} + 52.0\mathbf{k}\} \text{ m/s},$   
 $\mathbf{a}_B = \{-33.0\mathbf{i} - 159\mathbf{j} - 90\mathbf{k}\} \text{ m/s}^2$

- 20-53.  $\mathbf{v}_C = \{2.80\mathbf{j} - 5.60\mathbf{k}\} \text{ m/s}$ ,  
 $\mathbf{a}_C = \{-56.0\mathbf{i} + 2.10\mathbf{j} - 1.40\mathbf{k}\} \text{ m/s}^2$   
 20-54.  $\mathbf{v}_C = \{-2.7\mathbf{i} - 6\mathbf{k}\} \text{ m/s}$   
 $\mathbf{a}_C = \{-72\mathbf{i} - 13.5\mathbf{j} + 7.8\mathbf{k}\} \text{ m/s}^2$

## Chapter 21

- 21-2.  $I_y = \frac{3m}{80}(h^2 + 4a^2)$ ,  $I_{y'} = \frac{m}{20}(2h^2 + 3a^2)$   
 21-3.  $I_y = \frac{1}{3}mr^2$ ,  $I_x = \frac{m}{6}(r^2 + 3a^2)$   
 21-5.  $I_{xy} = \frac{m}{12}a^2$   
 21-6.  $\begin{pmatrix} \frac{2}{3}ma^2 & \frac{1}{4}ma^2 & \frac{1}{4}ma^2 \\ \frac{1}{4}ma^2 & \frac{2}{3}ma^2 & -\frac{1}{4}ma^2 \\ \frac{1}{4}ma^2 & -\frac{1}{4}ma^2 & \frac{2}{3}ma^2 \end{pmatrix}$   
 21-7.  $I_{xy} = 4.08 \text{ kg} \cdot \text{m}^2$ ,  $I_{yz} = 1.10 \text{ kg} \cdot \text{m}^2$ ,  
 $I_{xz} = 0.785 \text{ kg} \cdot \text{m}^2$   
 21-9.  $I_{z'} = 3.094 \text{ kg} \cdot \text{m}^2$   
 21-10.  $I_z = 0.0915 \text{ kg} \cdot \text{m}^2$   
 21-11.  $I_{aa} = \frac{m}{12}(3a^2 + 4h^2)$   
 21-13.  $I_{xx} = 0.626 \text{ kg} \cdot \text{m}^2$ ,  $I_{yy} = 0.547 \text{ kg} \cdot \text{m}^2$ ,  
 $I_{zz} = 1.09 \text{ kg} \cdot \text{m}^2$   
 21-14.  $I_{xy} = 0.32 \text{ kg} \cdot \text{m}^2$ ,  $I_{yz} = 0.08 \text{ kg} \cdot \text{m}^2$ ,  $I_{xz} = 0$   
 21-15.  $I_{z'} = 0.0595 \text{ kg} \cdot \text{m}^2$   
 21-17.  $x' = 0.2 \text{ m}$ ,  $y' = 0.15 \text{ m}$ ,  $I_{x'} = 0.1337 \text{ kg} \cdot \text{m}^2$ ,  
 $I_{y'} = 0.0764 \text{ kg} \cdot \text{m}^2$ ,  $I_{z'} = 0.2101 \text{ kg} \cdot \text{m}^2$   
 21-18.  $I_z = 0.429 \text{ kg} \cdot \text{m}^2$   
 21-19.  $I_x = 80 \text{ kg} \cdot \text{m}^2$ ,  $I_y = 128 \text{ kg} \cdot \text{m}^2$ ,  
 $I_z = 176 \text{ kg} \cdot \text{m}^2$ ,  $I_{xy} = 72 \text{ kg} \cdot \text{m}^2$ ,  
 $I_{yz} = -24 \text{ kg} \cdot \text{m}^2$ ,  $I_{xz} = -24 \text{ kg} \cdot \text{m}^2$   
 21-21.  $I_x = 4.50 \text{ kg} \cdot \text{m}^2$ ,  $I_y = 4.38 \text{ kg} \cdot \text{m}^2$ ,  
 $I_z = 0.125 \text{ kg} \cdot \text{m}^2$   
 21-22.  $I^2 - (I_{xx} + I_{yy} + I_{zz})I^2 + (I_{xx}I_{yy} + I_{yy}I_{zz} + I_{zz}I_{xx} - I_{xy}^2 - I_{yz}^2 - I_{zx}^2)I$   
 $- (I_{xx}I_{yy}I_{zz} - 2I_{xy}I_{yz}I_{zx} - I_{xx}I_{yz}^2 - I_{yy}I_{zx}^2 - I_{zz}I_{xy}^2) = 0$   
 21-23.  $\mathbf{H}_A = (\rho_{G/A} \times m\mathbf{v}_G) + \mathbf{H}_G$   
 21-25.  $\omega_{OA} = 20.2 \text{ rad/s}$   
 21-26.  $\omega_{OA} = 17.3 \text{ rad/s}$   
 21-27.  $\omega_x = 10.8 \text{ rad/s}$   
 21-29.  $\omega = \{8.73\mathbf{i} - 122\mathbf{j}\} \text{ rad/s}$   
 21-30.  $\omega_p = 4.82 \text{ rad/s}$   
 21-31.  $\omega = \{-0.0625\mathbf{i} - 0.119\mathbf{j} + 0.106\mathbf{k}\} \text{ rad/s}$   
 21-33.  $\omega_y = 26.2 \text{ rad/s}$   
 21-34.  $\mathbf{H}_O = [0.375\mathbf{i} + 7.575\mathbf{k}] \text{ kg} \cdot \text{m}^2/\text{s}$   
 $T = 25.5 \text{ J}$   
 21-35.  $\mathbf{H}_A = [-2000\mathbf{i} - 55000\mathbf{j} + 22500\mathbf{k}] \text{ kg} \cdot \text{m}^2/\text{s}$   
 21-37.  $T = 1.14 \text{ J}$   
 21-38.  $H_z = 0.4575 \text{ kg} \cdot \text{m}^2/\text{s}$   
 21-39.  $\omega_{AB} = 21.4 \text{ rad/s}$   
 21-41.  $\Sigma M_x = (I_x\dot{\omega}_x - I_{xy}\dot{\omega}_y - I_{xz}\dot{\omega}_z)$ ,  
 $- \Omega_z(I_y\omega_y - I_{yz}\omega_z - I_{yx}\omega_x)$ ,  
 $+ \Omega_y(I_z\omega_z - I_{zx}\omega_x - I_{zy}\omega_y)$   
 Similarly for  $\Sigma M_y$  and  $\Sigma M_z$ .  
 21-42.  $\Sigma M_x = I_x\dot{\omega}_x - I_y\Omega_z\omega_y + I_z\Omega_y\omega_z$   
 21-43.  $\theta = \cos^{-1}\left(\frac{3g}{2L\omega^2}\right)$   
 21-45.  $F_A = 277 \text{ N}$ ,  $F_B = 166 \text{ N}$   
 21-46.  $\dot{\omega}_x = -14.7 \text{ rad/s}^2$ ,  $B_z = 77.7 \text{ N}$ ,  $B_y = 3.33 \text{ N}$ ,  
 $A_x = 0$ ,  $A_y = 6.67 \text{ N}$ ,  $A_z = 81.75 \text{ N}$   
 21-47.  $\dot{\omega}_x = 9.285 \text{ rad/s}^2$ ,  $B_z = 97.7 \text{ N}$ ,  $B_y = 3.33 \text{ N}$ ,  
 $A_x = 0$ ,  $A_y = 6.67 \text{ N}$ ,  $A_z = 122 \text{ N}$   
 21-49.  $A_y = 0$ ,  $B_x = 3.75 \text{ kN}$   
 21-50.  $\dot{\omega}_z = 200 \text{ rad/s}^2$ ,  $D_y = -12.9 \text{ N}$ ,  $D_x = -37.5 \text{ N}$ ,  
 $C_x = -37.5 \text{ N}$ ,  $C_y = -11.1 \text{ N}$ ,  $C_z = 36.8 \text{ N}$   
 21-51.  $B_x = -250 \text{ N}$ ,  $A_z = B_z = 24.5 \text{ N}$   
 21-53.  $M_x = -\frac{4}{3}ml^2\omega_s\omega_p \cos \theta$ ,  
 $M_y = \frac{1}{3}ml^2\omega_p^2 \sin 2\theta$ ,  $M_z = 0$   
 21-54.  $T = 23.3 \text{ N}$ ,  $F_A = 41.3 \text{ N}$   
 21-55.  $\Sigma M_x = 0$ ,  $\Sigma M_y = (-0.036 \sin \theta) \text{ N} \cdot \text{m}$ ,  
 $\Sigma M_z = (0.003 \sin 2\theta) \text{ N} \cdot \text{m}$   
 21-57.  $F_A = F_B = 19.5 \text{ N}$   
 21-58.  $(M_0)_x = 72.0 \text{ N} \cdot \text{m}$ ,  $(M_0)_z = 0$   
 21-59.  $N = 148 \text{ N}$ ,  $F_f = 0$   
 21-61.  $\alpha = 69.3^\circ$ ,  $\beta = 128^\circ$ ,  $\gamma = 45^\circ$ .  
 No, the orientation will not be the same for any order. Finite rotations are not vectors.  
 21-62.  $\omega_p = 27.9 \text{ rad/s}$   
 21-63.  $\omega_R = 368 \text{ rad/s}$   
 21-65.  $\omega_p = 10.9 \text{ rad/s}$  or  $-1.76 \text{ rad/s}$   
 21-66.  $\omega_s = 105 \text{ rad/s}$   
 21-67.  $\dot{\phi} = \left(\frac{2g \cos \theta}{a + r \cos \theta}\right)^{1/2}$   
 21-69.  $\omega_s = 3.63(10^3) \text{ rad/s}$   
 21-70.  $H_G = 12.5 \text{ Mg} \cdot \text{m}^2/\text{s}$   
 21-71.  $\dot{\phi} = 81.7 \text{ rad/s}$ ,  $\dot{\psi} = 212 \text{ rad/s}$ ,  
 regular precession  
 21-73.  $\tan \theta = \frac{I}{I_z} \tan \beta$   
 21-74.  $\dot{\psi} = 2.35 \text{ rev/h}$   
 21-75.  $\alpha = 90^\circ$ ,  $\beta = 9.12^\circ$ ,  $\gamma = 80.9^\circ$   
 21-77.  $\beta = 0.673^\circ$   
 Regular Precession  
 Since  $I_z < I$ .  
 21-78.  $\dot{\phi} = 3.32 \text{ rad/s}$

## Chapter 22

22-1.  $\ddot{y} + 56.1y = 0, y|_{t=0.22\text{ s}} = 0.192\text{ m}$

22-2.  $x = -0.05 \cos(20t)$

22-3.  $y = 0.107 \sin(7.00t) + 0.100 \cos(7.00t),$   
 $\phi = 43.0^\circ$

22-5.  $\omega_n = 49.5\text{ rad/s}, \tau = 0.127\text{ s}$

22-6.  $x = \{-0.126 \sin(3.16t) - 0.09 \cos(3.16t)\}\text{ m},$   
 $C = 0.155\text{ m}$

22-7.  $\theta = A \sin(\omega_n t) + B \cos(\omega_n t),$   
 $A = -0.101\text{ rad}, B = 0.30\text{ rad}, \omega_n = 4.95\text{ rad/s}$

22-9.  $\omega_n = 8.16\text{ rad/s}, x = -0.05 \cos(8.16t), C = 50\text{ mm}$

22-10.  $\tau = 2\pi \sqrt{\frac{2mL}{3mg + 6kL}}$

22-11.  $\tau = 2\pi \sqrt{\frac{k_G^2 + d^2}{gd}}$

22-13.  $\tau = 2\pi \sqrt{\frac{m}{3k}}$

22-14.  $k = 1.36\text{ N/m}, m_B = 3.58\text{ kg}$

22-15.  $\tau = 2\pi \sqrt{\frac{2r}{g}}$

22-17.  $k_1 = 2067\text{ N/m}, k_2 = 302\text{ N/m},$  or vice versa

22-18.  $m_B = 21.2\text{ kg}, k = 609\text{ N/m}$

22-19.  $y = 503\text{ mm}$

22-21.  $\omega_n = 9.47\text{ rad/s}$

22-22.  $\omega_n = \sqrt{\frac{3g(4R^2 - l^2)^{1/2}}{6R^2 - l^2}}$

22-23.  $\tau = 1.66\text{ s}$

22-25.  $f = 0.900\text{ Hz}$

22-26.  $\tau = 2\pi k_O \sqrt{\frac{m}{C}}$

22-27.  $k_B = a \sqrt{\frac{m}{M} \left( \frac{\tau_2^2}{\tau_1^2 - \tau_2^2} \right)}$

$$k = \frac{4\pi^2}{\tau_1^2 - \tau_2^2} m$$

22-29.  $\tau = 2\pi \sqrt{\frac{l}{2g}}$

22-30.  $\ddot{x} + 333x = 0$

22-31.  $\tau = 2\pi \sqrt{\frac{m}{3k}}$

22-33.  $\tau = 0.774\text{ s}$

22-34.  $\ddot{\theta} + 468\theta = 0$

22-35.  $\tau = 0.487\text{ s}$

22-37.  $\tau = 2\pi \sqrt{\frac{m(r^2 + k_O^2)}{r^2(k_1 + k_2)}}$

22-38.  $\tau = \pi \sqrt{\frac{m}{k}}$

22-39.  $\theta'' + \omega_n^2 \theta = 0$  where  $\omega_n^2 = 468\text{ rad/s}^2$

22-41.  $y = A \sin \omega_n t + B \cos \omega_n t + \left( \frac{F_0}{(k - m\omega^2)} \right) \cos \omega t$

22-42.  $y = A \sin \omega_n t + B \cos \omega_n t + \frac{F_0}{k}$

22-43.  $c = 18.9\text{ N} \cdot \text{s/m}$

22-45.  $y = A \sin \omega_n t + B \cos \omega_n t + \frac{\delta_0}{1 - (\omega/\omega_n)^2} \sin \omega t$

22-46.  $y = (361 \sin 7.75t + 100 \cos 7.75t - 350 \sin 8t)\text{ mm}$

22-47.  $C = \frac{3F_O}{\frac{3}{2}(mg + Lk) - mL\omega^2}$

22-49.  $(x_p)_{\max} = 29.5\text{ mm}$

22-50.  $\ddot{\theta} + \frac{4c}{m} \dot{\theta} + \frac{k}{m} \theta = 0$

22-51.  $(v_p)_{\max} = 0.3125\text{ m/s}$

22-53.  $(x_p)_{\max} = 14.6\text{ mm}$

22-54.  $(x_p)_{\max} = 35.5\text{ mm}$

22-55.  $Y = \frac{m\omega^2 L^3}{48EI - M\omega^2 L^3}$

22-57.  $C = 21.92\text{ mm}$

22-58.  $\omega = 19.2\text{ rad/s}$

22-59.  $v_R = 1.20\text{ m/s}$

22-61.  $y_{\max} = 0.00357\text{ rad}$

22-62.  $\omega = 12.2\text{ rad/s}, \omega = 7.07\text{ rad/s}$

22-63.  $k = 417\text{ N/m}, k = 1250\text{ N/m}$

22-65.  $MF = 0.997$

22-66.  $y = \{-0.0702 e^{-3.57t} \sin(8.540t)\}\text{ m}$

22-67.  $x = 0.119 \cos(6t - 83.9^\circ)\text{ m}$

22-69.  $\ln\left(\frac{x_1}{x_2}\right) = \frac{2\pi\left(\frac{c}{c_c}\right)}{\sqrt{1 - \left(\frac{c}{c_c}\right)^2}}$

22-70.  $y = [33.8 e^{-7.5t} \sin(8.87t)]\text{ mm}$

22-71.  $\ddot{y} + 16\dot{y} + 12y = 0$

Since  $c > c_c$ , the system will not vibrate.

Therefore it is **overdamped**.

22-73.  $c_c = \sqrt{8(m + M)k}, x_{\max} = \left[ \frac{m}{e} \sqrt{\frac{1}{2k(m + M)}} \right] v_0$

22-74.  $x_{\max} = \frac{2mv_0}{\sqrt{8k(m + M) - c^2}} e^{-\pi c/(2\sqrt{8k(m + M) - c^2})}$

22-75.  $\ddot{y} + 16\dot{y} + 12y = 0$

Since  $c > c_c$ , the system will not vibrate.

Therefore it is **overdamped**.

22-77.  $L\ddot{q} + R\dot{q} + \left(\frac{2}{C}\right)q = 0$

22-78.  $L\ddot{q} + R\dot{q} + \frac{1}{C}q = 0$

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# Fundamental Equations of Dynamics

## KINEMATICS

### Particle Rectilinear Motion

| Variable $a$        | Constant $a = a_c$                      |
|---------------------|-----------------------------------------|
| $a = \frac{dv}{dt}$ | $v = v_0 + a_c t$                       |
| $v = \frac{ds}{dt}$ | $s = s_0 + v_0 t + \frac{1}{2} a_c t^2$ |
| $a ds = v dv$       | $v^2 = v_0^2 + 2a_c(s - s_0)$           |

### Particle Curvilinear Motion

| $x, y, z$ Coordinates                | $r, \theta, z$ Coordinates                                                        |
|--------------------------------------|-----------------------------------------------------------------------------------|
| $v_x = \dot{x} \quad a_x = \ddot{x}$ | $v_r = \dot{r} \quad a_r = \ddot{r} - r\dot{\theta}^2$                            |
| $v_y = \dot{y} \quad a_y = \ddot{y}$ | $v_\theta = r\dot{\theta} \quad a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$ |
| $v_z = \dot{z} \quad a_z = \ddot{z}$ | $v_z = \dot{z} \quad a_z = \ddot{z}$                                              |
| $n, t, b$ Coordinates                |                                                                                   |
| $v = \dot{s}$                        | $a_t = \dot{v} = v \frac{dv}{ds}$                                                 |
|                                      | $a_n = \frac{v^2}{\rho} \quad \rho = \frac{[1 + (dy/dx)^2]^{3/2}}{ d^2y/dx^2 }$   |

### Relative Motion

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \quad \mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

### Rigid Body Motion About a Fixed Axis

| Variable $\alpha$                 | Constant $\alpha = \alpha_c$                                |
|-----------------------------------|-------------------------------------------------------------|
| $\alpha = \frac{d\omega}{dt}$     | $\omega = \omega_0 + \alpha_c t$                            |
| $\omega = \frac{d\theta}{dt}$     | $\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha_c t^2$ |
| $\omega d\omega = \alpha d\theta$ | $\omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0)$      |

### For Point P

$$s = \theta r \quad v = \omega r \quad a_t = \alpha r \quad a_n = \omega^2 r$$

### Relative General Plane Motion—Translating Axes

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A(\text{pin})} \quad \mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A(\text{pin})}$$

### Relative General Plane Motion—Trans. and Rot. Axis

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{B/A} + (\mathbf{v}_{B/A})_{xy}$$

$$\mathbf{a}_B = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{B/A} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{B/A}) + 2\boldsymbol{\Omega} \times (\mathbf{v}_{B/A})_{xy} + (\mathbf{a}_{B/A})_{xy}$$

## KINETICS

### Mass Moment of Inertia

$$I = \int r^2 dm$$

### Parallel-Axis Theorem

$$I = I_G + md^2$$

### Radius of Gyration

$$k = \sqrt{\frac{I}{m}}$$

## Equations of Motion

| Particle       | $\Sigma \mathbf{F} = m\mathbf{a}$                                           |
|----------------|-----------------------------------------------------------------------------|
| Rigid Body     | $\Sigma F_x = m(a_G)_x$                                                     |
| (Plane Motion) | $\Sigma F_y = m(a_G)_y$                                                     |
|                | $\Sigma M_G = I_G \alpha \text{ or } \Sigma M_P = \Sigma (\mathcal{M}_k)_P$ |

### Principle of Work and Energy

$$T_1 + \Sigma U_{1-2} = T_2$$

### Kinetic Energy

| Particle                  | $T = \frac{1}{2}mv^2$                            |
|---------------------------|--------------------------------------------------|
| Rigid Body (Plane Motion) | $T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2$ |

### Work

### Variable force

$$U_F = \int F \cos \theta ds$$

### Constant force

$$U_F = (F_c \cos \theta) \Delta s$$

### Weight

$$U_W = -W \Delta y$$

### Spring

$$U_s = -\left(\frac{1}{2}ks_2^2 - \frac{1}{2}ks_1^2\right)$$

### Couple moment

$$U_M = M \Delta \theta$$

### Power and Efficiency

$$P = \frac{dU}{dt} = \mathbf{F} \cdot \mathbf{v} \quad \varepsilon = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{U_{\text{out}}}{U_{\text{in}}}$$

### Conservation of Energy Theorem

$$T_1 + V_1 = T_2 + V_2$$

### Potential Energy

$$V = V_g + V_e, \text{ where } V_g = \pm Wy, V_e = +\frac{1}{2}ks^2$$

### Principle of Linear Impulse and Momentum

| Particle   | $m\mathbf{v}_1 + \Sigma \int \mathbf{F} dt = m\mathbf{v}_2$         |
|------------|---------------------------------------------------------------------|
| Rigid Body | $m(\mathbf{v}_G)_1 + \Sigma \int \mathbf{F} dt = m(\mathbf{v}_G)_2$ |

### Conservation of Linear Momentum

$$\Sigma(\text{syst. } m\mathbf{v})_1 = \Sigma(\text{syst. } m\mathbf{v})_2$$

$$\text{Coefficient of Restitution } e = \frac{(v_B)_2 - (v_A)_2}{(v_A)_1 - (v_B)_1}$$

### Principle of Angular Impulse and Momentum

| Particle                     | $(\mathbf{H}_O)_1 + \Sigma \int \mathbf{M}_O dt = (\mathbf{H}_O)_2$<br>where $H_O = (d)(mv)$                                                                                                     |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rigid Body<br>(Plane motion) | $(\mathbf{H}_G)_1 + \Sigma \int \mathbf{M}_G dt = (\mathbf{H}_G)_2$<br>where $H_G = I_G\omega$<br>$(\mathbf{H}_O)_1 + \Sigma \int \mathbf{M}_O dt = (\mathbf{H}_O)_2$<br>where $H_O = I_O\omega$ |

### Conservation of Angular Momentum

$$\Sigma(\text{syst. } \mathbf{H})_1 = \Sigma(\text{syst. } \mathbf{H})_2$$

## SI Prefixes

| <i>Multiple</i>    | <i>Exponential Form</i> | <i>Prefix</i> | <i>SI Symbol</i> |
|--------------------|-------------------------|---------------|------------------|
| 1 000 000 000      | $10^9$                  | giga          | G                |
| 1 000 000          | $10^6$                  | mega          | M                |
| 1 000              | $10^3$                  | kilo          | k                |
| <i>Submultiple</i> |                         |               |                  |
| 0.001              | $10^{-3}$               | milli         | m                |
| 0.000 001          | $10^{-6}$               | micro         | $\mu$            |
| 0.000 000 001      | $10^{-9}$               | nano          | n                |

## Conversion Factors (FPS) to (SI)

| <i>Quantity</i> | <i>Unit of<br/>Measurement (FPS)</i> | <i>Equals</i> | <i>Unit of<br/>Measurement (SI)</i> |
|-----------------|--------------------------------------|---------------|-------------------------------------|
| Force           | lb                                   |               | 4.448 N                             |
| Mass            | slug                                 |               | 14.59 kg                            |
| Length          | ft                                   |               | 0.3048 m                            |

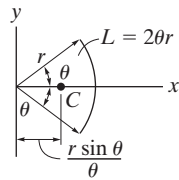
## Conversion Factors (FPS)

1 ft = 12 in. (inches)  
1 mi. (mile) = 5280 ft  
1 kip (kilopound) = 1000 lb  
1 ton = 2000 lb



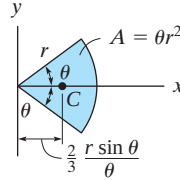
# Geometric Properties of Line and Area Elements

Centroid Location



Circular arc segment

Centroid Location

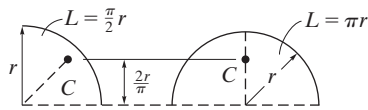


Circular sector area

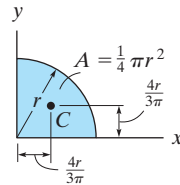
Area Moment of Inertia

$$I_x = \frac{1}{4} r^4 \left( \theta - \frac{1}{2} \sin 2\theta \right)$$

$$I_y = \frac{1}{4} r^4 \left( \theta + \frac{1}{2} \sin 2\theta \right)$$



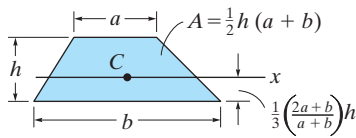
Quarter and semicircle arcs



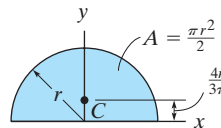
Quarter circle area

$$I_x = \frac{1}{16} \pi r^4$$

$$I_y = \frac{1}{16} \pi r^4$$



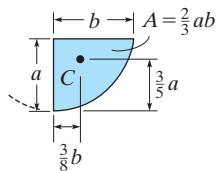
Trapezoidal area



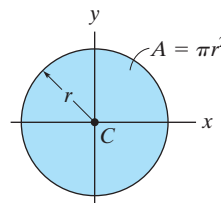
Semicircular area

$$I_x = \frac{1}{8} \pi r^4$$

$$I_y = \frac{1}{8} \pi r^4$$



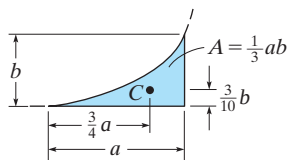
Semiparabolic area



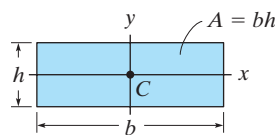
Circular area

$$I_x = \frac{1}{4} \pi r^4$$

$$I_y = \frac{1}{4} \pi r^4$$



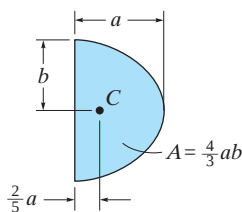
Exparabolic area



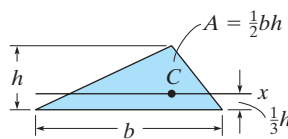
Rectangular area

$$I_x = \frac{1}{12} b h^3$$

$$I_y = \frac{1}{12} h b^3$$



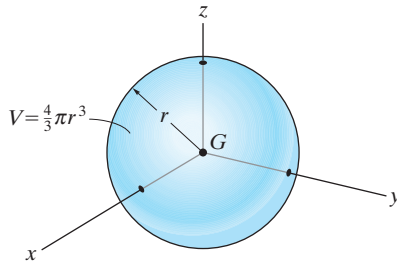
Parabolic area



Triangular area

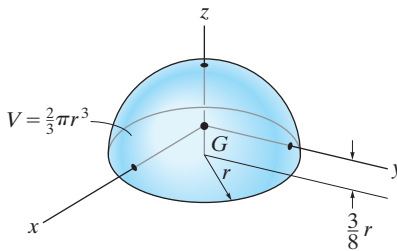
$$I_x = \frac{1}{36} b h^3$$

# Center of Gravity and Mass Moment of Inertia of Homogeneous Solids



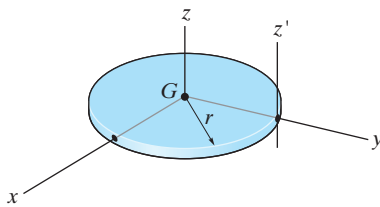
Sphere

$$I_{xx} = I_{yy} = I_{zz} = \frac{2}{5} mr^2$$



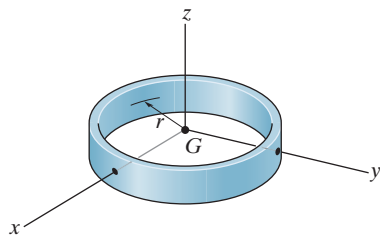
Hemisphere

$$I_{xx} = I_{yy} = 0.259 mr^2 \quad I_{zz} = \frac{2}{5} mr^2$$



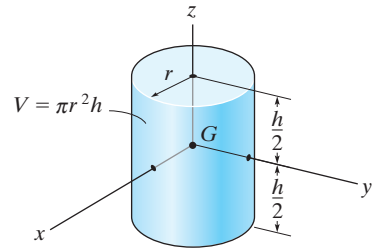
Thin Circular disk

$$I_{xx} = I_{yy} = \frac{1}{4} mr^2 \quad I_{zz} = \frac{1}{2} mr^2 \quad I_{z'z'} = \frac{3}{2} mr^2$$



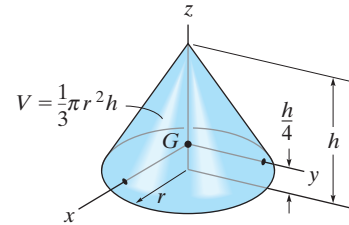
Thin ring

$$I_{xx} = I_{yy} = \frac{1}{2} mr^2 \quad I_{zz} = mr^2$$



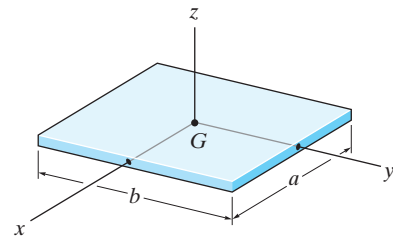
Cylinder

$$I_{xx} = I_{yy} = \frac{1}{12} m(3r^2 + h^2) \quad I_{zz} = \frac{1}{2} mr^2$$



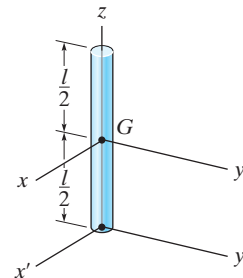
Cone

$$I_{xx} = I_{yy} = \frac{3}{80} m(4r^2 + h^2) \quad I_{zz} = \frac{3}{10} mr^2$$



Thin plate

$$I_{xx} = \frac{1}{12} mb^2 \quad I_{yy} = \frac{1}{12} ma^2 \quad I_{zz} = \frac{1}{12} m(a^2 + b^2)$$



Slender Rod

$$I_{xx} = I_{yy} = \frac{1}{12} ml^2 \quad I_{x'x'} = I_{y'y'} = \frac{1}{3} ml^2 \quad I_{z'z'} = 0$$